

Introduction

Trimmomatic이란?

⇒ Illumina NGS 데이터(FASTQ)에서 불량 염기, 어댑터 서열을 제거해서 downstream 분석 품질을 높이는 전처리 도구

Paired-end vs Single-end

- SRR 데이터가 PE(Paired-end)이면 → PE 모드, input file 2개
- SE(Single-end)이면 → SE 모드, input file 1개

FastQC → read quality 확인 → Trimmomatic으로 quality 기반 trimming

주요 Trimming Step

옵션(예시)	하는 일	공식 문서[1]
ILLUMINACLIP	adapter가 read 끝에 달려있는 경우 제거한다.	Remove adapters (ILLUMINACLIP:TruSeq3-PE.fa:2:30:10)
LEADING:3	read의 5' 끝에서부터 하나씩 기준 미만($Q < 3$)인 염기를 제거하다 기준을 넘는($Q > 3$) 염기를 만나는 순간 멈춘다.	Remove leading low quality or N bases (below quality 3) (LEADING:3)
TRAILING:3	read의 3' 끝에서부터 하나씩 기준 미만($Q < 3$)인 염기를 제거하다 기준을 넘는($Q > 3$) 염기를 만나는 순간 멈춘다.	Remove trailing low quality or N bases (below quality 3) (TRAILING:3)
SLIDINGWINDOW:4:15	4bp 단위로 평균 phred score가 15 미만이면 그 지점부터 뒷부분을 전부 삭제	Scan the read with a 4-base wide sliding window, cutting when the average quality per base drops below 15 (SLIDINGWINDOW:4:15)
MINLEN:36	36bp 미만 read 제거	Drop reads below 36 bases long (MINLEN:36)

SLIDINGWINDOW의 작동 원리 추가 정리

Illumina sequencing은 read가 길어질수록 phred score가 점점 떨어지는 경향이 있다. 그래서 LEADING/TRAILING처럼 끝에서 하나씩 보는 것보다 중간에 quality drop이 일어나는 지점을 찾아서 거기서 통째로 자르는 게 더 효과적인 경우가 있다.

read: [1][2][3][4][5][6][7][8][9][10]...

↑___창___↑
평균 Q 계산

→ 기준(15) 이상이면 창을 한 칸 오른쪽으로 이동

↑__창__↑
평균 Q 계산

→ 기준 미만이면 여기서 자름

이 지점 이후 전부 제거 (따라서 이 경우 read는 [1][2][3][4]만 남는다)

분석할 데이터(Paired end)

- 1) SRR37843240
- 2) SRR37843242
- 3) SRR37843243
- 4) SRR37843246

ILLUMINACLIP:<adapter_fasta>:<seedMismatches>:<palindromeClipThreshold>:<simpleClipThreshold>[:<minAdapterLengthPalindrome>:<keepBothReads>]

Fragment ⇒ 5'—[Adapter A]—[DNA 조각]—[Adapter B]—3'

Read ⇒ Sequencer가 실제로 읽어낸 염기서열 결과물, 기계가 읽을 수 있는 길이

Study

[SRX32708345](#): WES of Homo sapiens

Design: DNA was sequenced on the Illumina HiSeq platform using the Agilent SureSelect Human All Exon V6 kit

Submitted by: Universitat Autònoma de Barcelona

Study: Male Infertility Exome Analysis

Abstract

Whole-exome sequencing data from 61 males from infertile couples (ages 25-56) with abnormal semen parameters. All participants had normal karyotypes and, if applicable, were negative for Y-chromosome microdeletions and CBAVD. DNA was extracted from blood or semen and sequenced on the Illumina HiSeq platform using the Agilent SureSelect Human All Exon V6 kit.

SRR37843240

Description of Data

SRR37843240_1.fastq

The number of read sequences is 61945280

Total length of read sequences is 9353737280

<Text distribution of read lengths>

Length	Count	Abundance
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151 61945280 100.00%

<Graphic distribution of read lengths>

Length Distribution Abundance

151 ■■■■■■■■■■ 100.00%

SRR37843240_2.fastq

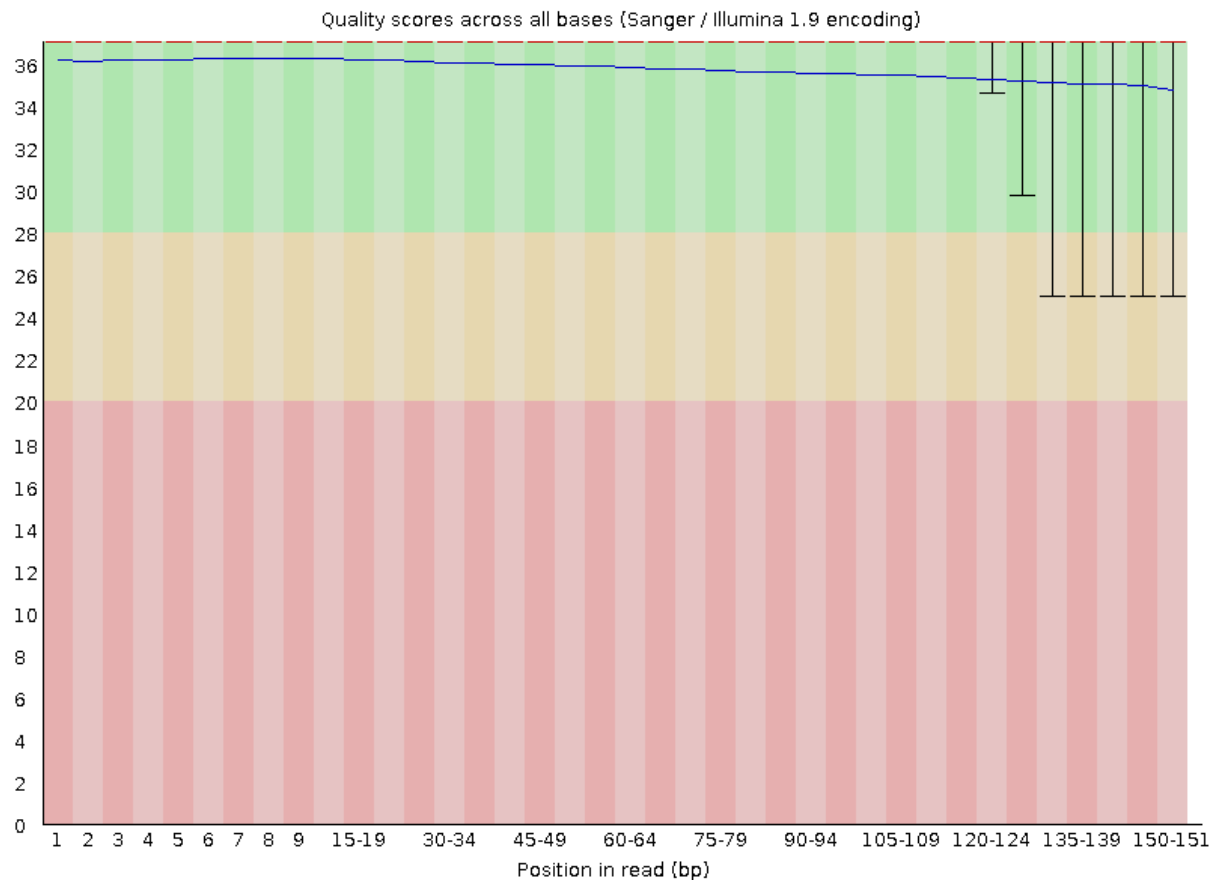
Basic Statistics

SRR37843240_1.fastq.gz	
Measure	Value
Filename	SRR37843240_1.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	61,945,280
Total Bases	9.3 Gbp
Sequences flagged as poor quality	0
Sequence length	151
%GC	51

SRR37843240_2.fastq.gz	
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Filename	SRR37843240_2.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	61,945,280
Total Bases	9.3 Gbp
Sequences flagged as poor quality	0
Sequence length	151
%GC	51

- Conventional base calls ⇒ DNA sequencing에서 기계가 얻는 것은 형광 세기, 전기 신호 등의 raw signal이고, 이를 A/T/G/C로 변환하는 과정이 base calling. 즉 표준 4개 염기에 대한 일반적인 base calling 결과 파일
- Sanger / Illumina 1.9 ⇒ Sanger FASTQ(표준 FASTQ 형식)와 Illumina 1.8+ 이후 데이터가 공통으로 쓰는 Phred+33 인코딩(Phred score에 33을 더해 ASCII 문자로 저장하는 인코딩).
- Total Sequences ⇒ FASTQ 파일에서 FastQC가 분석한 read의 개수, 즉 기계가 생성한 read 수.
- Total Bases ⇒ 모든 read에 포함된 염기(A/T/G/C/N)의 총 개수, 각 read 길이의 합.
- Sequences flagged as poor quality ⇒ Total sequences 중에서 FastQC 기준 overall quality가 낮다고 판정된 read의 수
- Sequence length ⇒ 이 파일에서 나타나는 read 길이의 최소 길이와 최대 길이. 만약 모든 read가 같은 length를 가진다면 하나의 value만 나타남.
- %GC ⇒ 전체 read의 염기에서 GC의 비율

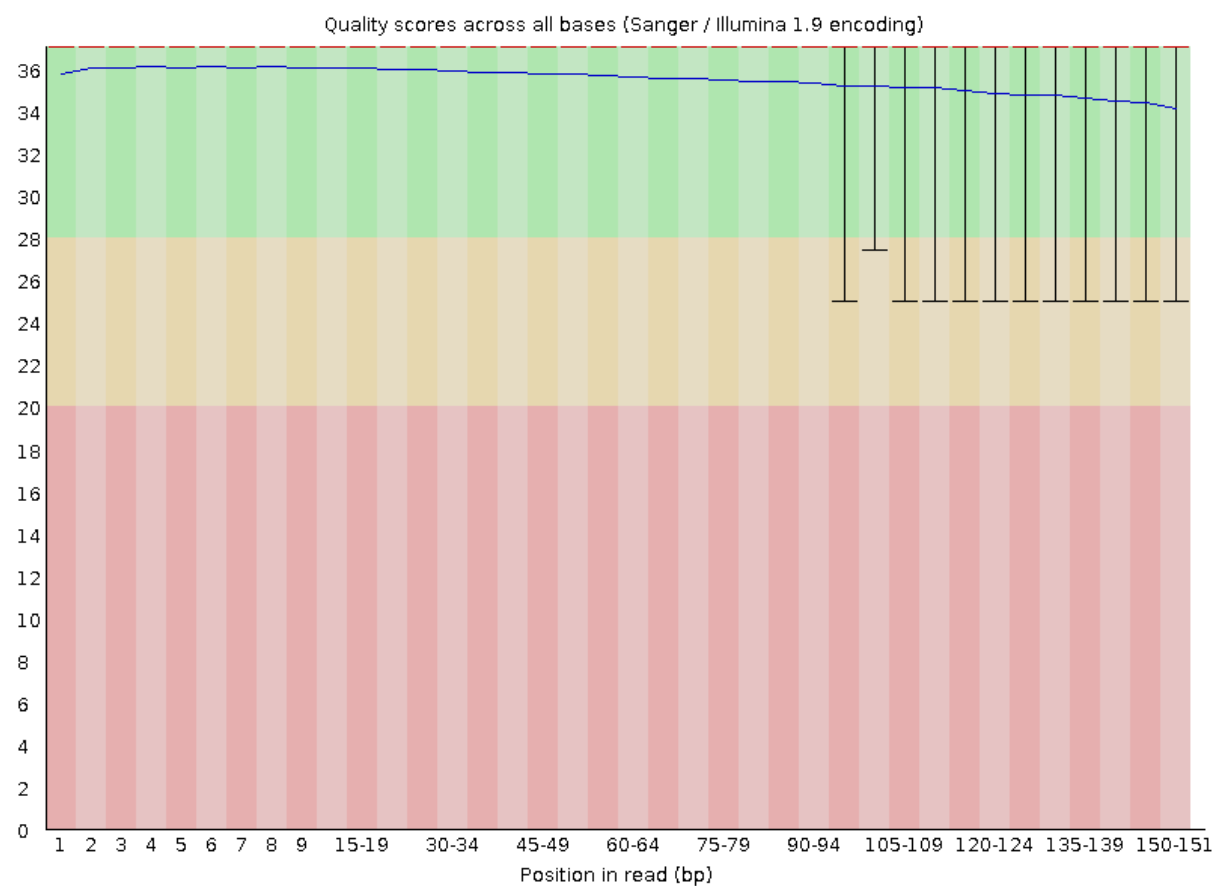
Per base sequence quality



SRR37843240_1.fastq.gz. 파란 선 : mean quality, 붉은 선 : median value, 검은 error bar : 10-90% percentiles[2]

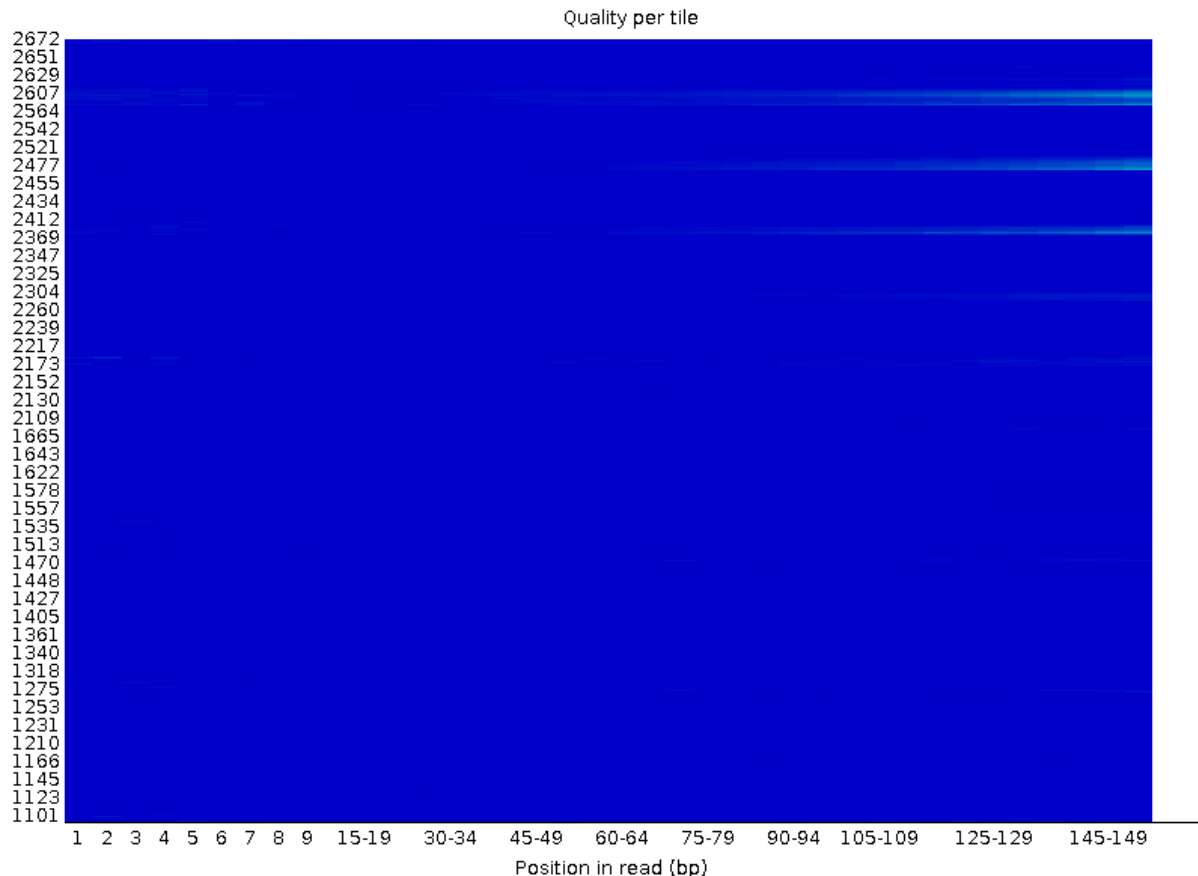
모든 bp에서 Phred score 평균이 30을 넘는(>99.9% accuracy) 모습을 보인다.

120bp 이후에 error bar가 아래로 길게 내려간다, 이는 해당 position에서 일부 read들의 Phred 점수가 낮아진다는 의미다. 하지만 90%의 경우에도 Phred score가 20을 안정적으로 넘기 때문에(>99% accuracy) 특별한 Trimming은 필요 없다고 해석할 수 있다.



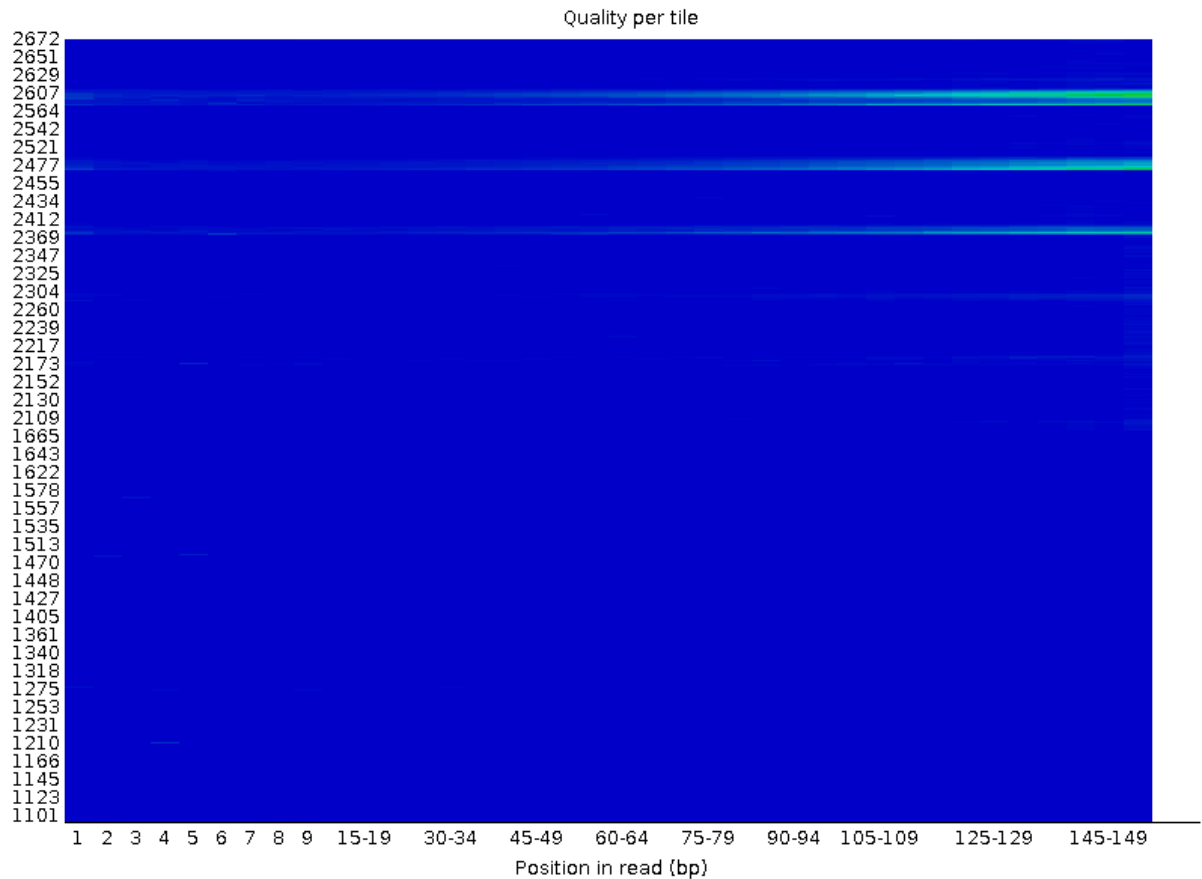
SRR37843240_2.fastq.gz. 이하 동문.

Per tile sequence quality



SRR37843240_1.fastq.gz. original sequence identifier를 가지고 있는 Illumina library를 사용할 경우, 각 read가 어느 flowcell tile에서 왔는지에 대한 정보를 알 수 있다. 전체 염기들에 대해 각 tile의 quality score를 보여주기 때문에, 특정 flowcell의 부분에서 품질 저하가 있는지 판단할 수 있다. flowcell은 sequencing 반응이 일어나는 유리 기판이고, flowcell tile은 이 유리 기판을 카메라가 읽기 위해 잘게 나눈 작은 구역 하나하나를 뜻한다. 한 문장으로 요약하면 flowcell tile 하나 안에 수많은 cluster(=read의 집합)가 있고, 그 tile에서 나온 모든 read들의 base position당 quality score를 평균을 내서 <특정 tile의 특정 base의 평균 Phred score>로 대표시킨 뒤, FastQC는 이 값을 전체 평균과 비교해서 색으로 보여준다. 특정 base position에 대해 다른 tile들보다 quality가 비슷하거나 높으면 cold colors, quality가 나쁘면 hotter colors를 나타낸다.

전체적으로 거의 모든 tile이 cold colors로 분포한다. 상단 일부 tile에서 약간의 품질 저하가 보이지만, sequencing quality에 큰 영향을 줄 만한 요소는 아니라고 판단했다.

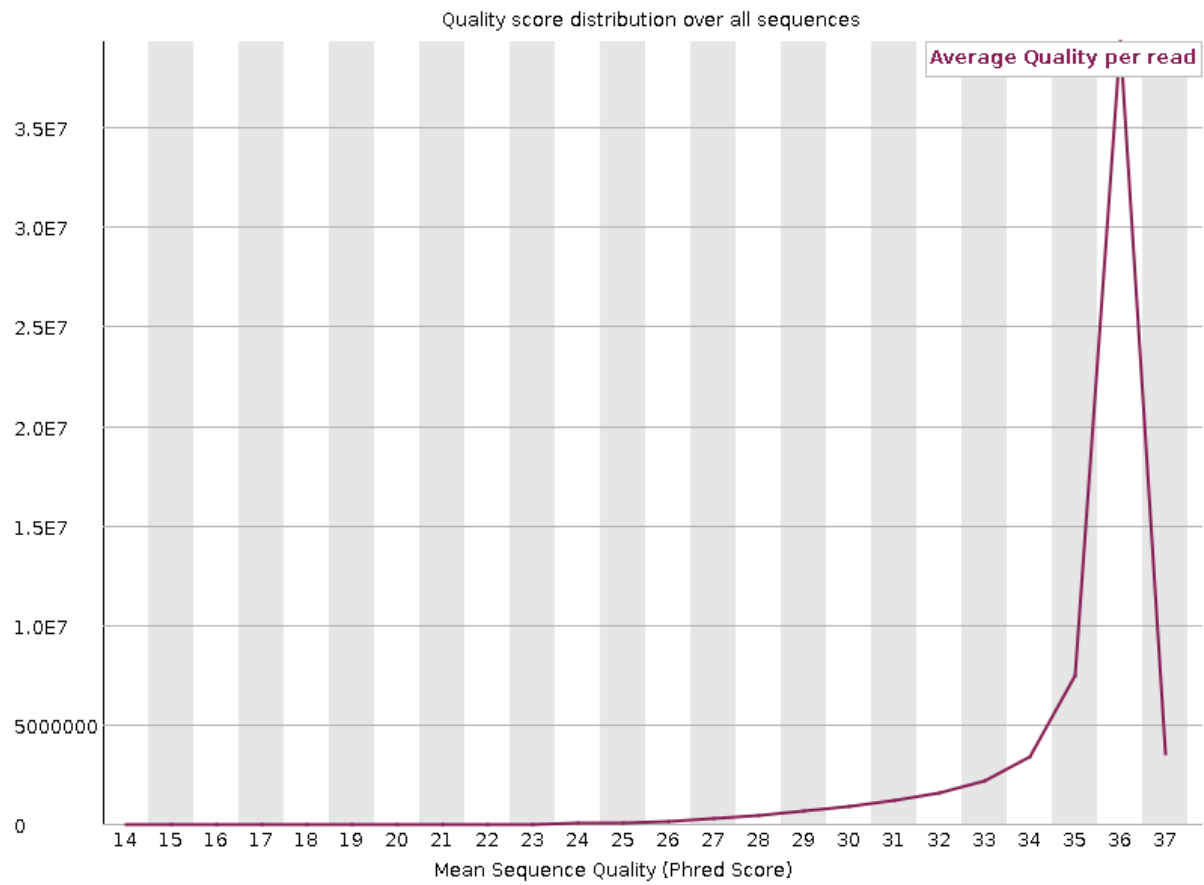


SRR37843240_2.fastq.gz.

하나의 tile이라도 특정 base position에 대한 평균보다 2이상 낮은 Phred score를 가진다면 Warning이 표시된다.

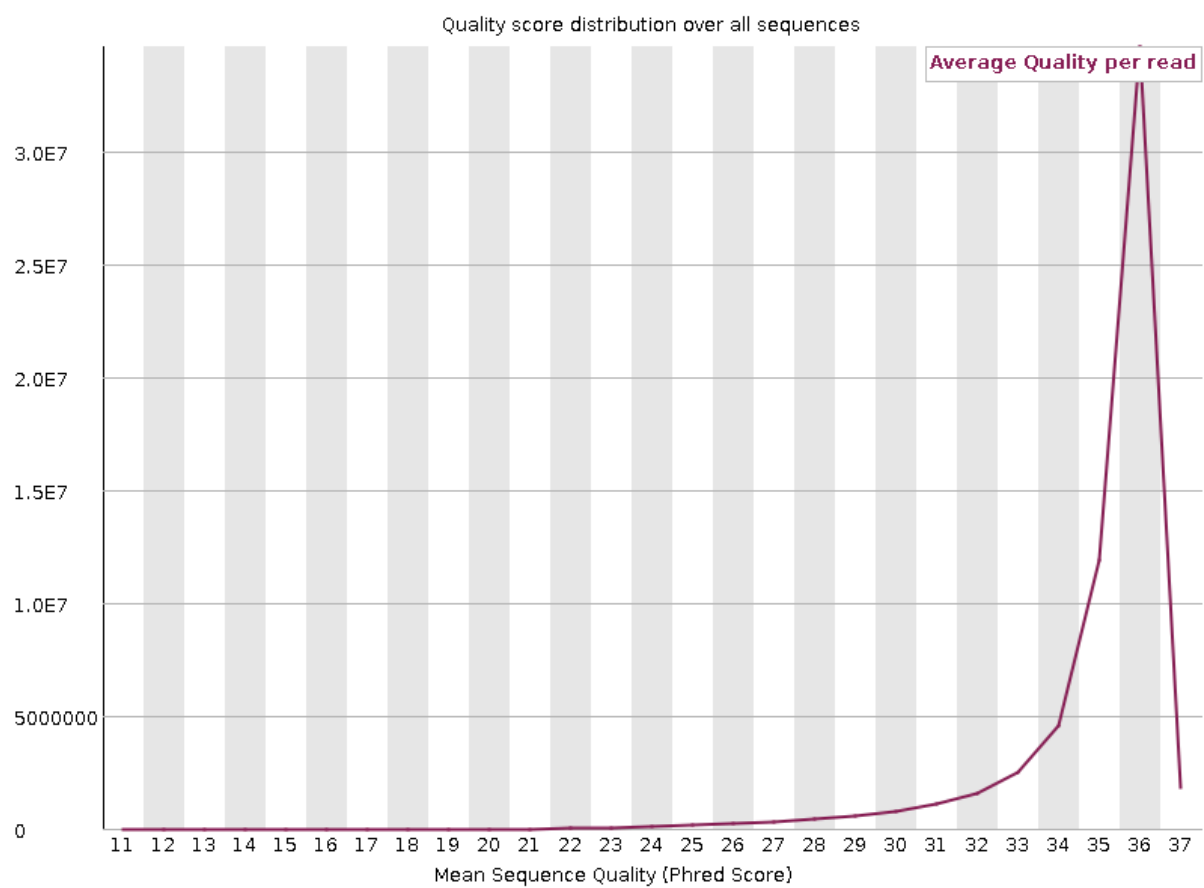
전반적으로 tile들은 양호한 quality를 보이나, 일부 tile에서 100bp 이후에 지속적인 품질 저하 패턴이 관찰되어 Warning이 발생했다. 해당 패턴은 특정 타일에 국한된 국소적인 시퀀싱 문제(예: flowcell 상단 영역의 물리적 결함, 기포·얼룩 등)에 더 가깝기[2] 때문에 Trimmomatic을 통한 trimming으로는 완전히 해결되기는 어려운 문제로 판단했다.

Per sequence quality scores



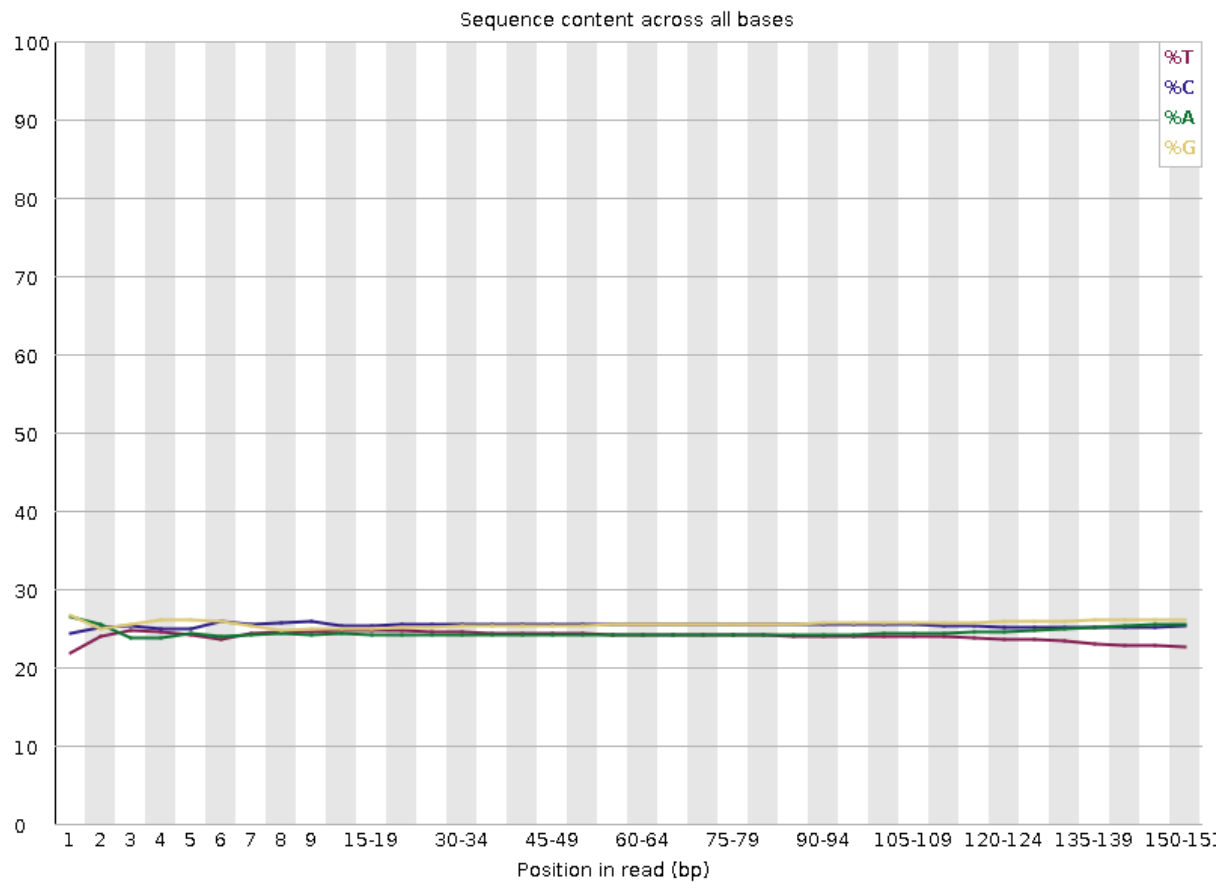
SRR37843240_1.fastq.gz. Phred Score별 read의 수 distribution

Per base sequence quality와 동일하게 quality에 이상 없음.



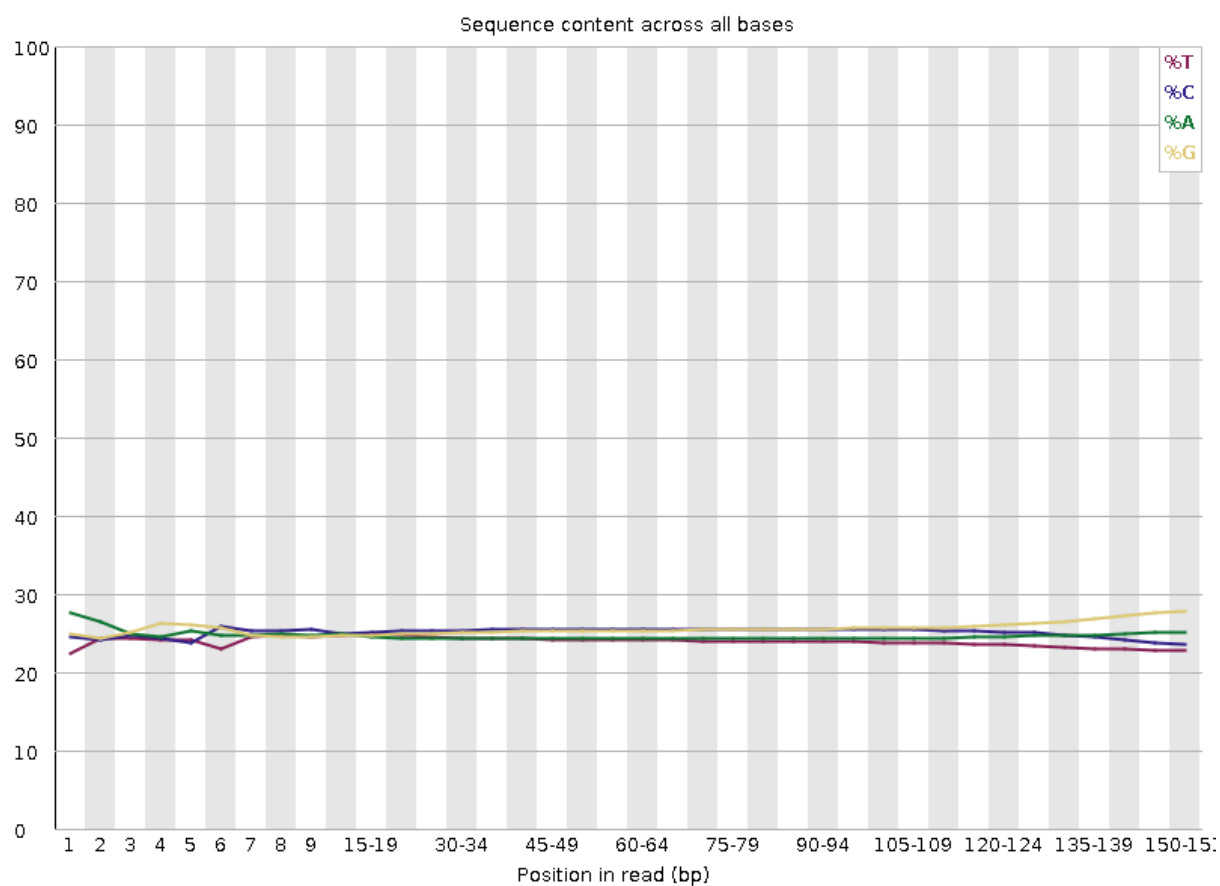
SRR37843240_2.fastq.gz. 이하 동문.

Per base sequence content



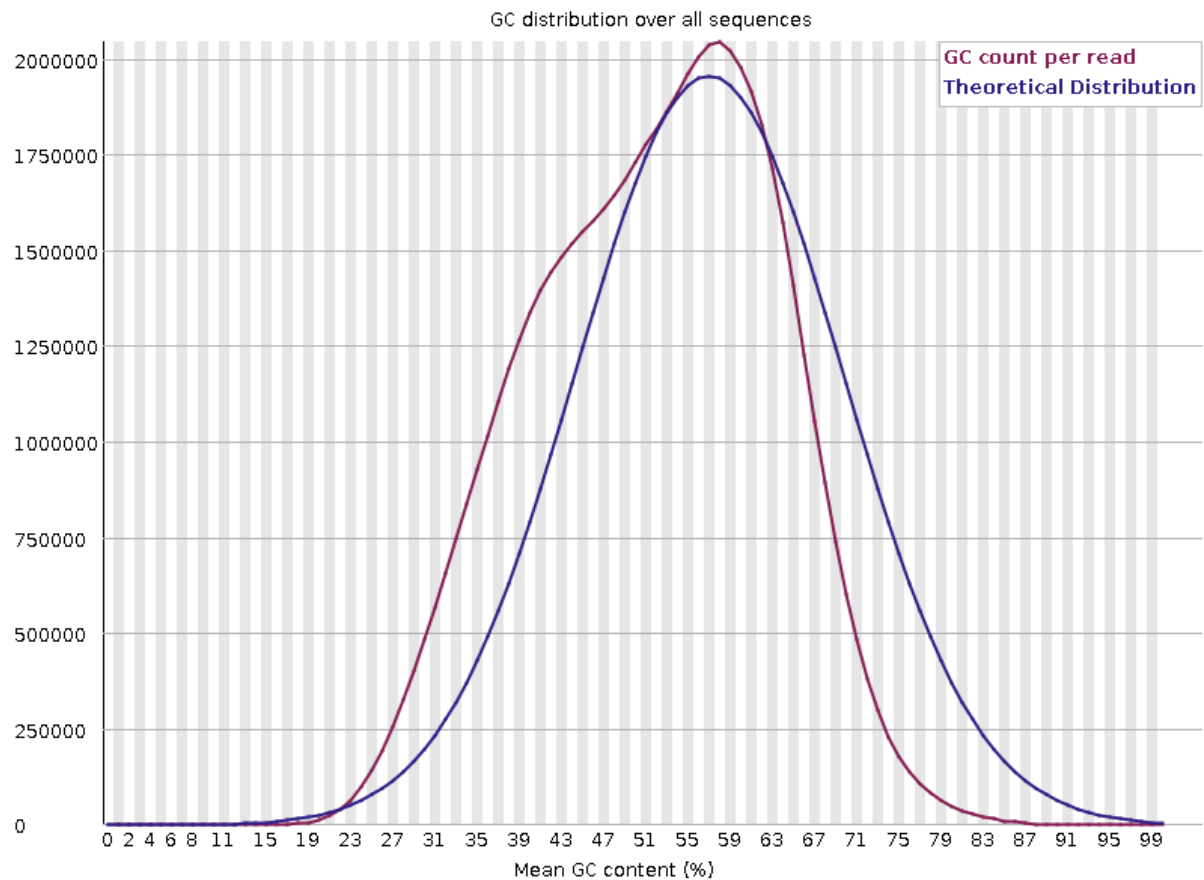
SRR37843240_1.fastq.gz. 각 read 내 base position별 염기의 비율을 나타낸 그래프.

이상 없는 것으로 판단된다.



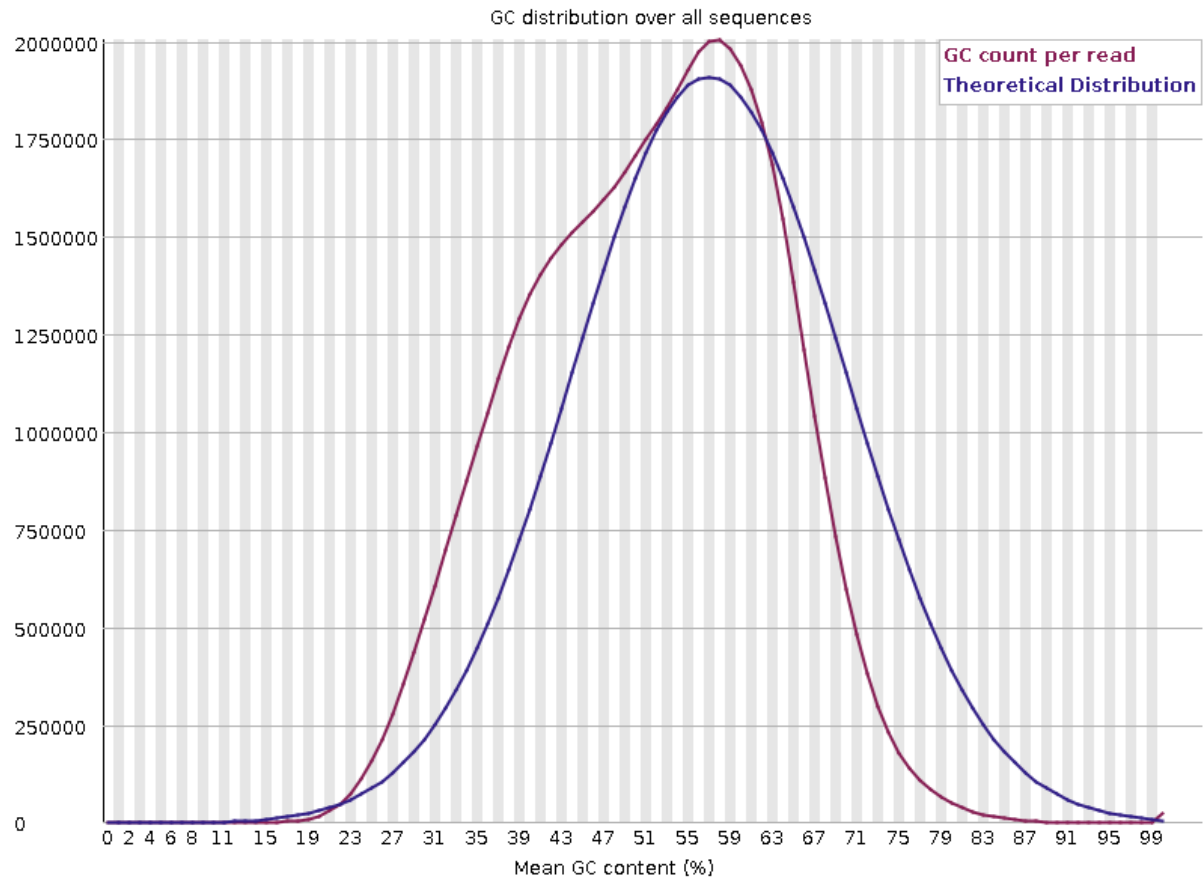
SRR37843240_2.fastq.gz. 이하 동문.

Per sequence GC content



SRR37843240_1.fastq.gz. 각 read의 전체 길이에 대한 GC 비율 distribution과, 이를 바탕으로 정규분포로 모델링한 GC distribution을 비교하는 그래프이다.

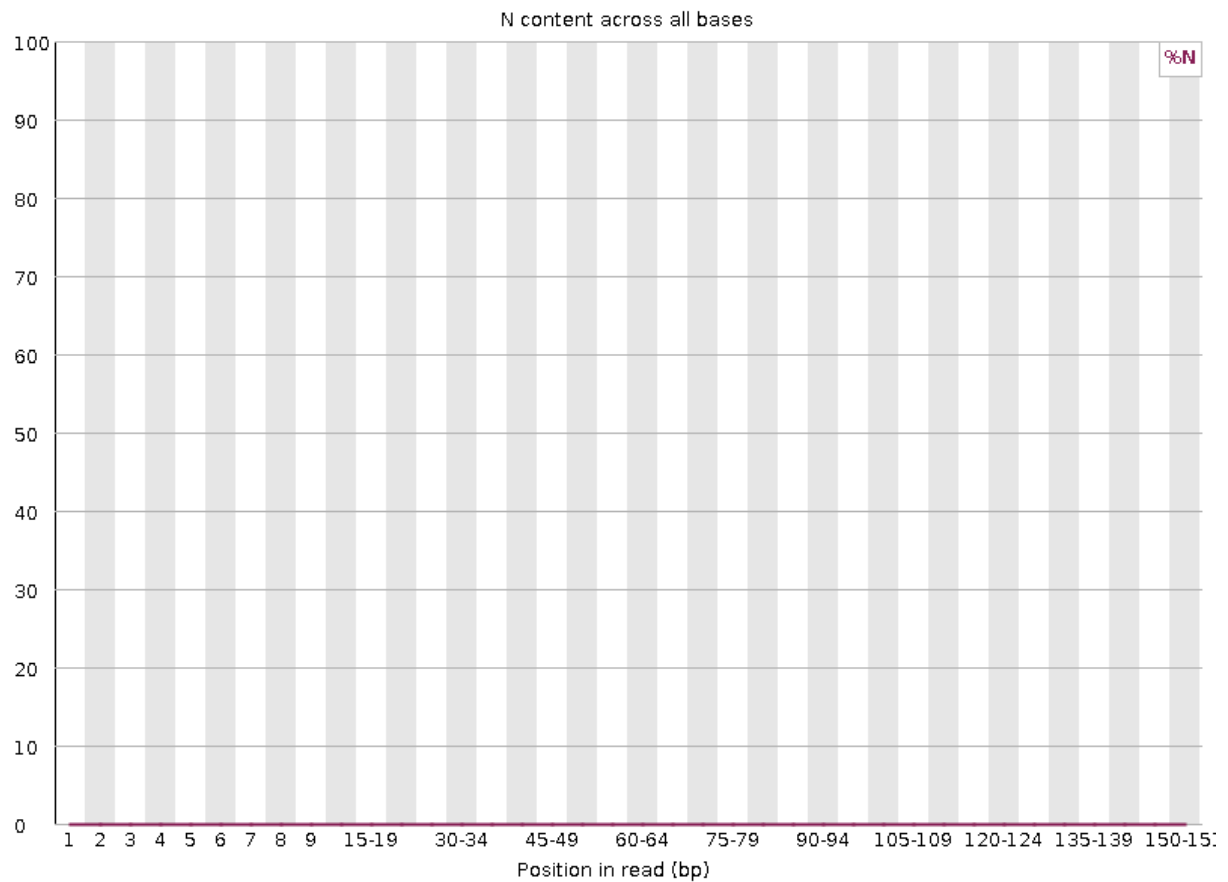
두 분포의 편차 합이 전체 read의 15%를 초과해 Warning이 표시되었다. 일반적인 genome sequencing이라면 contamination을 의심해볼 수 있겠으나, 해당 데이터는 Whole exome sequencing을 통해 만들어졌기 때문에 자연스러운 것으로 판단해 trimming option을 결정할 때 특별히 고려하지 않았다.



SRR37843240_2.fastq.gz.

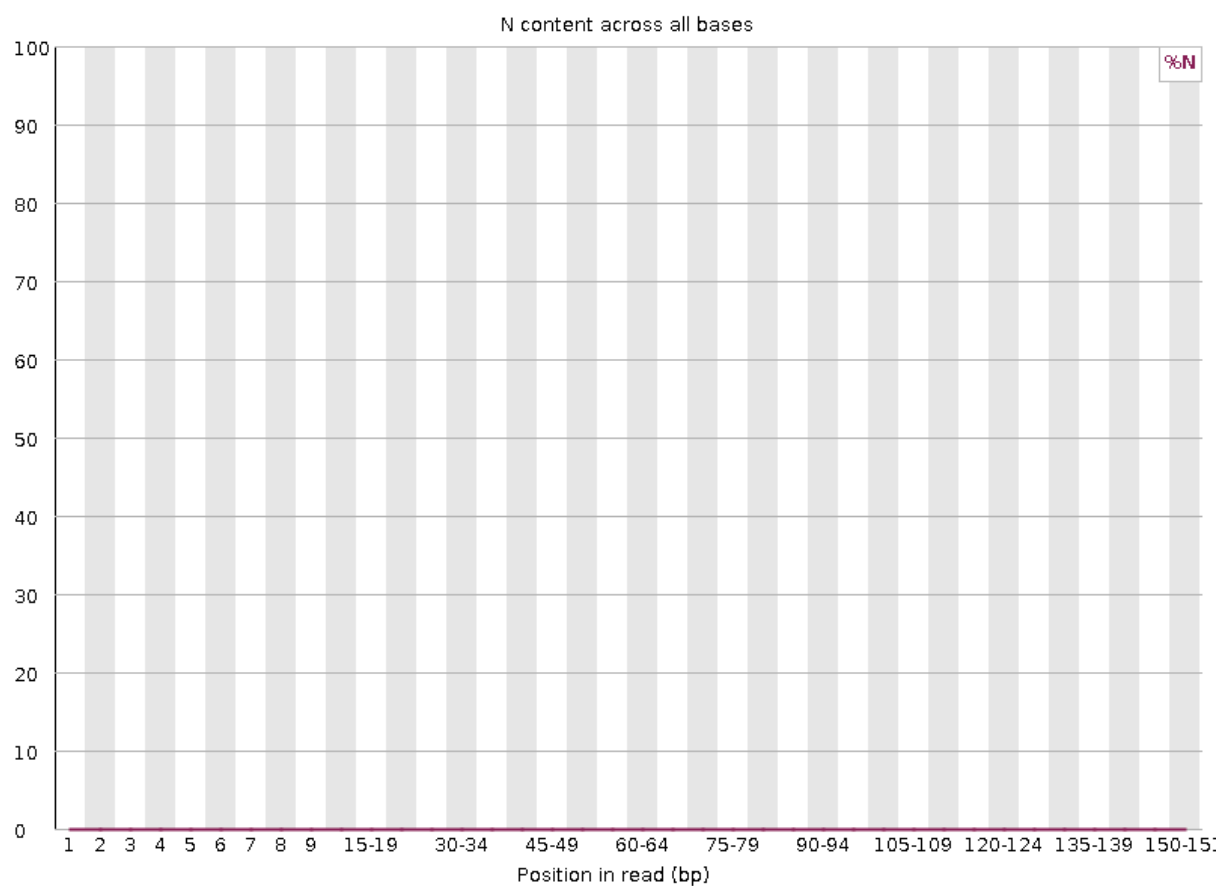
두 분포의 편차 합이 전체 read의 30%를 초과해 Failure이 표시되었다. 이하 동문.

Per base N content



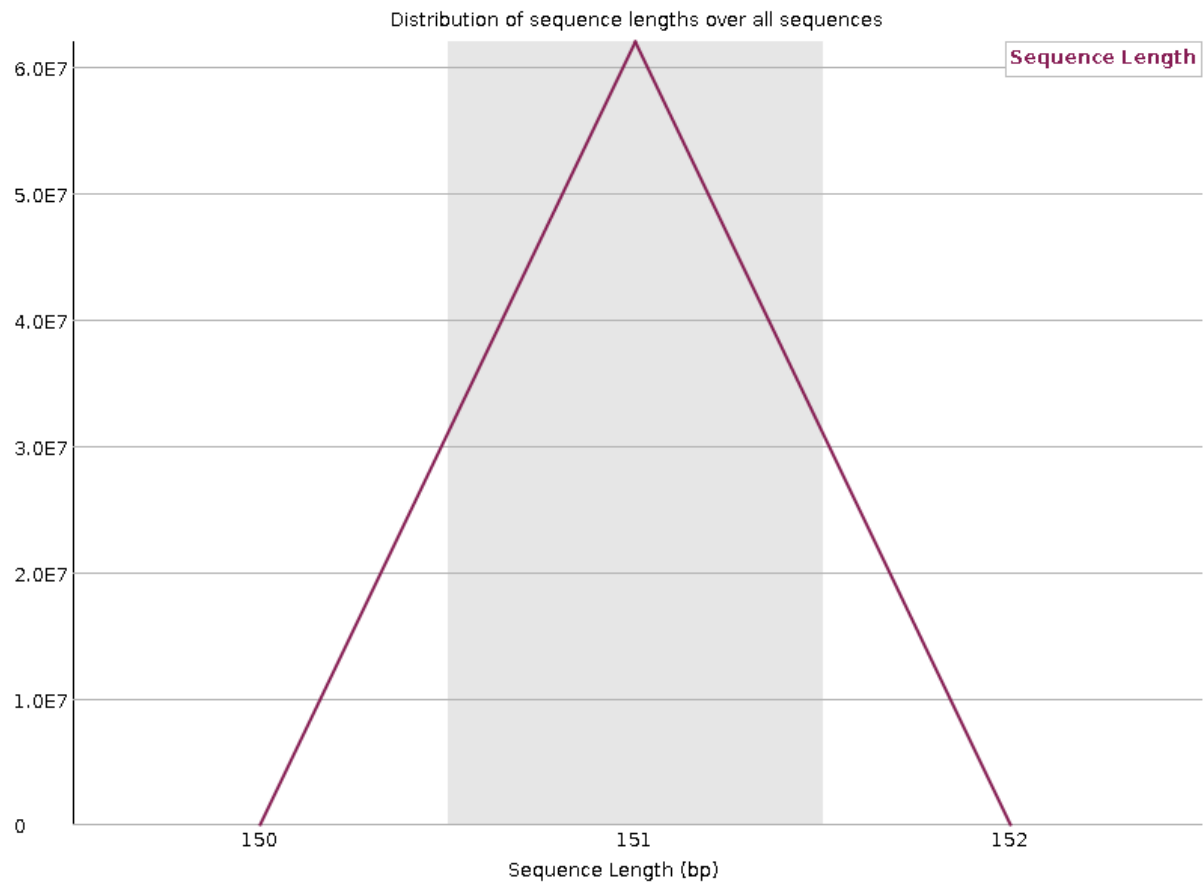
SRR37843240_1.fastq.gz sequencer가 base call에 실패할 경우 N을 반환한다. 각 read별 base position당 N의 수.

N이 전혀 없어 양호하다.

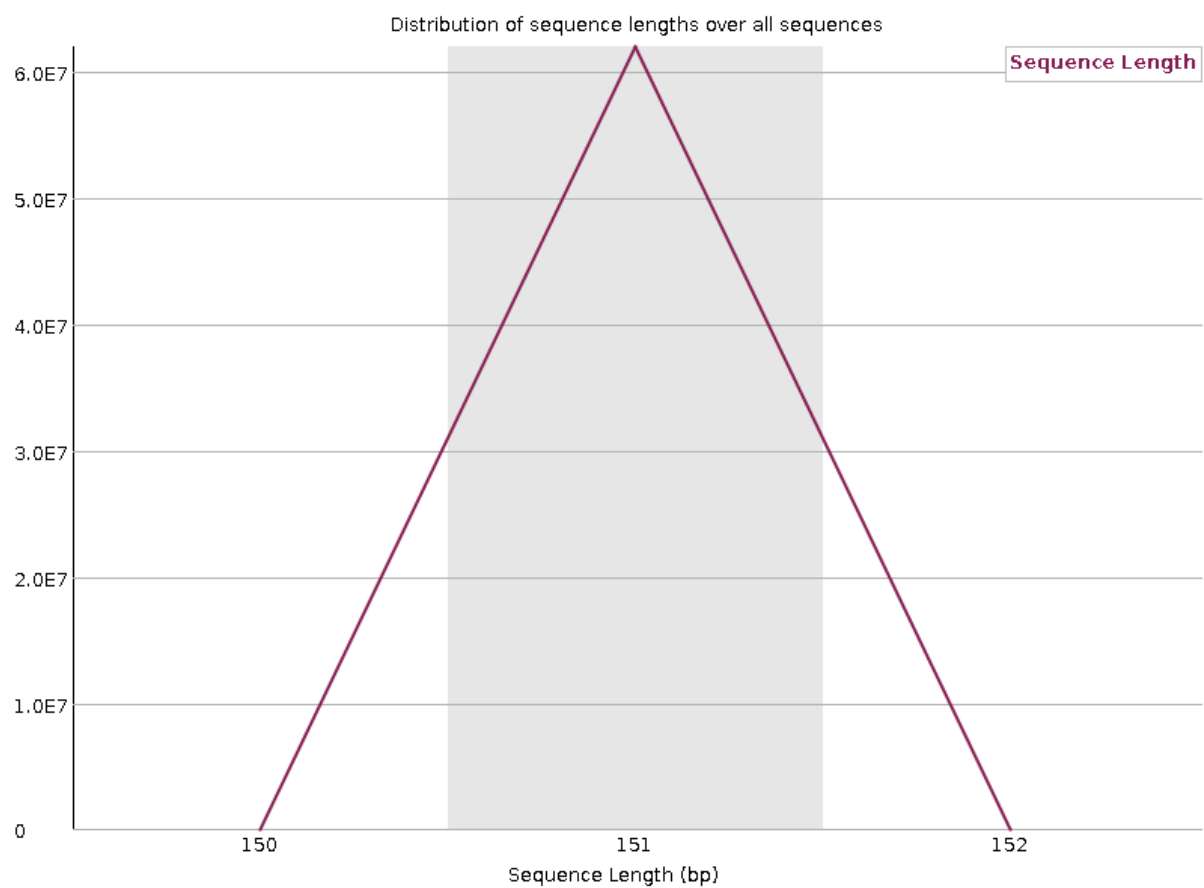


SRR37843240_2.fastq.gz. 이하 동문.

Sequence Length Distribution

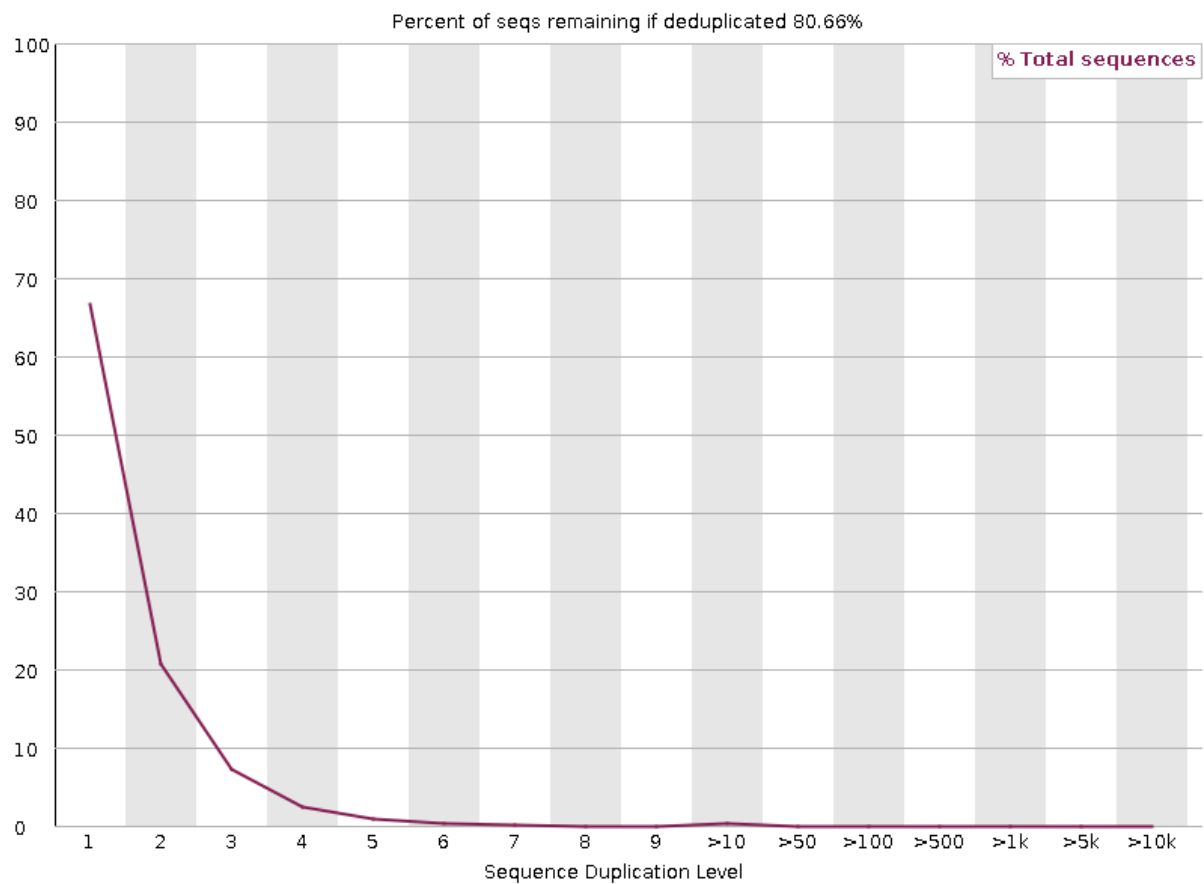


SRR37843240_1.fastq.gz Sequence length별 비율.
151bp의 read밖에 없기 때문에 단일 값만 나왔다.



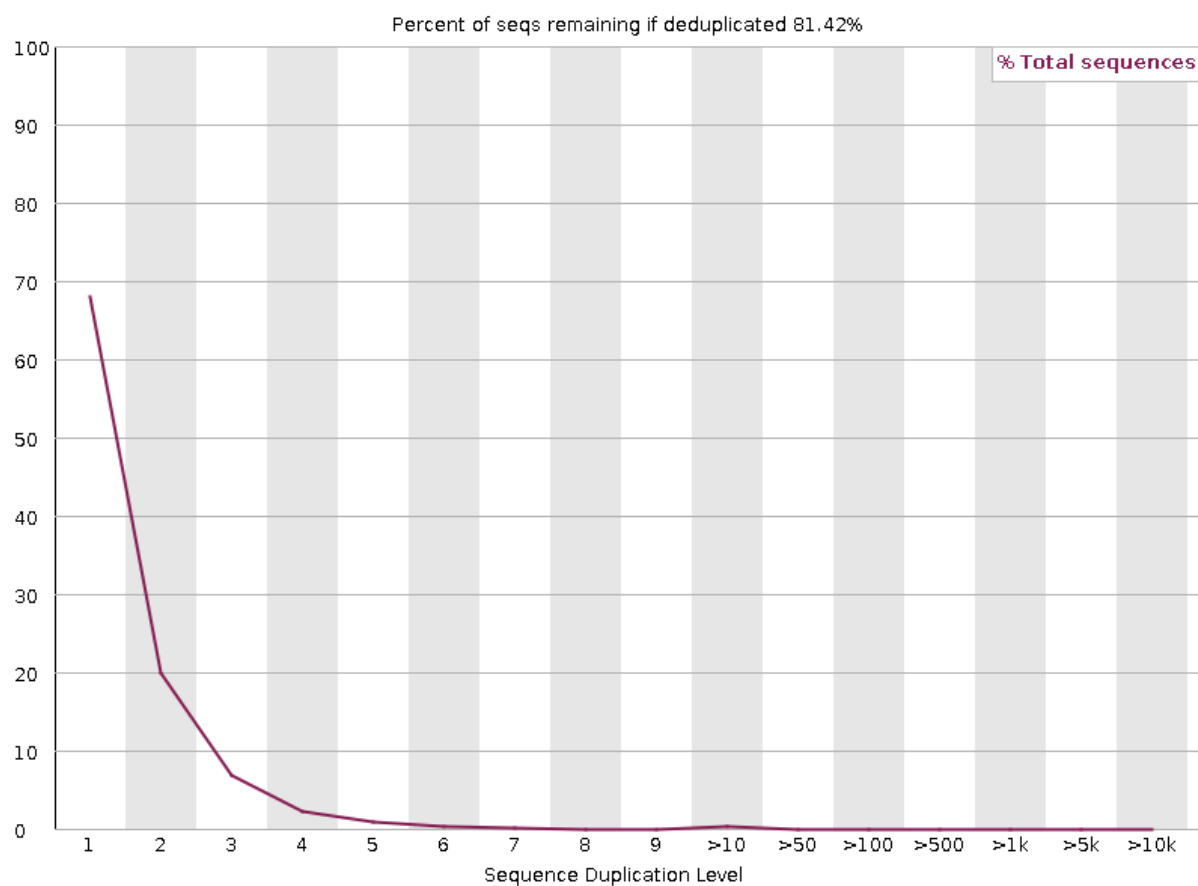
SRR37843240_2.fastq.gz 이하 동문.

Sequence Duplication Levels



SRR37843240_1.fastq.gz. 동일한 read가 나온 횟수에 따라 read의 비율을 나타낸 그래프.

이상 없다.

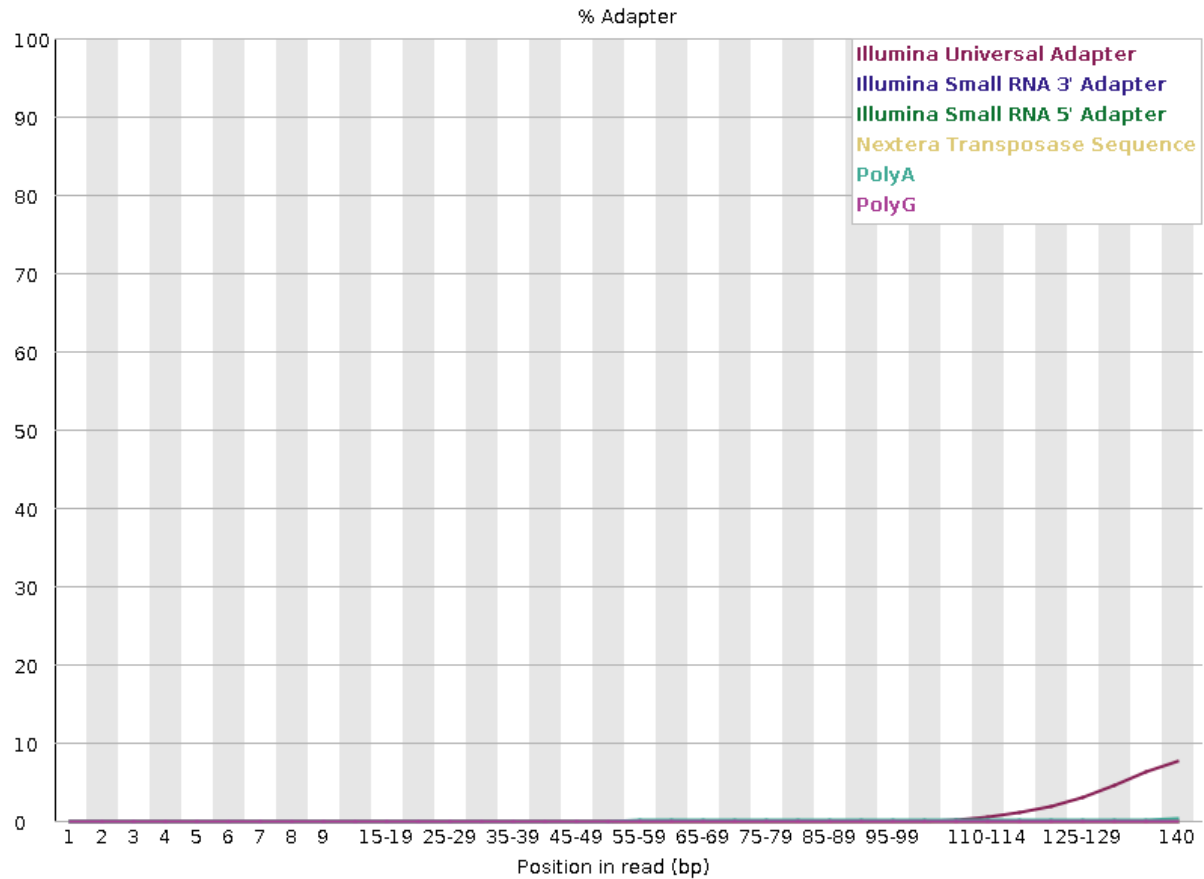


SRR37843240_2.fastq.gz. 이하 동문.

Overrepresented sequences

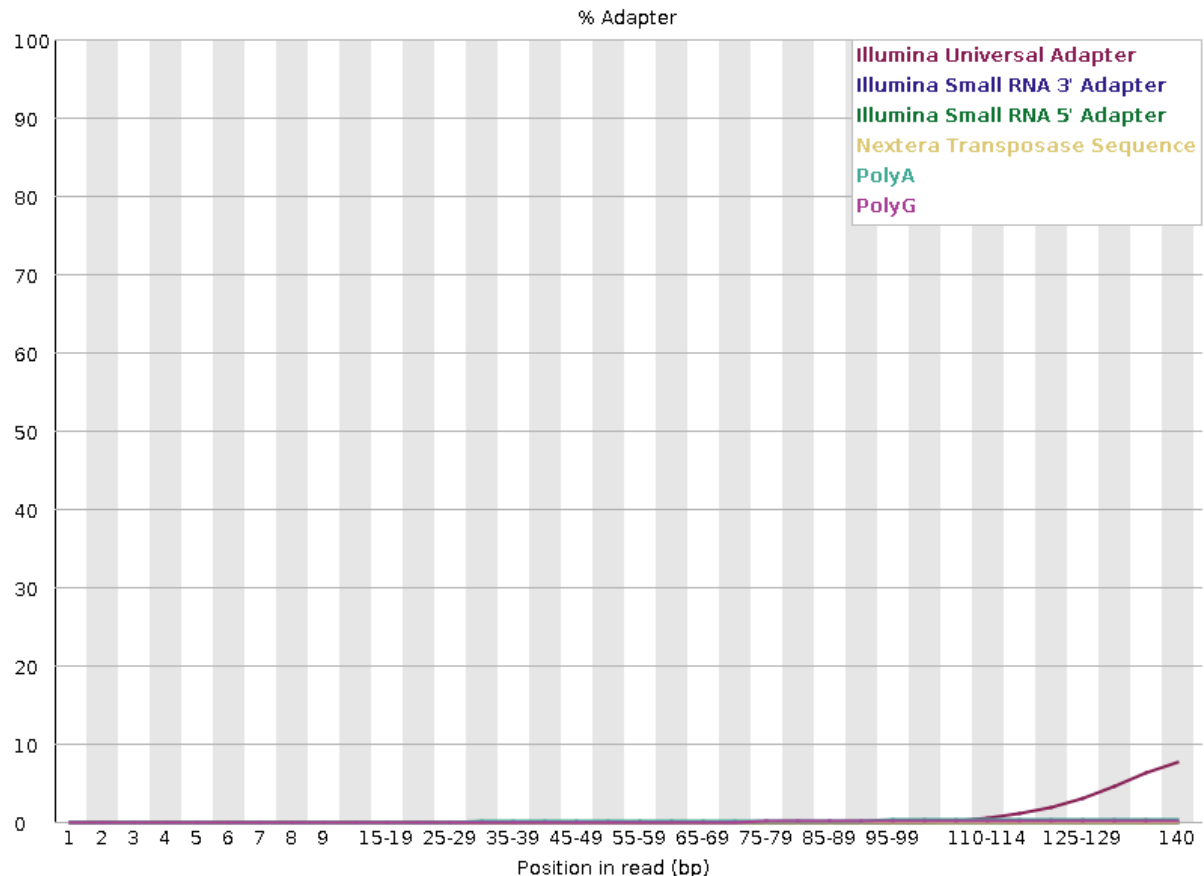
SRR37843240_1.fastq.gz과 SRR37843240_2.fastq.gz 모두 No overrepresented sequences

Adapter Content



SRR37843240_1.fastq.gz. read의 base position별 특정 Adapter들의 비율을 나타낸 그래프.

특정 Adapter의 비율이 전체 read의 5% 이상의 비율로 존재해 Warning이 표시되었다.
read 끝부분에 Illumina Universal Adapter가 감지되고, 해당 Study에서 사용한 Illumina NovaSeq 6000의 Adapter가 TruSeq3-PE라는 정보를 바탕으로 Trimming option을 결정했다.



SRR37843240_2.fastq.gz. 이하 동문.

After Trimming

TrimmomaticPE: Started with arguments:

SRR37843240_1.fastq SRR37843240_2.fastq SRR37843240_1.p.fastq SRR37843240_1.up.fastq
SRR37843240_2.p.fastq SRR37843240_2.up.fastq

ILLUMINACLIP:/mss/home_student/202112379/programs/Trimmomatic-0.39/adapters/TruSeq3-PE.

fa:2:30:10:2:True SLIDINGWINDOW:4:20 LEADING:10 TRAILING:10 MINLEN:60

Using PrefixPair: 'TACACTCTTTCCTACACGACGCTCTTCCGATCT' and

'GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT'

ILLUMINACLIP: Using 1 prefix pairs, 0 forward/reverse sequences, 0 forward only sequences, 0 reverse only sequences

Quality encoding detected as phred33

Input Read Pairs: 61945280 **Both Surviving: 57459044 (92.76%)** Forward Only Surviving:

1966982 (3.18%) Reverse Only Surviving: 1622948 (2.62%) Dropped: 896306 (1.45%)

TrimmomaticPE: Completed successfully

SRR37843240_1.p.fastq

==== PA1.py for SRR37843240_1.p ====

The number of read sequences is 57459044

Total length of read sequences is 8406475700

<Text distribution of read lengths>

Length	Count	Abundance
60	32867	0.06%
61	33390	0.06%
62	32522	0.06%
63	33334	0.06%
64	32634	0.06%
65	34129	0.06%
66	32743	0.06%
67	33503	0.06%
68	32450	0.06%
69	33087	0.06%
70	32864	0.06%
71	32743	0.06%
72	32742	0.06%
73	33487	0.06%
74	33578	0.06%
75	32762	0.06%
76	33283	0.06%
77	32294	0.06%
78	34372	0.06%
79	34774	0.06%
80	34196	0.06%
81	35215	0.06%
82	35948	0.06%
83	34218	0.06%
84	34107	0.06%
85	34816	0.06%
86	34355	0.06%
87	34624	0.06%
88	34991	0.06%
89	35185	0.06%
90	36377	0.06%
91	35978	0.06%
92	35959	0.06%
93	36187	0.06%
94	38182	0.07%
95	37466	0.07%
96	38672	0.07%
97	40354	0.07%
98	39600	0.07%
99	42321	0.07%
100	43850	0.08%
101	46240	0.08%
102	48834	0.08%
103	50976	0.09%
104	52905	0.09%

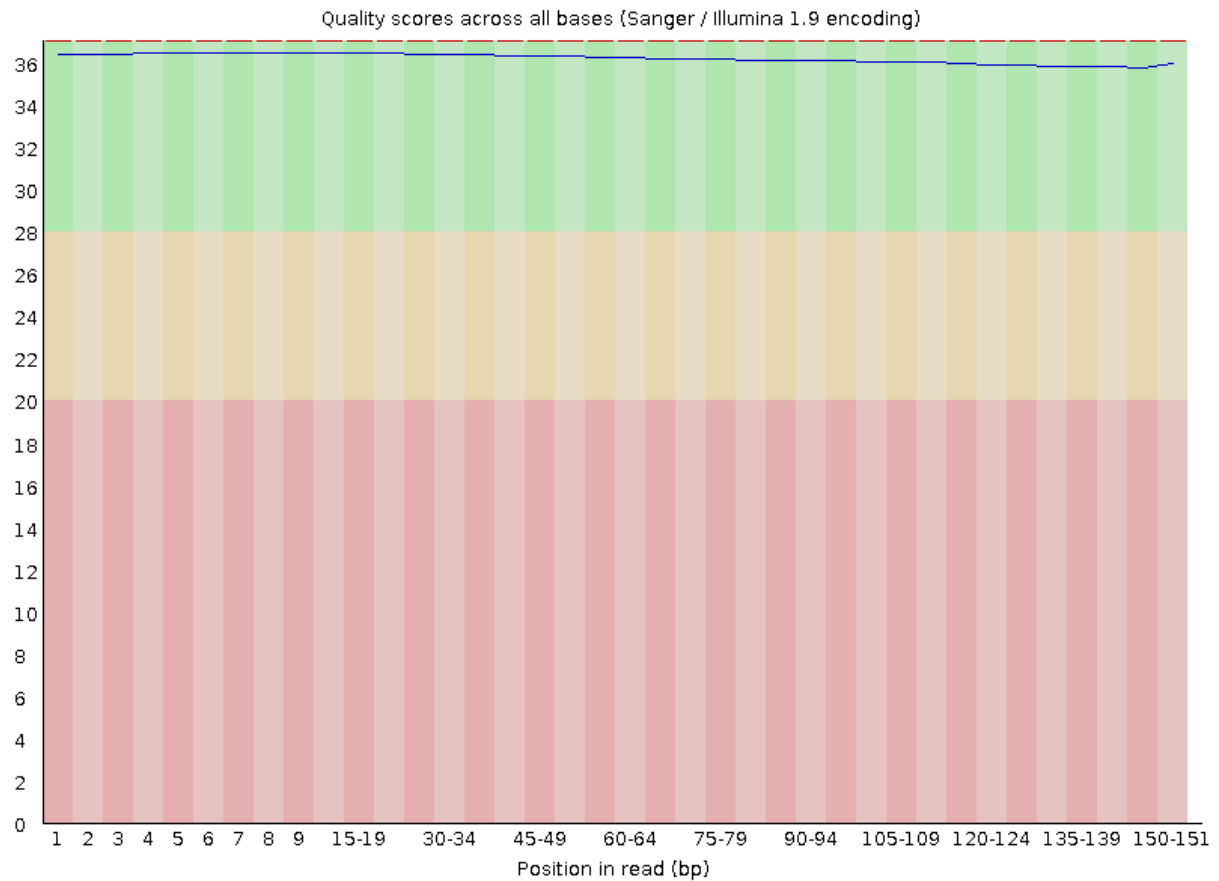
105	55240	0.10%
106	58541	0.10%
107	62284	0.11%
108	66266	0.12%
109	70735	0.12%
110	73773	0.13%
111	78756	0.14%
112	84097	0.15%
113	90095	0.16%
114	95740	0.17%
115	102756	0.18%
116	109286	0.19%
117	115721	0.20%
118	124257	0.22%
119	131935	0.23%
120	140900	0.25%
121	146160	0.25%
122	155143	0.27%
123	163301	0.28%
124	173291	0.30%
125	180006	0.31%
126	187516	0.33%
127	194668	0.34%
128	205686	0.36%
129	213498	0.37%
130	223137	0.39%
131	229935	0.40%
132	239087	0.42%
133	245626	0.43%
134	253312	0.44%
135	259580	0.45%
136	268057	0.47%
137	272160	0.47%
138	284301	0.49%
139	282846	0.49%
140	292700	0.51%
141	298920	0.52%
142	303219	0.53%
143	311144	0.54%
144	317436	0.55%
145	320223	0.56%
146	323171	0.56%
147	328618	0.57%
148	332376	0.58%
149	407156	0.71%
150	1462349	2.55%
151	45488926	79.17%

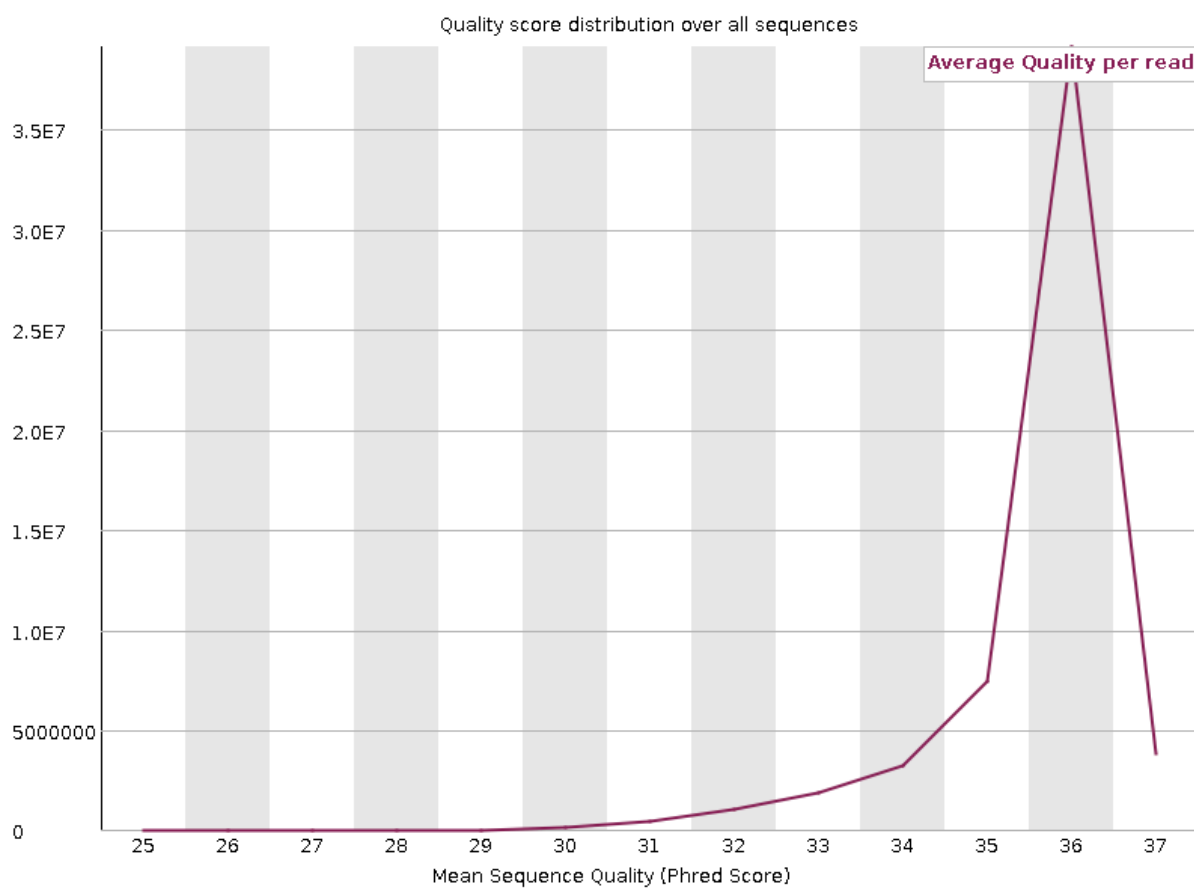
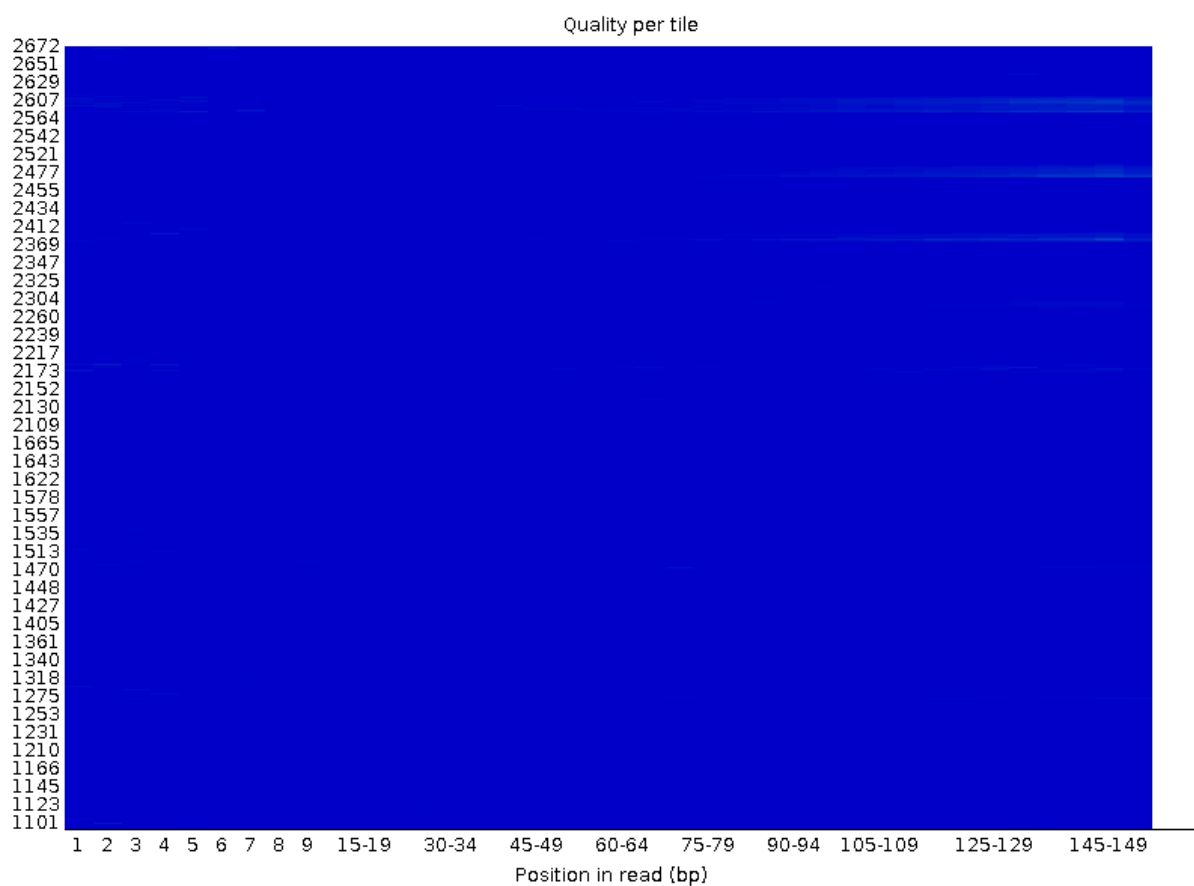
<Graphic distribution of read lengths>

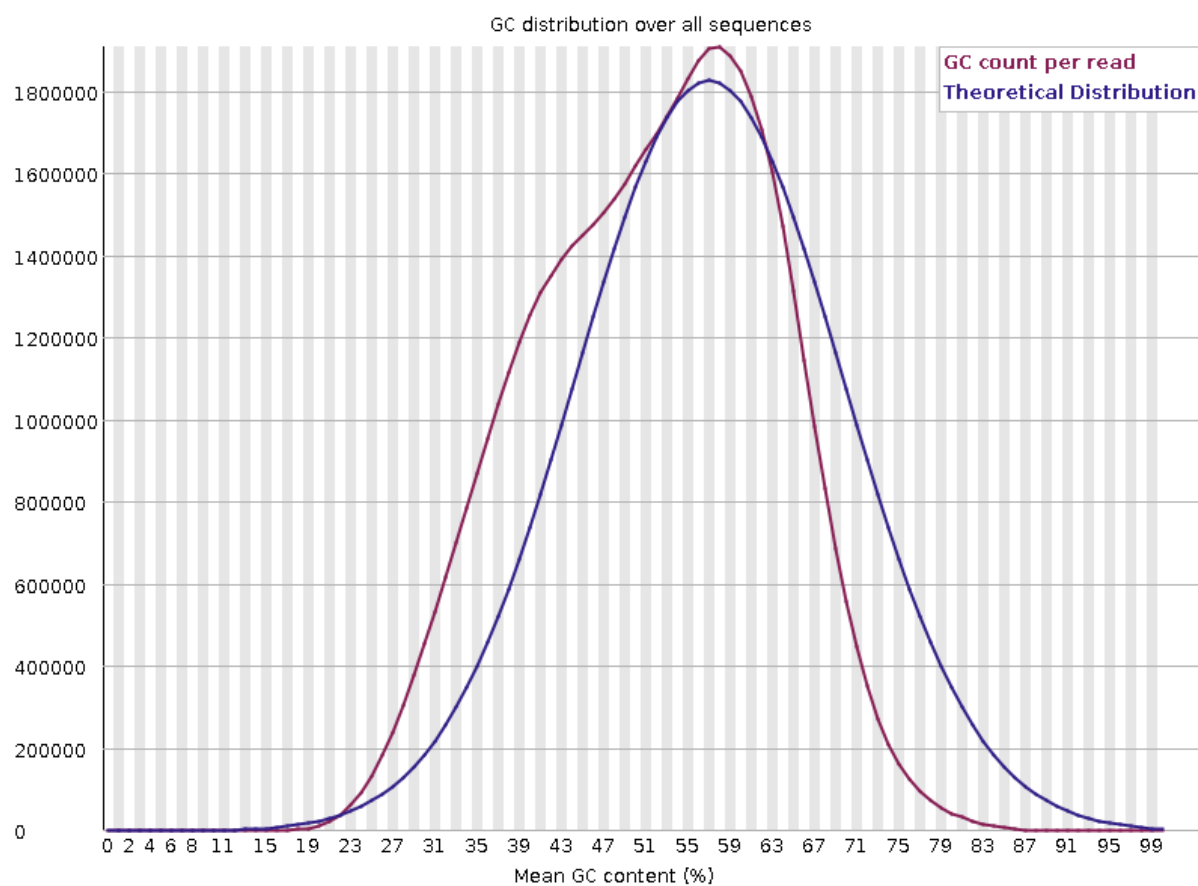
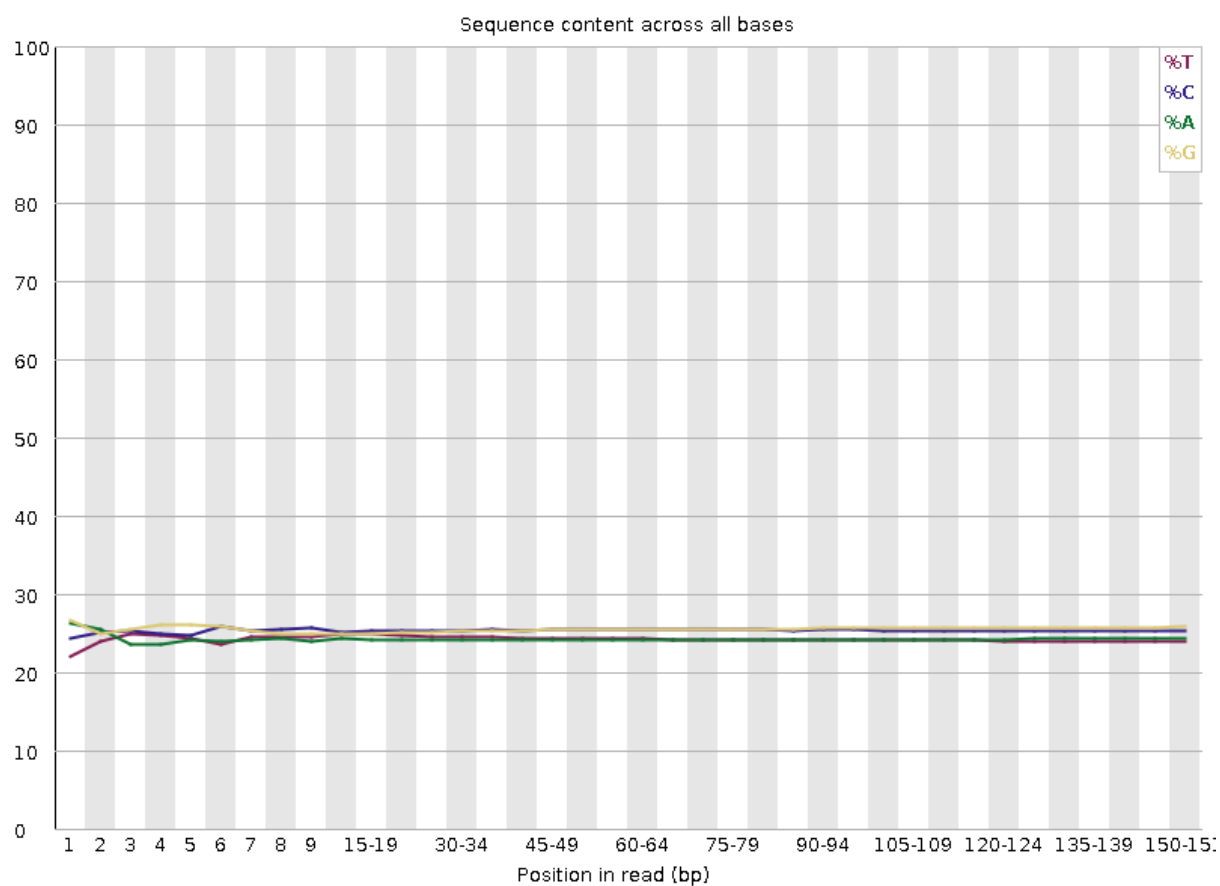
Length	Distribution	Abundance
60	□□□□□□□□□□	0.06%
61	□□□□□□□□□□	0.06%
62	□□□□□□□□□□	0.06%
63	□□□□□□□□□□	0.06%
64	□□□□□□□□□□	0.06%
65	□□□□□□□□□□	0.06%
66	□□□□□□□□□□	0.06%
67	□□□□□□□□□□	0.06%
68	□□□□□□□□□□	0.06%
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72	□□□□□□□□□□	0.06%
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76	□□□□□□□□□□	0.06%
77	□□□□□□□□□□	0.06%
78	□□□□□□□□□□	0.06%
79	□□□□□□□□□□	0.06%
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94	□□□□□□□□□□	0.07%
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96	□□□□□□□□□□	0.07%
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100	□□□□□□□□□□	0.08%
101	□□□□□□□□□□	0.08%
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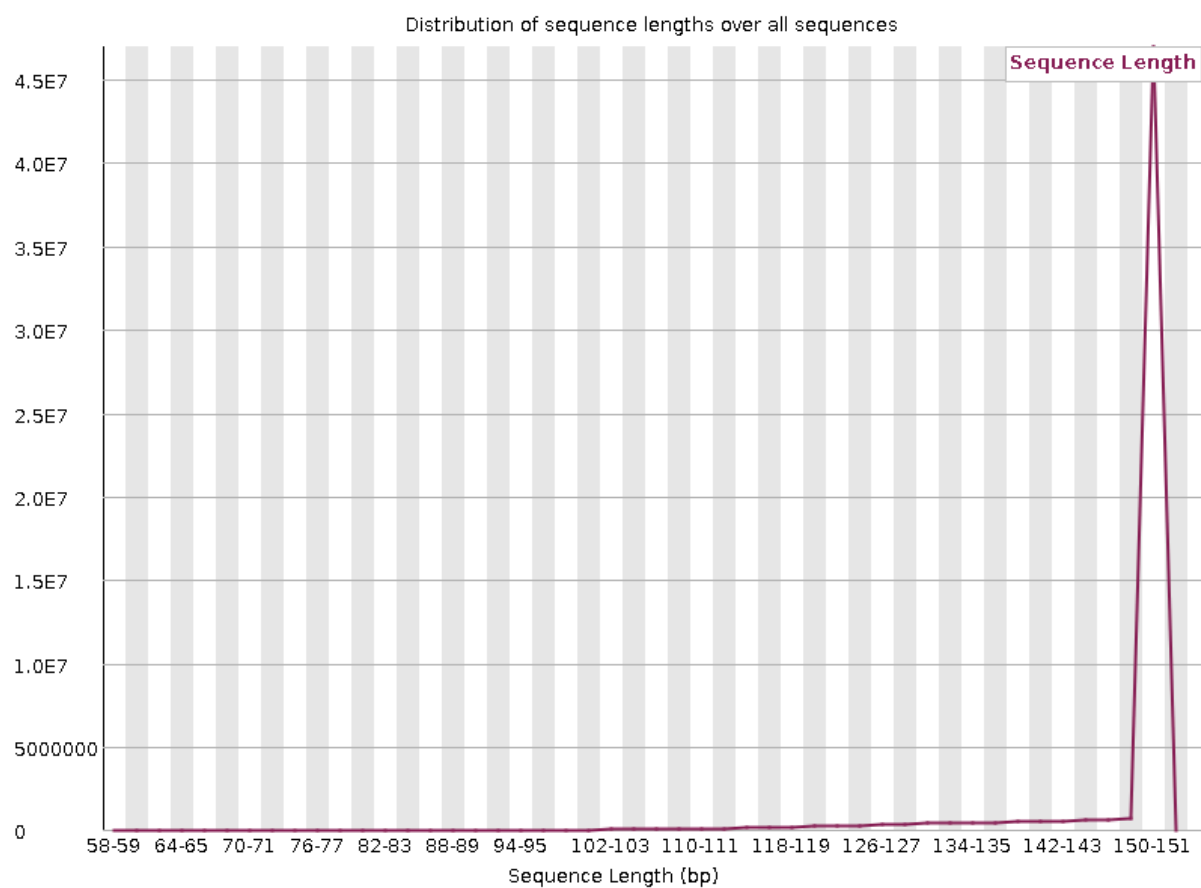
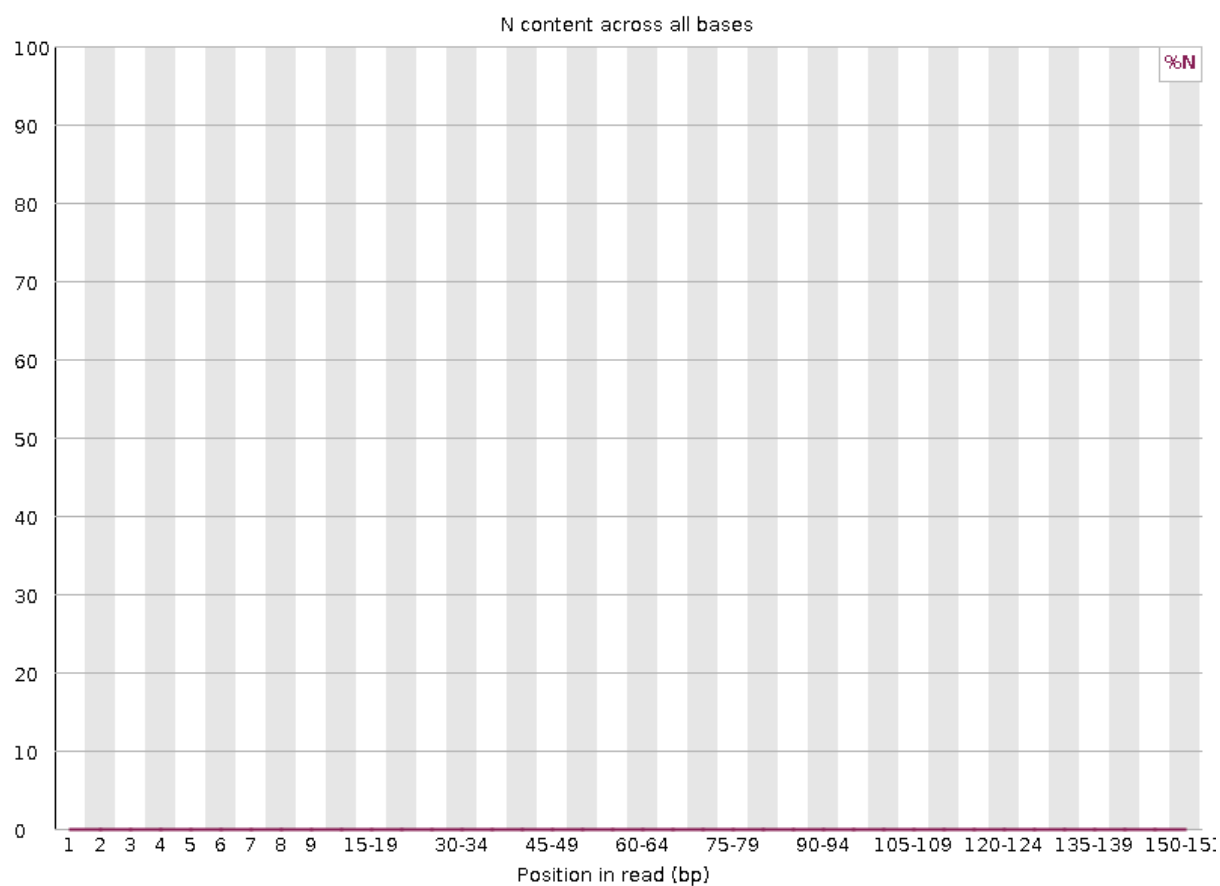
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149	□□□□□□□□	0.71%
150	□□□□□□□□	2.55%
151	■ ■ ■ ■ ■ ■ ■ □ □ □	79.17%

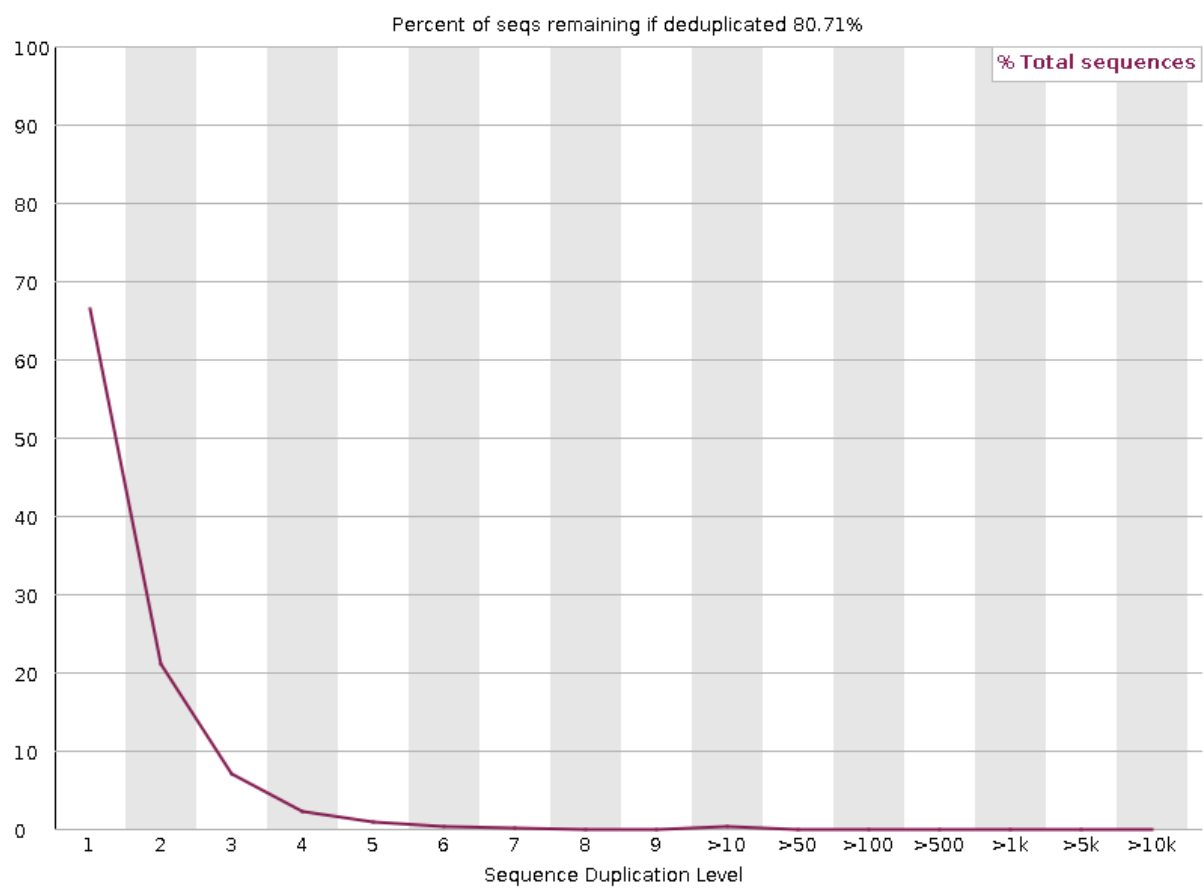
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Measure	Value
Filename	SRR37843240_1.p.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	57,459,044
Total Bases	8.4 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51

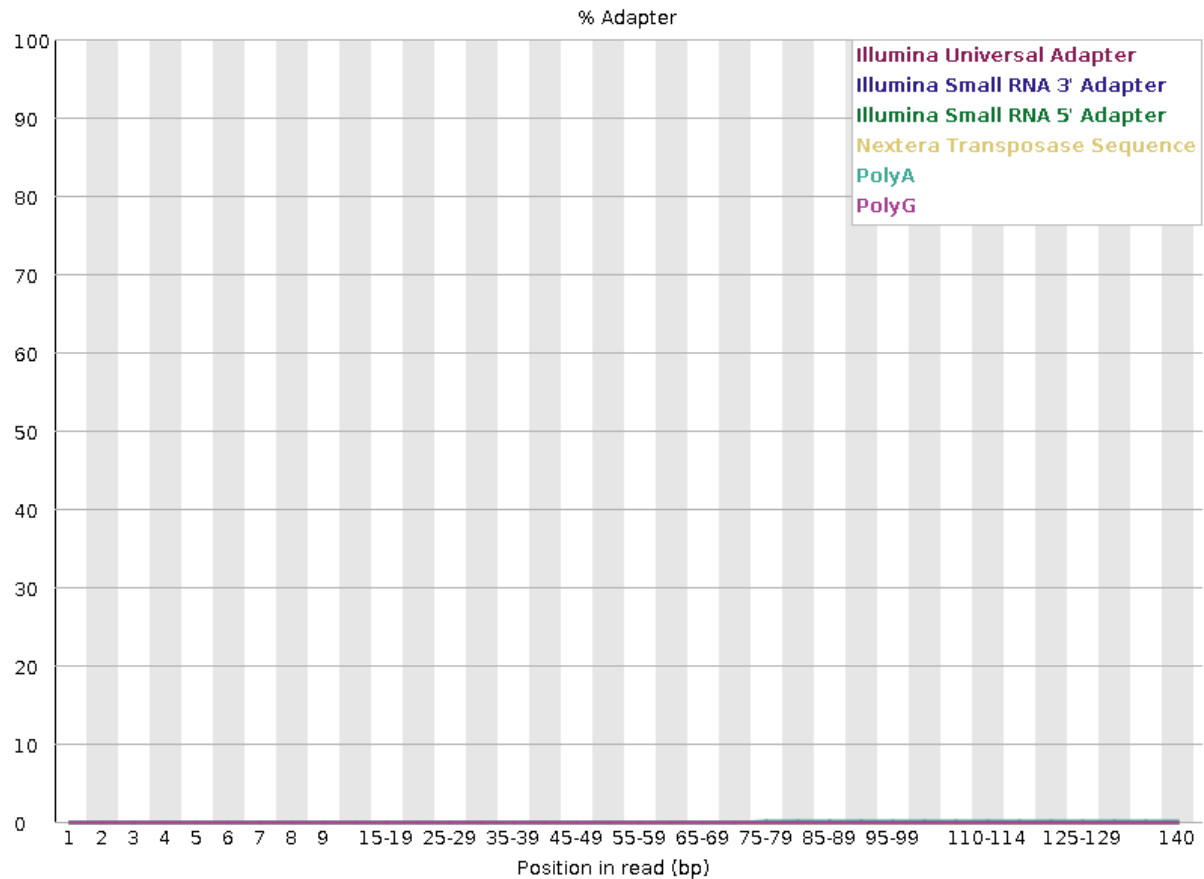












SRR37843240_2.p.fastq

The number of read sequences is 57459044

Total length of read sequences is 8396597683

<Text distribution of read lengths>

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62	30900	0.05%
63	30000	0.05%
64	31671	0.06%
65	30165	0.05%
66	31956	0.06%
67	34330	0.06%
68	32007	0.06%
69	32177	0.06%
70	36491	0.06%
71	31489	0.05%
72	31735	0.06%
73	30988	0.05%
74	32614	0.06%
75	31348	0.05%
76	34322	0.06%

77	34011	0.06%
78	33198	0.06%
79	33494	0.06%
80	36505	0.06%
81	34718	0.06%
82	35289	0.06%
83	33803	0.06%
84	33894	0.06%
85	34148	0.06%
86	34484	0.06%
87	33795	0.06%
88	32687	0.06%
89	35373	0.06%
90	35508	0.06%
91	36319	0.06%
92	38160	0.07%
93	36520	0.06%
94	37874	0.07%
95	42279	0.07%
96	42652	0.07%
97	45000	0.08%
98	44305	0.08%
99	44422	0.08%
100	50853	0.09%
101	48935	0.09%
102	50343	0.09%
103	53664	0.09%
104	56972	0.10%
105	58174	0.10%
106	64382	0.11%
107	67577	0.12%
108	69808	0.12%
109	74797	0.13%
110	76765	0.13%
111	82494	0.14%
112	89945	0.16%
113	95316	0.17%
114	102881	0.18%
115	108872	0.19%
116	115345	0.20%
117	123112	0.21%
118	131789	0.23%
119	138422	0.24%
120	147278	0.26%
121	156621	0.27%
122	165909	0.29%
123	178400	0.31%
124	185583	0.32%

125	192501	0.34%
126	197858	0.34%
127	208411	0.36%
128	215691	0.38%
129	224867	0.39%
130	230629	0.40%
131	240393	0.42%
132	251517	0.44%
133	257542	0.45%
134	268147	0.47%
135	269814	0.47%
136	282778	0.49%
137	286122	0.50%
138	294280	0.51%
139	300004	0.52%
140	311835	0.54%
141	318725	0.55%
142	324032	0.56%
143	327667	0.57%
144	336360	0.59%
145	332971	0.58%
146	341268	0.59%
147	345977	0.60%
148	353190	0.61%
149	466078	0.81%
150	1978177	3.44%
151	44417978	77.30%

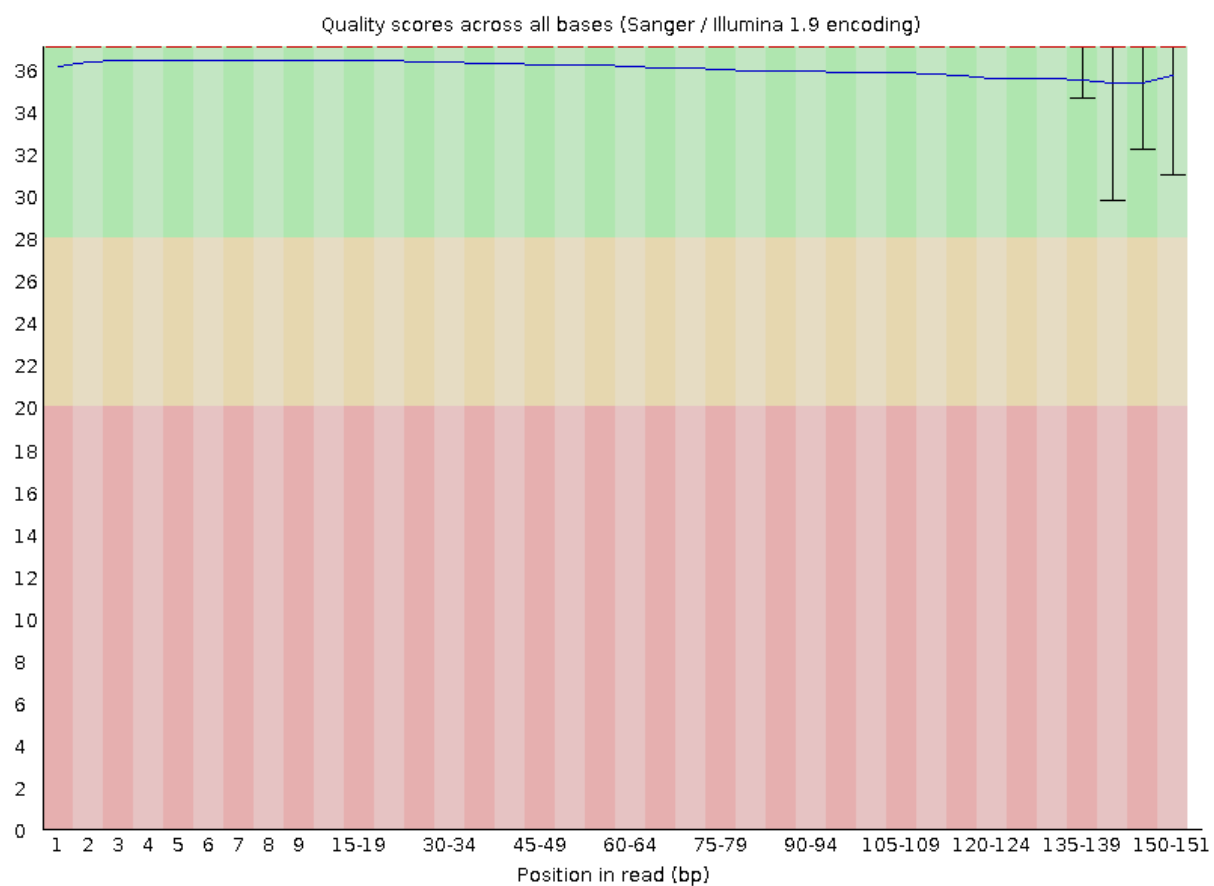
<Graphic distribution of read lengths>

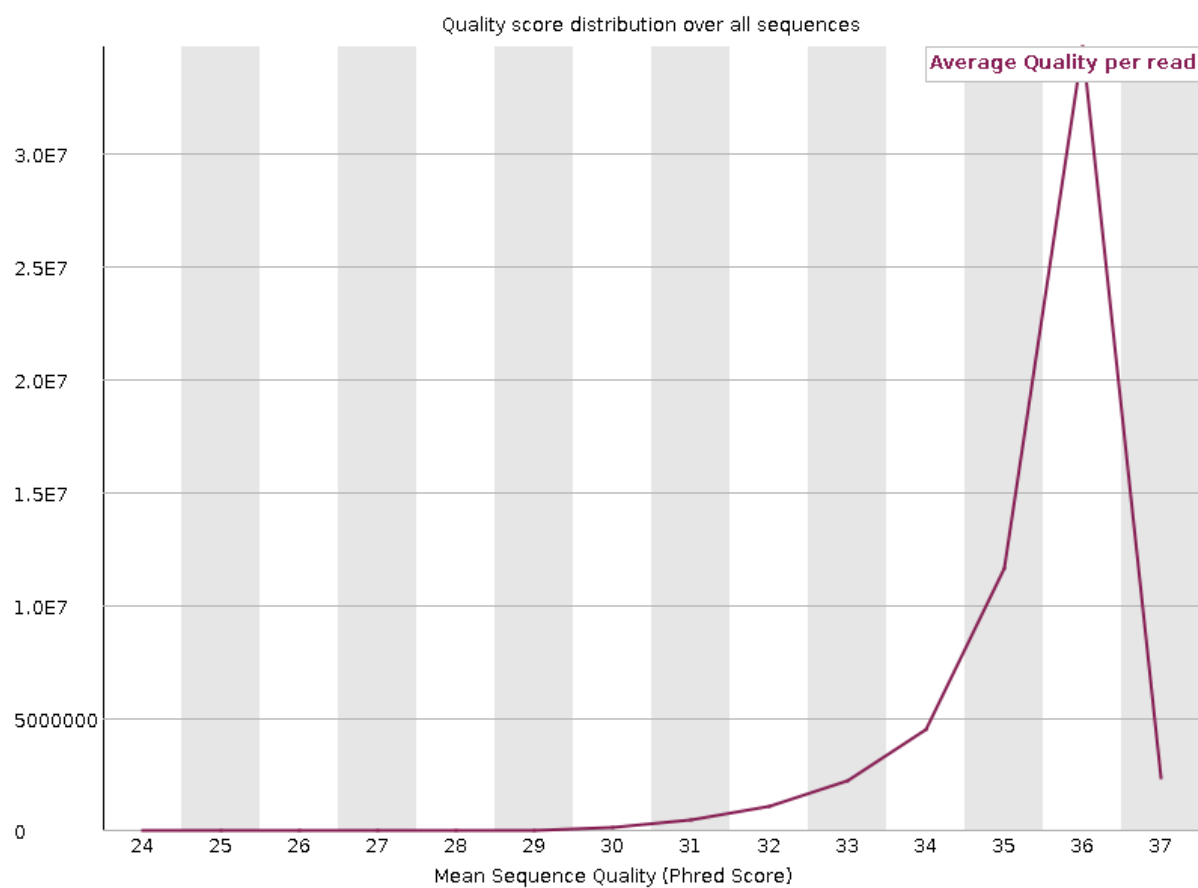
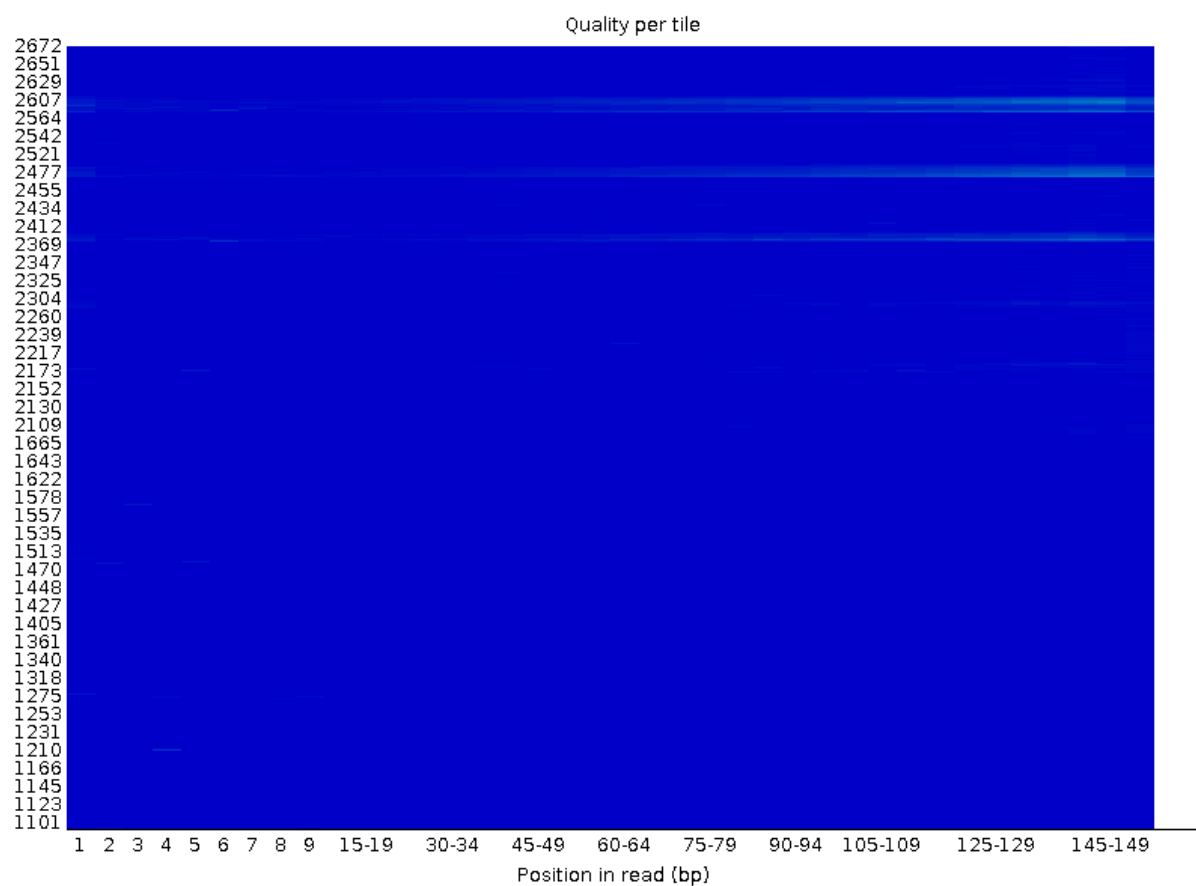
Length	Distribution	Abundance
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61	□□□□□□□□□□	0.05%
62	□□□□□□□□□□	0.05%
63	□□□□□□□□□□	0.05%
64	□□□□□□□□□□	0.06%
65	□□□□□□□□□□	0.05%
66	□□□□□□□□□□	0.06%
67	□□□□□□□□□□	0.06%
68	□□□□□□□□□□	0.06%
69	□□□□□□□□□□	0.06%
70	□□□□□□□□□□	0.06%
71	□□□□□□□□□□	0.05%
72	□□□□□□□□□□	0.06%
73	□□□□□□□□□□	0.05%
74	□□□□□□□□□□	0.06%
75	□□□□□□□□□□	0.05%
76	□□□□□□□□□□	0.06%
77	□□□□□□□□□□	0.06%

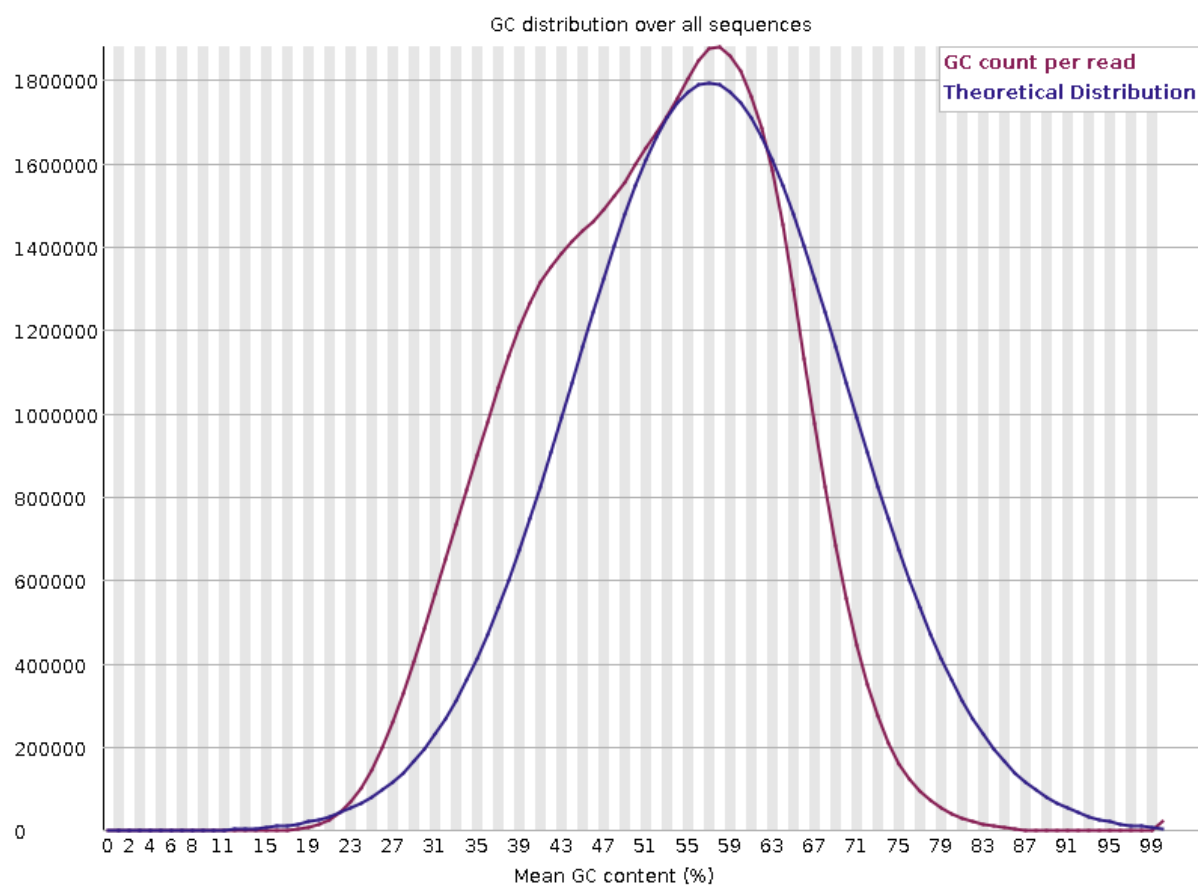
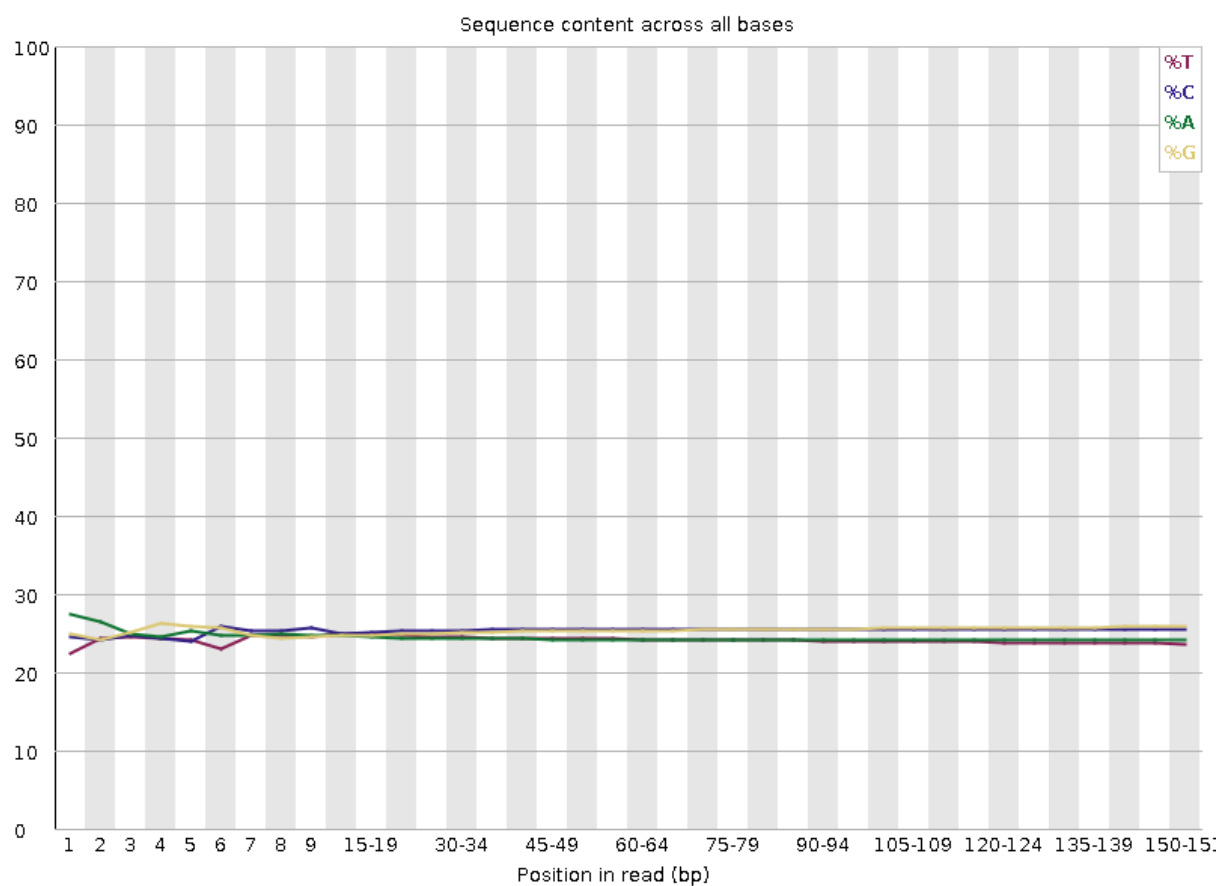
78	□□□□□□□□	0.06%
79	□□□□□□□□	0.06%
80	□□□□□□□□	0.06%
81	□□□□□□□□	0.06%
82	□□□□□□□□	0.06%
83	□□□□□□□□	0.06%
84	□□□□□□□□	0.06%
85	□□□□□□□□	0.06%
86	□□□□□□□□	0.06%
87	□□□□□□□□	0.06%
88	□□□□□□□□	0.06%
89	□□□□□□□□	0.06%
90	□□□□□□□□	0.06%
91	□□□□□□□□	0.06%
92	□□□□□□□□	0.07%
93	□□□□□□□□	0.06%
94	□□□□□□□□	0.07%
95	□□□□□□□□	0.07%
96	□□□□□□□□	0.07%
97	□□□□□□□□	0.08%
98	□□□□□□□□	0.08%
99	□□□□□□□□	0.08%
100	□□□□□□□□	0.09%
101	□□□□□□□□	0.09%
102	□□□□□□□□	0.09%
103	□□□□□□□□	0.09%
104	□□□□□□□□	0.10%
105	□□□□□□□□	0.10%
106	□□□□□□□□	0.11%
107	□□□□□□□□	0.12%
108	□□□□□□□□	0.12%
109	□□□□□□□□	0.13%
110	□□□□□□□□	0.13%
111	□□□□□□□□	0.14%
112	□□□□□□□□	0.16%
113	□□□□□□□□	0.17%
114	□□□□□□□□	0.18%
115	□□□□□□□□	0.19%
116	□□□□□□□□	0.20%
117	□□□□□□□□	0.21%
118	□□□□□□□□	0.23%
119	□□□□□□□□	0.24%
120	□□□□□□□□	0.26%
121	□□□□□□□□	0.27%
122	□□□□□□□□	0.29%
123	□□□□□□□□	0.31%
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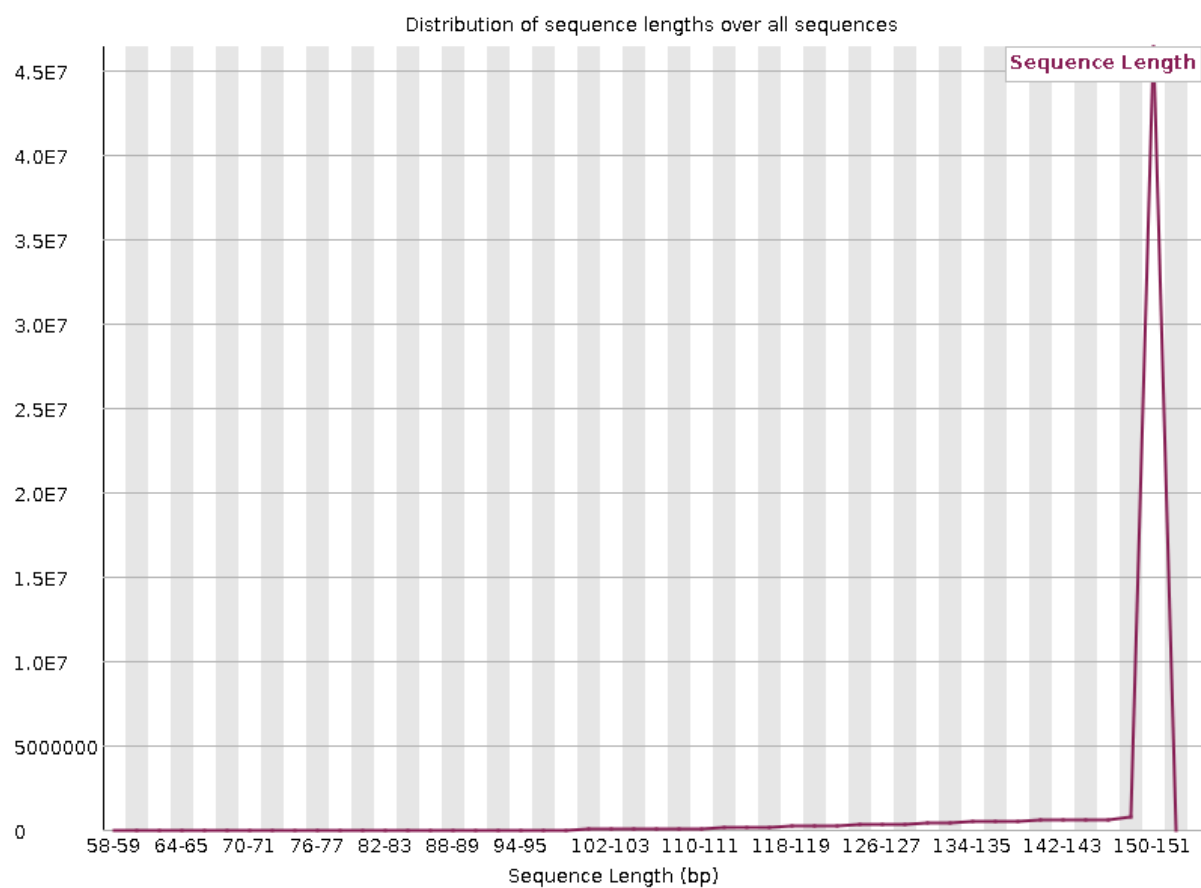
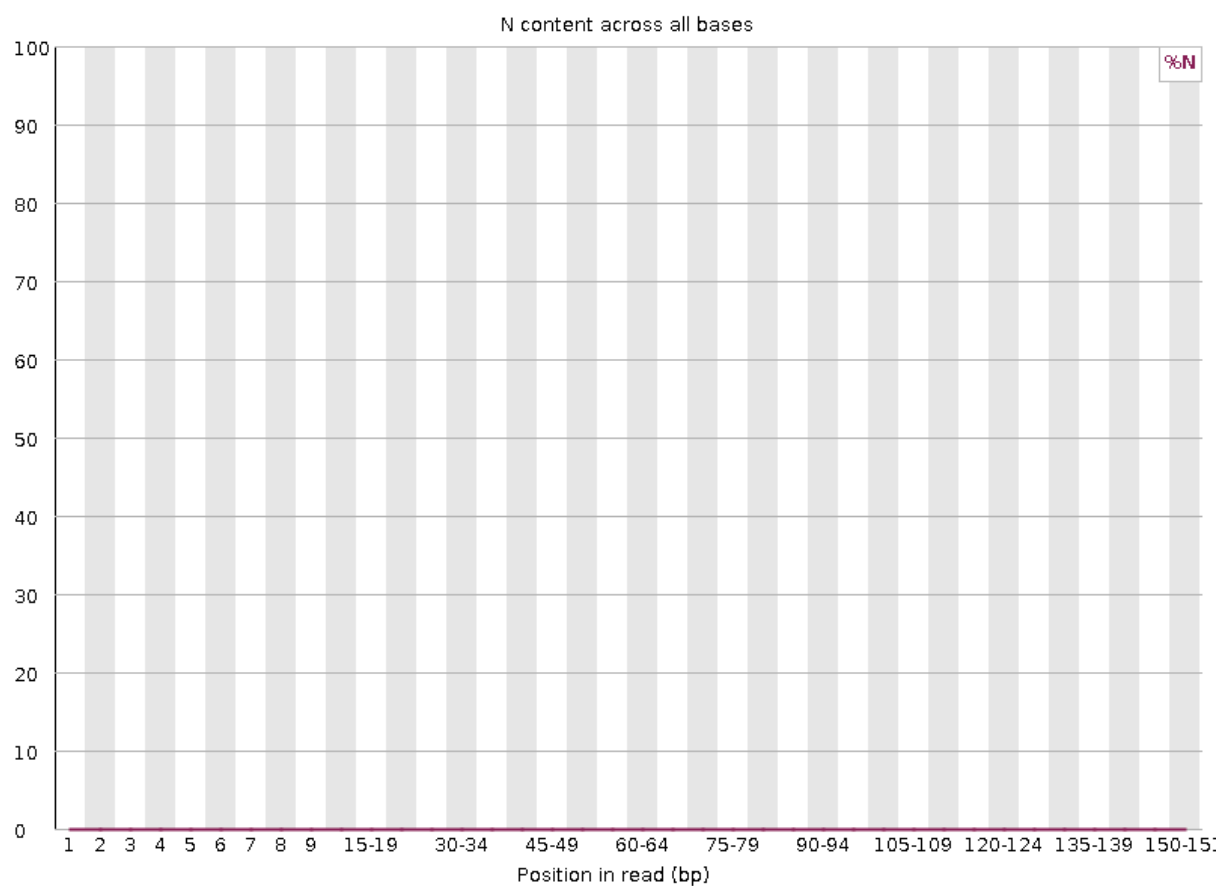
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133	□□□□□□□□□□	0.45%
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137	□□□□□□□□□□	0.50%
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139	□□□□□□□□□□	0.52%
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149	□□□□□□□□□□	0.81%
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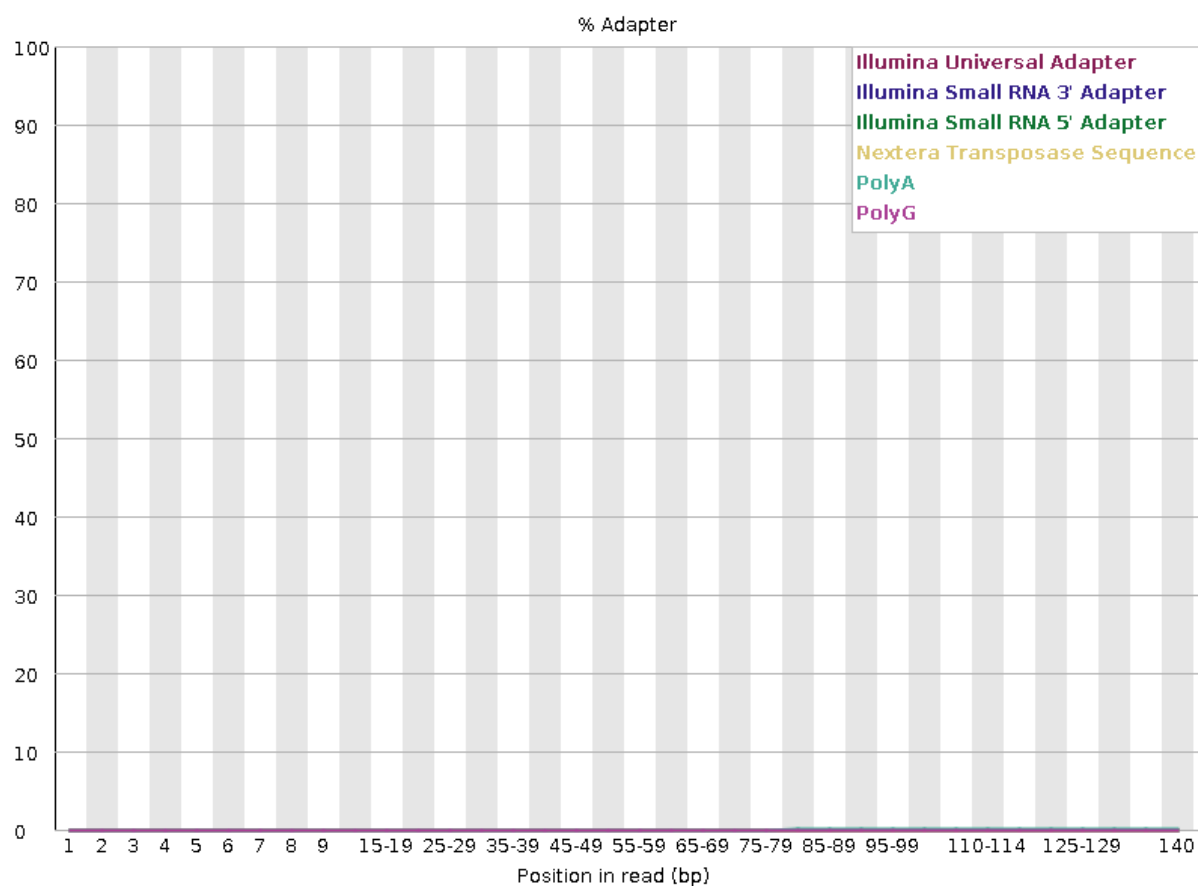
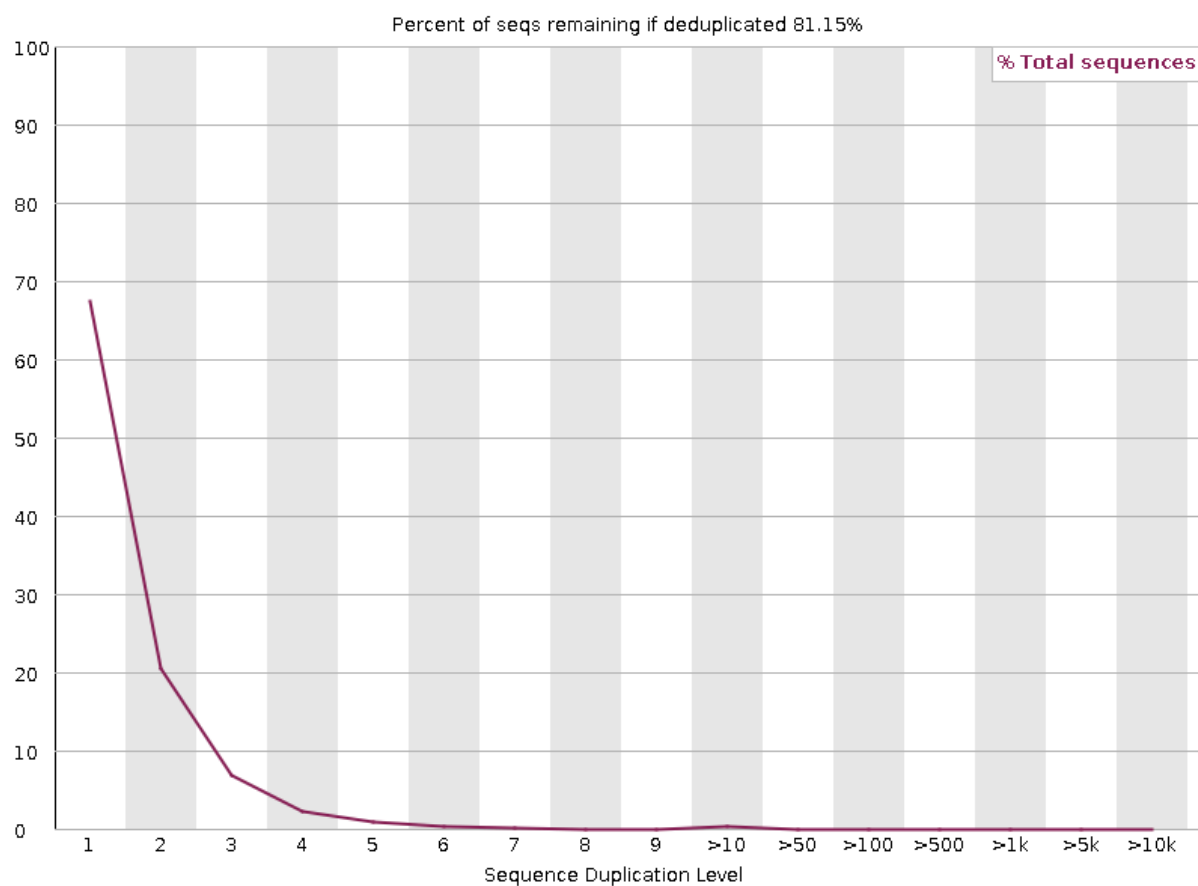
SRR37843240_2.p.fastq	
Measure	Value
Filename	SRR37843240_2.p.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	57,459,044
Total Bases	8.3 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51











SRR37843240_1.up.fastq

The number of read sequences is 1966982

Total length of read sequences is 261104684

<Text distribution of read lengths>

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63	10521	0.53%
64	10235	0.52%
65	10331	0.53%
66	9920	0.50%
67	9970	0.51%
68	9761	0.50%
69	9485	0.48%
70	9242	0.47%
71	9410	0.48%
72	9124	0.46%
73	9060	0.46%
74	9034	0.46%
75	8943	0.45%
76	8973	0.46%
77	8758	0.45%
78	8816	0.45%
79	8810	0.45%
80	8809	0.45%
81	8910	0.45%
82	8894	0.45%
83	8552	0.43%
84	8126	0.41%
85	8156	0.41%
86	8176	0.42%
87	8041	0.41%
88	7851	0.40%
89	7949	0.40%
90	8086	0.41%
91	7903	0.40%
92	7870	0.40%
93	7538	0.38%
94	7842	0.40%
95	7585	0.39%
96	7545	0.38%
97	7736	0.39%
98	7362	0.37%
99	7252	0.37%
100	7459	0.38%

101	7412	0.38%
102	7630	0.39%
103	7459	0.38%
104	7550	0.38%
105	7116	0.36%
106	7108	0.36%
107	7566	0.38%
108	7256	0.37%
109	7343	0.37%
110	7143	0.36%
111	7257	0.37%
112	7258	0.37%
113	7398	0.38%
114	7474	0.38%
115	7501	0.38%
116	7613	0.39%
117	7603	0.39%
118	7611	0.39%
119	7752	0.39%
120	7983	0.41%
121	7515	0.38%
122	7834	0.40%
123	7625	0.39%
124	8120	0.41%
125	7866	0.40%
126	7642	0.39%
127	7538	0.38%
128	7954	0.40%
129	8138	0.41%
130	8209	0.42%
131	8042	0.41%
132	8462	0.43%
133	8054	0.41%
134	7974	0.41%
135	8201	0.42%
136	8434	0.43%
137	8122	0.41%
138	8748	0.44%
139	7951	0.40%
140	8379	0.43%
141	8283	0.42%
142	8207	0.42%
143	8546	0.43%
144	8509	0.43%
145	8373	0.43%
146	8405	0.43%
147	8391	0.43%
148	8396	0.43%

149	15982	0.81%
150	92365	4.70%
151	1121468	57.01%

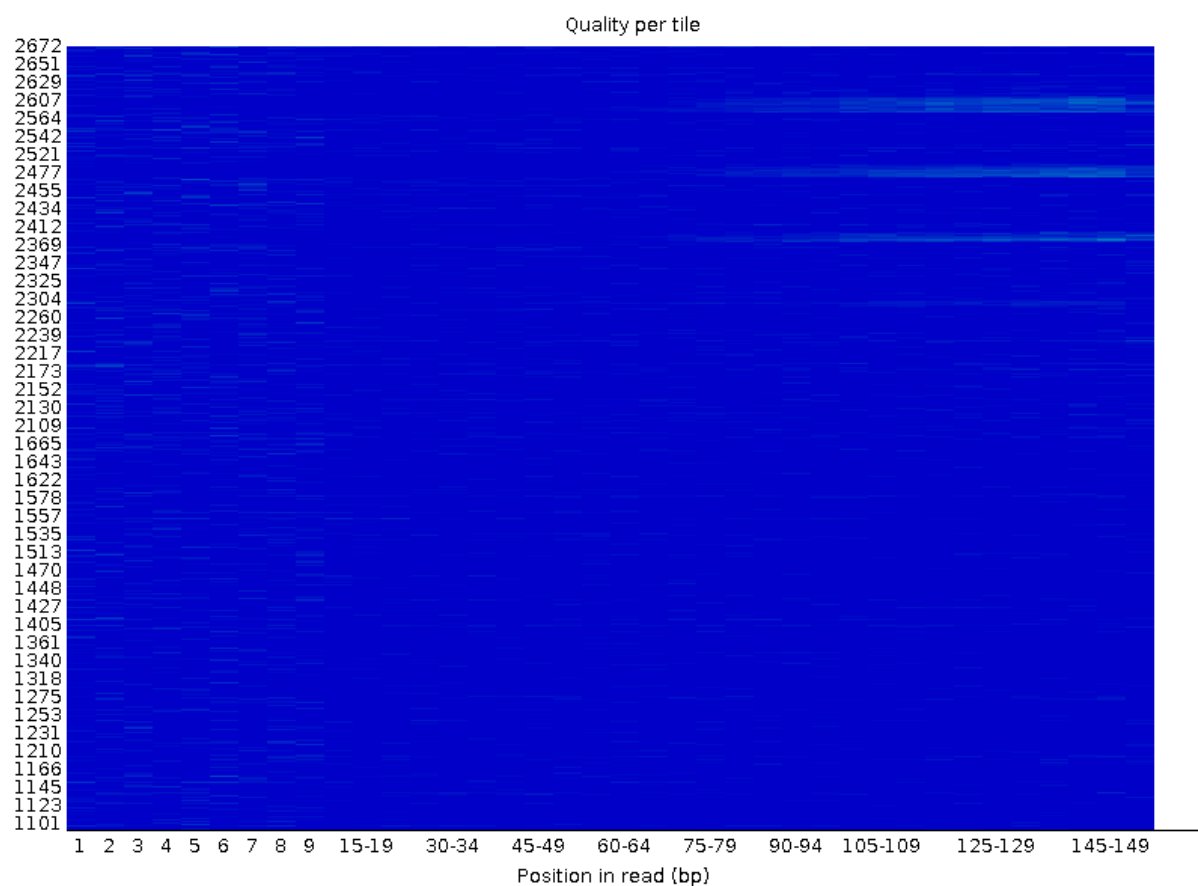
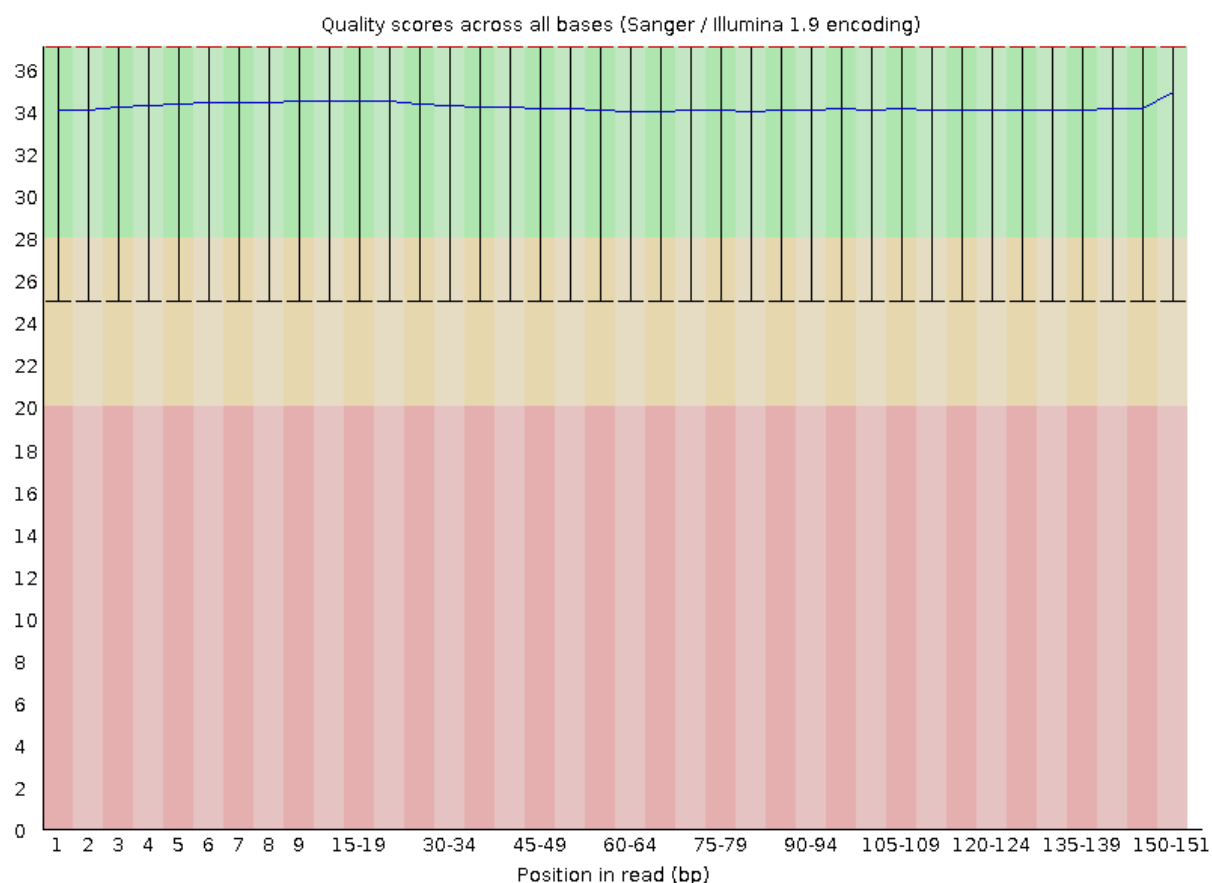
<Graphic distribution of read lengths>

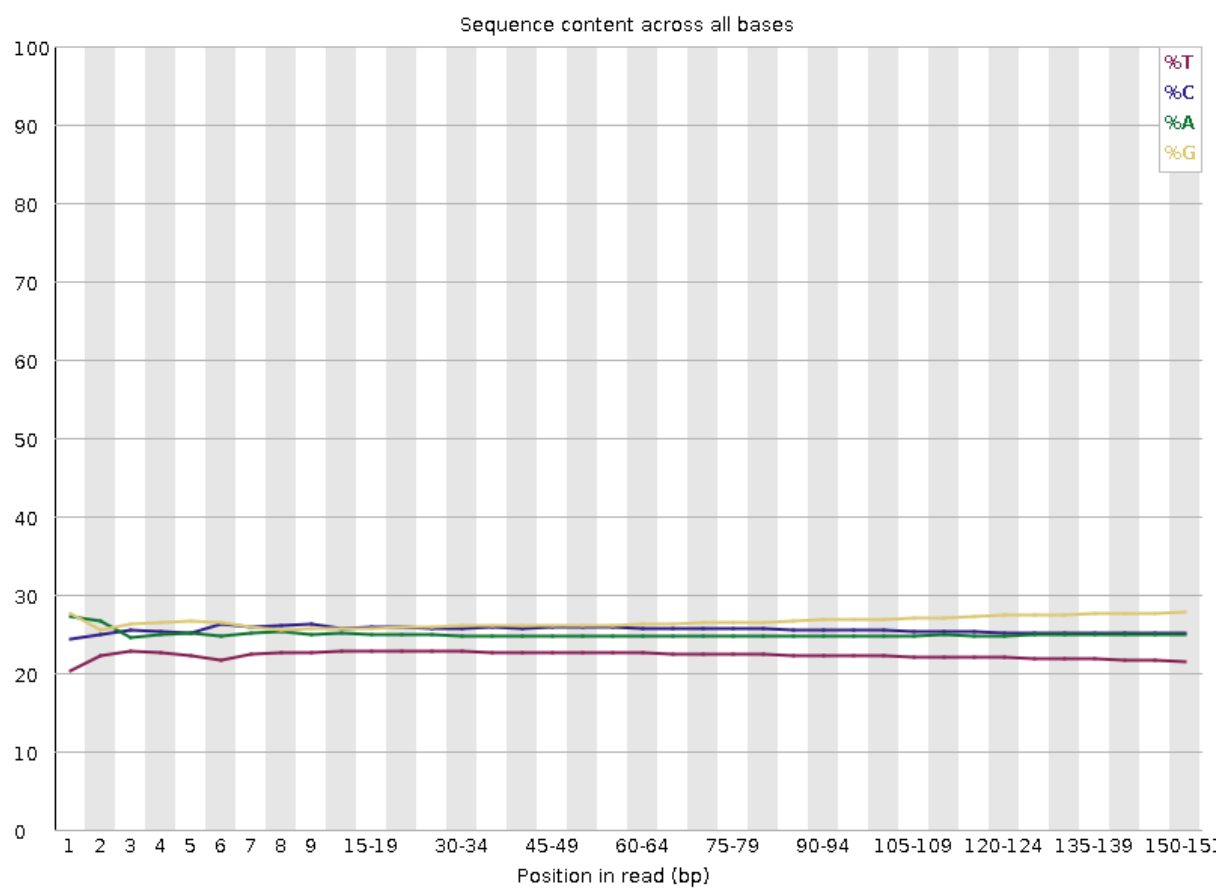
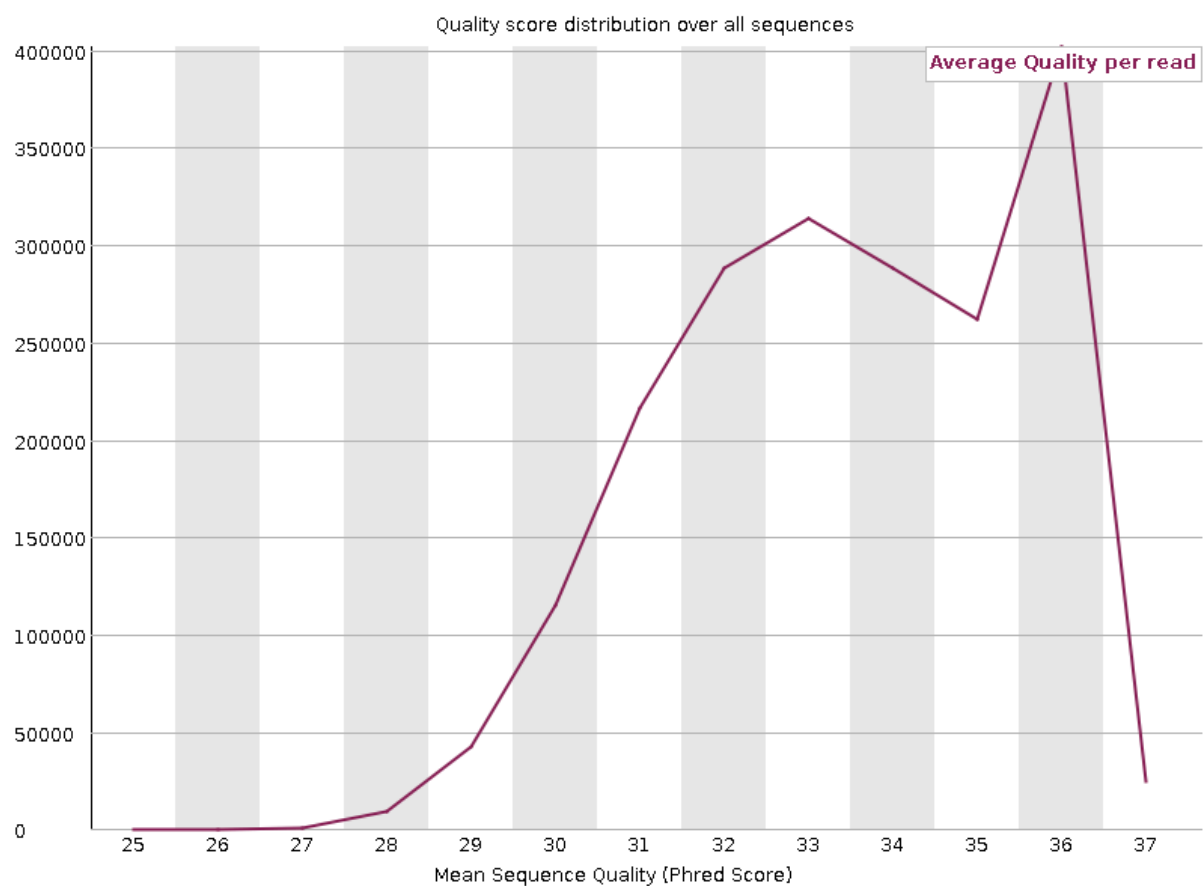
Length	Distribution	Abundance
60	□□□□□□□□□□	0.56%
61	□□□□□□□□□□	0.55%
62	□□□□□□□□□□	0.53%
63	□□□□□□□□□□	0.53%
64	□□□□□□□□□□	0.52%
65	□□□□□□□□□□	0.53%
66	□□□□□□□□□□	0.50%
67	□□□□□□□□□□	0.51%
68	□□□□□□□□□□	0.50%
69	□□□□□□□□□□	0.48%
70	□□□□□□□□□□	0.47%
71	□□□□□□□□□□	0.48%
72	□□□□□□□□□□	0.46%
73	□□□□□□□□□□	0.46%
74	□□□□□□□□□□	0.46%
75	□□□□□□□□□□	0.45%
76	□□□□□□□□□□	0.46%
77	□□□□□□□□□□	0.45%
78	□□□□□□□□□□	0.45%
79	□□□□□□□□□□	0.45%
80	□□□□□□□□□□	0.45%
81	□□□□□□□□□□	0.45%
82	□□□□□□□□□□	0.45%
83	□□□□□□□□□□	0.43%
84	□□□□□□□□□□	0.41%
85	□□□□□□□□□□	0.41%
86	□□□□□□□□□□	0.42%
87	□□□□□□□□□□	0.41%
88	□□□□□□□□□□	0.40%
89	□□□□□□□□□□	0.40%
90	□□□□□□□□□□	0.41%
91	□□□□□□□□□□	0.40%
92	□□□□□□□□□□	0.40%
93	□□□□□□□□□□	0.38%
94	□□□□□□□□□□	0.40%
95	□□□□□□□□□□	0.39%
96	□□□□□□□□□□	0.38%
97	□□□□□□□□□□	0.39%
98	□□□□□□□□□□	0.37%
99	□□□□□□□□□□	0.37%
100	□□□□□□□□□□	0.38%
101	□□□□□□□□□□	0.38%

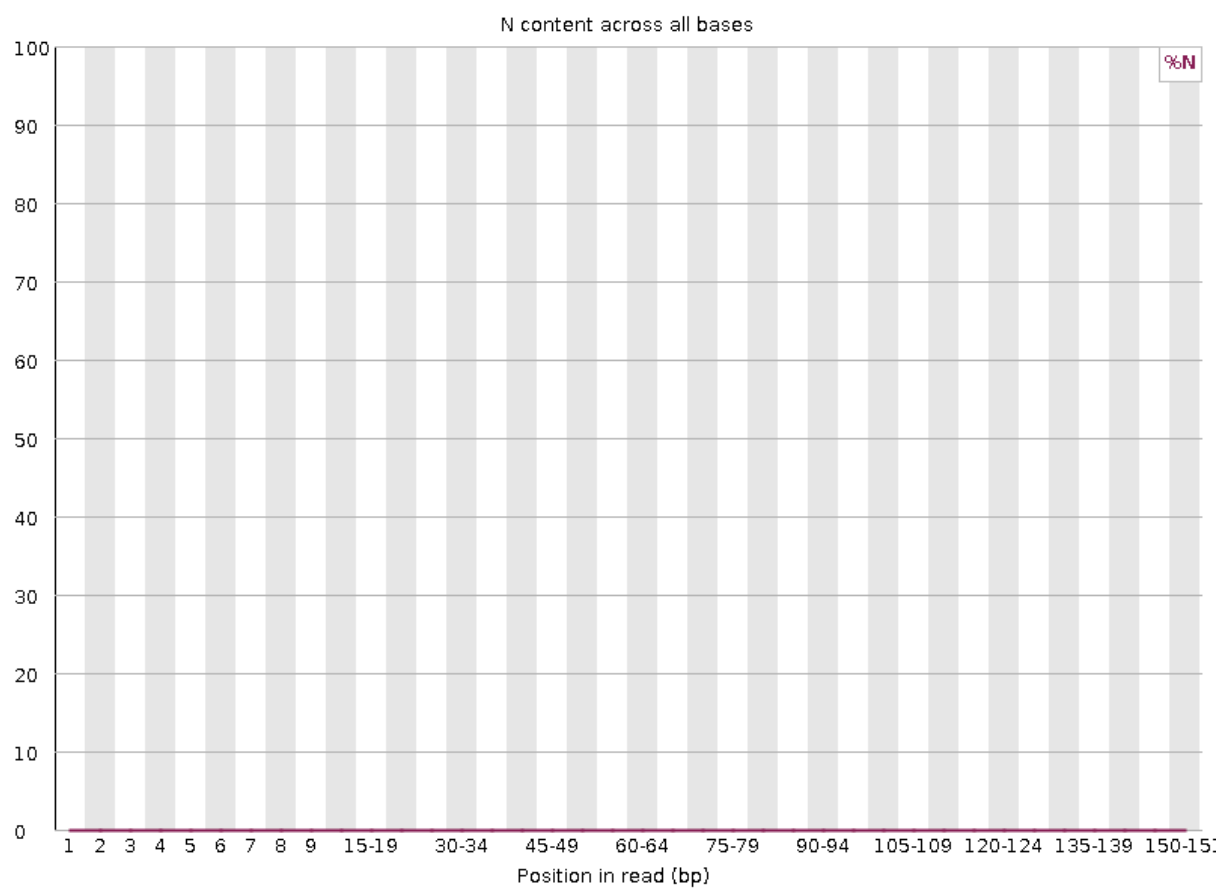
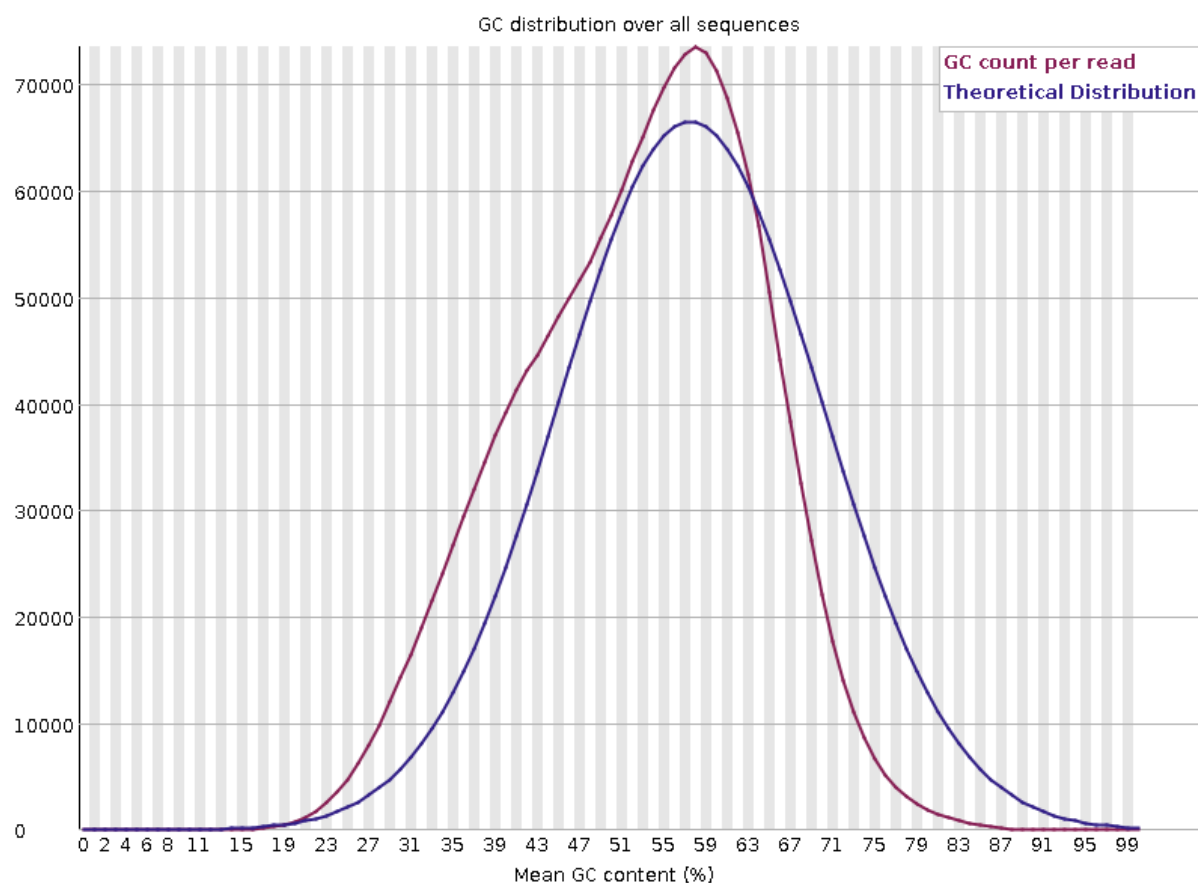
102	□□□□□□□□	0.39%
103	□□□□□□□□	0.38%
104	□□□□□□□□	0.38%
105	□□□□□□□□	0.36%
106	□□□□□□□□	0.36%
107	□□□□□□□□	0.38%
108	□□□□□□□□	0.37%
109	□□□□□□□□	0.37%
110	□□□□□□□□	0.36%
111	□□□□□□□□	0.37%
112	□□□□□□□□	0.37%
113	□□□□□□□□	0.38%
114	□□□□□□□□	0.38%
115	□□□□□□□□	0.38%
116	□□□□□□□□	0.39%
117	□□□□□□□□	0.39%
118	□□□□□□□□	0.39%
119	□□□□□□□□	0.39%
120	□□□□□□□□	0.41%
121	□□□□□□□□	0.38%
122	□□□□□□□□	0.40%
123	□□□□□□□□	0.39%
124	□□□□□□□□	0.41%
125	□□□□□□□□	0.40%
126	□□□□□□□□	0.39%
127	□□□□□□□□	0.38%
128	□□□□□□□□	0.40%
129	□□□□□□□□	0.41%
130	□□□□□□□□	0.42%
131	□□□□□□□□	0.41%
132	□□□□□□□□	0.43%
133	□□□□□□□□	0.41%
134	□□□□□□□□	0.41%
135	□□□□□□□□	0.42%
136	□□□□□□□□	0.43%
137	□□□□□□□□	0.41%
138	□□□□□□□□	0.44%
139	□□□□□□□□	0.40%
140	□□□□□□□□	0.43%
141	□□□□□□□□	0.42%
142	□□□□□□□□	0.42%
143	□□□□□□□□	0.43%
144	□□□□□□□□	0.43%
145	□□□□□□□□	0.43%
146	□□□□□□□□	0.43%
147	□□□□□□□□	0.43%
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149	□□□□□□□□	0.81%

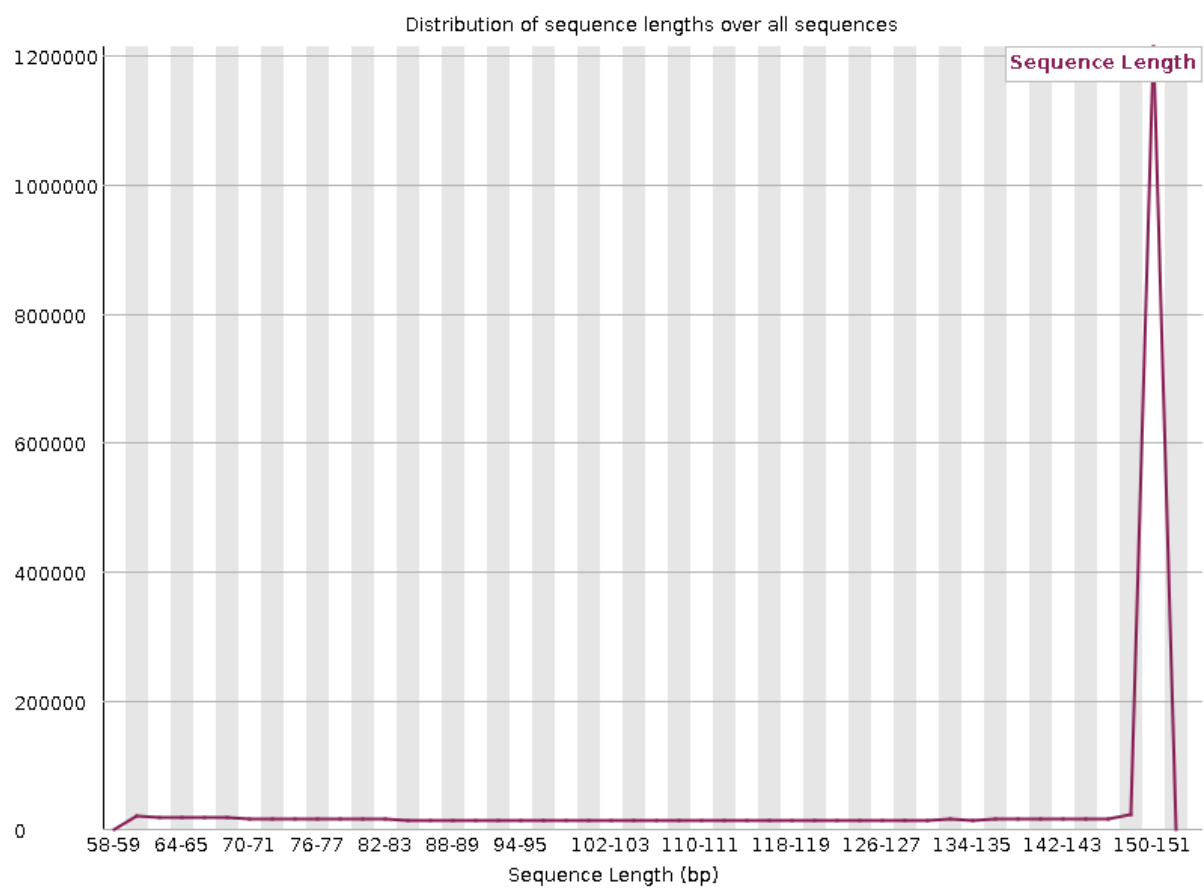
150 □□□□□□□□ 4.70%
 151 ■■■■■□□□□□ 57.01%

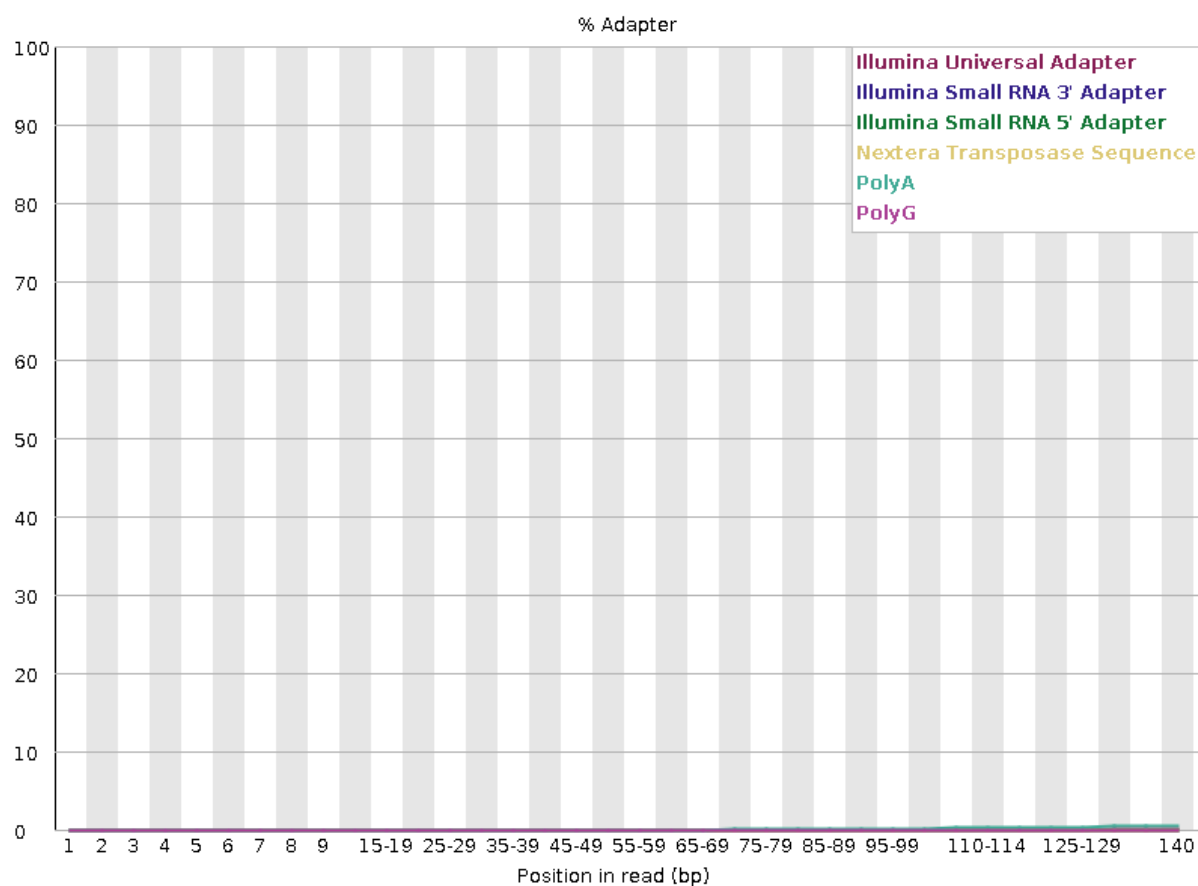
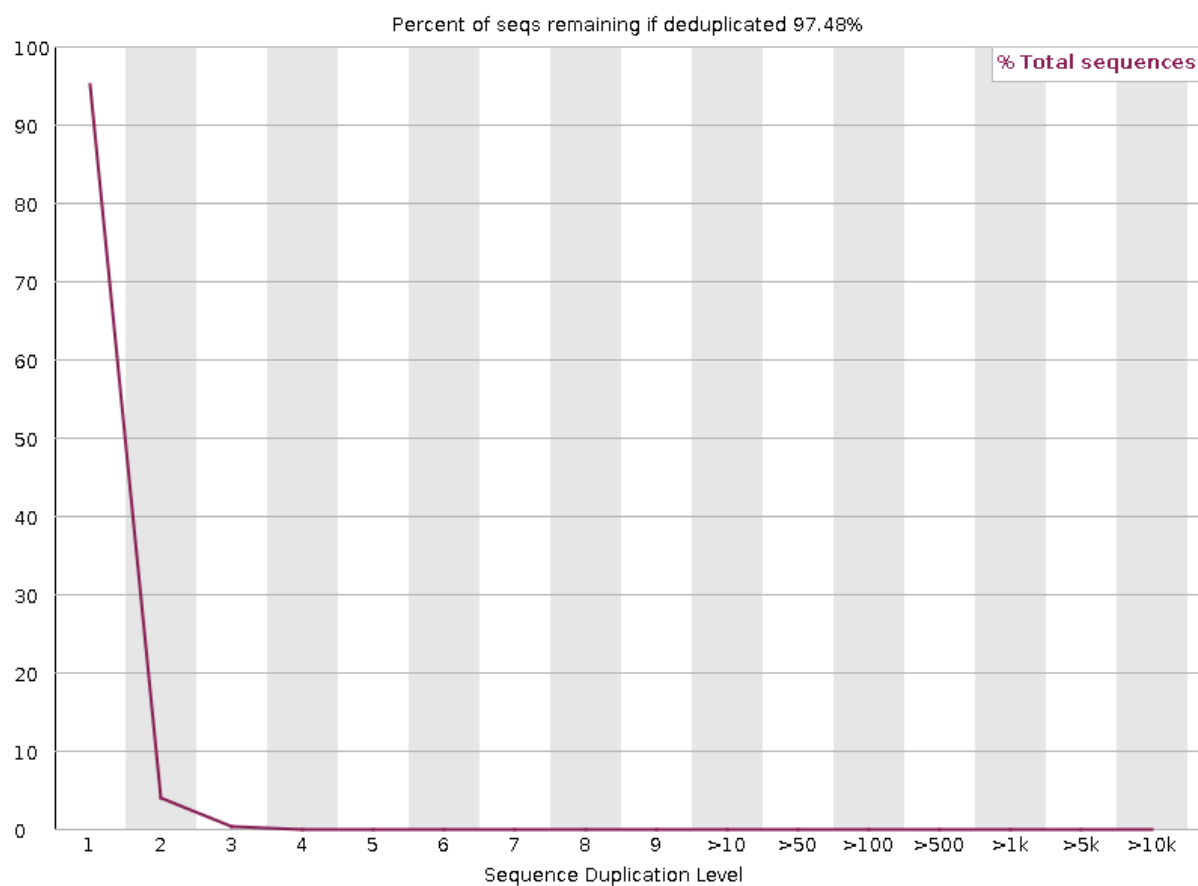
SRR37843240_1.up.fastq	
Measure	Value
Filename	SRR37843240_1.up.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1,966,982
Total Bases	261.1 Mbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	52











SRR37843240_2.up.fastq

The number of read sequences is 1622948

Total length of read sequences is 218480806

<Text distribution of read lengths>

Length	Count	Abundance
60	6768	0.42%
61	6805	0.42%
62	6819	0.42%
63	6666	0.41%
64	6951	0.43%
65	6549	0.40%
66	6709	0.41%
67	6719	0.41%
68	6489	0.40%
69	6244	0.38%
70	6588	0.41%
71	6045	0.37%
72	5989	0.37%
73	5837	0.36%
74	5933	0.37%
75	5816	0.36%
76	5962	0.37%
77	5945	0.37%
78	5753	0.35%
79	5673	0.35%
80	6212	0.38%
81	5832	0.36%
82	5662	0.35%
83	5484	0.34%
84	5464	0.34%
85	5442	0.34%
86	5270	0.32%
87	5000	0.31%
88	4997	0.31%
89	5256	0.32%
90	5176	0.32%
91	5193	0.32%
92	5031	0.31%
93	5052	0.31%
94	5127	0.32%
95	5344	0.33%
96	5207	0.32%
97	5543	0.34%
98	5245	0.32%
99	5186	0.32%
100	5726	0.35%

101	5413	0.33%
102	5264	0.32%
103	5297	0.33%
104	5465	0.34%
105	5289	0.33%
106	5521	0.34%
107	5538	0.34%
108	5553	0.34%
109	5663	0.35%
110	5518	0.34%
111	5851	0.36%
112	6162	0.38%
113	6204	0.38%
114	6511	0.40%
115	6506	0.40%
116	6667	0.41%
117	6979	0.43%
118	7198	0.44%
119	7434	0.46%
120	7695	0.47%
121	7948	0.49%
122	8365	0.52%
123	8633	0.53%
124	8912	0.55%
125	8720	0.54%
126	9051	0.56%
127	9047	0.56%
128	9247	0.57%
129	9351	0.58%
130	9430	0.58%
131	9542	0.59%
132	10177	0.63%
133	9874	0.61%
134	10131	0.62%
135	10141	0.62%
136	10616	0.65%
137	10476	0.65%
138	10722	0.66%
139	10676	0.66%
140	10851	0.67%
141	11083	0.68%
142	11184	0.69%
143	11309	0.70%
144	10984	0.68%
145	10893	0.67%
146	11164	0.69%
147	11274	0.69%
148	11184	0.69%

149	16755	1.03%
150	74768	4.61%
151	890003	54.84%

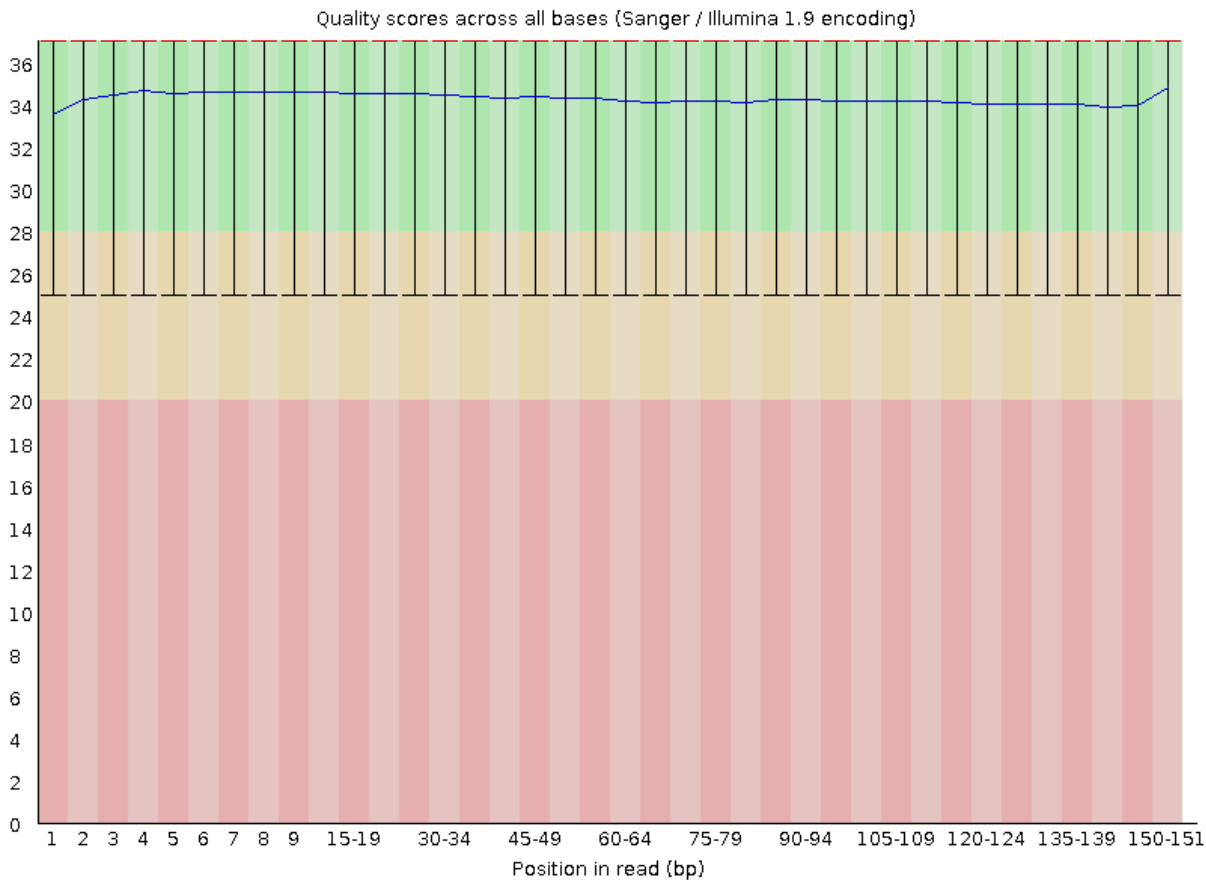
<Graphic distribution of read lengths>

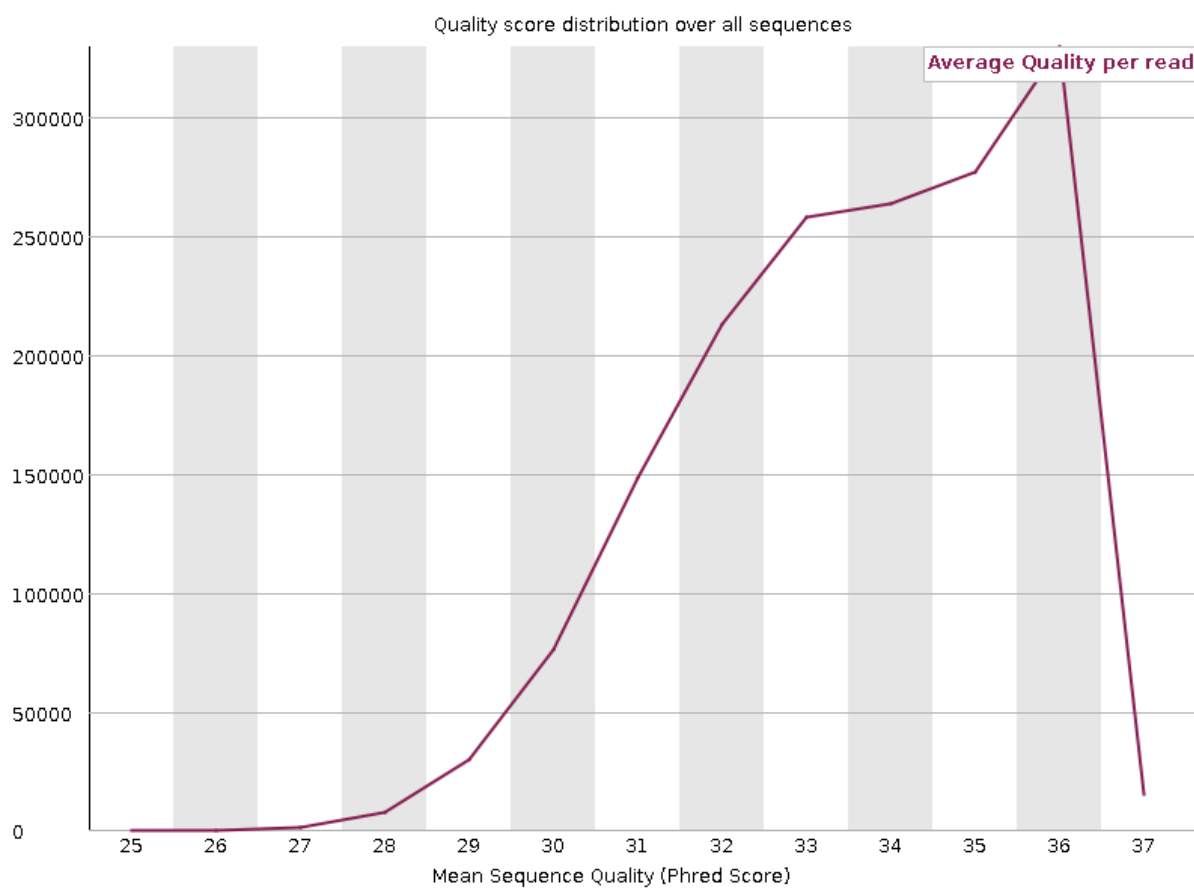
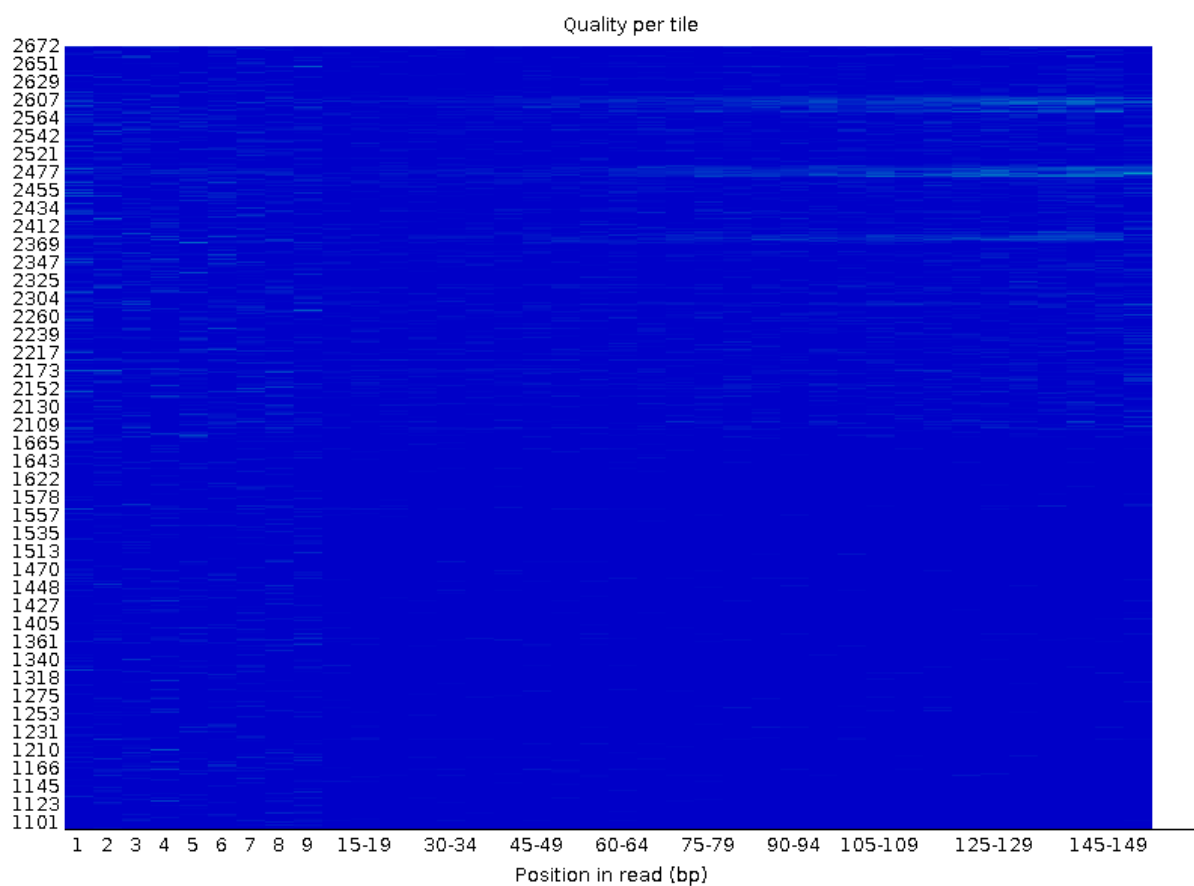
Length	Distribution	Abundance
60	□□□□□□□□□□	0.42%
61	□□□□□□□□□□	0.42%
62	□□□□□□□□□□	0.42%
63	□□□□□□□□□□	0.41%
64	□□□□□□□□□□	0.43%
65	□□□□□□□□□□	0.40%
66	□□□□□□□□□□	0.41%
67	□□□□□□□□□□	0.41%
68	□□□□□□□□□□	0.40%
69	□□□□□□□□□□	0.38%
70	□□□□□□□□□□	0.41%
71	□□□□□□□□□□	0.37%
72	□□□□□□□□□□	0.37%
73	□□□□□□□□□□	0.36%
74	□□□□□□□□□□	0.37%
75	□□□□□□□□□□	0.36%
76	□□□□□□□□□□	0.37%
77	□□□□□□□□□□	0.37%
78	□□□□□□□□□□	0.35%
79	□□□□□□□□□□	0.35%
80	□□□□□□□□□□	0.38%
81	□□□□□□□□□□	0.36%
82	□□□□□□□□□□	0.35%
83	□□□□□□□□□□	0.34%
84	□□□□□□□□□□	0.34%
85	□□□□□□□□□□	0.34%
86	□□□□□□□□□□	0.32%
87	□□□□□□□□□□	0.31%
88	□□□□□□□□□□	0.31%
89	□□□□□□□□□□	0.32%
90	□□□□□□□□□□	0.32%
91	□□□□□□□□□□	0.32%
92	□□□□□□□□□□	0.31%
93	□□□□□□□□□□	0.31%
94	□□□□□□□□□□	0.32%
95	□□□□□□□□□□	0.33%
96	□□□□□□□□□□	0.32%
97	□□□□□□□□□□	0.34%
98	□□□□□□□□□□	0.32%
99	□□□□□□□□□□	0.32%
100	□□□□□□□□□□	0.35%
101	□□□□□□□□□□	0.33%

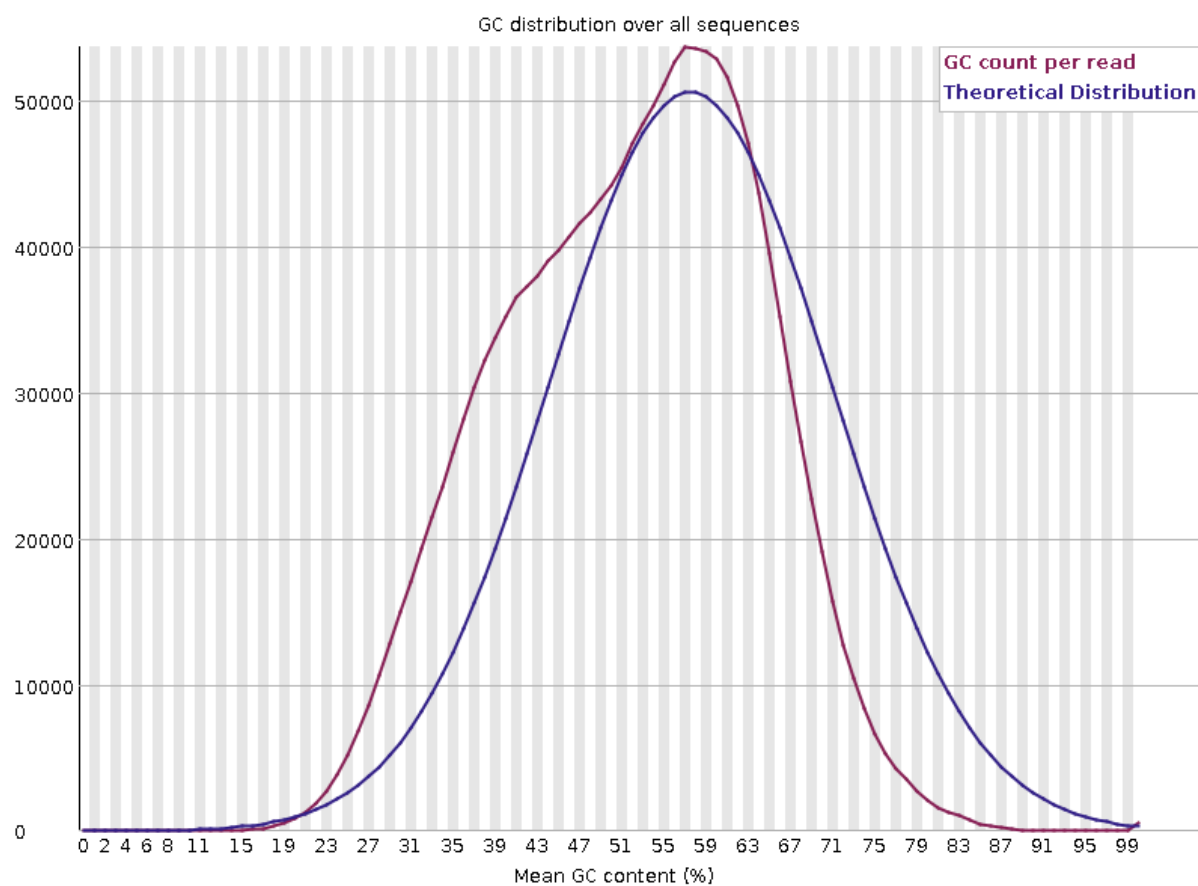
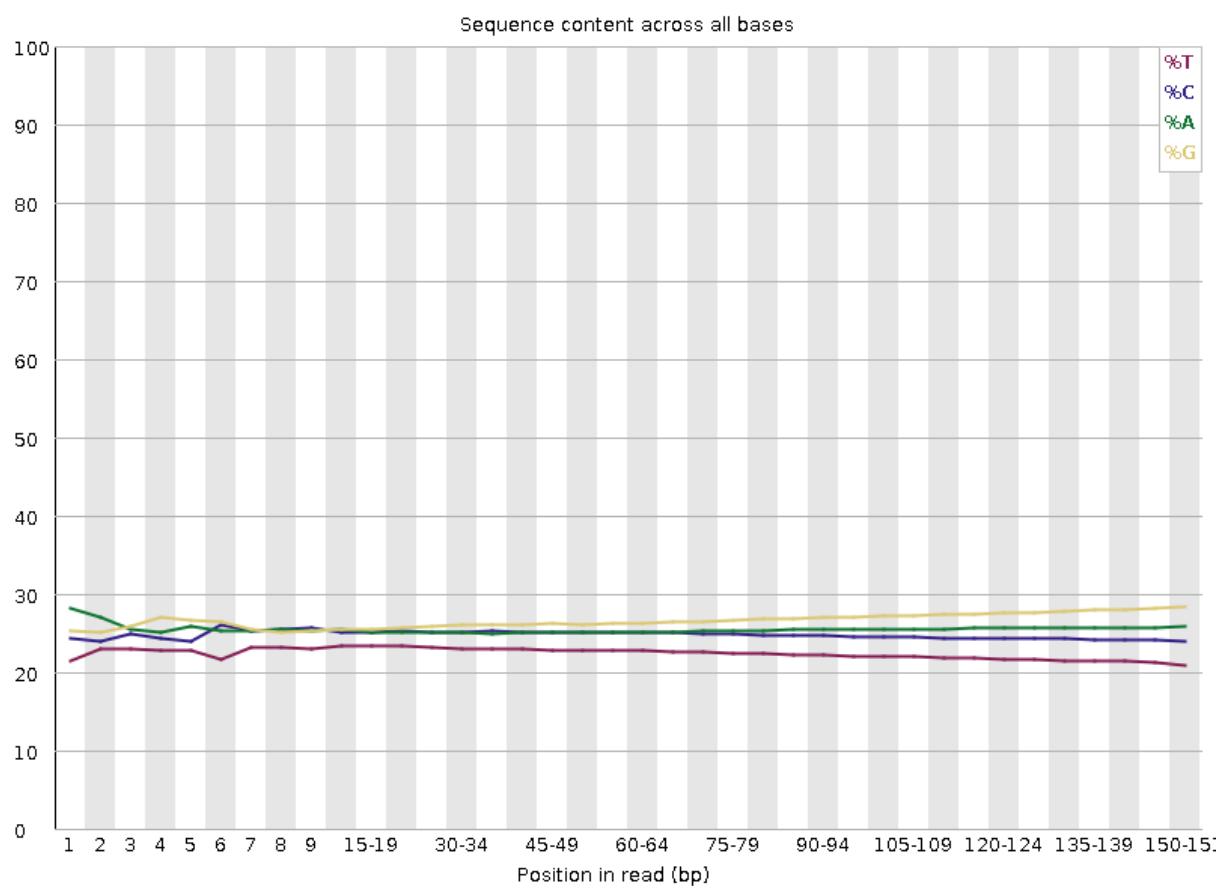
102	□□□□□□□□	0.32%
103	□□□□□□□□	0.33%
104	□□□□□□□□	0.34%
105	□□□□□□□□	0.33%
106	□□□□□□□□	0.34%
107	□□□□□□□□	0.34%
108	□□□□□□□□	0.34%
109	□□□□□□□□	0.35%
110	□□□□□□□□	0.34%
111	□□□□□□□□	0.36%
112	□□□□□□□□	0.38%
113	□□□□□□□□	0.38%
114	□□□□□□□□	0.40%
115	□□□□□□□□	0.40%
116	□□□□□□□□	0.41%
117	□□□□□□□□	0.43%
118	□□□□□□□□	0.44%
119	□□□□□□□□	0.46%
120	□□□□□□□□	0.47%
121	□□□□□□□□	0.49%
122	□□□□□□□□	0.52%
123	□□□□□□□□	0.53%
124	□□□□□□□□	0.55%
125	□□□□□□□□	0.54%
126	□□□□□□□□	0.56%
127	□□□□□□□□	0.56%
128	□□□□□□□□	0.57%
129	□□□□□□□□	0.58%
130	□□□□□□□□	0.58%
131	□□□□□□□□	0.59%
132	□□□□□□□□	0.63%
133	□□□□□□□□	0.61%
134	□□□□□□□□	0.62%
135	□□□□□□□□	0.62%
136	□□□□□□□□	0.65%
137	□□□□□□□□	0.65%
138	□□□□□□□□	0.66%
139	□□□□□□□□	0.66%
140	□□□□□□□□	0.67%
141	□□□□□□□□	0.68%
142	□□□□□□□□	0.69%
143	□□□□□□□□	0.70%
144	□□□□□□□□	0.68%
145	□□□□□□□□	0.67%
146	□□□□□□□□	0.69%
147	□□□□□□□□	0.69%
148	□□□□□□□□	0.69%
149	□□□□□□□□	1.03%

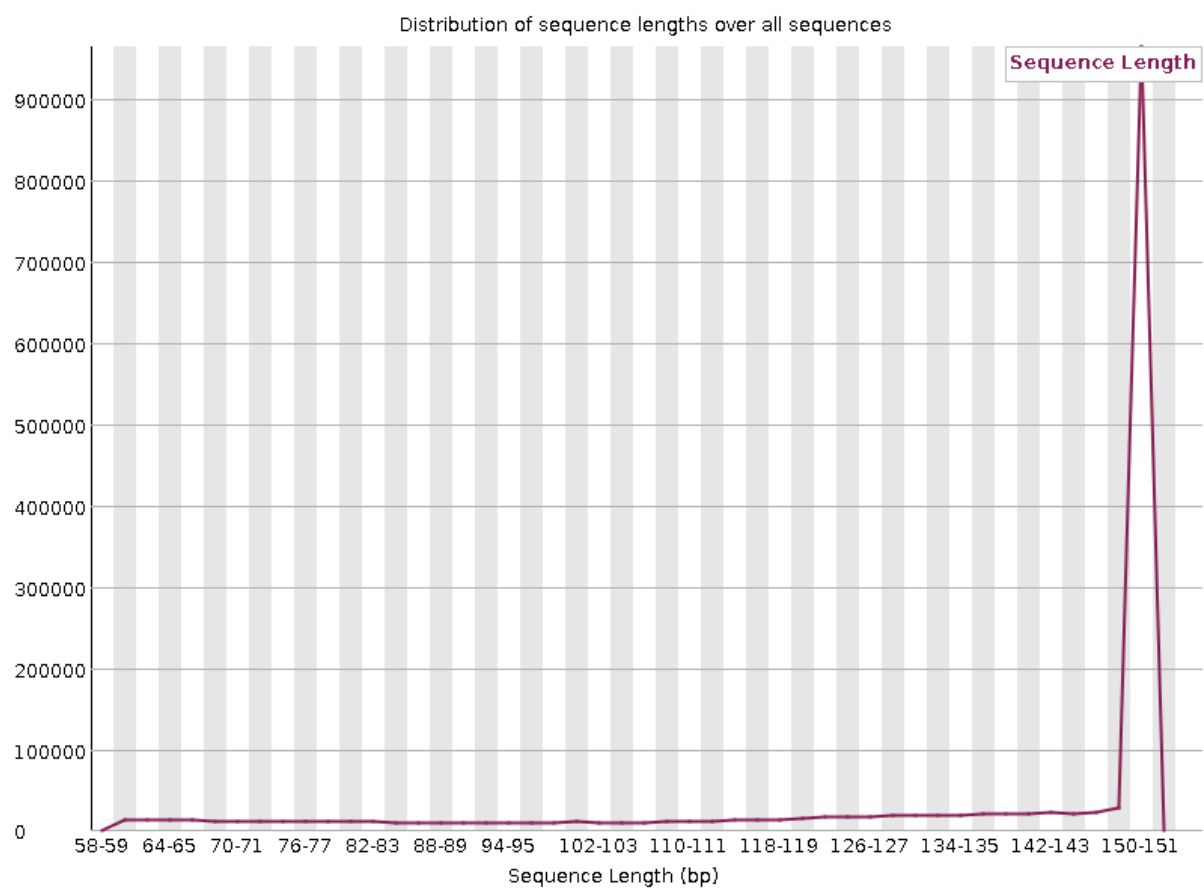
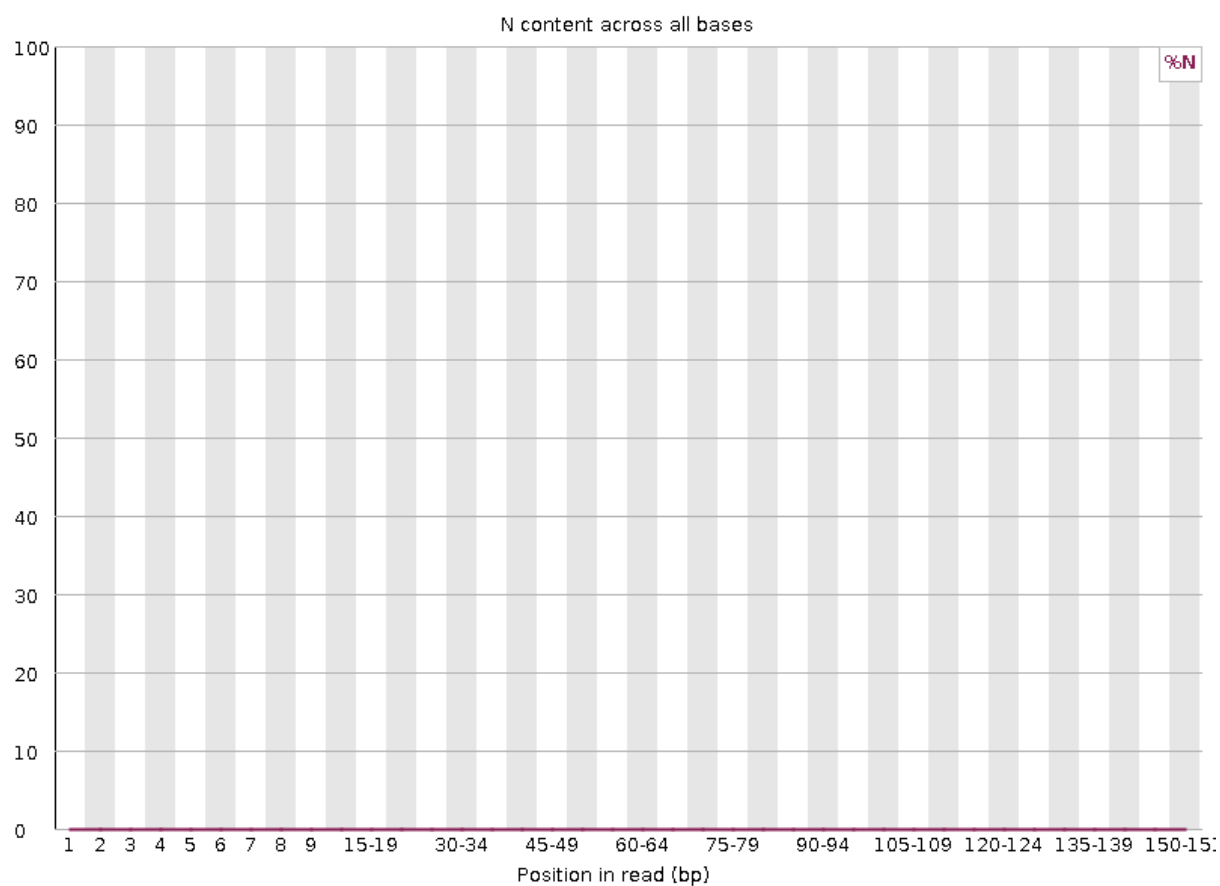
150 □□□□□□□□ 4.61%
151 ■■■■■□□□□ 54.84%

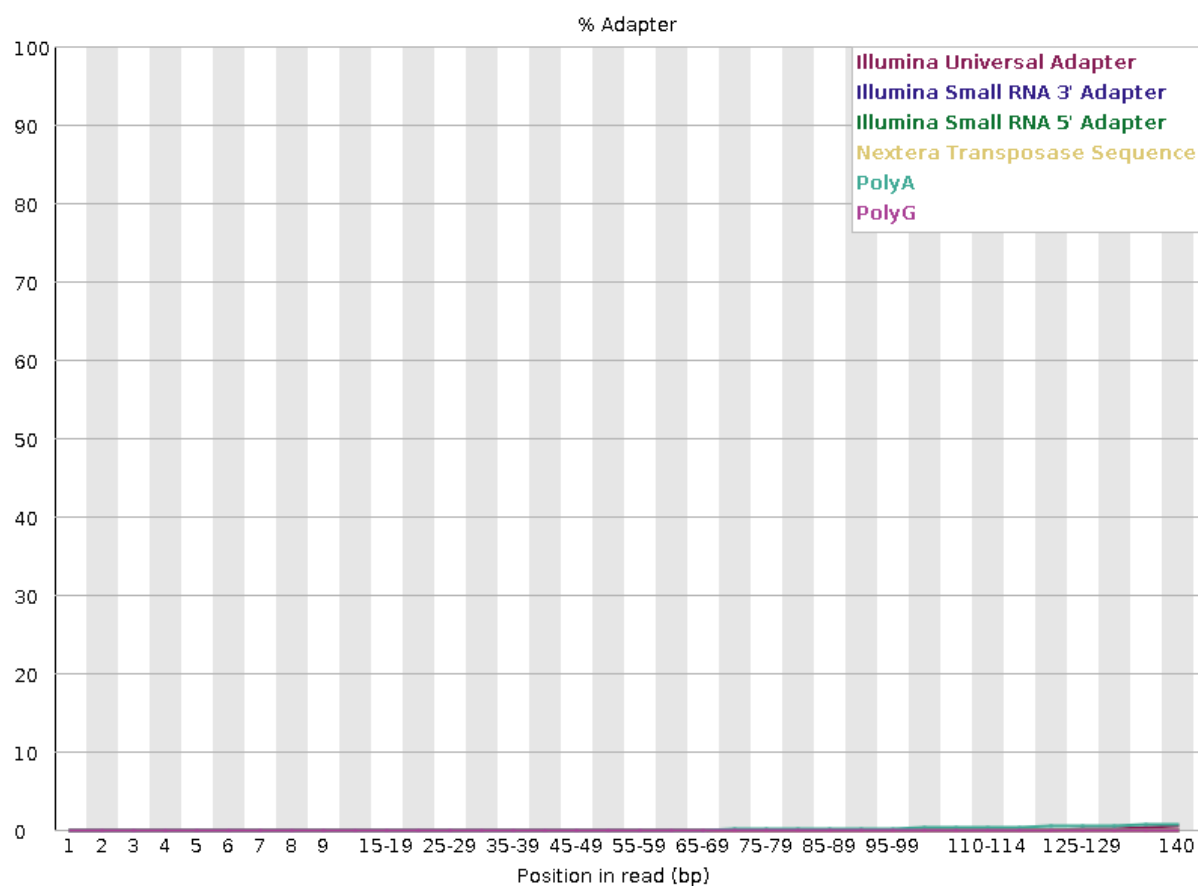
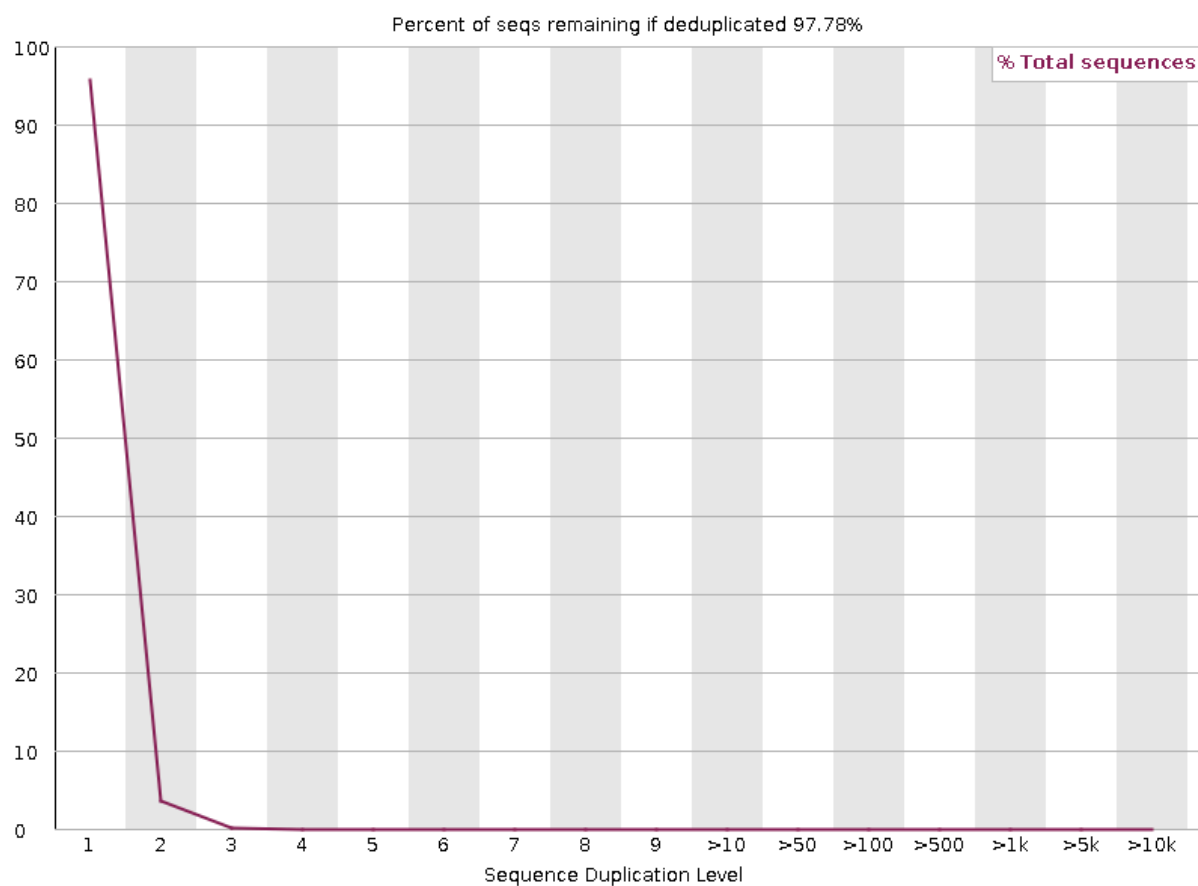
SRR37843240_2.up.fastq	
Measure	Value
Filename	SRR37843240_2.up.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1,622,948
Total Bases	218.4 Mbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51











No overrepresented sequences in all of SRR37843240 fastq.
Adapter Content는 모두 Warning 문제가 해결되었다. 이 과정에서 Sequence Length Distribution에 Warning이 표시되었지만 이는 자연스러운 현상이다.
Per sequence GC content의 경우 1.up file에서 Warning, 나머지 모든 file에서 Fail이 표시되었다. 해당 데이터가 생성된 Study가 WES(Whole Exome Sequencing)이기 때문에 문제되지 않는다고 판단했다.
따라서 trimming options를 재조정하거나 추가 trimming을 거치지 않아도 이후 downstream 분석을 위한 준비는 충분하다고 볼 수 있다.

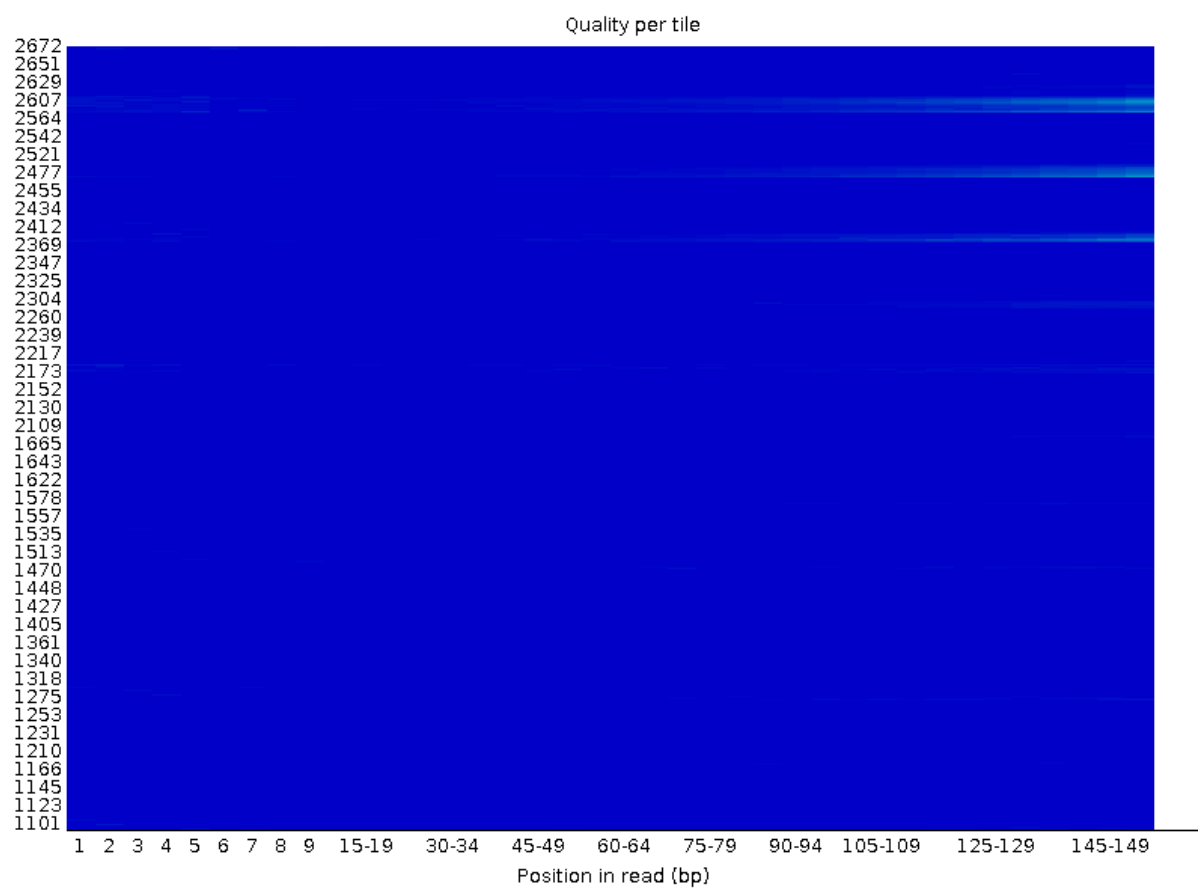
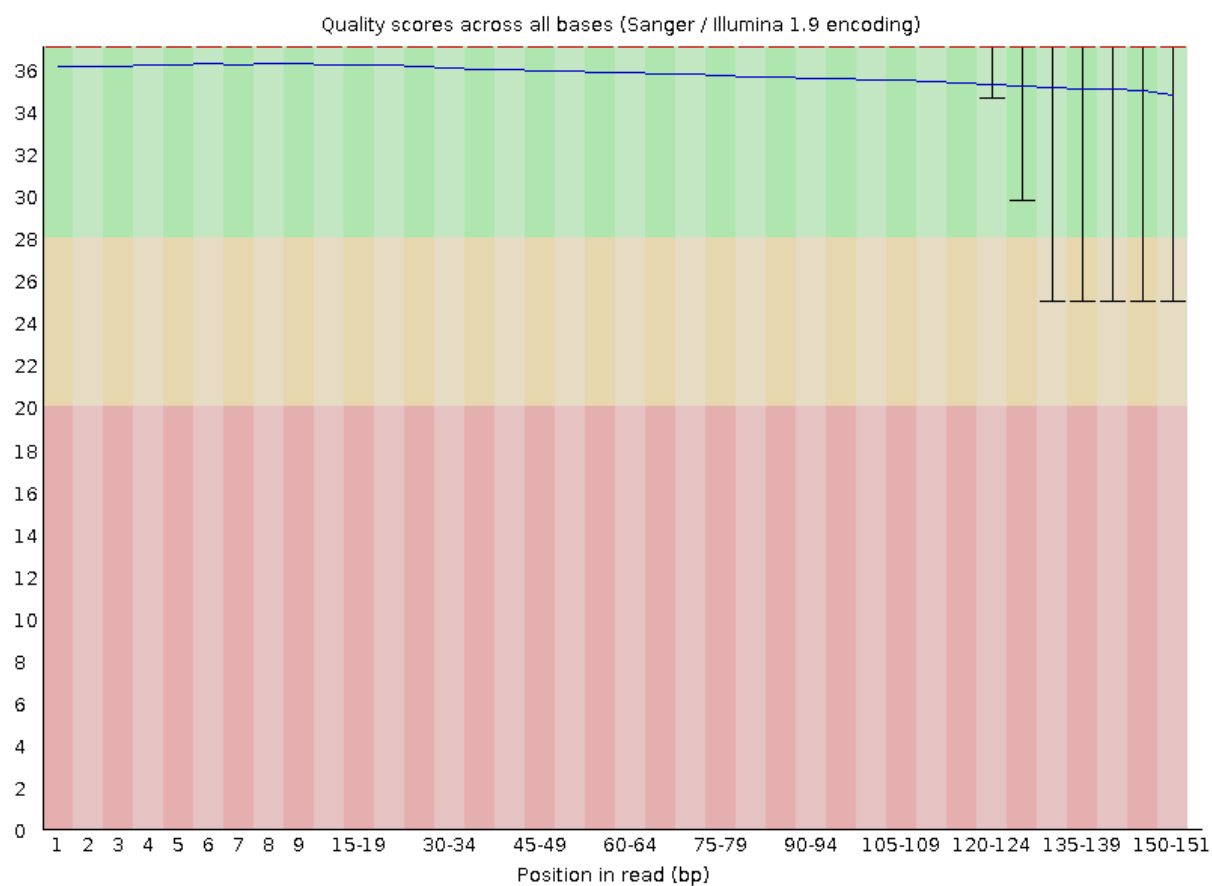
Description of Data

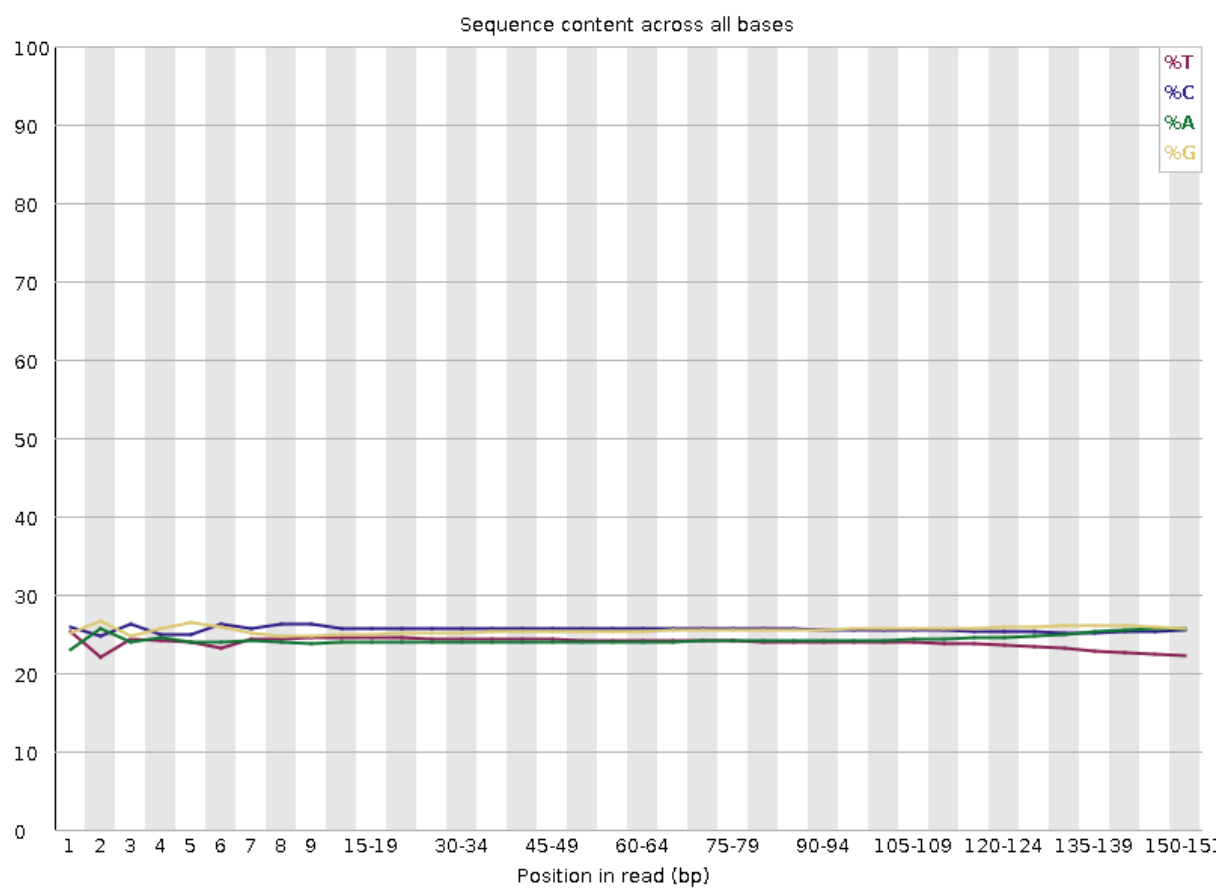
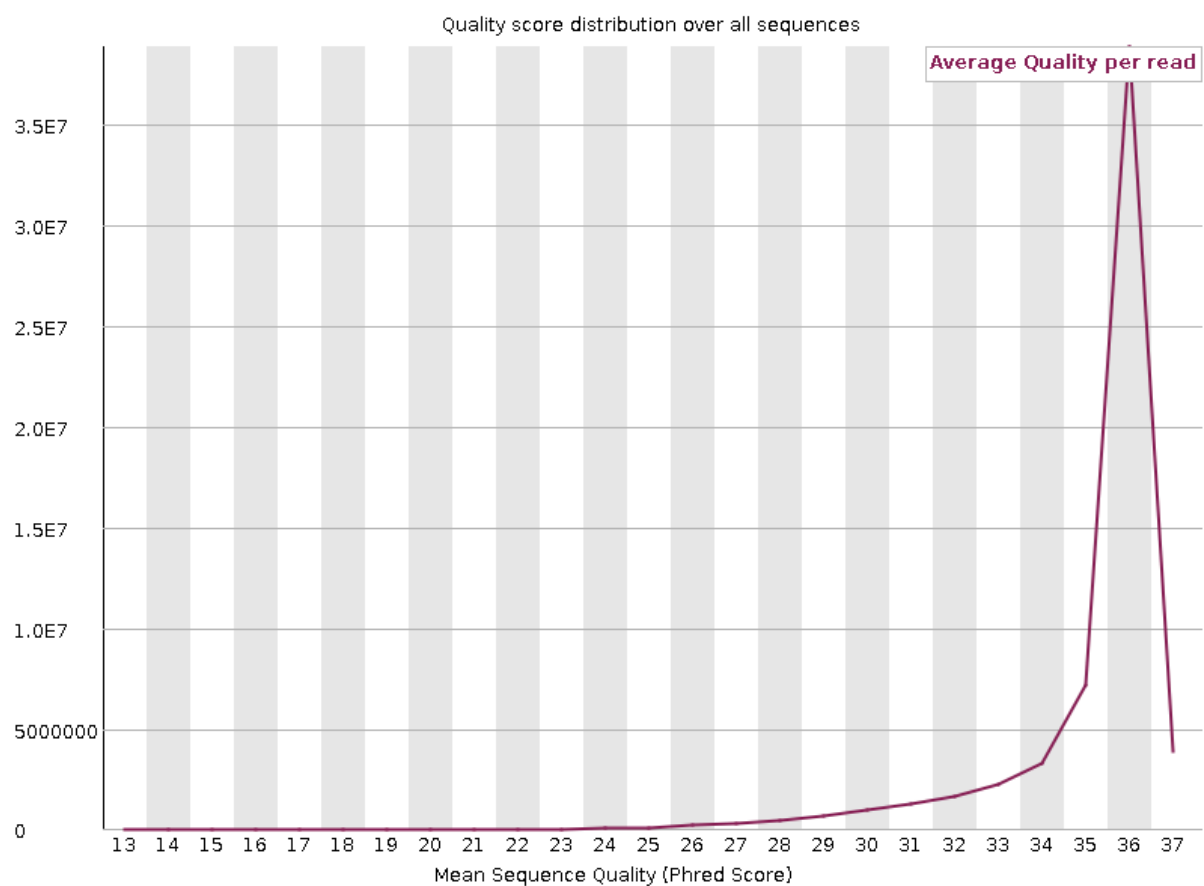
The number of read sequences is 61666767
Total length of read sequences is 9311681817

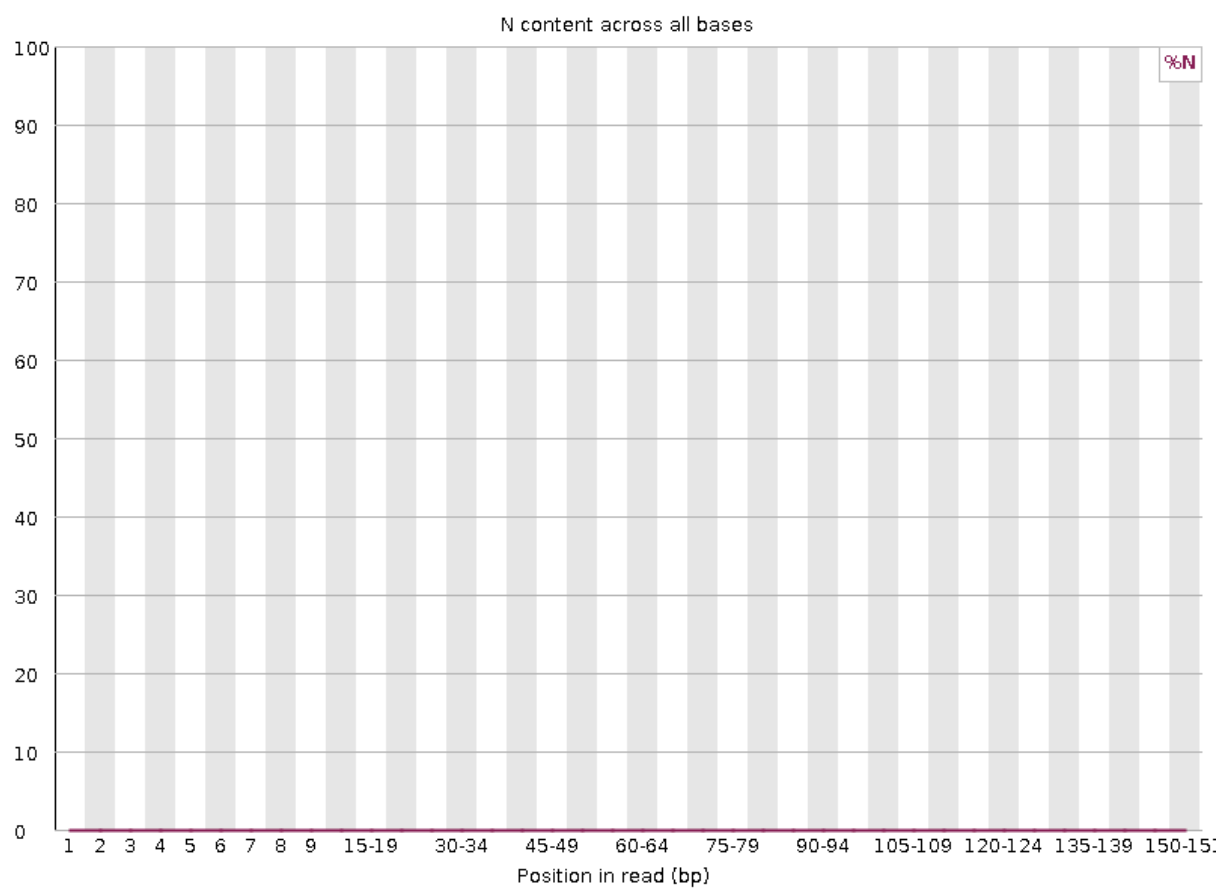
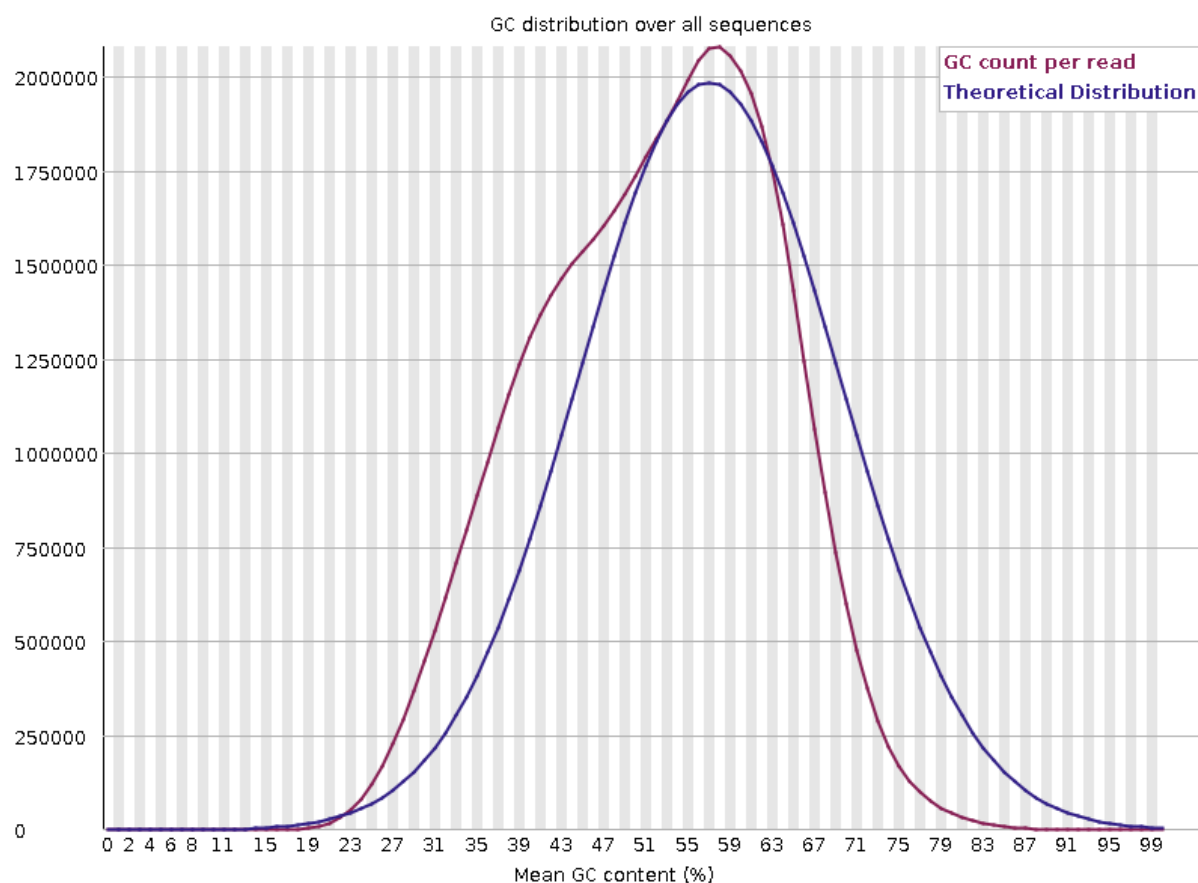
Length	Count	Abundance
151	61666767	100.00%

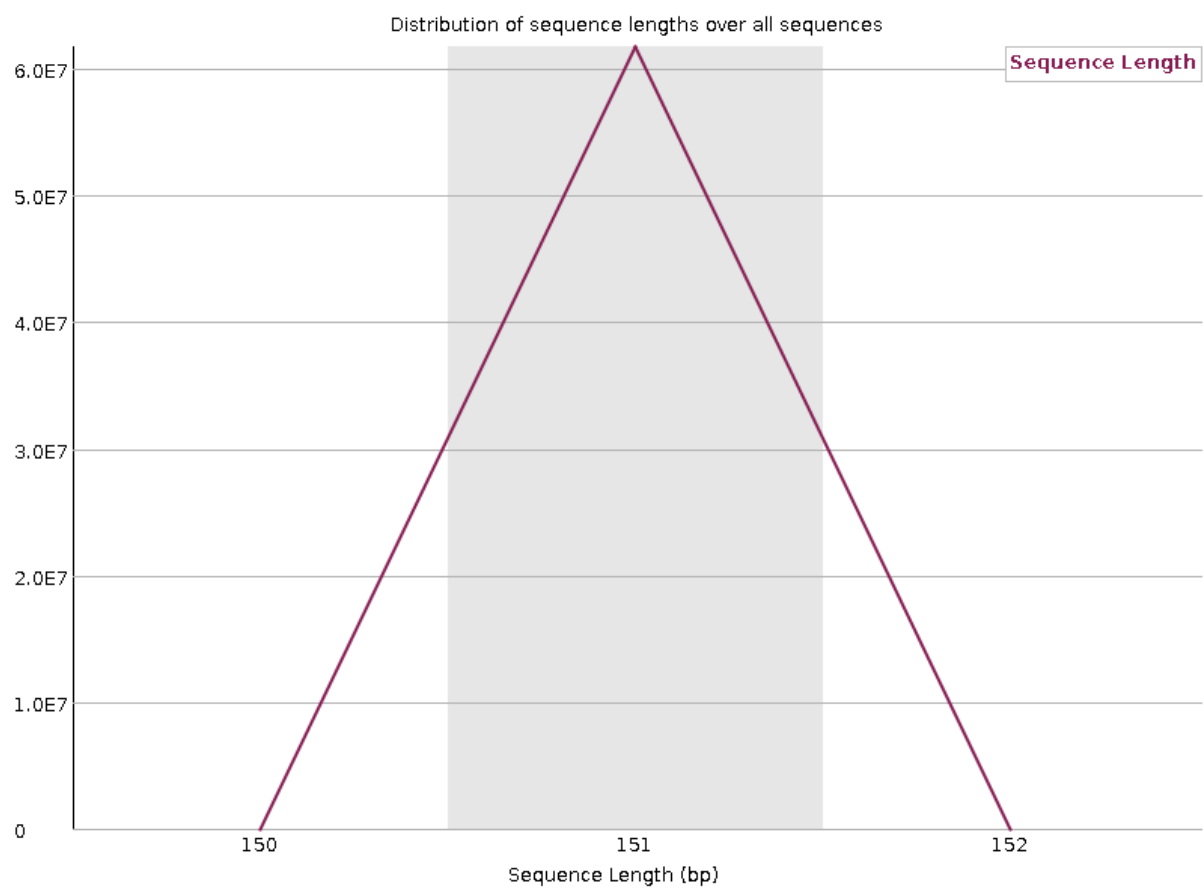
Length	Distribution	Abundance
151	■■■■■■■■■■	100.00%

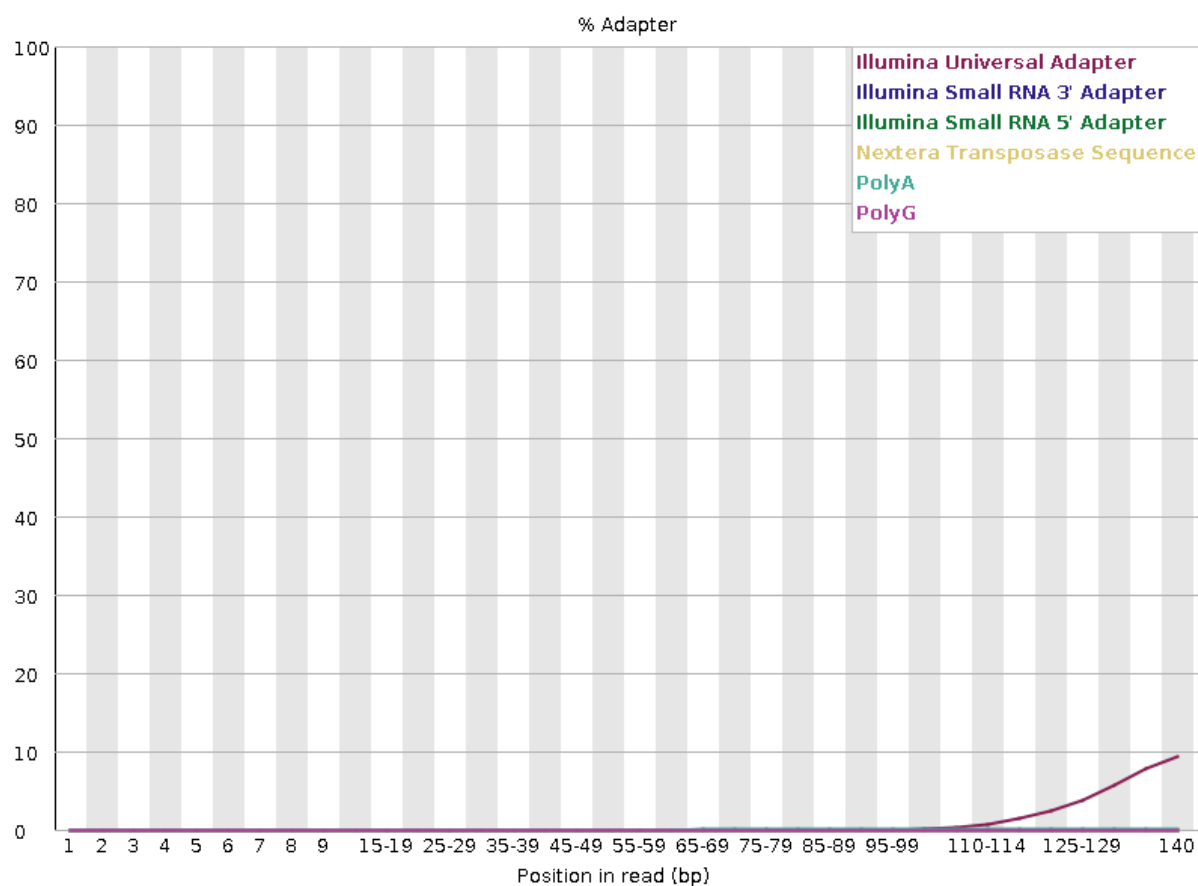
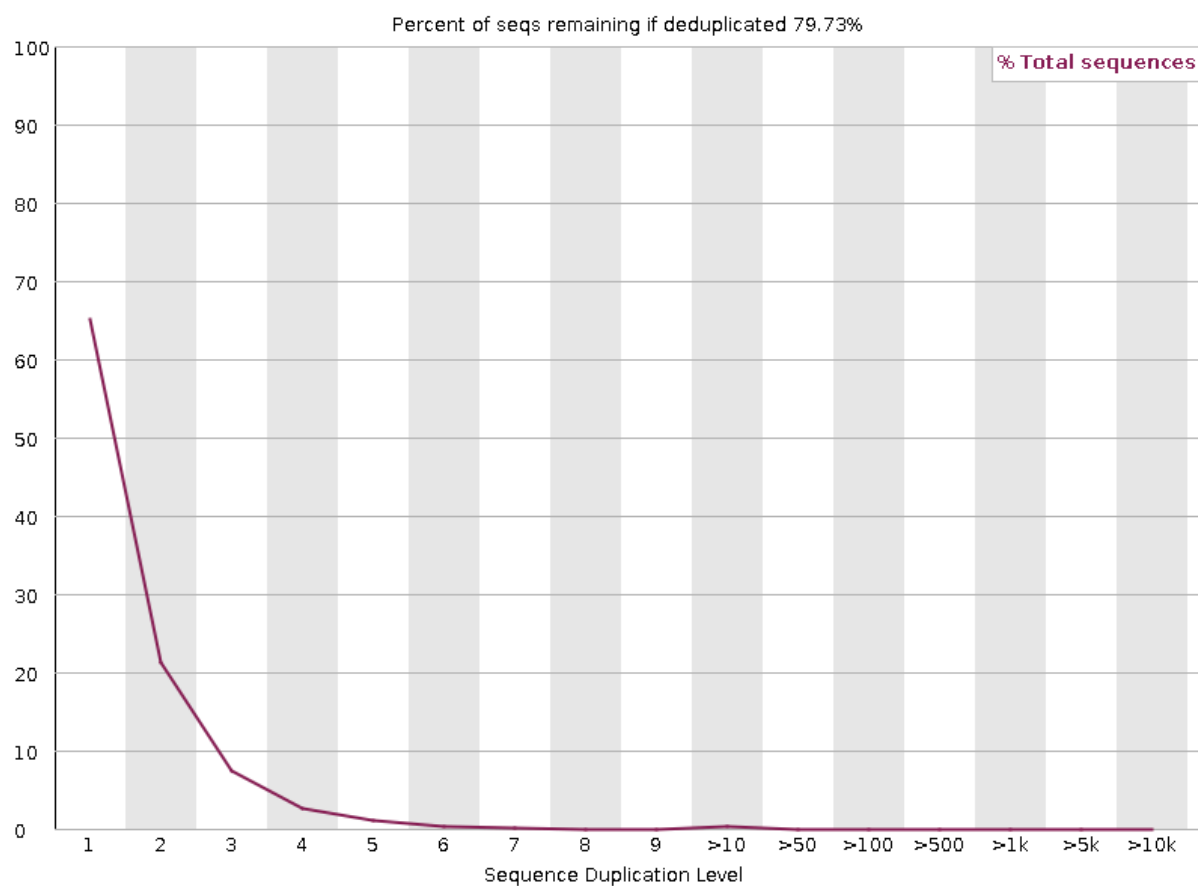
SRR37843242_1.fastq.gz	
Measure	Value
Filename	SRR37843242_1.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	61,666,767
Total Bases	9.3 Gbp
Sequences flagged as poor quality	0
Sequence length	151
%GC	51











SRR37843242_2.fastq

The number of read sequences is 61666767

Total length of read sequences is 9311681817

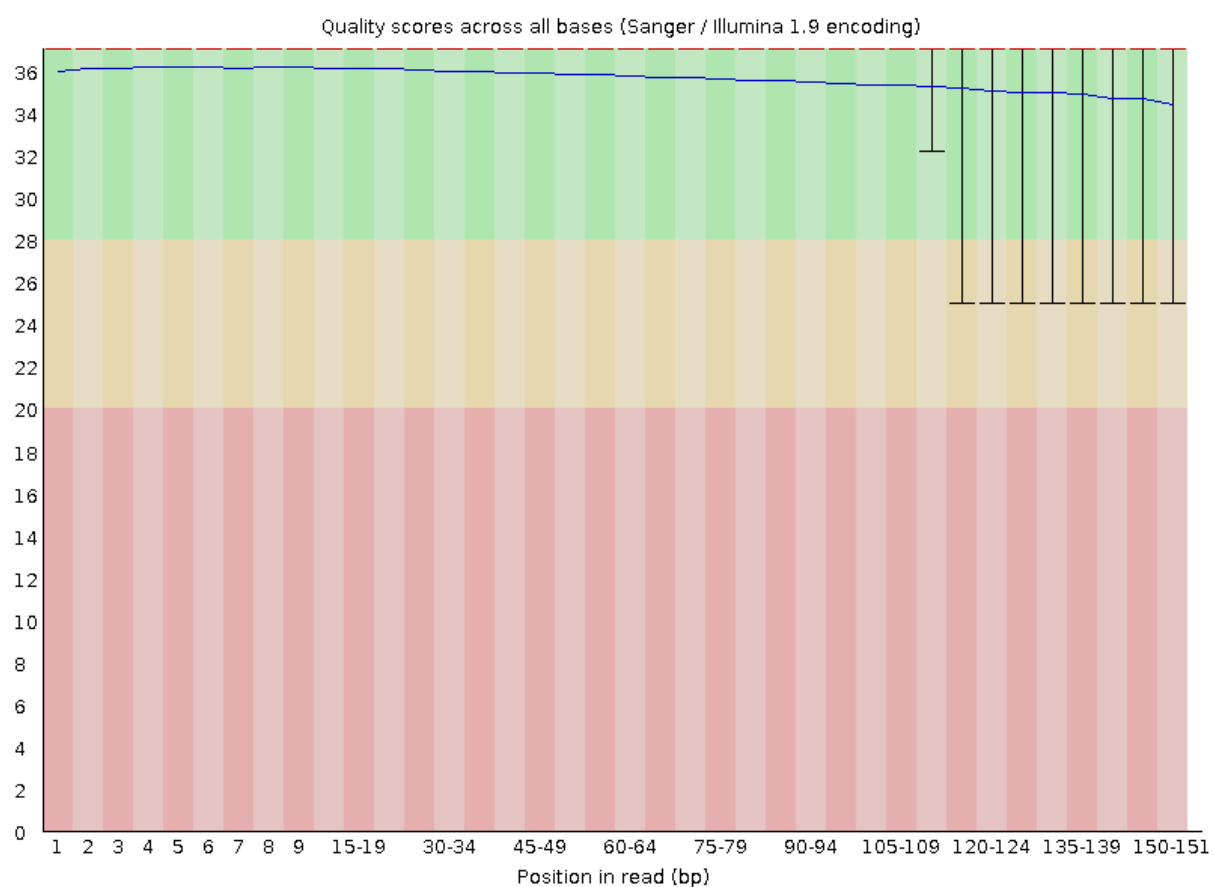
<Text distribution of read lengths>

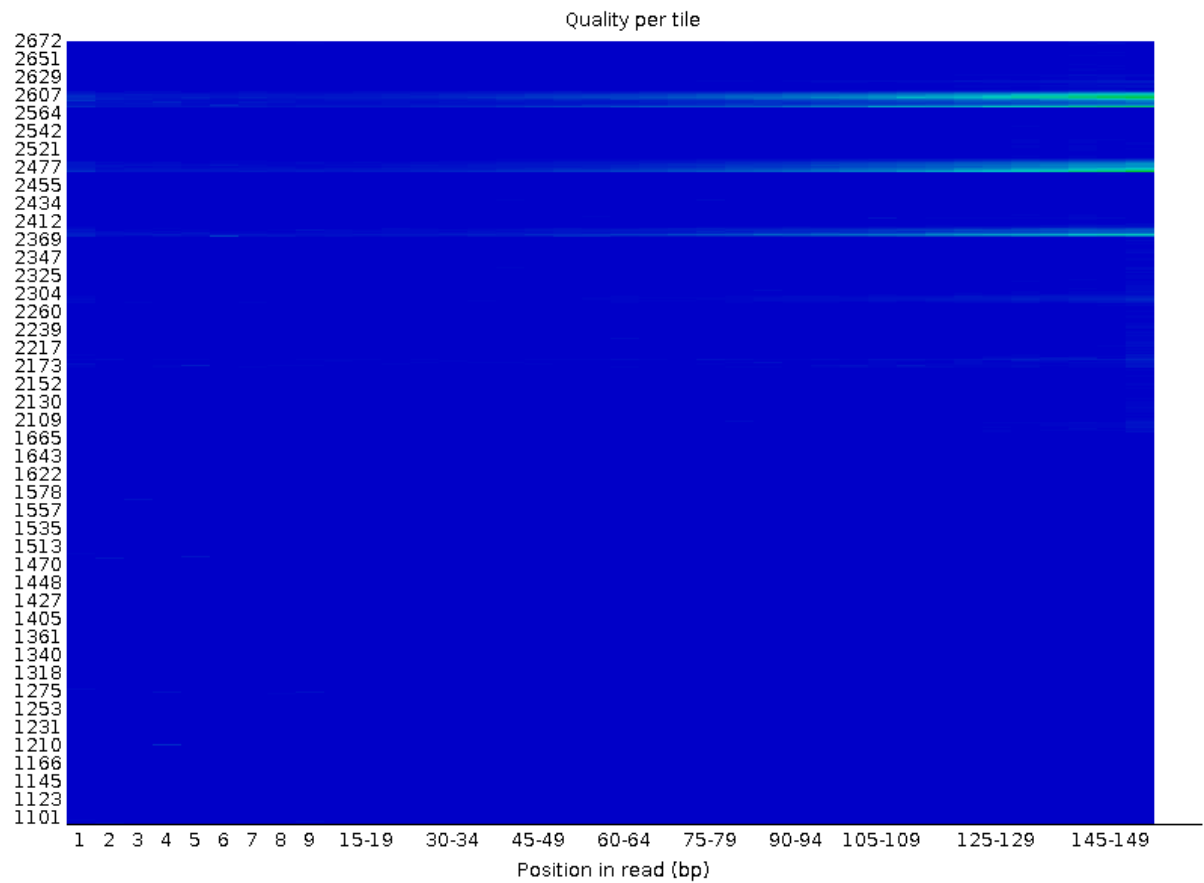
Length	Count	Abundance
151	61666767	100.00%

<Graphic distribution of read lengths>

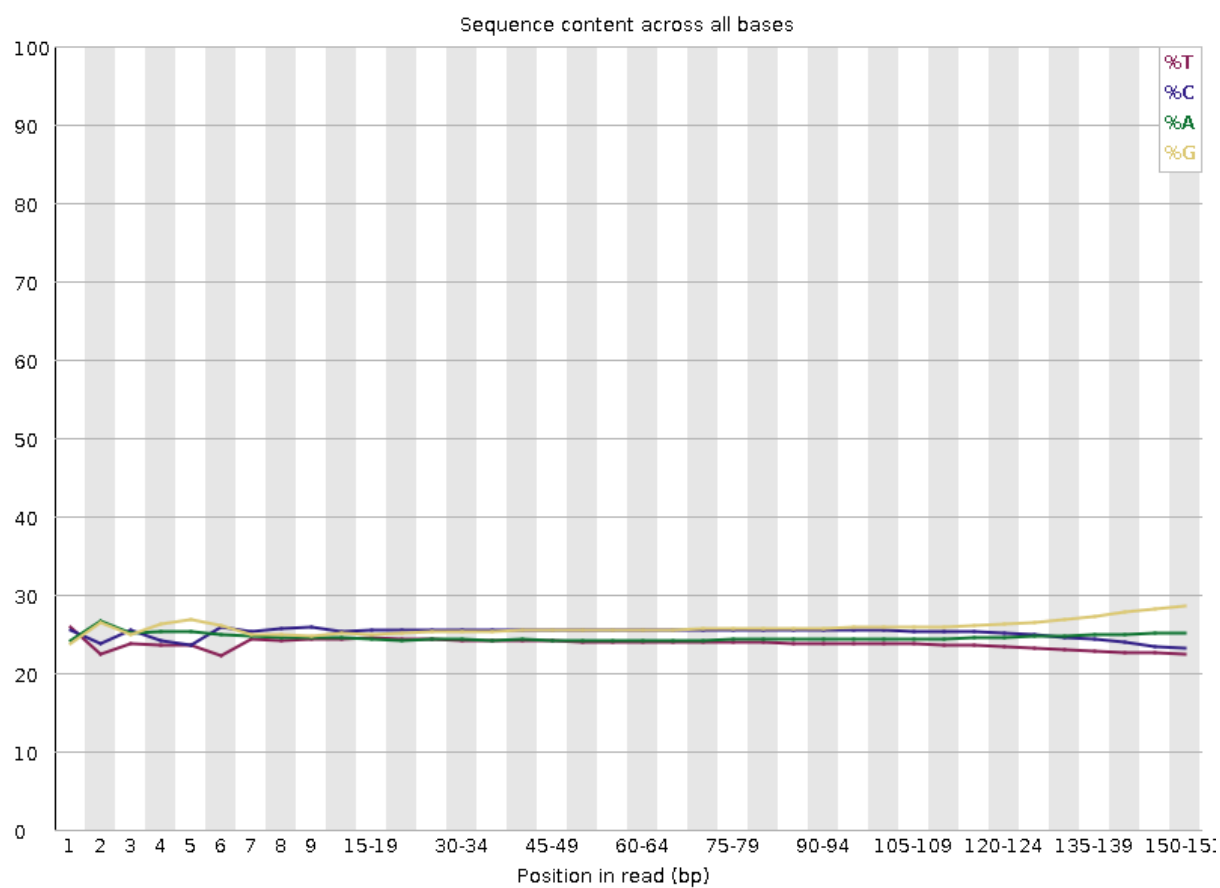
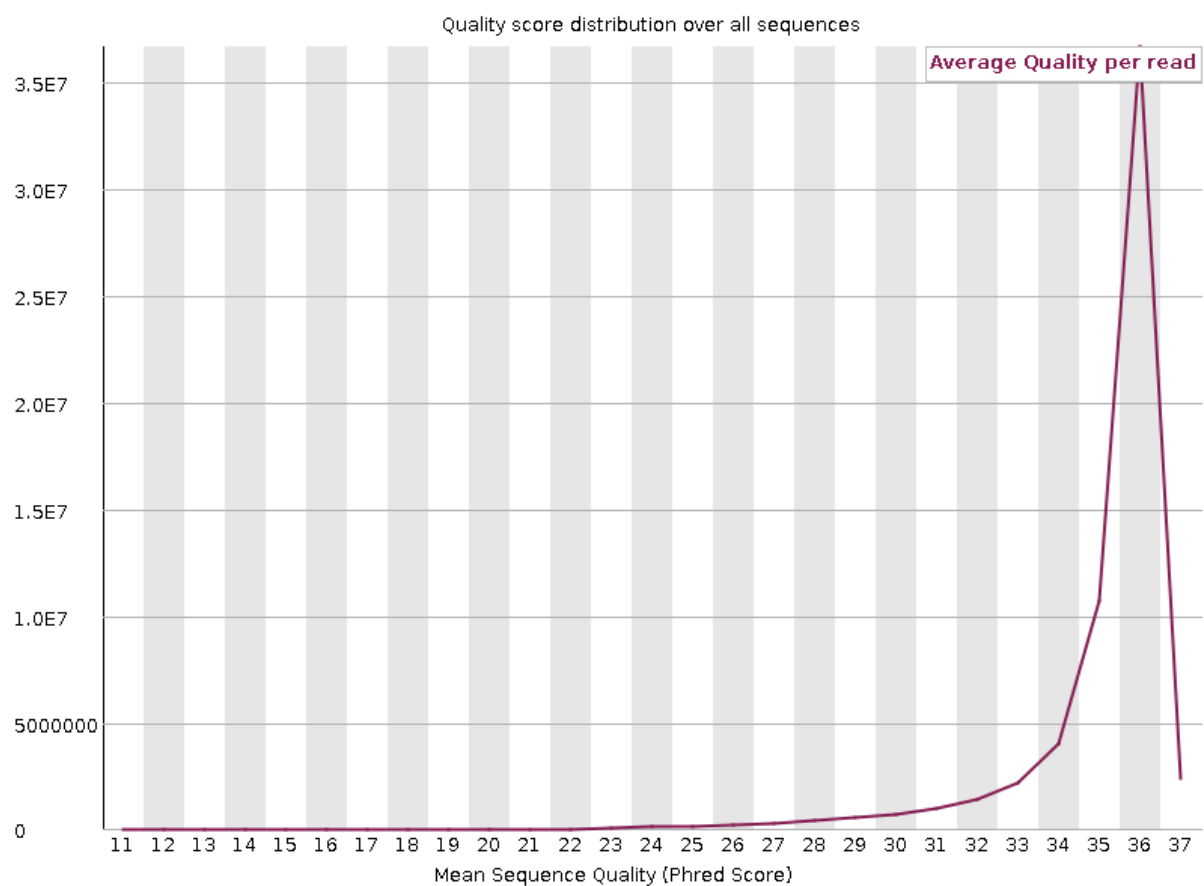
Length	Distribution	Abundance
151	■■■■■■■■■■	100.00%

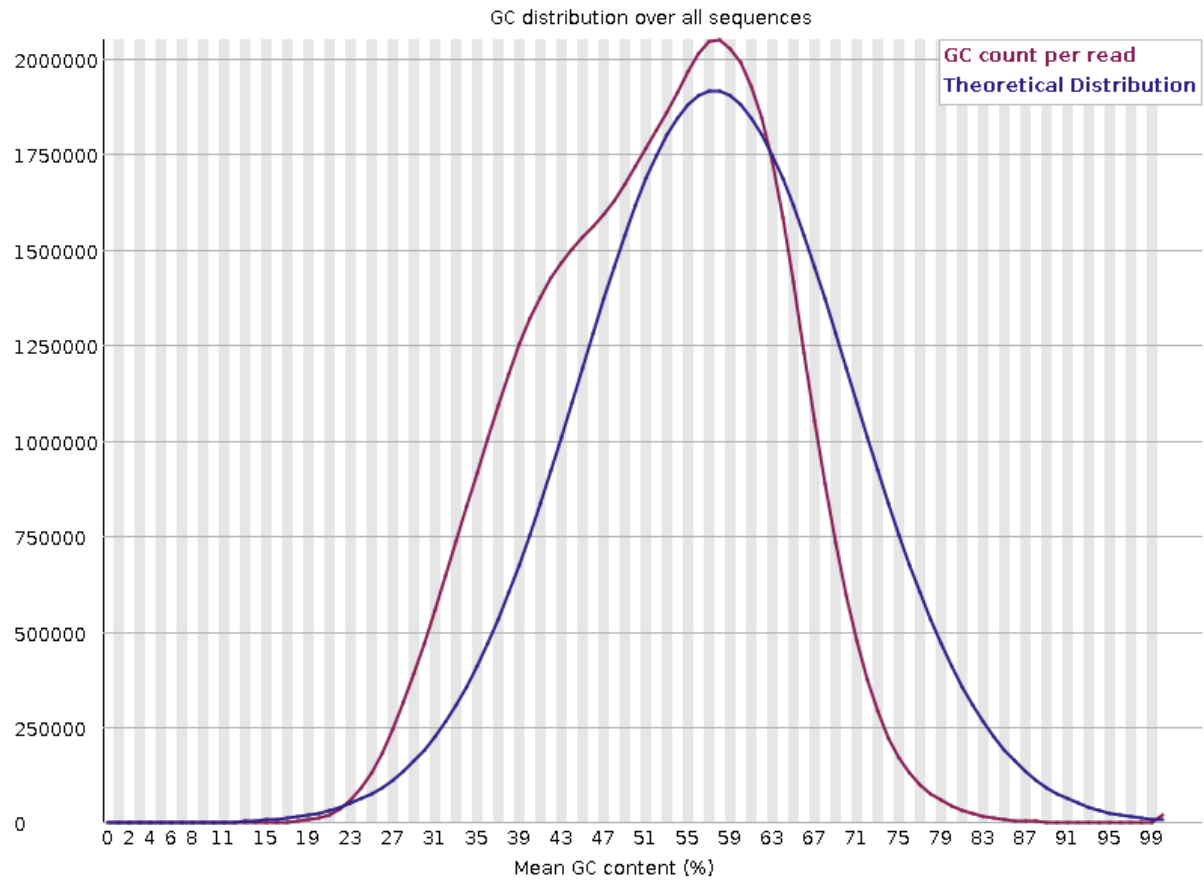
SRR37843242_2.fastq.gz	
Measure	Value
Filename	SRR37843242_2.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	61,666,767
Total Bases	9.3 Gbp
Sequences flagged as poor quality	0
Sequence length	151
%GC	51



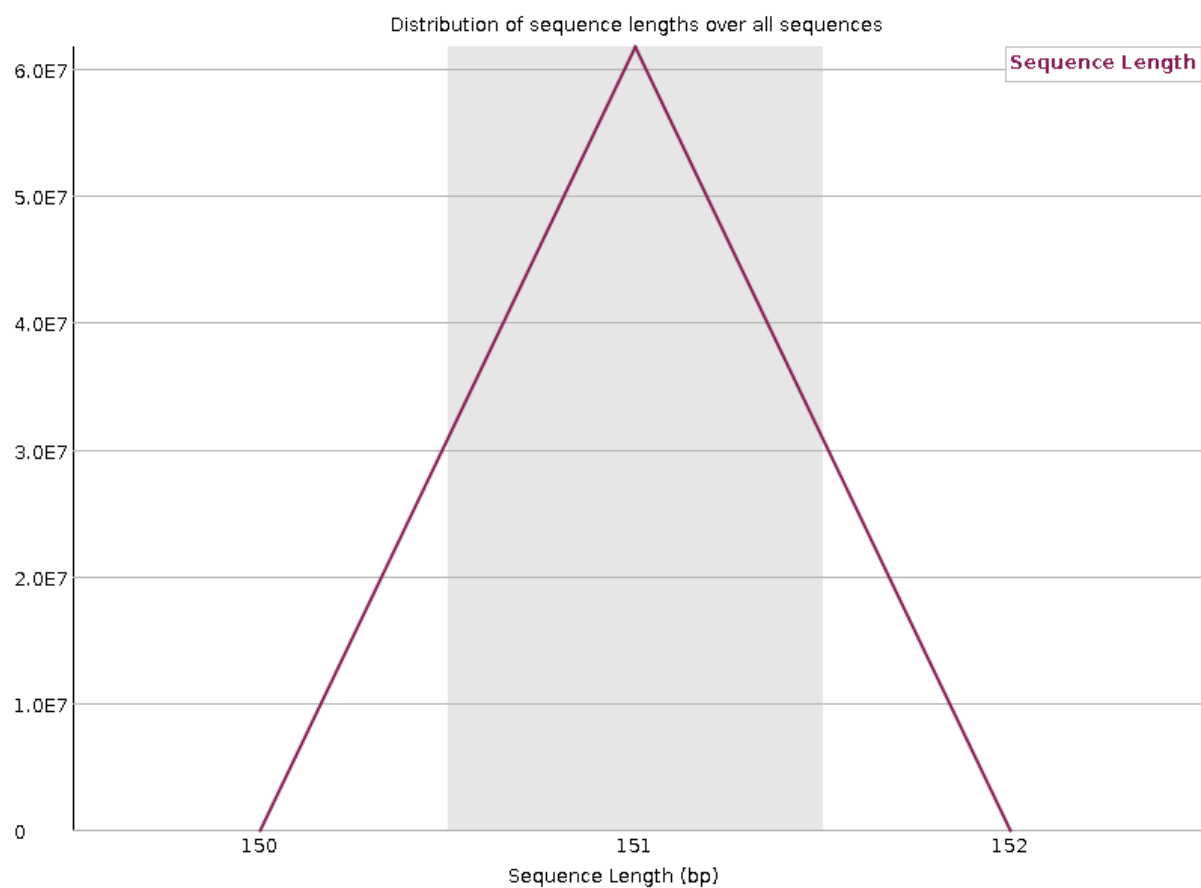
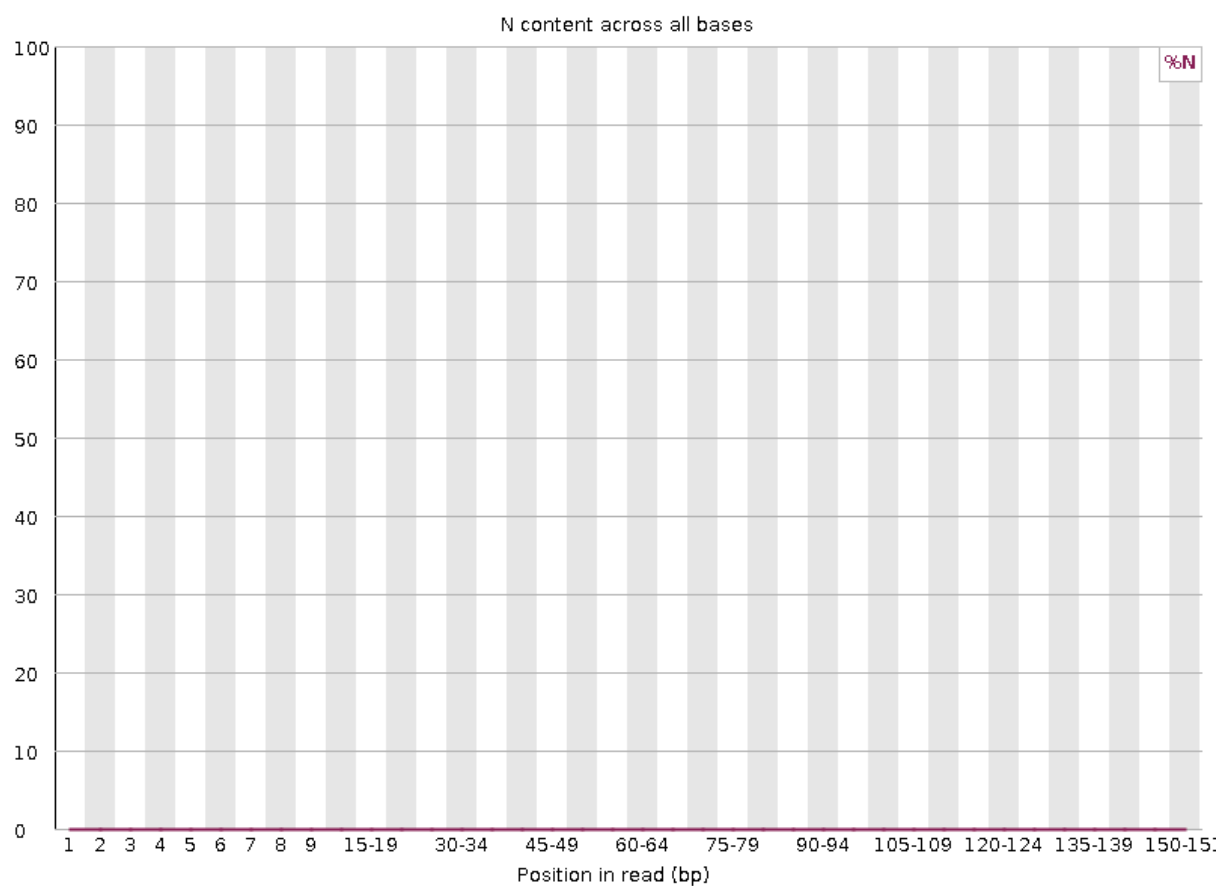


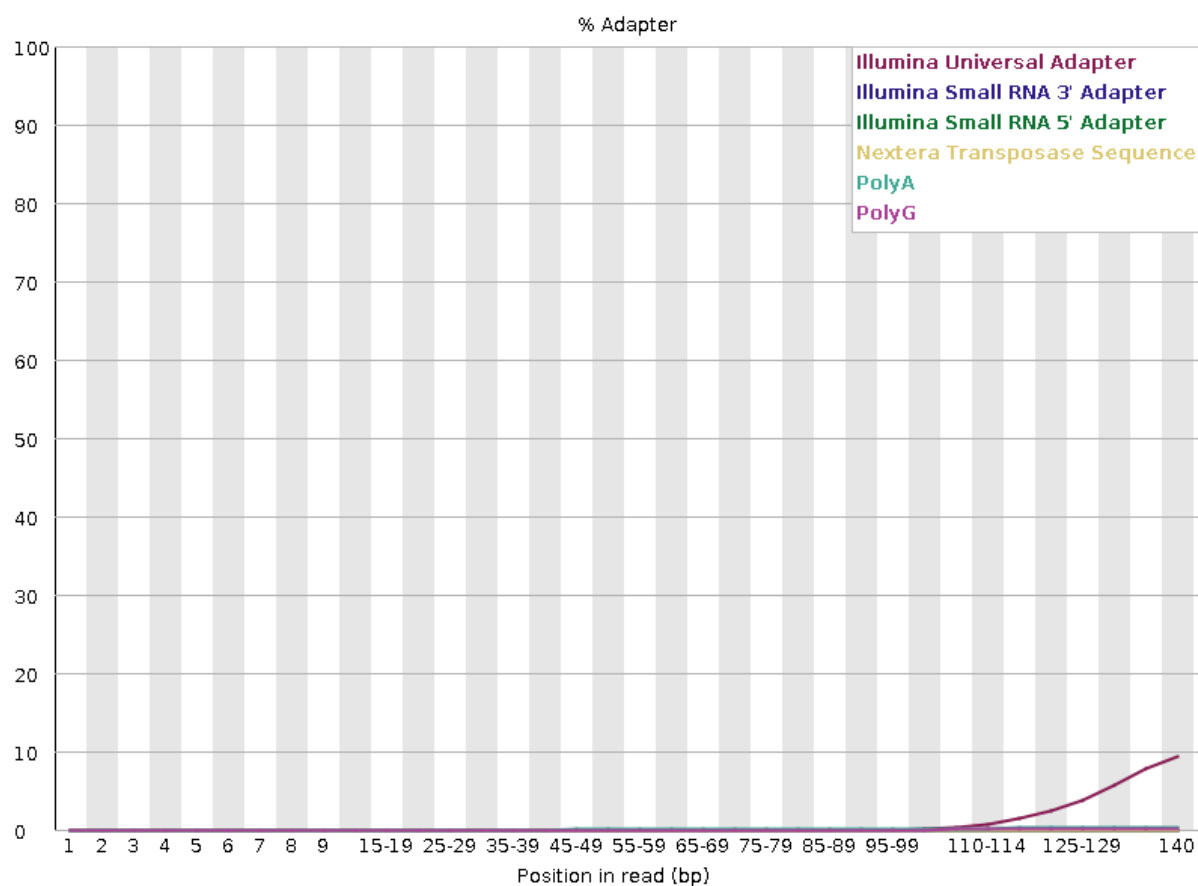
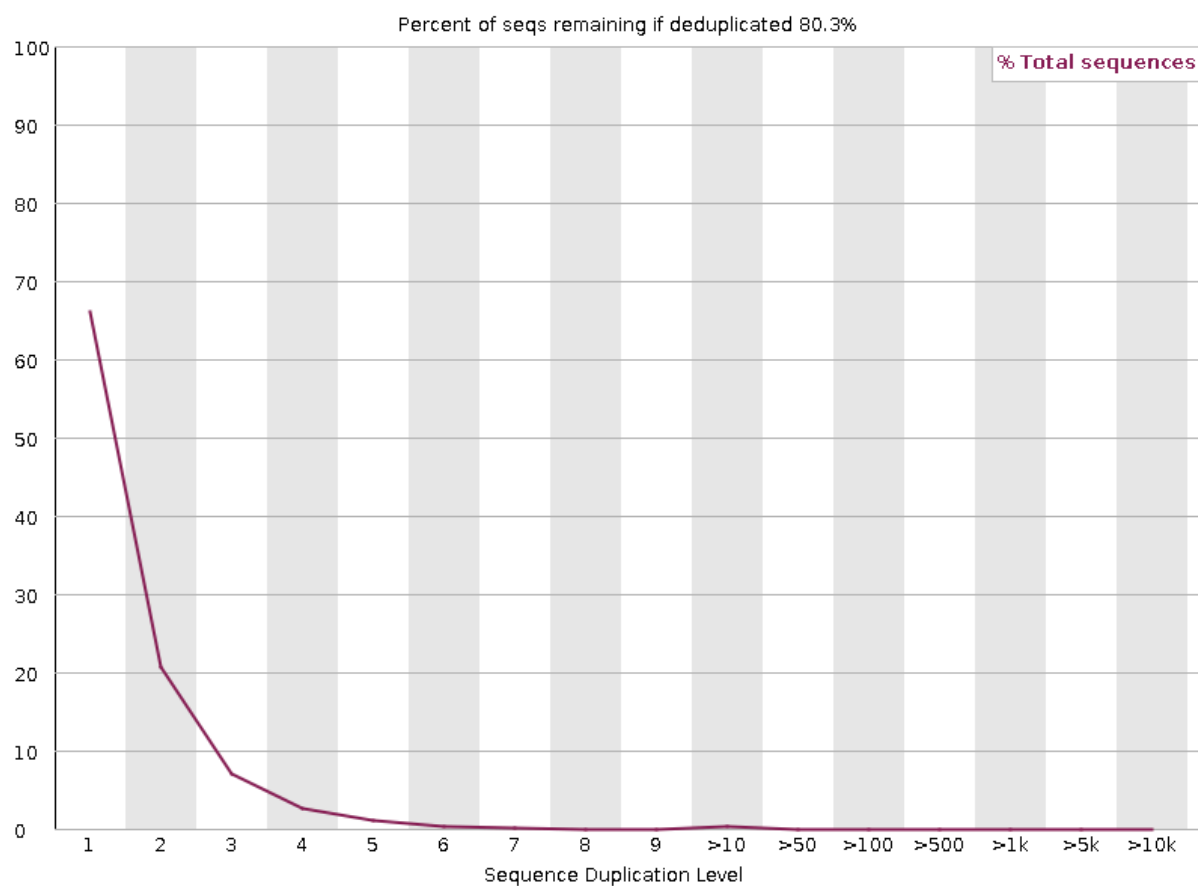
Warning으로 표시되었음. 이는 read 끝부분에 대한 Trimming을 더 엄격하게 진행하는 근거가 되었음.





Failure로 표시되었지만 앞에서와 같은 이유로 문제 삼지 않았음.





Overrepresented sequences

No overrepresented sequences in both 1 and 2.

After Trimming

<log>

TrimmomaticPE: Started with arguments:

SRR37843242_1.fastq.gz SRR37843242_2.fastq.gz SRR37843242_1.p.fastq.gz

SRR37843242_1.up.fastq.gz SRR37843243_2.p.fastq.gz SRR37843242_2.up.fastq.gz

ILLUMINACLIP:/mss/home_student/202112379/programs/Trimmomatic-0.39/adapters/TruSeq3-PE.

fa:2:30:10 SLIDINGWINDOW:4:20 LEADING:10 TRAILING:10 MINLEN:60

Using PrefixPair: 'TACACTCTTCCCTACACGACGCTCTTCCGATCT' and

'GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT'

ILLUMINACLIP: Using 1 prefix pairs, 0 forward/reverse sequences, 0 forward only sequences, 0 reverse only sequences

Quality encoding detected as phred33

Input Read Pairs: 61666767 **Both Surviving: 50307973 (81.58%)** Forward Only Surviving:

8677068 (14.07%) Reverse Only Surviving: 1508577 (2.45%) Dropped: 1173149 (1.90%)

TrimmomaticPE: Completed successfully

SRR37843242_1.p.fastq

The number of read sequences is 50307973

Total length of read sequences is 7470949395

<Text distribution of read lengths>

Length	Count	Abundance
60	29374	0.06%
61	28965	0.06%
62	28802	0.06%
63	29367	0.06%
64	28440	0.06%
65	29796	0.06%
66	28526	0.06%
67	29122	0.06%
68	28764	0.06%
69	28853	0.06%
70	28415	0.06%
71	28677	0.06%
72	28295	0.06%
73	28238	0.06%
74	28312	0.06%
75	28260	0.06%
76	28588	0.06%
77	27660	0.05%

78	29566	0.06%
79	29614	0.06%
80	29533	0.06%
81	29513	0.06%
82	30427	0.06%
83	28723	0.06%
84	28960	0.06%
85	28599	0.06%
86	28921	0.06%
87	28688	0.06%
88	28648	0.06%
89	28382	0.06%
90	29537	0.06%
91	28179	0.06%
92	28450	0.06%
93	28178	0.06%
94	29215	0.06%
95	27885	0.06%
96	27848	0.06%
97	29435	0.06%
98	27791	0.06%
99	28418	0.06%
100	29143	0.06%
101	29441	0.06%
102	30247	0.06%
103	29640	0.06%
104	29573	0.06%
105	28781	0.06%
106	28582	0.06%
107	29765	0.06%
108	30151	0.06%
109	29825	0.06%
110	29050	0.06%
111	29532	0.06%
112	29624	0.06%
113	30113	0.06%
114	29603	0.06%
115	30340	0.06%
116	30857	0.06%
117	30718	0.06%
118	31782	0.06%
119	32032	0.06%
120	33230	0.07%
121	30990	0.06%
122	32360	0.06%
123	32122	0.06%
124	34123	0.07%
125	32971	0.07%

126	32097	0.06%
127	30995	0.06%
128	33848	0.07%
129	34644	0.07%
130	35142	0.07%
131	34814	0.07%
132	35913	0.07%
133	34765	0.07%
134	34710	0.07%
135	34988	0.07%
136	35125	0.07%
137	34498	0.07%
138	38763	0.08%
139	32886	0.07%
140	35059	0.07%
141	35263	0.07%
142	34996	0.07%
143	37082	0.07%
144	37977	0.08%
145	36817	0.07%
146	36643	0.07%
147	37229	0.07%
148	38077	0.08%
149	111773	0.22%
150	1445213	2.87%
151	45993097	91.42%

<Graphic distribution of read lengths>

Length	Distribution	Abundance
60	□□□□□□□□□□	0.06%
61	□□□□□□□□□□	0.06%
62	□□□□□□□□□□	0.06%
63	□□□□□□□□□□	0.06%
64	□□□□□□□□□□	0.06%
65	□□□□□□□□□□	0.06%
66	□□□□□□□□□□	0.06%
67	□□□□□□□□□□	0.06%
68	□□□□□□□□□□	0.06%
69	□□□□□□□□□□	0.06%
70	□□□□□□□□□□	0.06%
71	□□□□□□□□□□	0.06%
72	□□□□□□□□□□	0.06%
73	□□□□□□□□□□	0.06%
74	□□□□□□□□□□	0.06%
75	□□□□□□□□□□	0.06%
76	□□□□□□□□□□	0.06%
77	□□□□□□□□□□	0.05%
78	□□□□□□□□□□	0.06%

[illegible]

127	□□□□□□□□□□	0.06%
128	□□□□□□□□□□	0.07%
129	□□□□□□□□□□	0.07%
130	□□□□□□□□□□	0.07%
131	□□□□□□□□□□	0.07%
132	□□□□□□□□□□	0.07%
133	□□□□□□□□□□	0.07%
134	□□□□□□□□□□	0.07%
135	□□□□□□□□□□	0.07%
136	□□□□□□□□□□	0.07%
137	□□□□□□□□□□	0.07%
138	□□□□□□□□□□	0.08%
139	□□□□□□□□□□	0.07%
140	□□□□□□□□□□	0.07%
141	□□□□□□□□□□	0.07%
142	□□□□□□□□□□	0.07%
143	□□□□□□□□□□	0.07%
144	□□□□□□□□□□	0.08%
145	□□□□□□□□□□	0.07%
146	□□□□□□□□□□	0.07%
147	□□□□□□□□□□	0.07%
148	□□□□□□□□□□	0.08%
149	□□□□□□□□□□	0.22%
150	□□□□□□□□□□	2.87%
151	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ □	91.42%

SRR37843242_1.p.fastq	
Measure	Value
Filename	SRR37843242_1.p.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	50,307,973
Total Bases	7.4 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51

SRR37843242_2.p.fastq

The number of read sequences is 50307973

Total length of read sequences is 7478449870

<Text distribution of read lengths>

Length	Count	Abundance
60	23359	0.05%
61	22779	0.05%
62	24134	0.05%
63	23059	0.05%
64	24529	0.05%
65	23246	0.05%
66	24361	0.05%
67	26406	0.05%
68	24781	0.05%
69	24651	0.05%
70	28369	0.06%
71	23828	0.05%
72	24424	0.05%
73	23691	0.05%
74	25356	0.05%
75	23629	0.05%
76	26035	0.05%
77	25973	0.05%
78	25275	0.05%
79	25490	0.05%
80	28292	0.06%
81	26194	0.05%
82	26393	0.05%
83	25522	0.05%
84	25433	0.05%
85	25530	0.05%
86	25822	0.05%
87	24684	0.05%
88	24045	0.05%
89	25894	0.05%
90	25862	0.05%
91	26447	0.05%
92	27541	0.05%
93	25961	0.05%
94	25747	0.05%
95	29144	0.06%
96	28526	0.06%
97	29633	0.06%
98	28619	0.06%
99	26945	0.05%
100	31172	0.06%
101	28372	0.06%
102	28041	0.06%
103	28122	0.06%
104	29179	0.06%

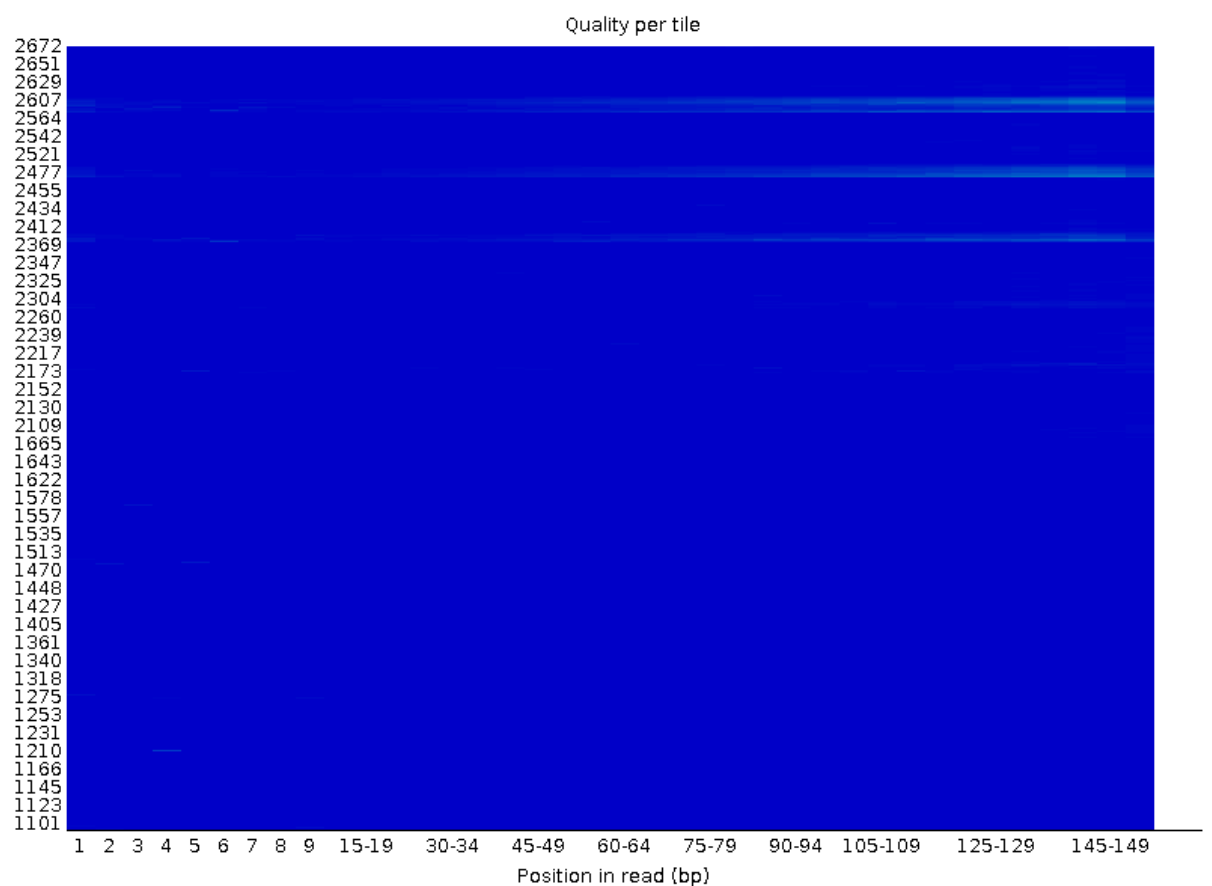
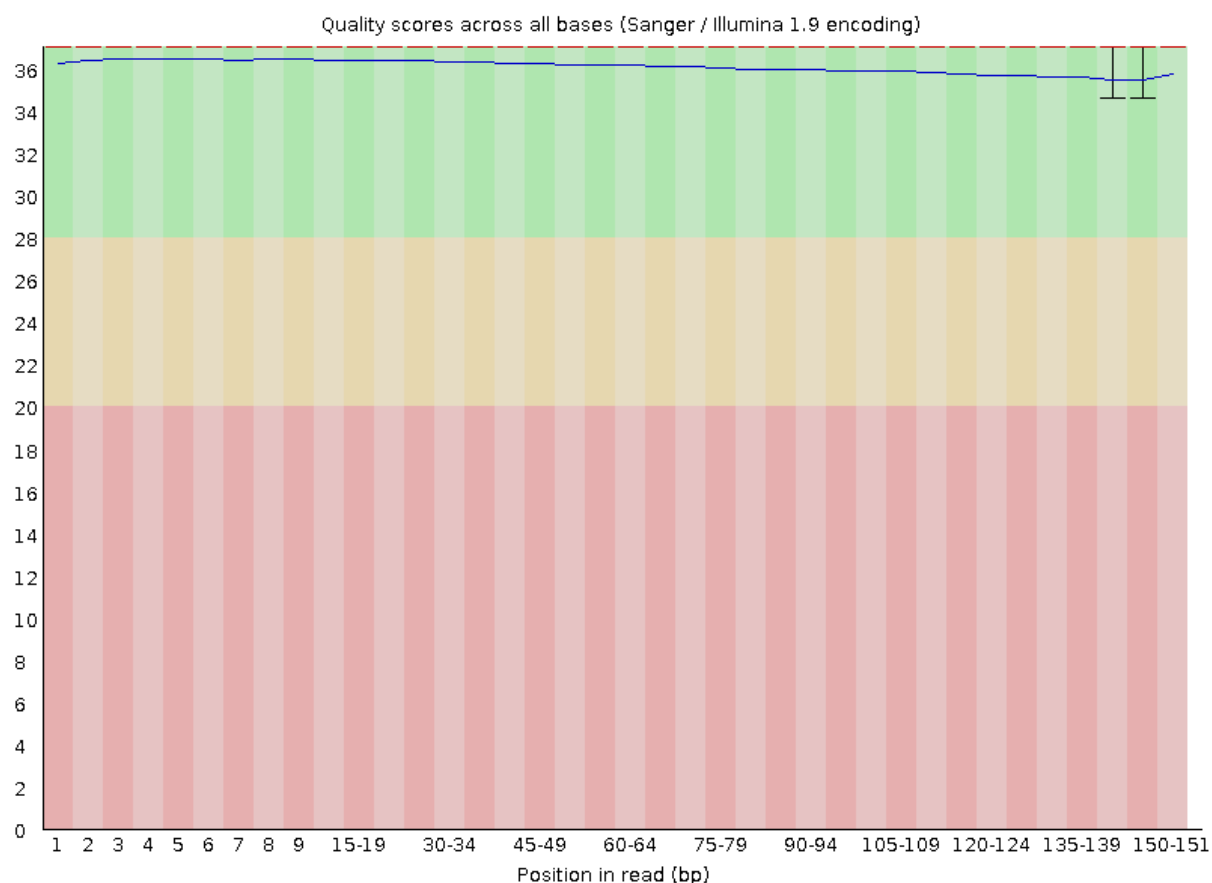
105	28247	0.06%
106	30485	0.06%
107	30521	0.06%
108	29289	0.06%
109	29920	0.06%
110	28106	0.06%
111	28776	0.06%
112	30152	0.06%
113	29892	0.06%
114	31683	0.06%
115	31864	0.06%
116	31014	0.06%
117	31768	0.06%
118	32587	0.06%
119	32125	0.06%
120	32926	0.07%
121	34072	0.07%
122	35501	0.07%
123	38033	0.08%
124	37191	0.07%
125	36743	0.07%
126	35656	0.07%
127	35837	0.07%
128	35224	0.07%
129	36683	0.07%
130	34839	0.07%
131	35829	0.07%
132	38010	0.08%
133	37566	0.07%
134	39802	0.08%
135	35869	0.07%
136	38835	0.08%
137	36897	0.07%
138	38488	0.08%
139	38654	0.08%
140	42818	0.09%
141	42125	0.08%
142	42489	0.08%
143	42495	0.08%
144	45238	0.09%
145	39090	0.08%
146	41430	0.08%
147	41387	0.08%
148	46494	0.09%
149	149088	0.30%
150	1828709	3.64%
151	45587031	90.62%

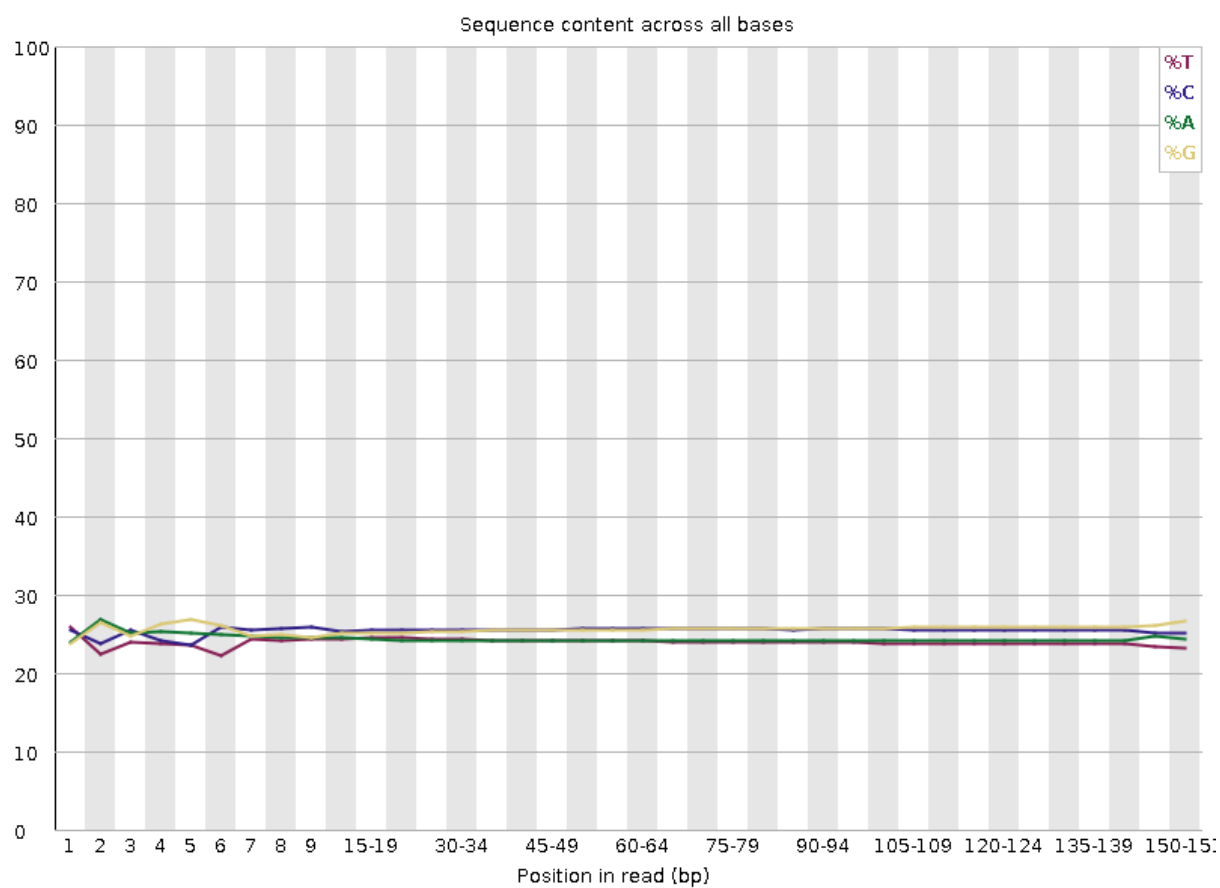
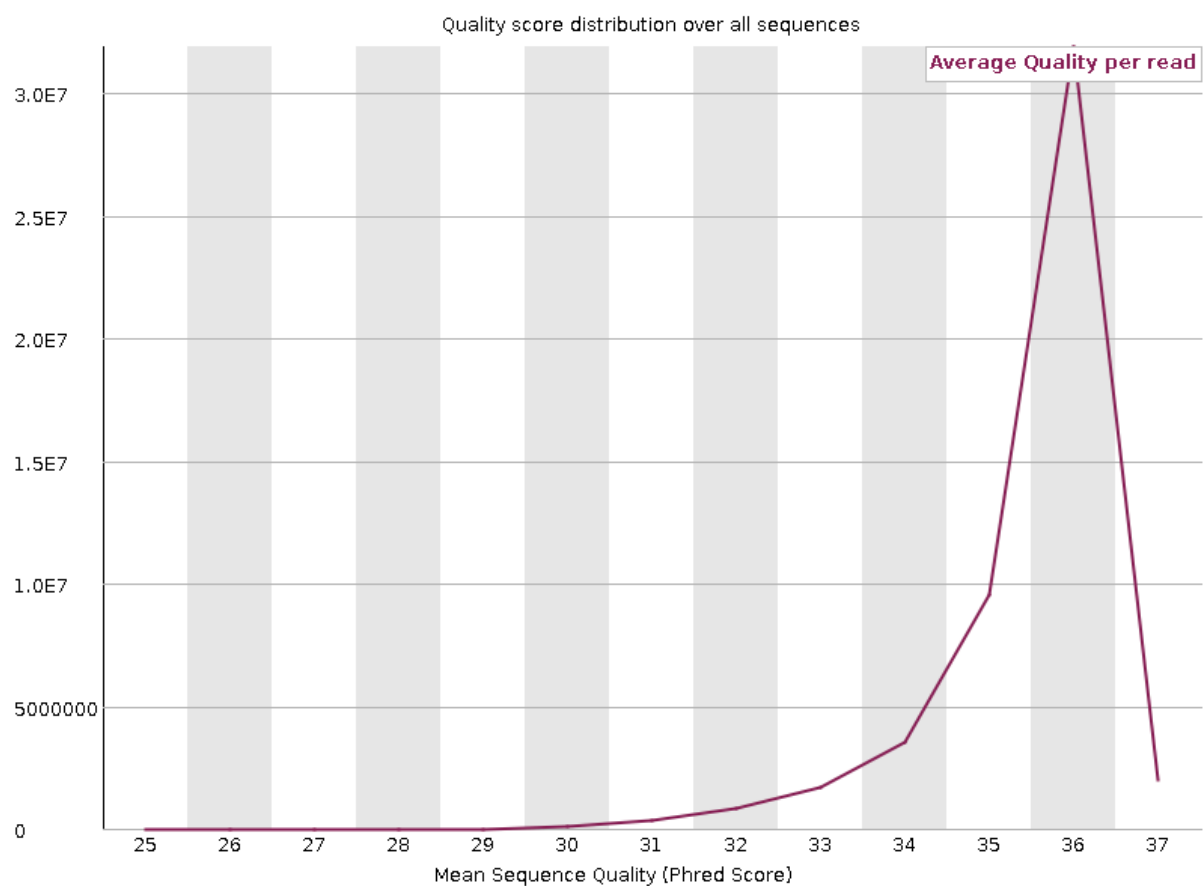
<Graphic distribution of read lengths>

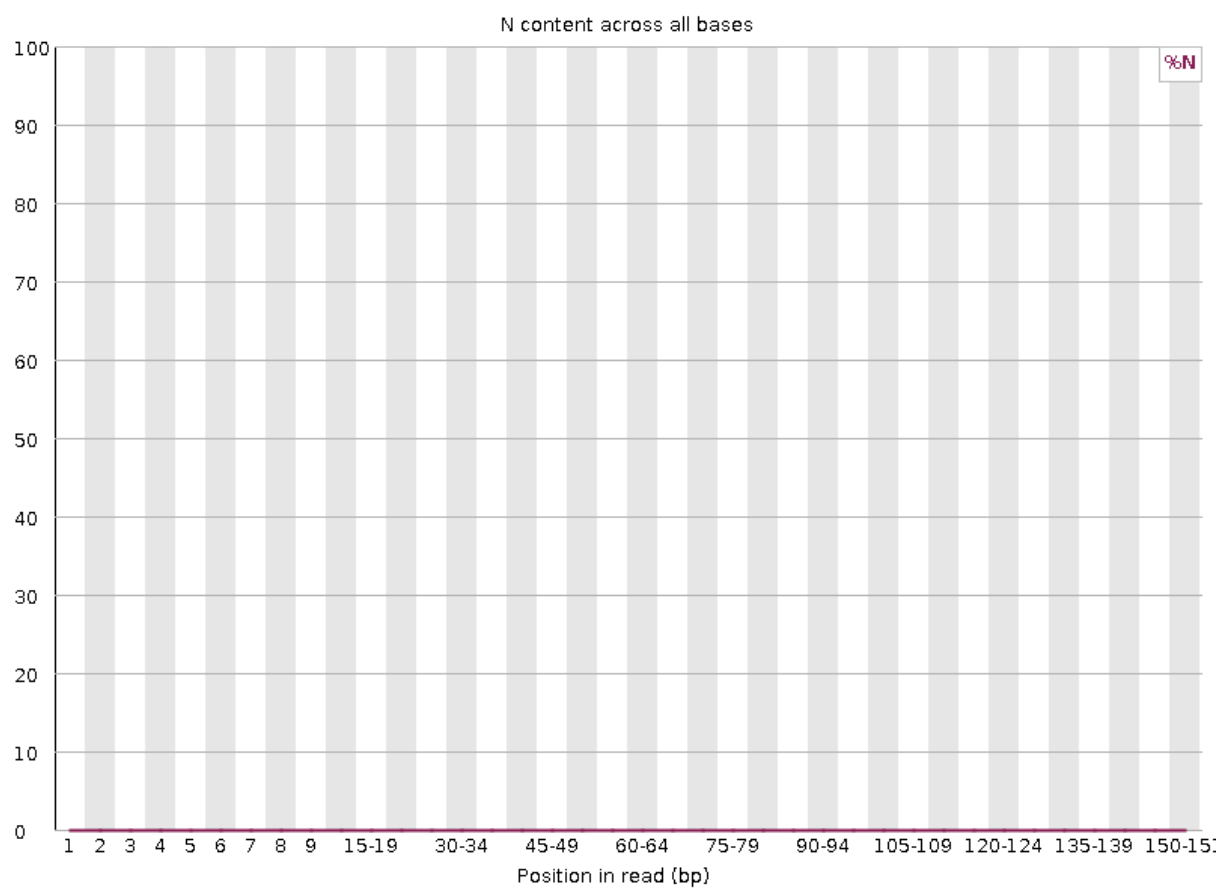
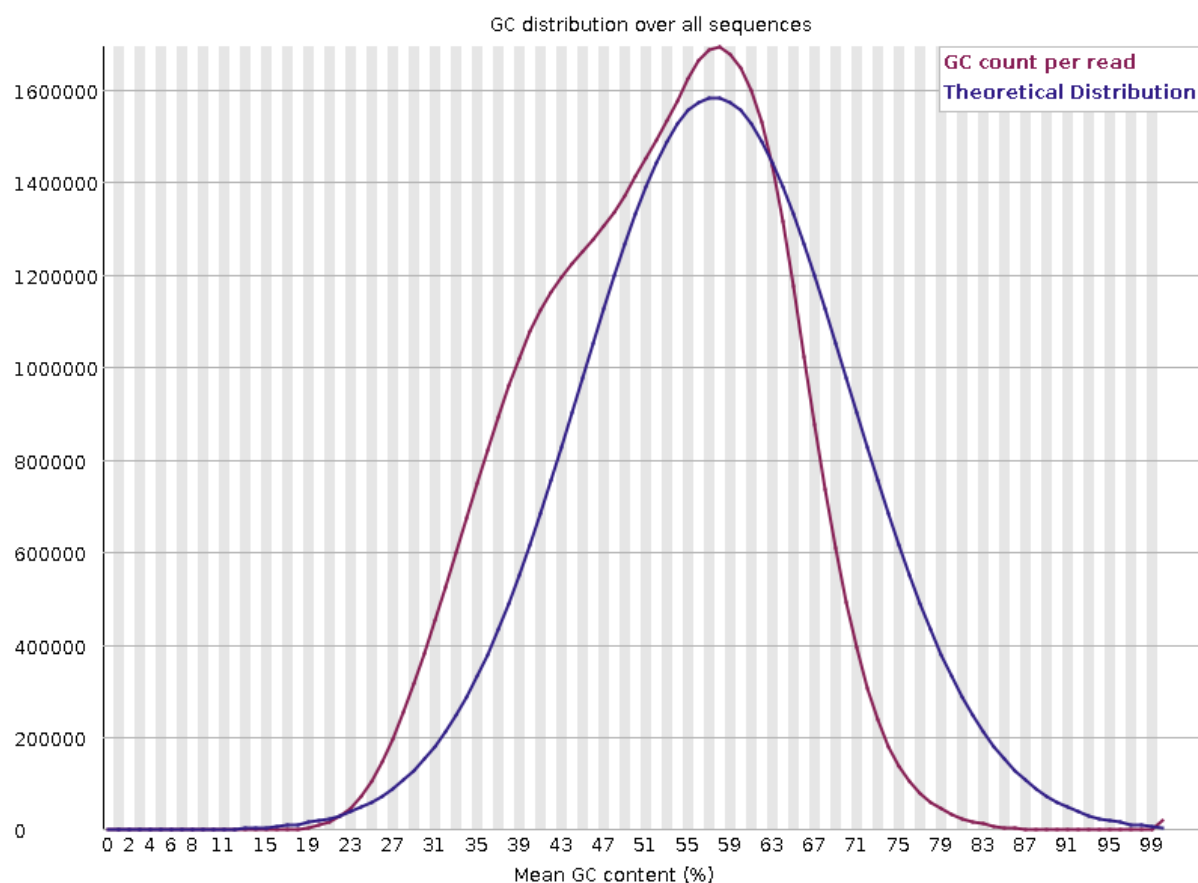
Length	Distribution	Abundance
60	□□□□□□□□□□	0.05%
61	□□□□□□□□□□	0.05%
62	□□□□□□□□□□	0.05%
63	□□□□□□□□□□	0.05%
64	□□□□□□□□□□	0.05%
65	□□□□□□□□□□	0.05%
66	□□□□□□□□□□	0.05%
67	□□□□□□□□□□	0.05%
68	□□□□□□□□□□	0.05%
69	□□□□□□□□□□	0.05%
70	□□□□□□□□□□	0.06%
71	□□□□□□□□□□	0.05%
72	□□□□□□□□□□	0.05%
73	□□□□□□□□□□	0.05%
74	□□□□□□□□□□	0.05%
75	□□□□□□□□□□	0.05%
76	□□□□□□□□□□	0.05%
77	□□□□□□□□□□	0.05%
78	□□□□□□□□□□	0.05%
79	□□□□□□□□□□	0.05%
80	□□□□□□□□□□	0.06%
81	□□□□□□□□□□	0.05%
82	□□□□□□□□□□	0.05%
83	□□□□□□□□□□	0.05%
84	□□□□□□□□□□	0.05%
85	□□□□□□□□□□	0.05%
86	□□□□□□□□□□	0.05%
87	□□□□□□□□□□	0.05%
88	□□□□□□□□□□	0.05%
89	□□□□□□□□□□	0.05%
90	□□□□□□□□□□	0.05%
91	□□□□□□□□□□	0.05%
92	□□□□□□□□□□	0.05%
93	□□□□□□□□□□	0.05%
94	□□□□□□□□□□	0.05%
95	□□□□□□□□□□	0.06%
96	□□□□□□□□□□	0.06%
97	□□□□□□□□□□	0.06%
98	□□□□□□□□□□	0.06%
99	□□□□□□□□□□	0.05%
100	□□□□□□□□□□	0.06%
101	□□□□□□□□□□	0.06%
102	□□□□□□□□□□	0.06%
103	□□□□□□□□□□	0.06%
104	□□□□□□□□□□	0.06%
105	□□□□□□□□□□	0.06%

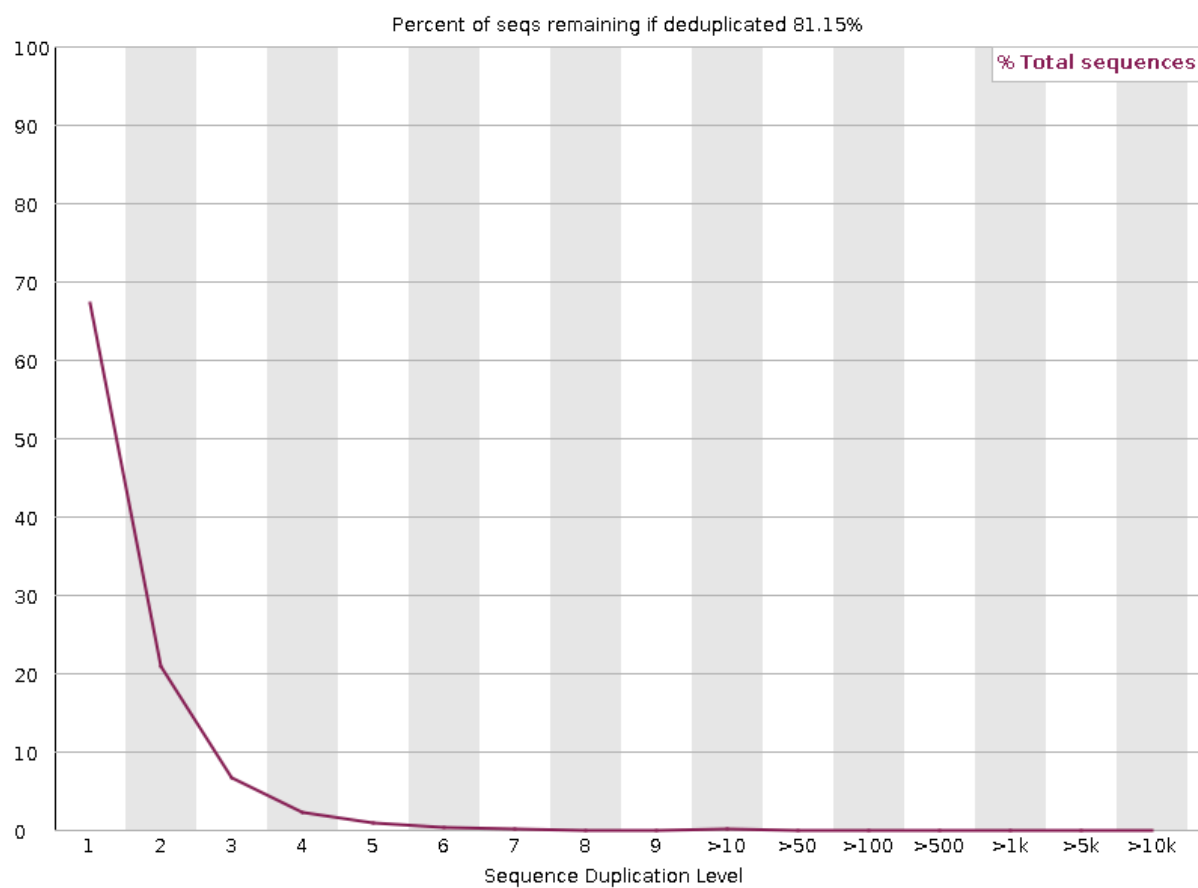
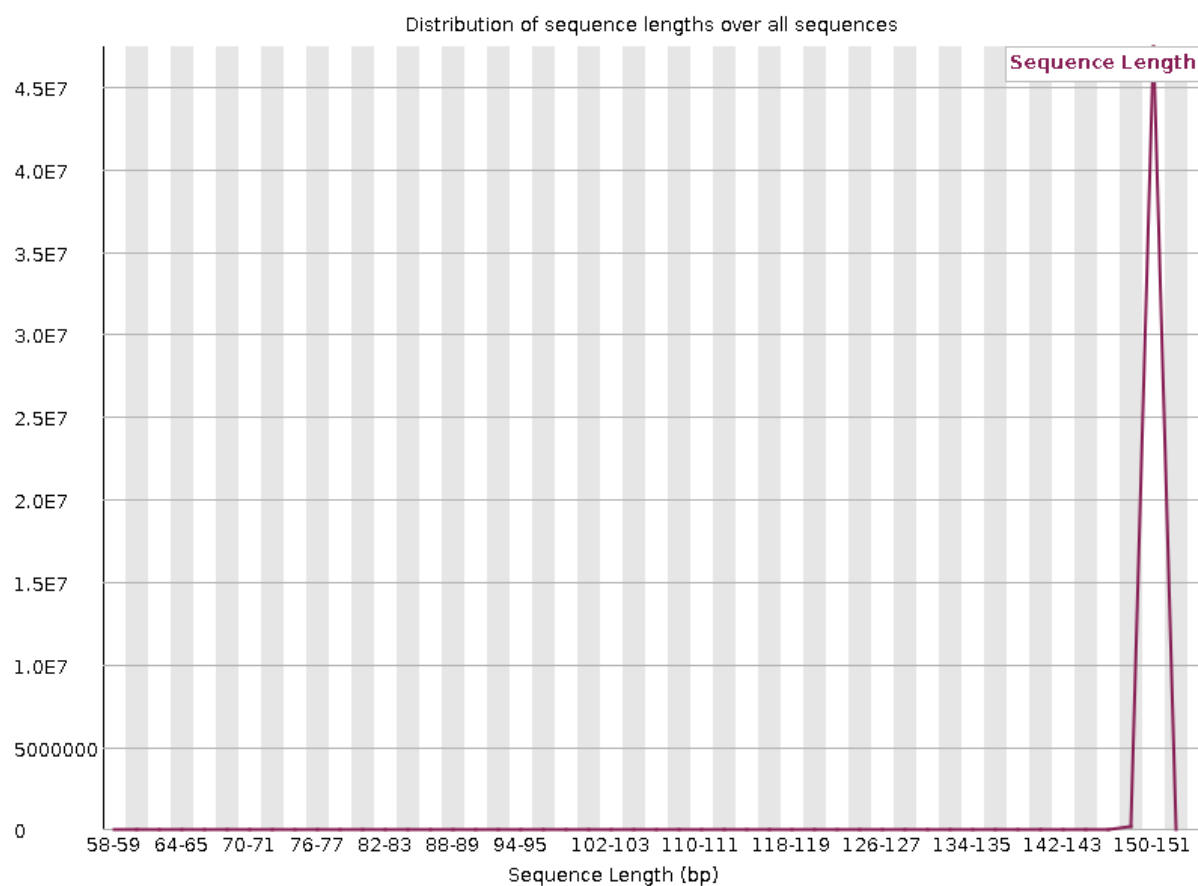
106	□□□□□□□□	0.06%
107	□□□□□□□□	0.06%
108	□□□□□□□□	0.06%
109	□□□□□□□□	0.06%
110	□□□□□□□□	0.06%
111	□□□□□□□□	0.06%
112	□□□□□□□□	0.06%
113	□□□□□□□□	0.06%
114	□□□□□□□□	0.06%
115	□□□□□□□□	0.06%
116	□□□□□□□□	0.06%
117	□□□□□□□□	0.06%
118	□□□□□□□□	0.06%
119	□□□□□□□□	0.06%
120	□□□□□□□□	0.07%
121	□□□□□□□□	0.07%
122	□□□□□□□□	0.07%
123	□□□□□□□□	0.08%
124	□□□□□□□□	0.07%
125	□□□□□□□□	0.07%
126	□□□□□□□□	0.07%
127	□□□□□□□□	0.07%
128	□□□□□□□□	0.07%
129	□□□□□□□□	0.07%
130	□□□□□□□□	0.07%
131	□□□□□□□□	0.07%
132	□□□□□□□□	0.08%
133	□□□□□□□□	0.07%
134	□□□□□□□□	0.08%
135	□□□□□□□□	0.07%
136	□□□□□□□□	0.08%
137	□□□□□□□□	0.07%
138	□□□□□□□□	0.08%
139	□□□□□□□□	0.08%
140	□□□□□□□□	0.09%
141	□□□□□□□□	0.08%
142	□□□□□□□□	0.08%
143	□□□□□□□□	0.08%
144	□□□□□□□□	0.09%
145	□□□□□□□□	0.08%
146	□□□□□□□□	0.08%
147	□□□□□□□□	0.08%
148	□□□□□□□□	0.09%
149	□□□□□□□□	0.30%
150	□□□□□□□□	3.64%
151	■ ■ ■ ■ ■ ■ ■ ■ □	90.62%

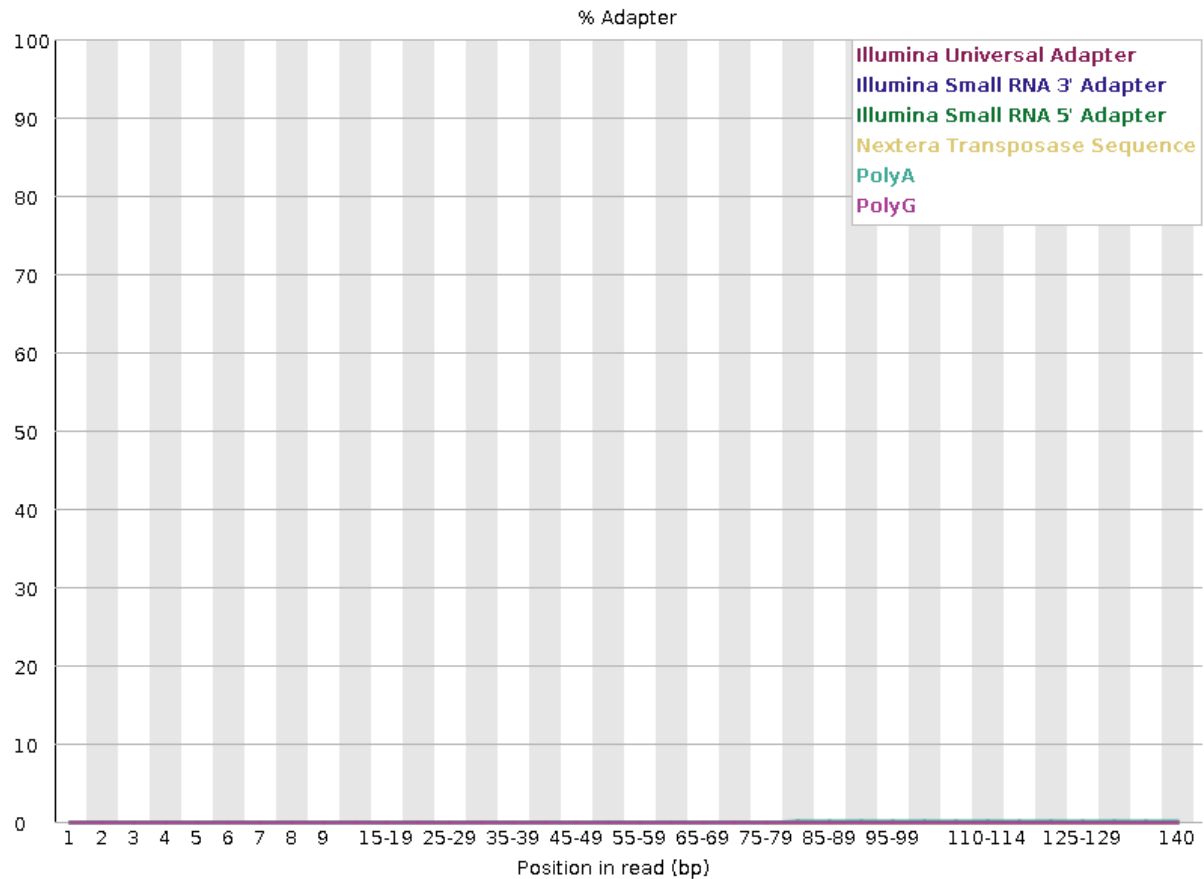
SRR37843242_2.p.fastq	
Measure	Value
Filename	SRR37843242_2.p.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	50,307,973
Total Bases	7.4 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51











SRR37843242_1.up.fastq

The number of read sequences is 8677068

Total length of read sequences is 1111675740

<Text distribution of read lengths>

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62	15385	0.18%
63	15395	0.18%
64	14738	0.17%
65	15258	0.18%
66	14935	0.17%
67	14906	0.17%
68	14429	0.17%
69	14594	0.17%
70	14505	0.17%
71	14297	0.16%
72	14023	0.16%
73	14128	0.16%

74	14228	0.16%
75	14045	0.16%
76	13792	0.16%
77	14140	0.16%
78	14442	0.17%
79	14577	0.17%
80	14496	0.17%
81	14517	0.17%
82	15017	0.17%
83	14231	0.16%
84	14246	0.16%
85	14421	0.17%
86	14547	0.17%
87	14769	0.17%
88	15004	0.17%
89	15643	0.18%
90	16146	0.19%
91	16406	0.19%
92	16374	0.19%
93	17261	0.20%
94	18491	0.21%
95	18927	0.22%
96	20091	0.23%
97	21775	0.25%
98	22494	0.26%
99	24613	0.28%
100	26683	0.31%
101	28591	0.33%
102	31624	0.36%
103	33953	0.39%
104	36498	0.42%
105	40470	0.47%
106	43640	0.50%
107	48228	0.56%
108	52539	0.61%
109	58146	0.67%
110	62416	0.72%
111	68687	0.79%
112	74632	0.86%
113	81482	0.94%
114	88022	1.01%
115	95143	1.10%
116	104488	1.20%
117	111742	1.29%
118	120426	1.39%
119	128507	1.48%
120	137233	1.58%
121	146677	1.69%

122	156135	1.80%
123	165670	1.91%
124	175201	2.02%
125	184013	2.12%
126	193964	2.24%
127	201994	2.33%
128	212537	2.45%
129	221365	2.55%
130	229998	2.65%
131	238744	2.75%
132	247175	2.85%
133	255839	2.95%
134	264127	3.04%
135	270961	3.12%
136	279254	3.22%
137	287673	3.32%
138	293360	3.38%
139	297944	3.43%
140	305986	3.53%
141	312104	3.60%
142	315919	3.64%
143	316363	3.65%
144	4763	0.05%
145	4627	0.05%
146	4659	0.05%
147	4615	0.05%
148	4636	0.05%
149	11156	0.13%
150	74958	0.86%
151	889153	10.25%

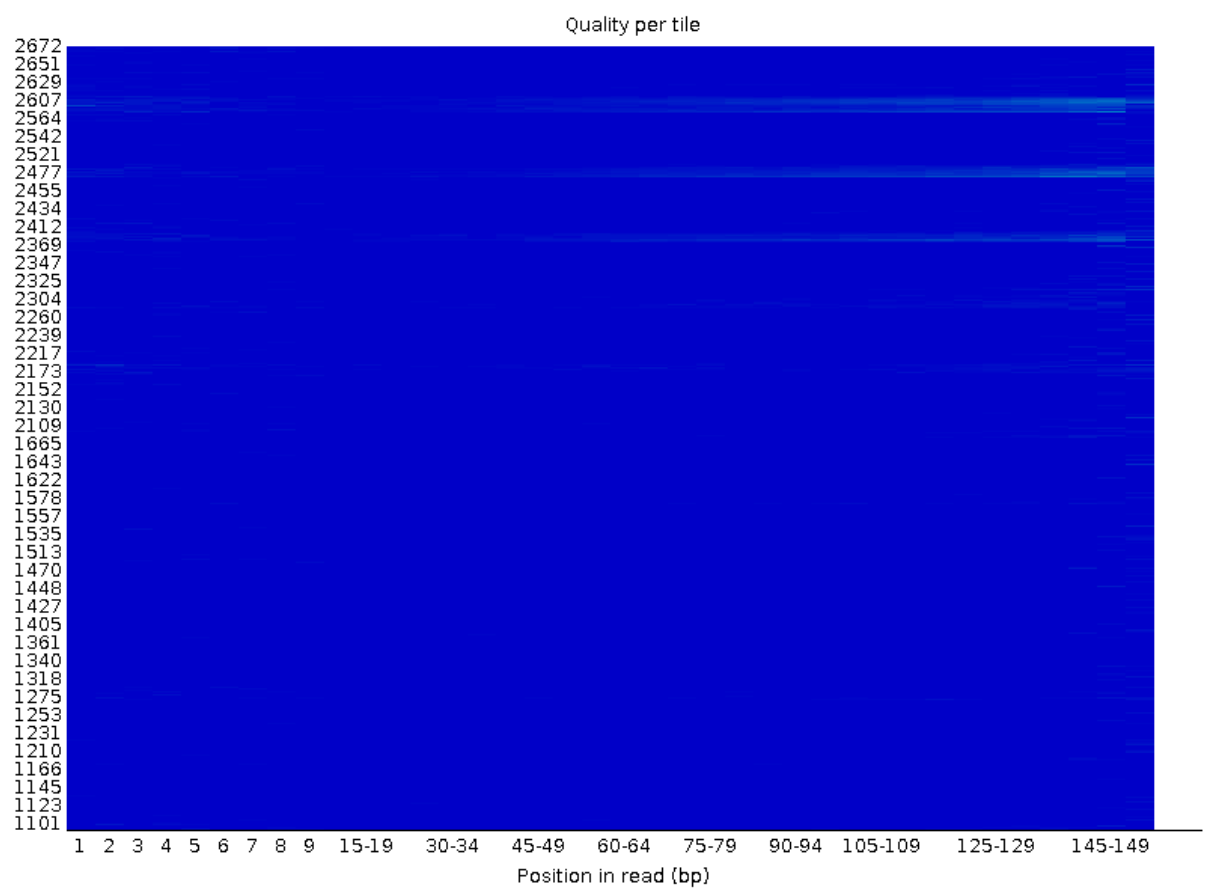
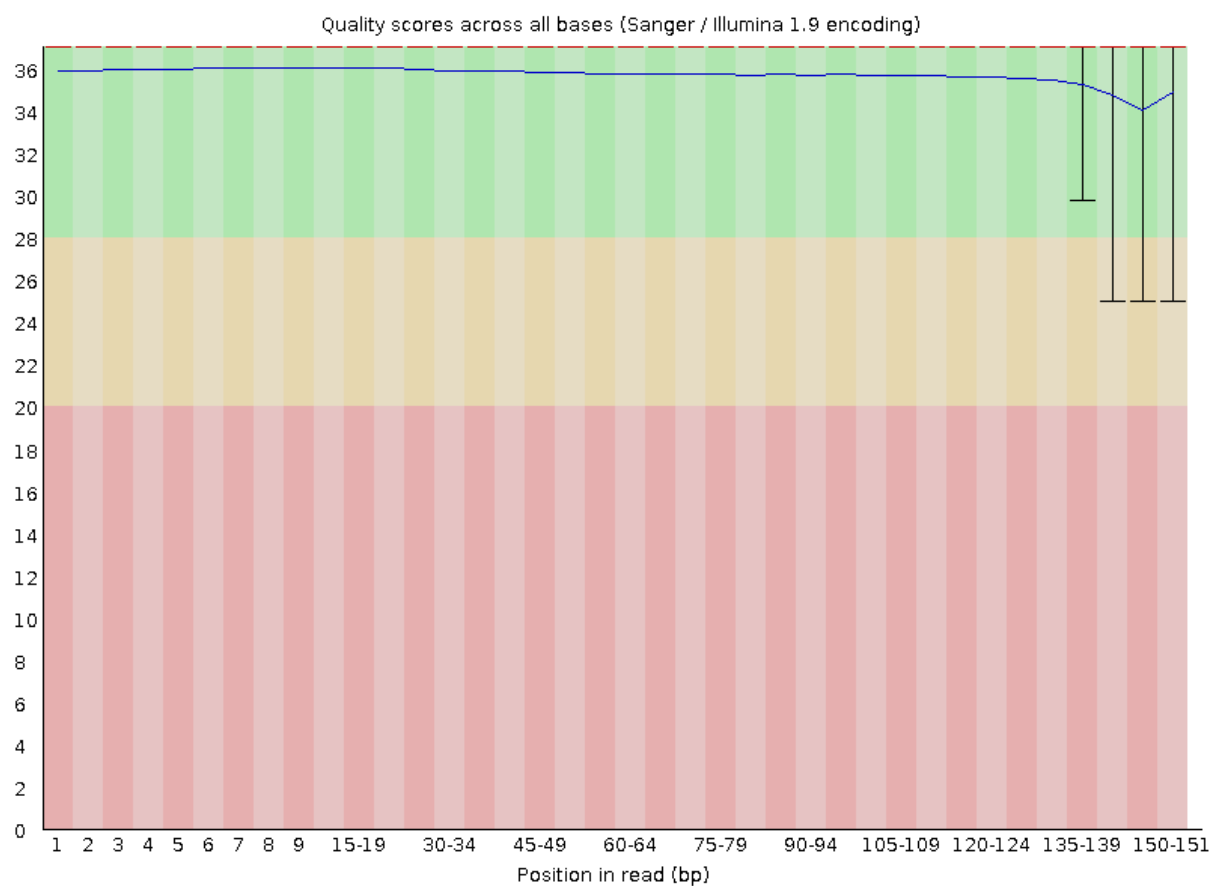
<Graphic distribution of read lengths>

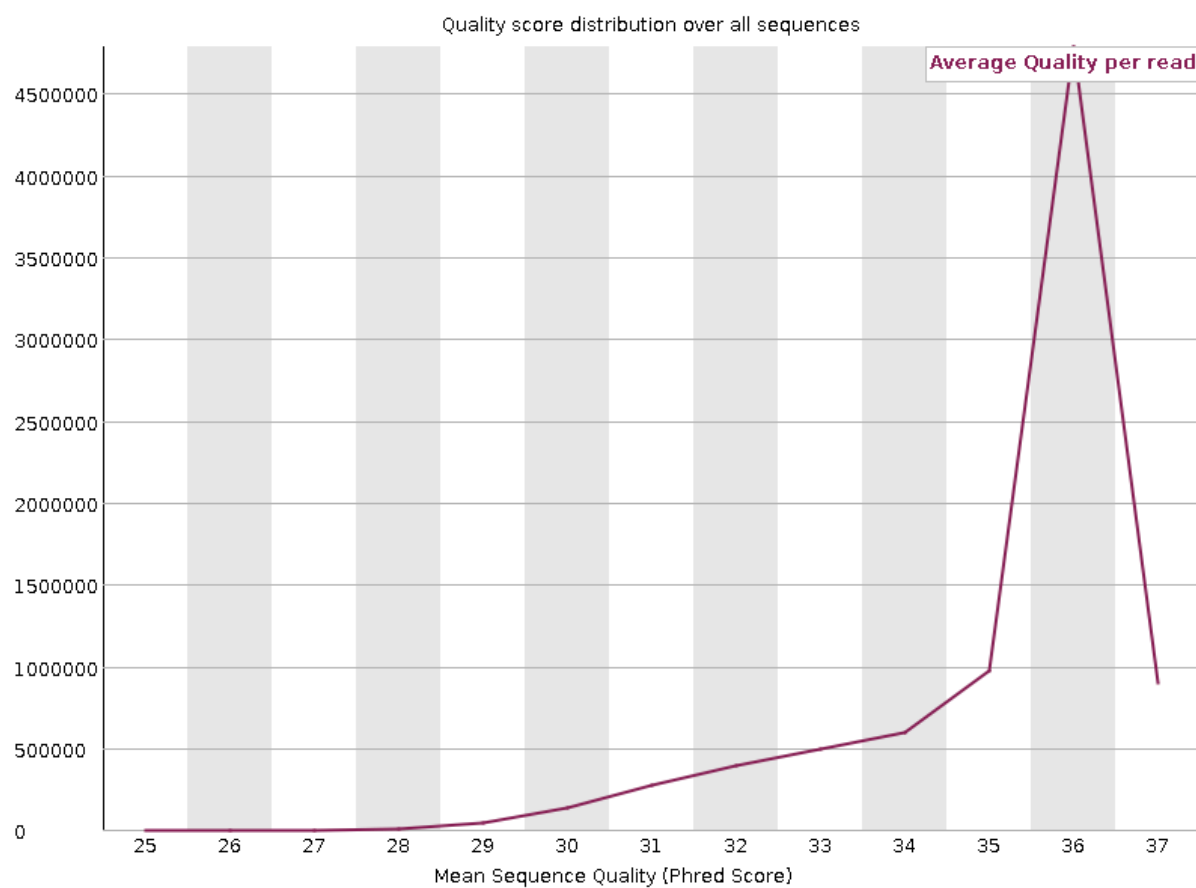
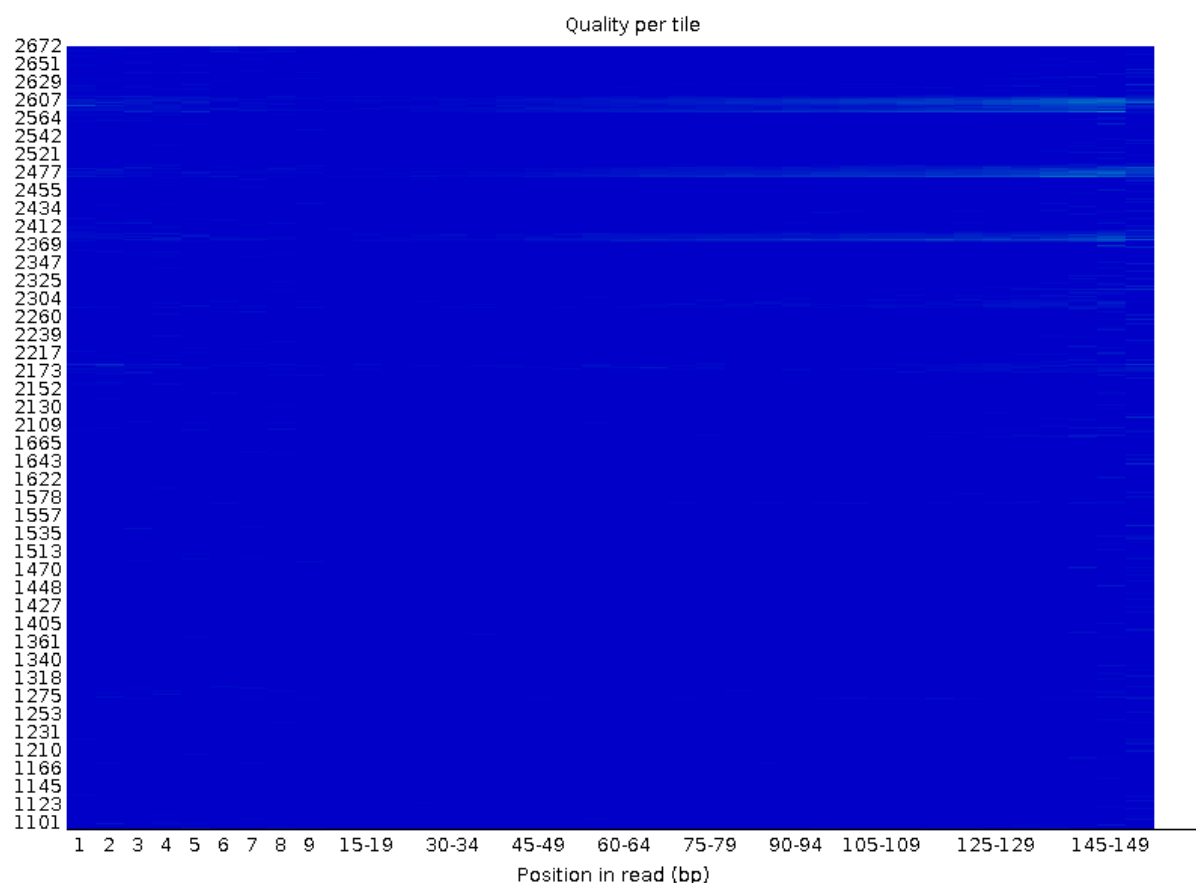
Length	Distribution	Abundance
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61	□□□□□□□□□□	0.18%
62	□□□□□□□□□□	0.18%
63	□□□□□□□□□□	0.18%
64	□□□□□□□□□□	0.17%
65	□□□□□□□□□□	0.18%
66	□□□□□□□□□□	0.17%
67	□□□□□□□□□□	0.17%
68	□□□□□□□□□□	0.17%
69	□□□□□□□□□□	0.17%
70	□□□□□□□□□□	0.17%
71	□□□□□□□□□□	0.16%
72	□□□□□□□□□□	0.16%
73	□□□□□□□□□□	0.16%
74	□□□□□□□□□□	0.16%

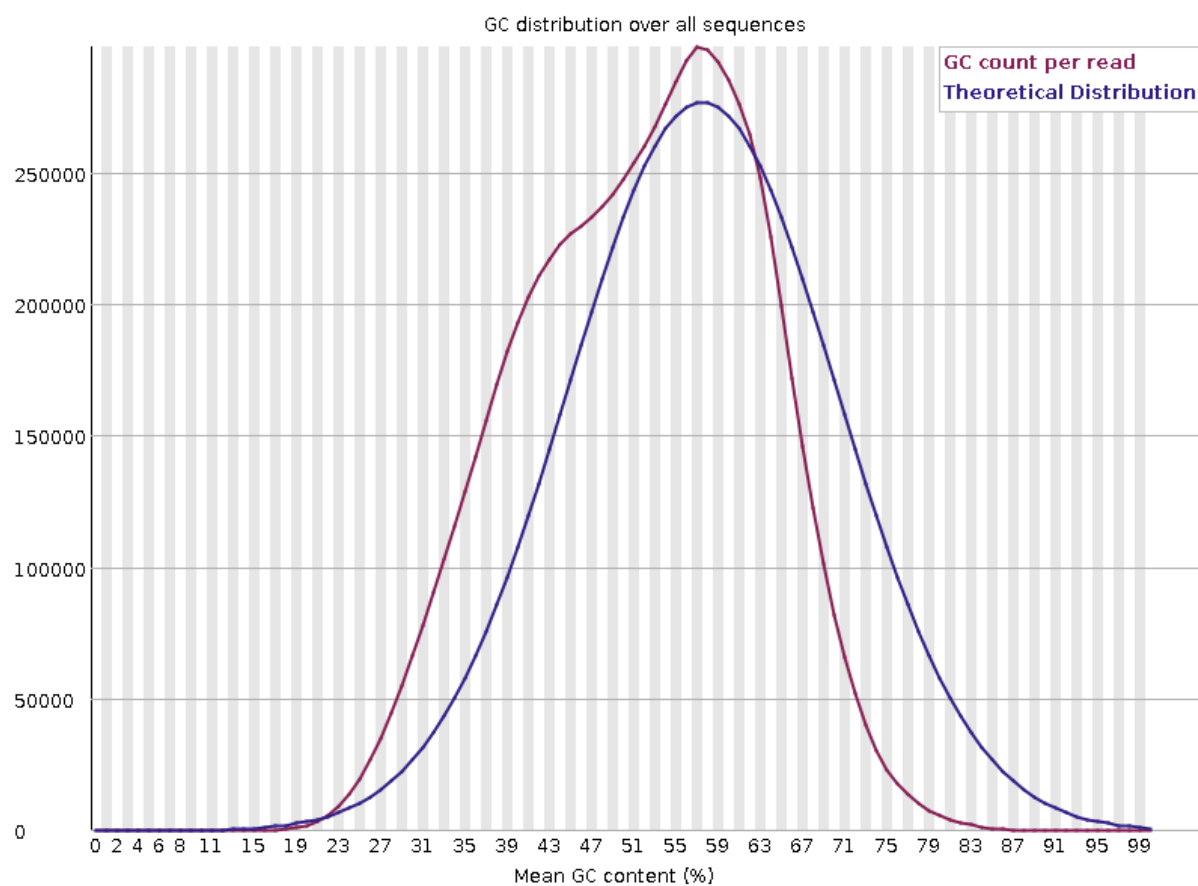
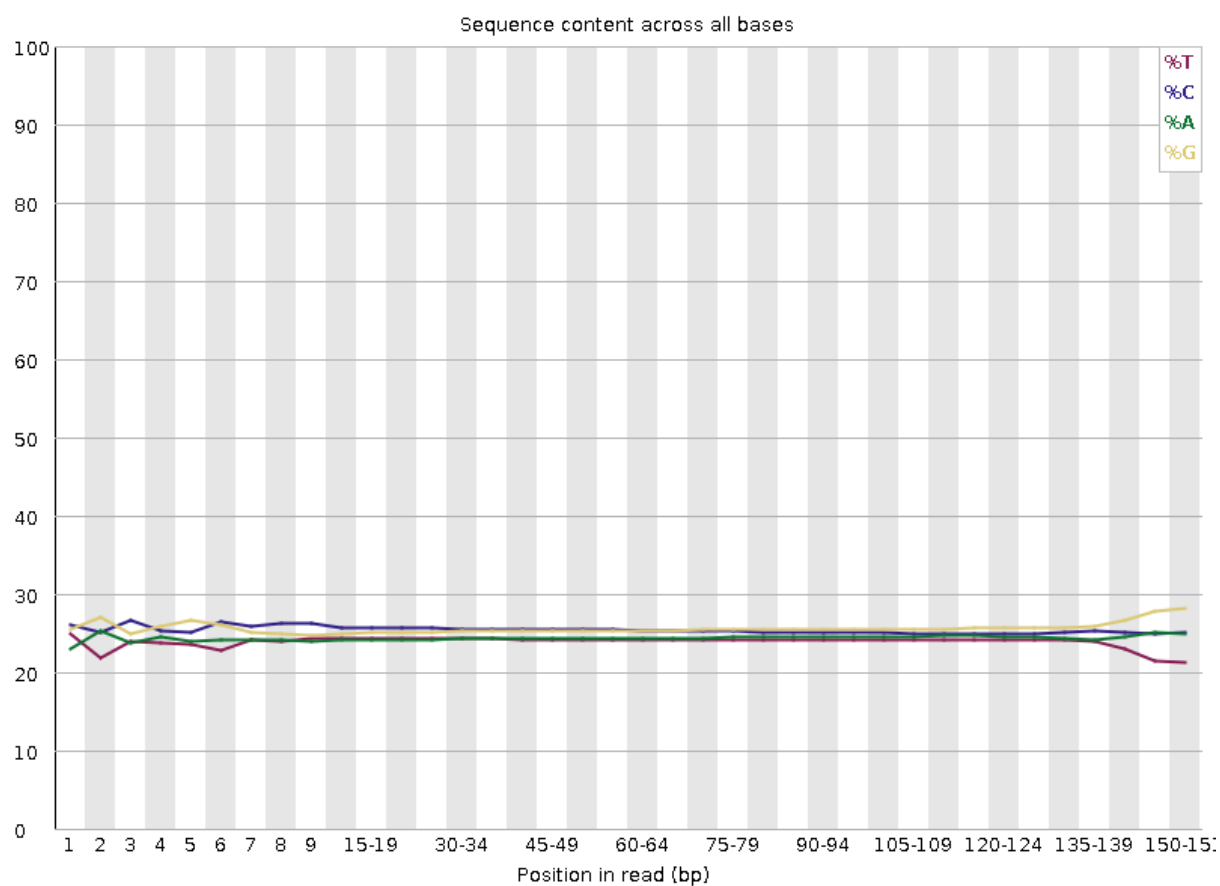
75	□□□□□□□□	0.16%
76	□□□□□□□□	0.16%
77	□□□□□□□□	0.16%
78	□□□□□□□□	0.17%
79	□□□□□□□□	0.17%
80	□□□□□□□□	0.17%
81	□□□□□□□□	0.17%
82	□□□□□□□□	0.17%
83	□□□□□□□□	0.16%
84	□□□□□□□□	0.16%
85	□□□□□□□□	0.17%
86	□□□□□□□□	0.17%
87	□□□□□□□□	0.17%
88	□□□□□□□□	0.17%
89	□□□□□□□□	0.18%
90	□□□□□□□□	0.19%
91	□□□□□□□□	0.19%
92	□□□□□□□□	0.19%
93	□□□□□□□□	0.20%
94	□□□□□□□□	0.21%
95	□□□□□□□□	0.22%
96	□□□□□□□□	0.23%
97	□□□□□□□□	0.25%
98	□□□□□□□□	0.26%
99	□□□□□□□□	0.28%
100	□□□□□□□□	0.31%
101	□□□□□□□□	0.33%
102	□□□□□□□□	0.36%
103	□□□□□□□□	0.39%
104	□□□□□□□□	0.42%
105	□□□□□□□□	0.47%
106	□□□□□□□□	0.50%
107	□□□□□□□□	0.56%
108	□□□□□□□□	0.61%
109	□□□□□□□□	0.67%
110	□□□□□□□□	0.72%
111	□□□□□□□□	0.79%
112	□□□□□□□□	0.86%
113	□□□□□□□□	0.94%
114	□□□□□□□□	1.01%
115	□□□□□□□□	1.10%
116	□□□□□□□□	1.20%
117	□□□□□□□□	1.29%
118	□□□□□□□□	1.39%
119	□□□□□□□□	1.48%
120	□□□□□□□□	1.58%
121	□□□□□□□□	1.69%
122	□□□□□□□□	1.80%

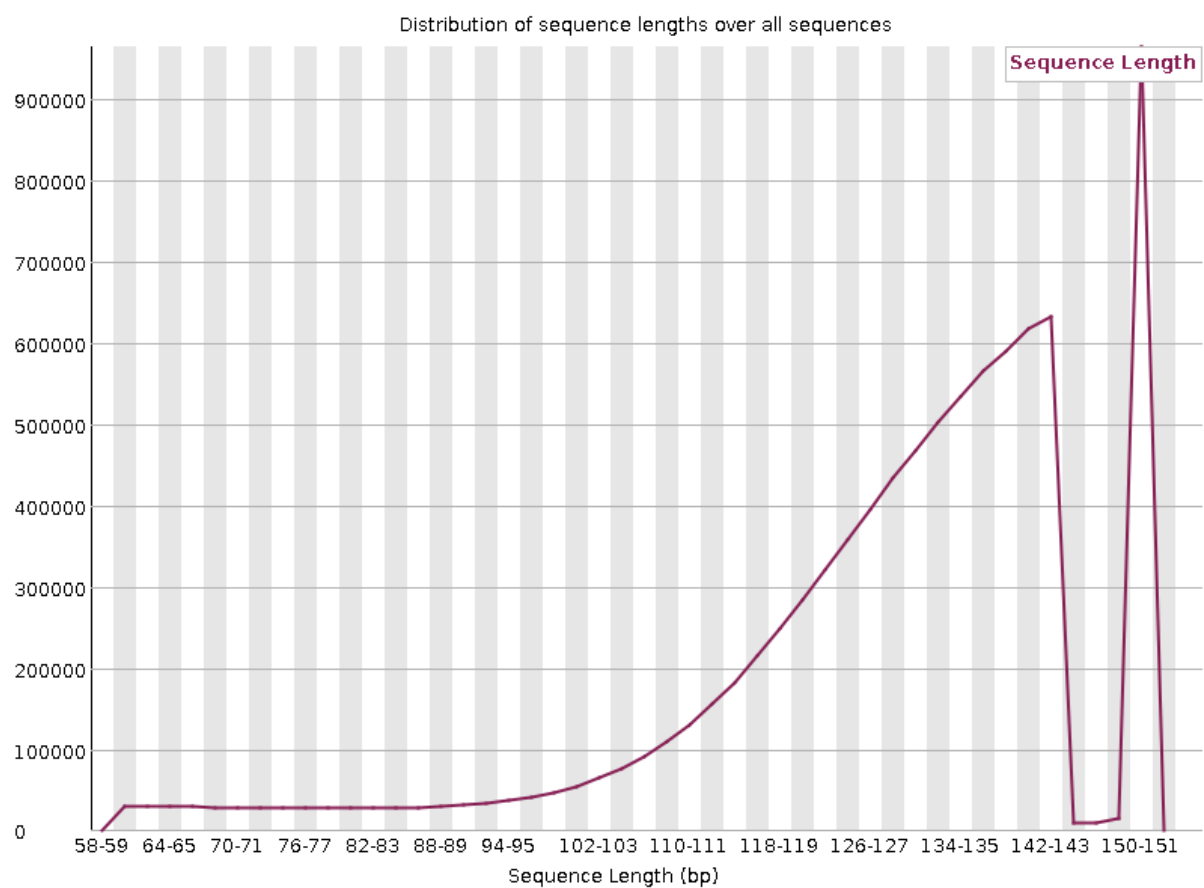
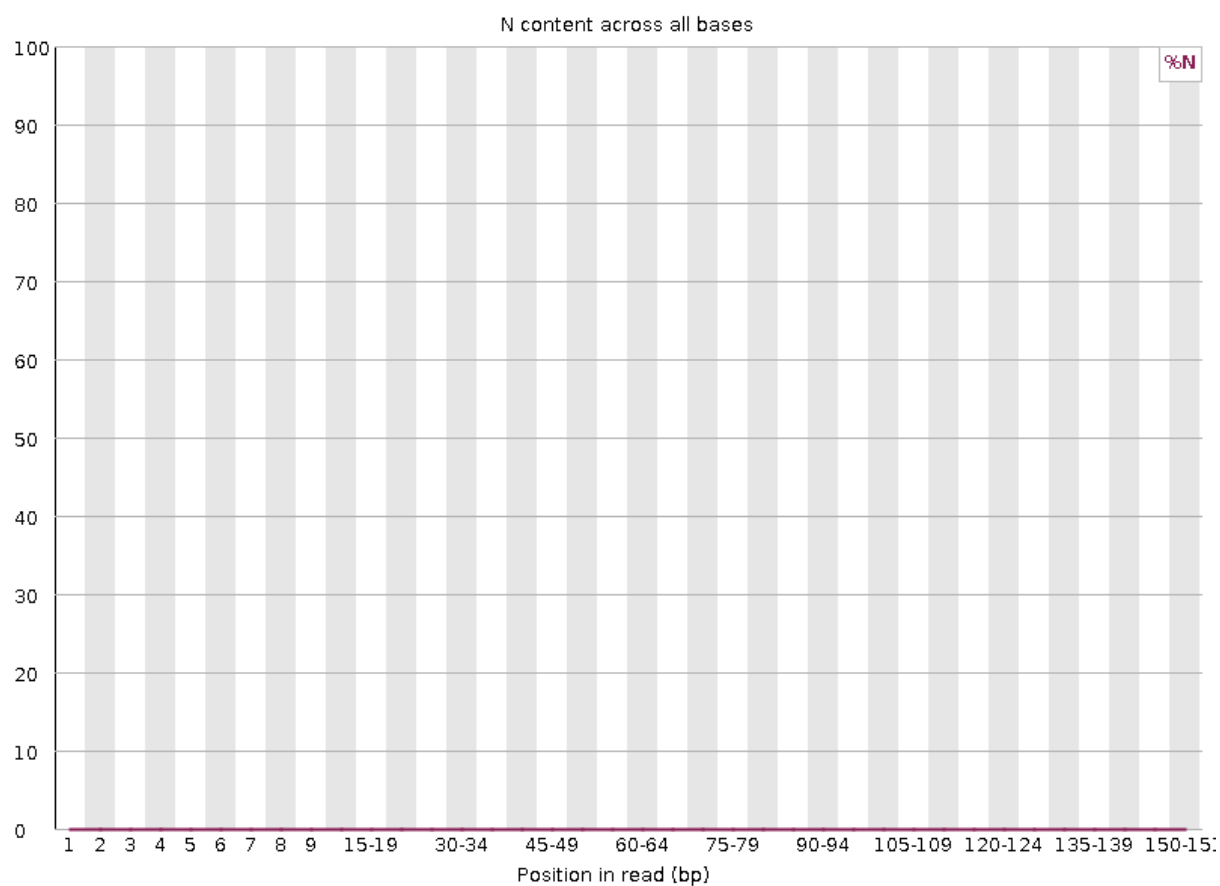
123	□□□□□□□□	1.91%
124	□□□□□□□□	2.02%
125	□□□□□□□□	2.12%
126	□□□□□□□□	2.24%
127	□□□□□□□□	2.33%
128	□□□□□□□□	2.45%
129	□□□□□□□□	2.55%
130	□□□□□□□□	2.65%
131	□□□□□□□□	2.75%
132	□□□□□□□□	2.85%
133	□□□□□□□□	2.95%
134	□□□□□□□□	3.04%
135	□□□□□□□□	3.12%
136	□□□□□□□□	3.22%
137	□□□□□□□□	3.32%
138	□□□□□□□□	3.38%
139	□□□□□□□□	3.43%
140	□□□□□□□□	3.53%
141	□□□□□□□□	3.60%
142	□□□□□□□□	3.64%
143	□□□□□□□□	3.65%
144	□□□□□□□□	0.05%
145	□□□□□□□□	0.05%
146	□□□□□□□□	0.05%
147	□□□□□□□□	0.05%
148	□□□□□□□□	0.05%
149	□□□□□□□□	0.13%
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151	■□□□□□□□	10.25%

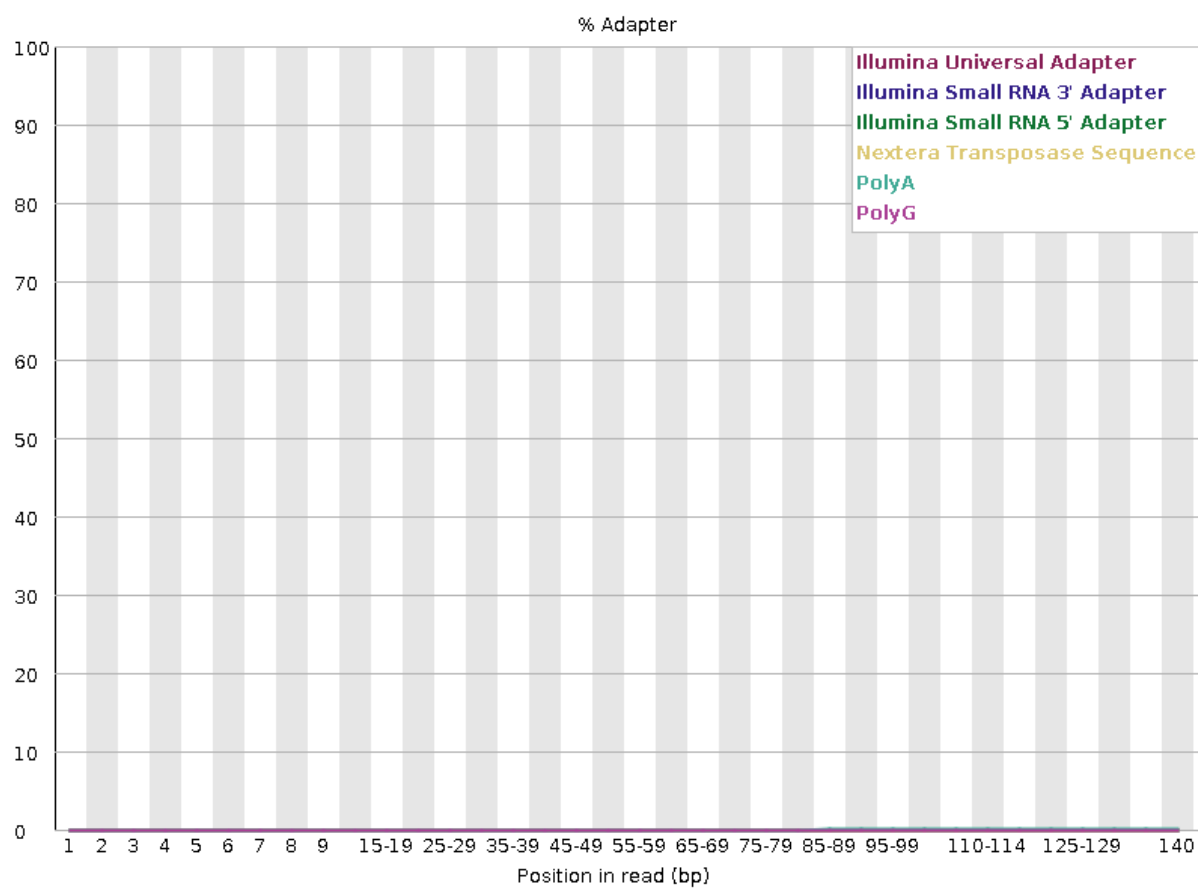
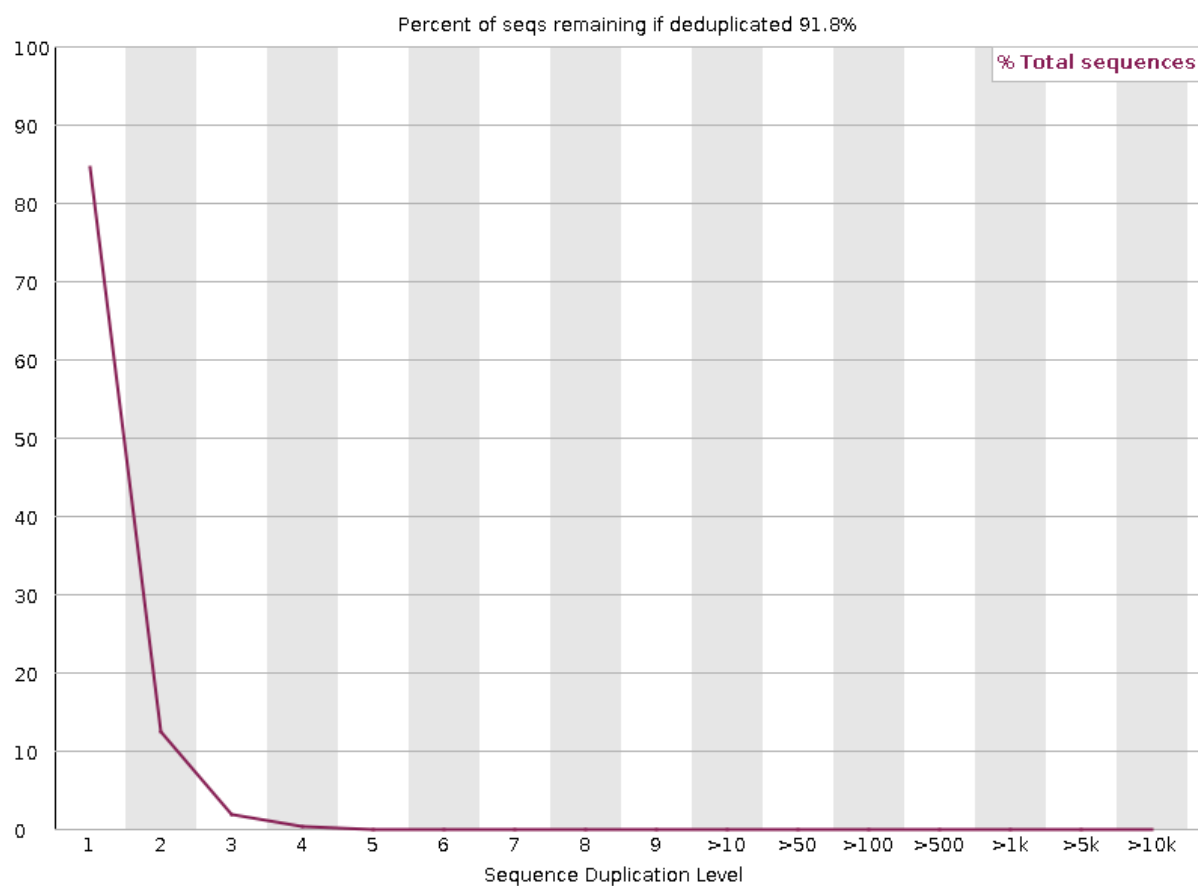
SRR37843242_1.up.fastq	
Measure	Value
Filename	SRR37843242_1.up.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	8,677,068
Total Bases	1.1 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51











SRR37843242_2.up.fastq

The number of read sequences is 1508577

Total length of read sequences is 206376159

<Text distribution of read lengths>

Length	Count	Abundance
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63	6199	0.41%
64	6414	0.43%
65	5993	0.40%
66	6179	0.41%
67	6549	0.43%
68	5979	0.40%
69	5959	0.40%
70	6160	0.41%
71	5493	0.36%
72	5621	0.37%
73	5549	0.37%
74	5675	0.38%
75	5303	0.35%
76	5474	0.36%
77	5486	0.36%
78	5421	0.36%
79	5286	0.35%
80	5743	0.38%
81	5281	0.35%
82	5279	0.35%
83	5031	0.33%
84	5114	0.34%
85	5089	0.34%
86	4871	0.32%
87	4855	0.32%
88	4650	0.31%
89	4608	0.31%
90	4860	0.32%
91	4665	0.31%
92	4656	0.31%
93	4660	0.31%
94	4627	0.31%
95	4841	0.32%
96	4655	0.31%
97	4849	0.32%
98	4600	0.30%
99	4358	0.29%
100	4923	0.33%

101	4416	0.29%
102	4391	0.29%
103	4410	0.29%
104	4592	0.30%
105	4217	0.28%
106	4350	0.29%
107	4442	0.29%
108	4112	0.27%
109	4153	0.28%
110	3976	0.26%
111	4108	0.27%
112	4267	0.28%
113	4167	0.28%
114	4321	0.29%
115	4262	0.28%
116	4263	0.28%
117	4139	0.27%
118	4019	0.27%
119	4275	0.28%
120	4185	0.28%
121	4275	0.28%
122	4543	0.30%
123	4612	0.31%
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125	4405	0.29%
126	4169	0.28%
127	4349	0.29%
128	4287	0.28%
129	4108	0.27%
130	4140	0.27%
131	4078	0.27%
132	4170	0.28%
133	4087	0.27%
134	4153	0.28%
135	3986	0.26%
136	4277	0.28%
137	4176	0.28%
138	4171	0.28%
139	4185	0.28%
140	4314	0.29%
141	4282	0.28%
142	4344	0.29%
143	4262	0.28%
144	4401	0.29%
145	3914	0.26%
146	4162	0.28%
147	4119	0.27%
148	4365	0.29%

149	11098	0.74%
150	84112	5.58%
151	989633	65.60%

<Graphic distribution of read lengths>

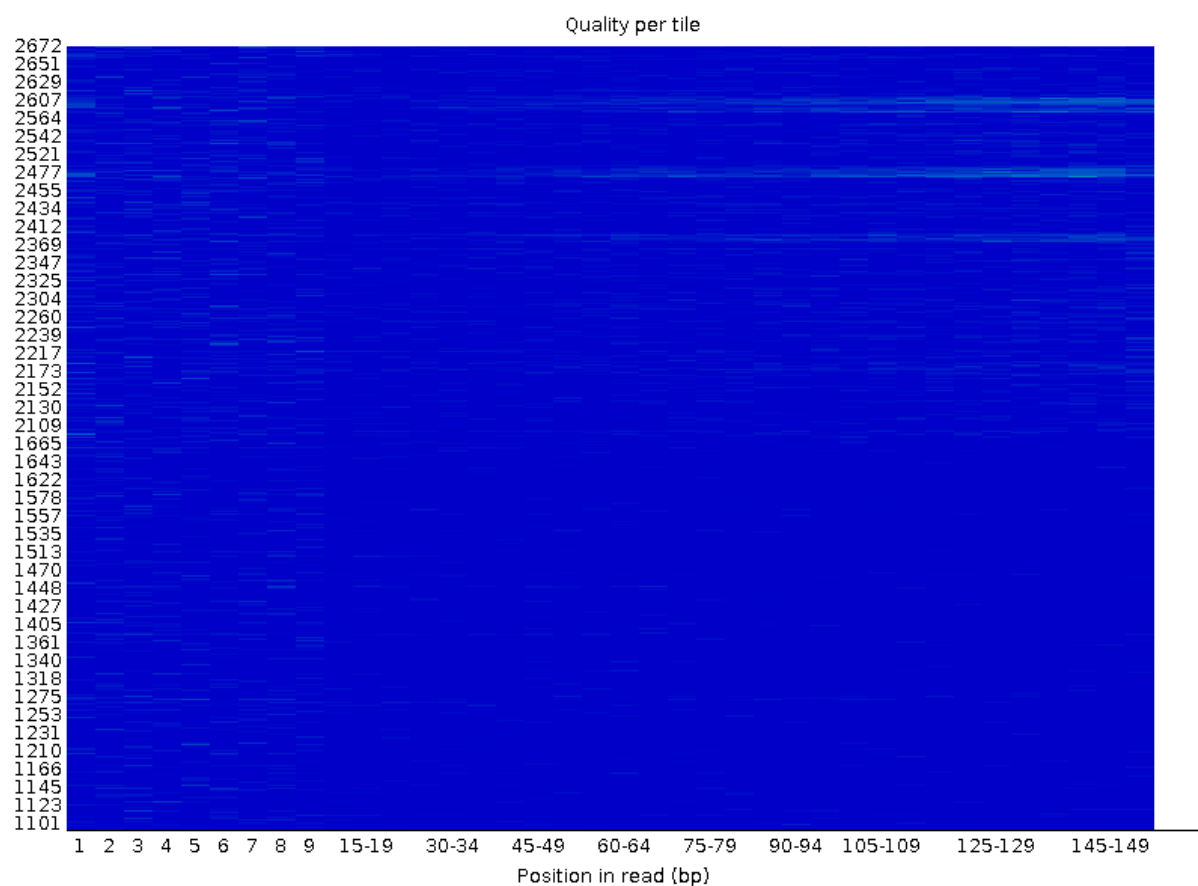
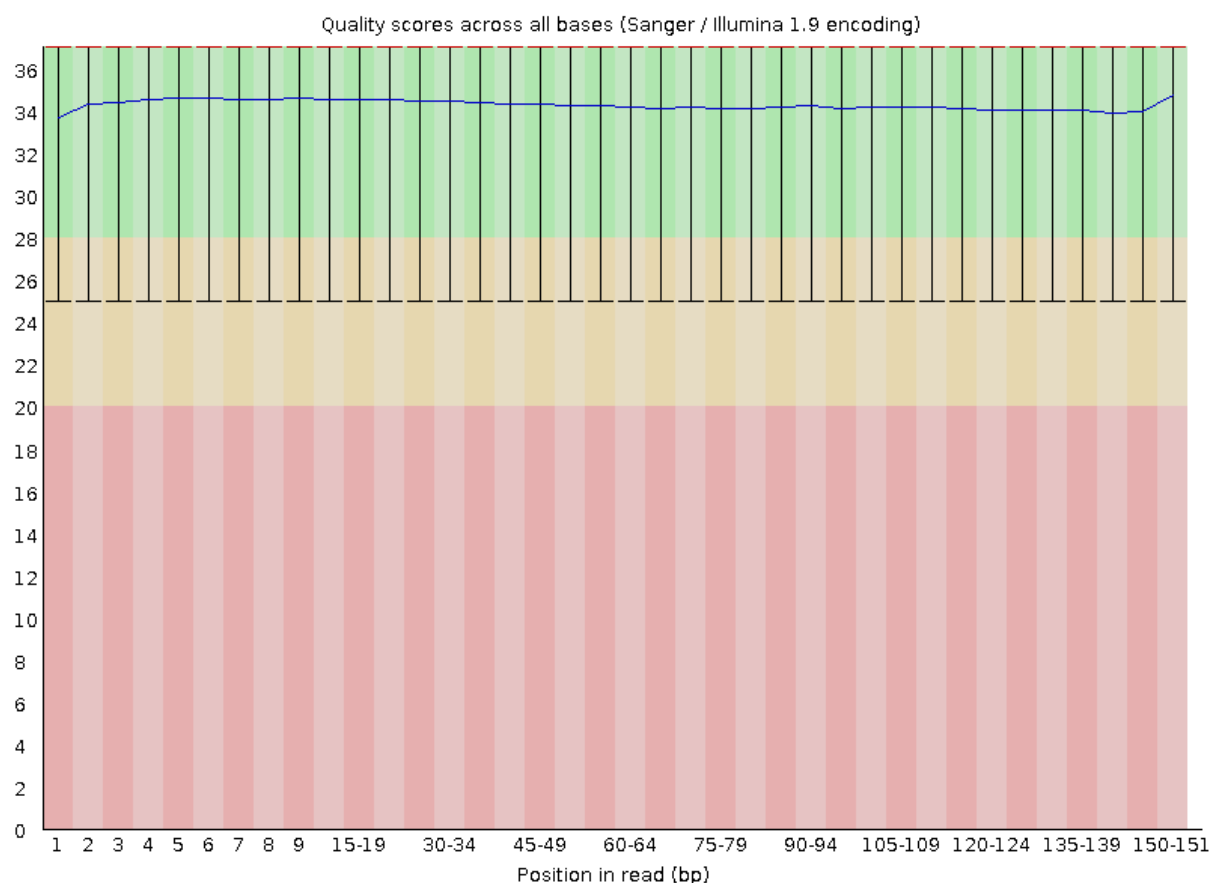
Length	Distribution	Abundance
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61	□□□□□□□□□□	0.41%
62	□□□□□□□□□□	0.42%
63	□□□□□□□□□□	0.41%
64	□□□□□□□□□□	0.43%
65	□□□□□□□□□□	0.40%
66	□□□□□□□□□□	0.41%
67	□□□□□□□□□□	0.43%
68	□□□□□□□□□□	0.40%
69	□□□□□□□□□□	0.40%
70	□□□□□□□□□□	0.41%
71	□□□□□□□□□□	0.36%
72	□□□□□□□□□□	0.37%
73	□□□□□□□□□□	0.37%
74	□□□□□□□□□□	0.38%
75	□□□□□□□□□□	0.35%
76	□□□□□□□□□□	0.36%
77	□□□□□□□□□□	0.36%
78	□□□□□□□□□□	0.36%
79	□□□□□□□□□□	0.35%
80	□□□□□□□□□□	0.38%
81	□□□□□□□□□□	0.35%
82	□□□□□□□□□□	0.35%
83	□□□□□□□□□□	0.33%
84	□□□□□□□□□□	0.34%
85	□□□□□□□□□□	0.34%
86	□□□□□□□□□□	0.32%
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88	□□□□□□□□□□	0.31%
89	□□□□□□□□□□	0.31%
90	□□□□□□□□□□	0.32%
91	□□□□□□□□□□	0.31%
92	□□□□□□□□□□	0.31%
93	□□□□□□□□□□	0.31%
94	□□□□□□□□□□	0.31%
95	□□□□□□□□□□	0.32%
96	□□□□□□□□□□	0.31%
97	□□□□□□□□□□	0.32%
98	□□□□□□□□□□	0.30%
99	□□□□□□□□□□	0.29%
100	□□□□□□□□□□	0.33%
101	□□□□□□□□□□	0.29%

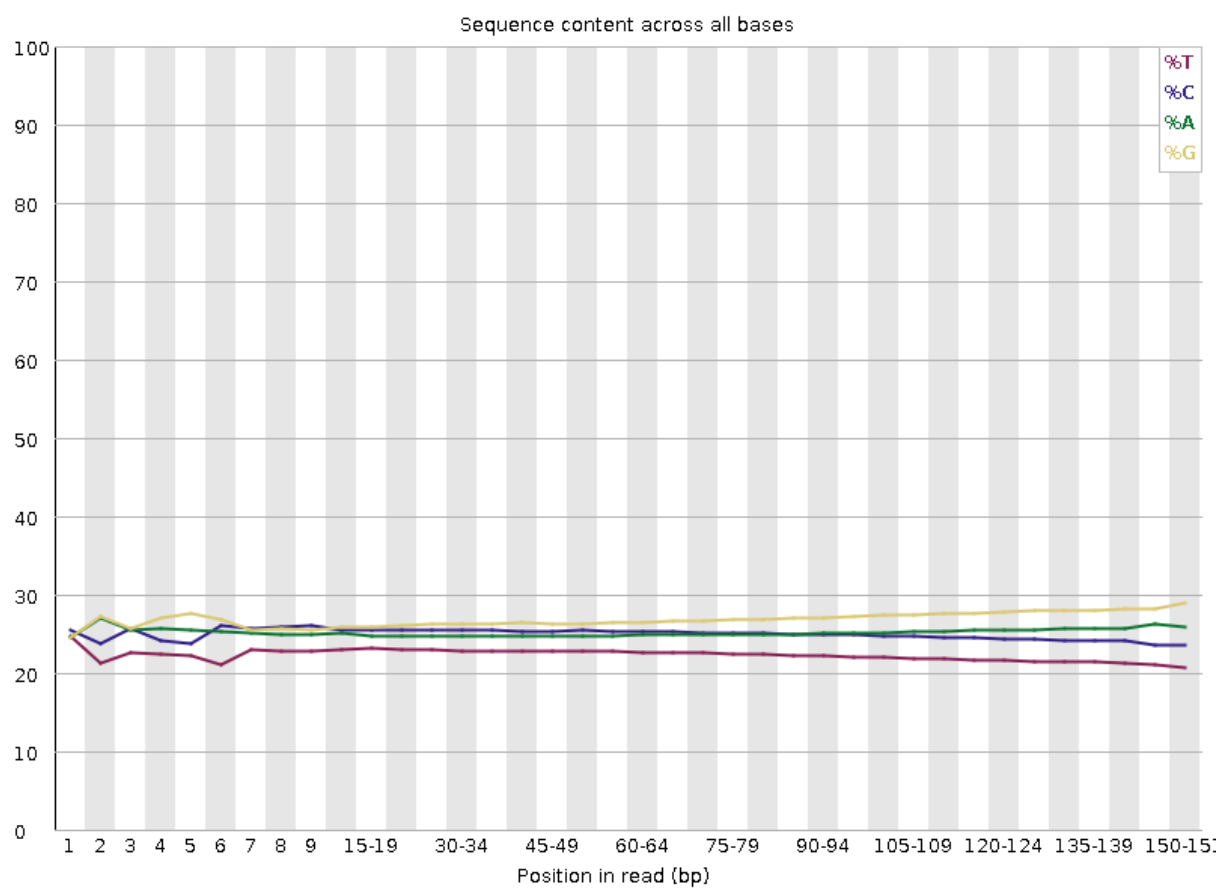
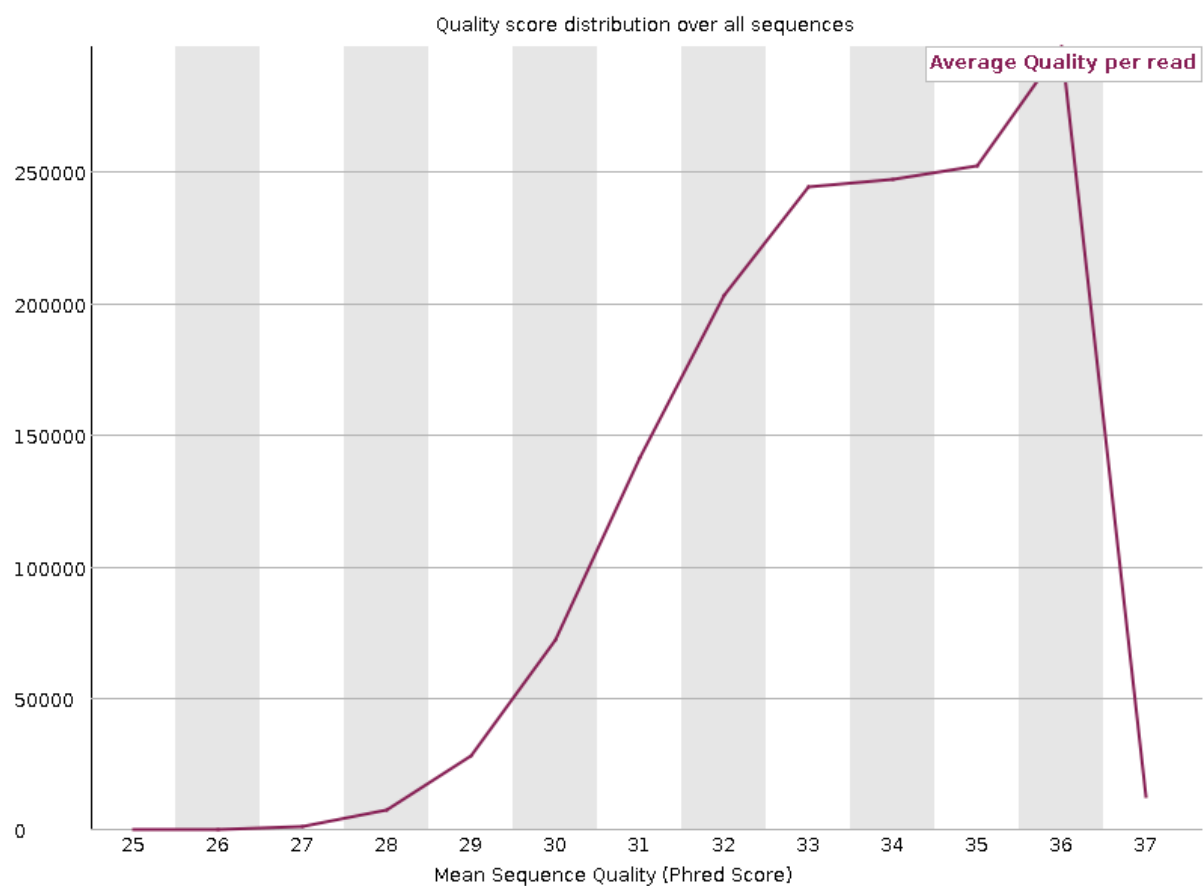
102	□□□□□□□□	0.29%
103	□□□□□□□□	0.29%
104	□□□□□□□□	0.30%
105	□□□□□□□□	0.28%
106	□□□□□□□□	0.29%
107	□□□□□□□□	0.29%
108	□□□□□□□□	0.27%
109	□□□□□□□□	0.28%
110	□□□□□□□□	0.26%
111	□□□□□□□□	0.27%
112	□□□□□□□□	0.28%
113	□□□□□□□□	0.28%
114	□□□□□□□□	0.29%
115	□□□□□□□□	0.28%
116	□□□□□□□□	0.28%
117	□□□□□□□□	0.27%
118	□□□□□□□□	0.27%
119	□□□□□□□□	0.28%
120	□□□□□□□□	0.28%
121	□□□□□□□□	0.28%
122	□□□□□□□□	0.30%
123	□□□□□□□□	0.31%
124	□□□□□□□□	0.30%
125	□□□□□□□□	0.29%
126	□□□□□□□□	0.28%
127	□□□□□□□□	0.29%
128	□□□□□□□□	0.28%
129	□□□□□□□□	0.27%
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135	□□□□□□□□	0.26%
136	□□□□□□□□	0.28%
137	□□□□□□□□	0.28%
138	□□□□□□□□	0.28%
139	□□□□□□□□	0.28%
140	□□□□□□□□	0.29%
141	□□□□□□□□	0.28%
142	□□□□□□□□	0.29%
143	□□□□□□□□	0.28%
144	□□□□□□□□	0.29%
145	□□□□□□□□	0.26%
146	□□□□□□□□	0.28%
147	□□□□□□□□	0.27%
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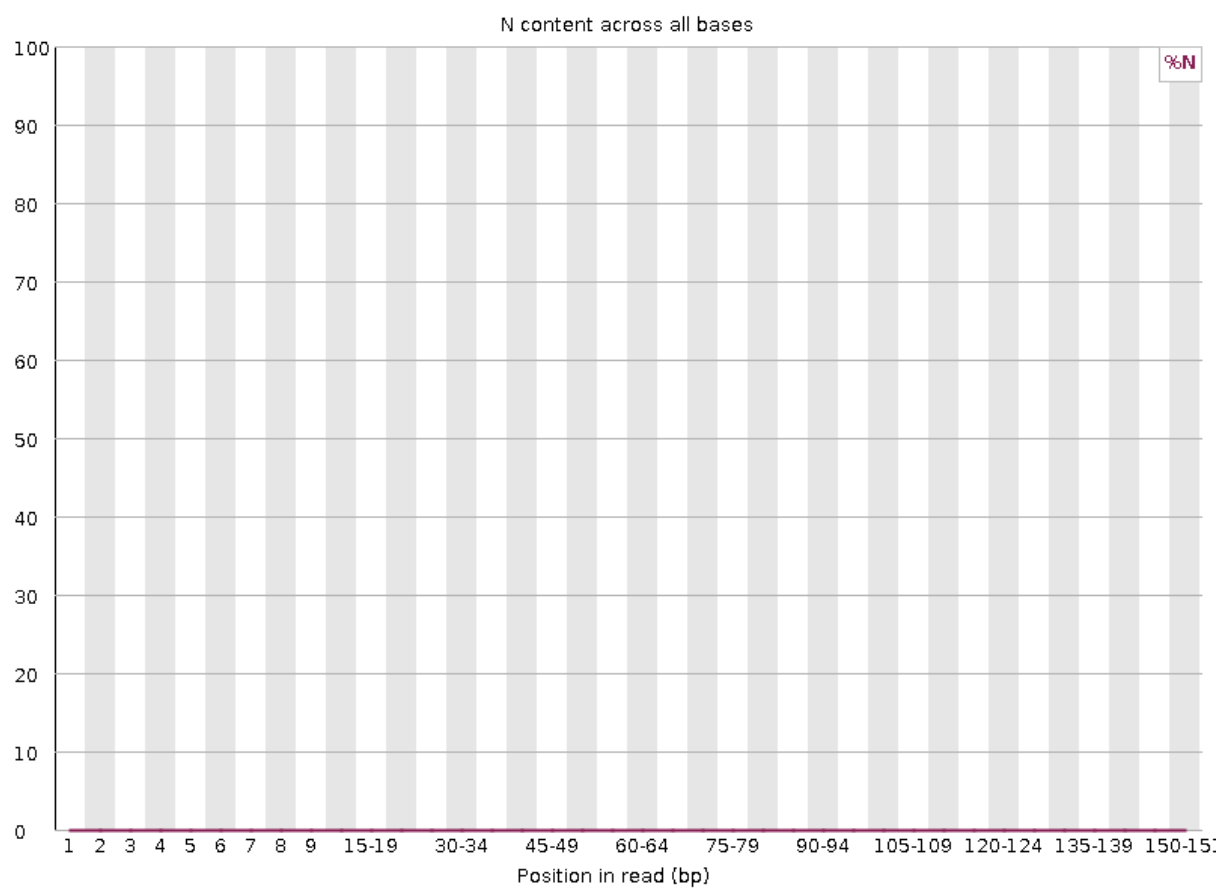
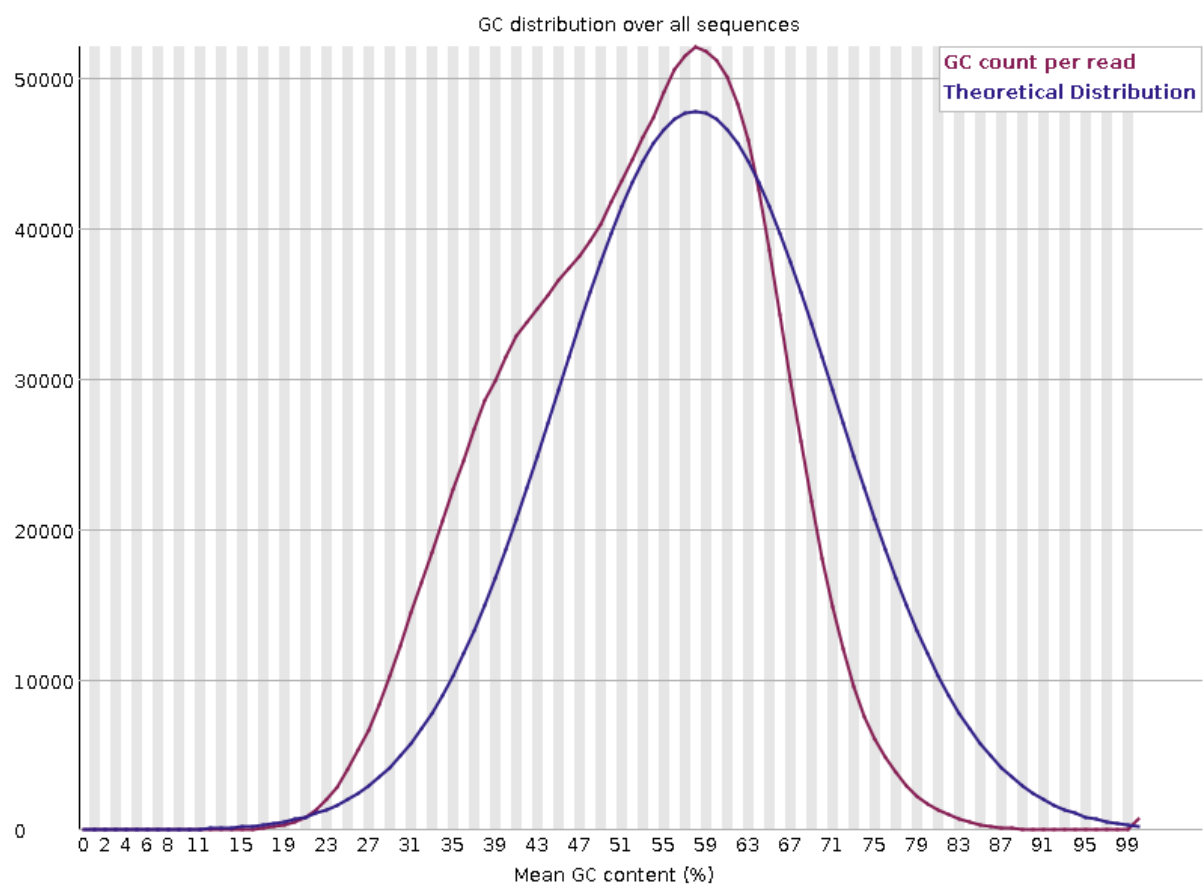
150 □□□□□□□□ 5.58%

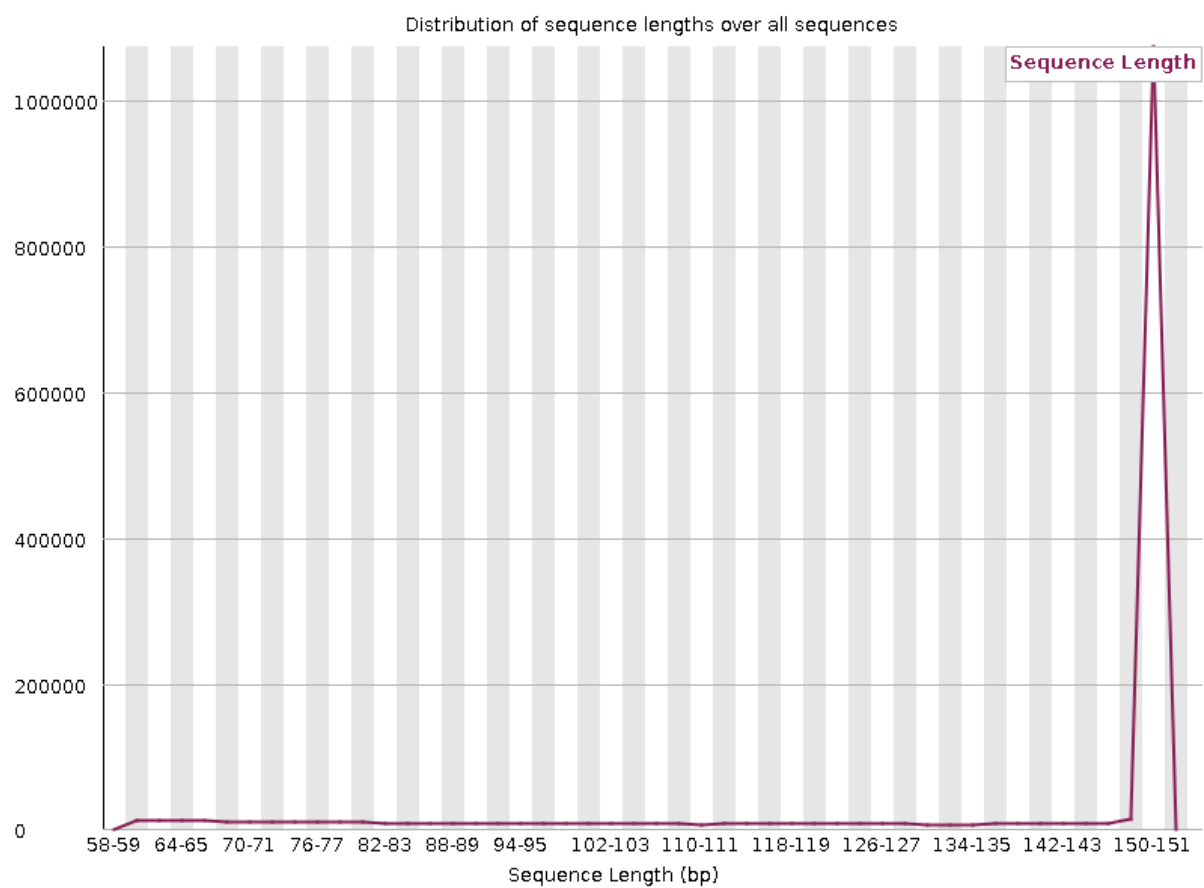
151 ■■■■■■□□□□ 65.60%

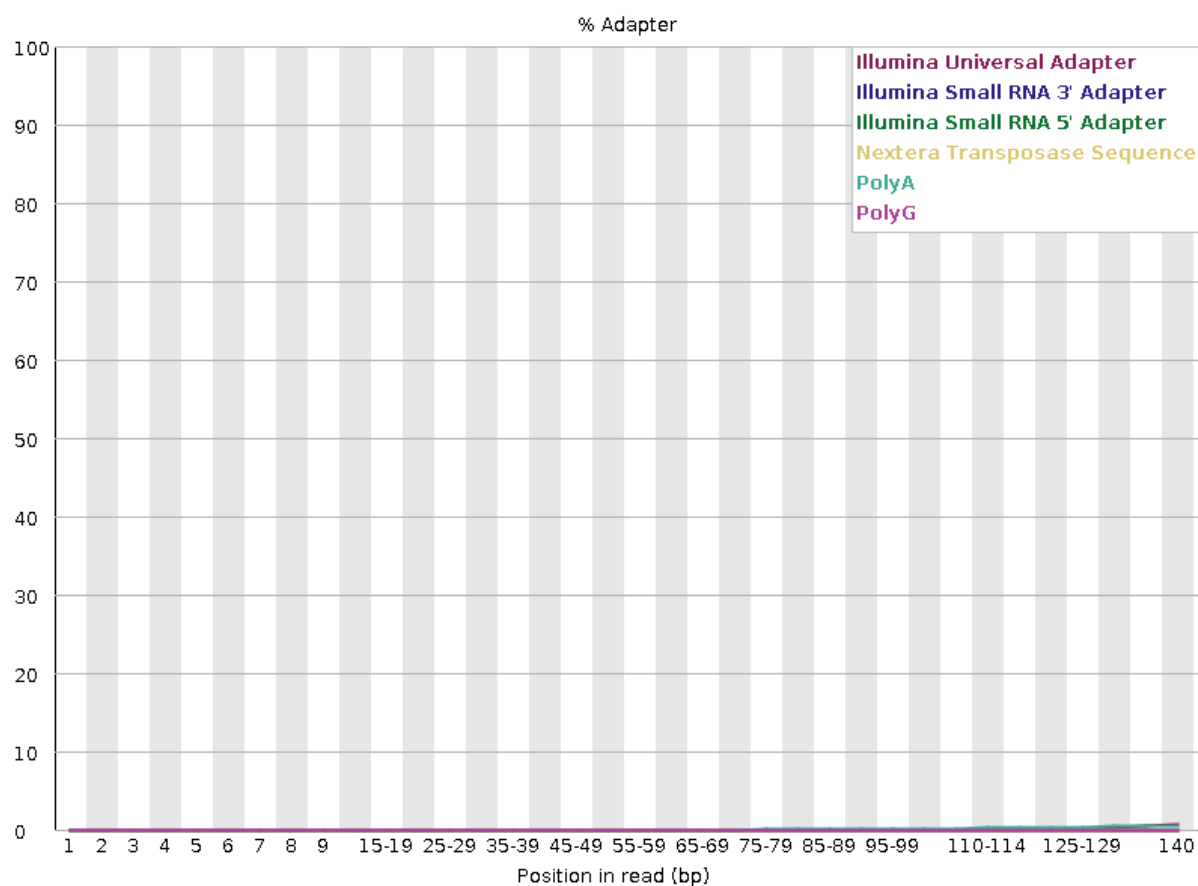
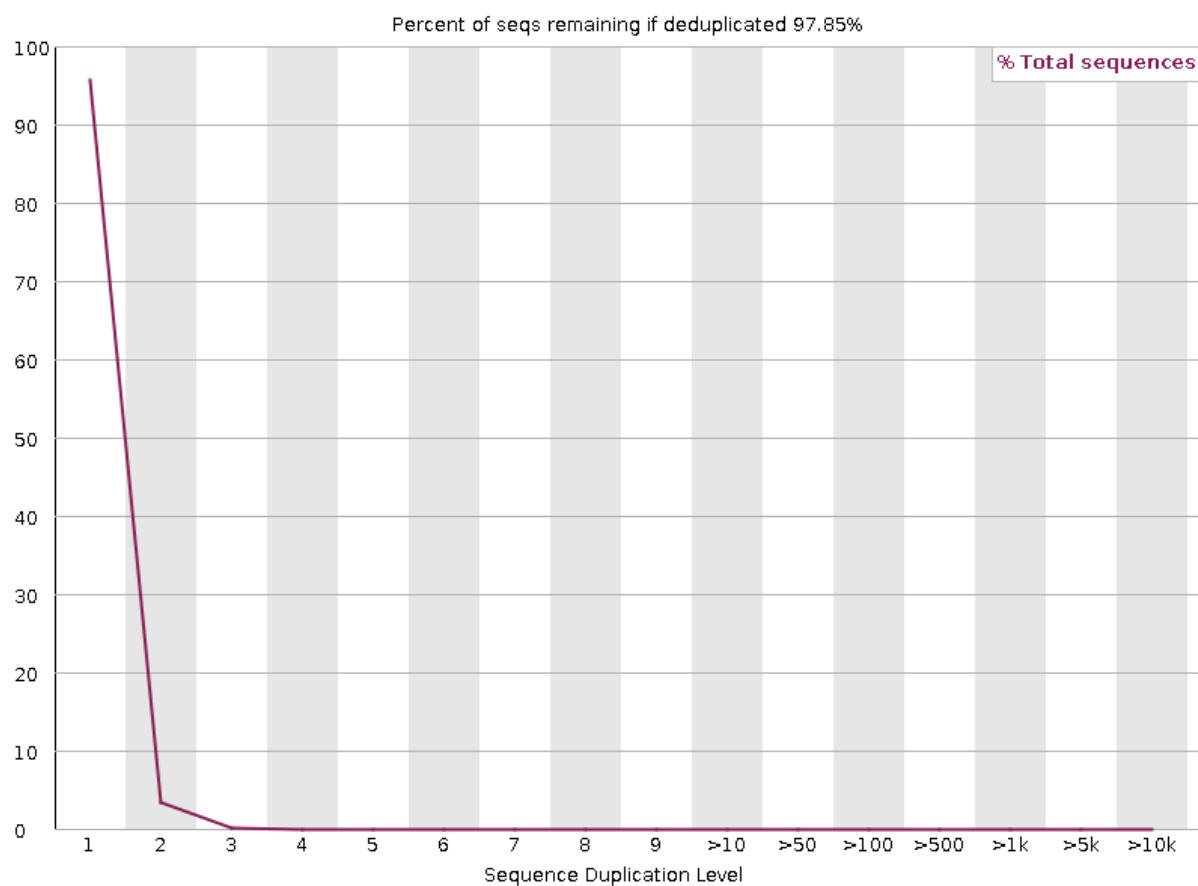
SRR37843242_2.up.fastq	
Measure	Value
Filename	SRR37843242_2.up.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1,508,577
Total Bases	206.3 Mbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	52











Discussion

Overrepresented sequences 모든 파일에서 검출되지 않았다. 2.fastq에서 감지되던 Per tile sequence quality의 Warning도 해결되었음. Adapter content 문제도 해결되었음. downstream 분석에 넘어가도 문제 없다고 판단.

SRR37843243

Description of Data

SRR37843243.1.fastq

The number of read sequences is 62721090

Total length of read sequences is 9470884590

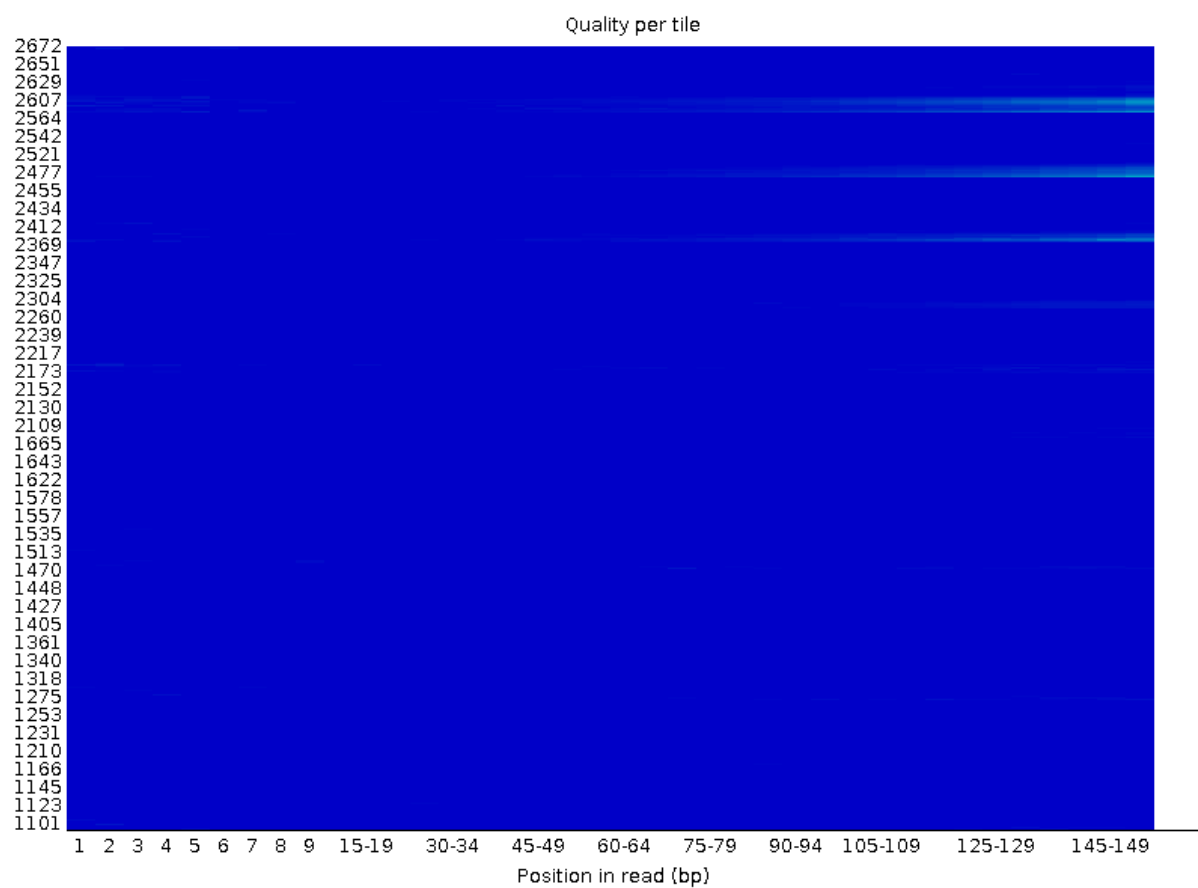
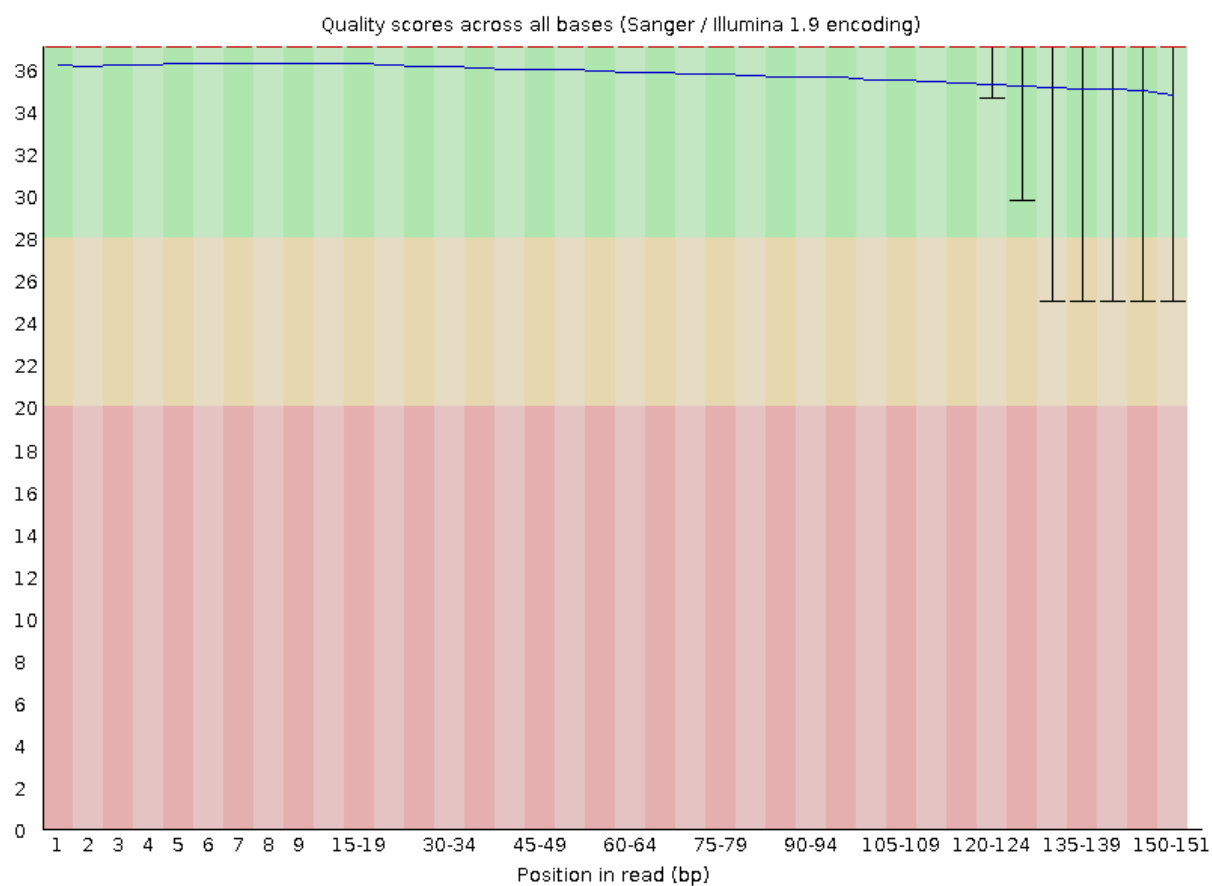
<Text distribution of read lengths>

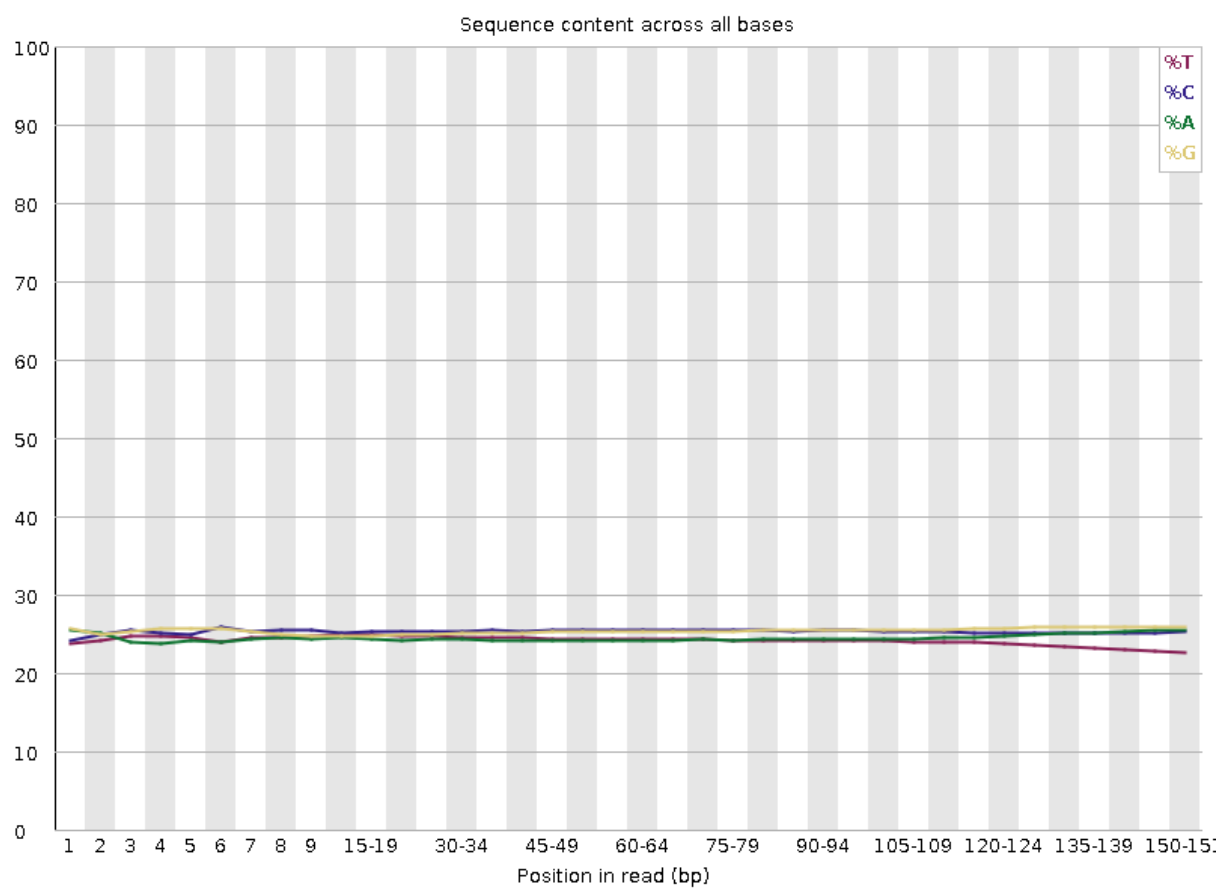
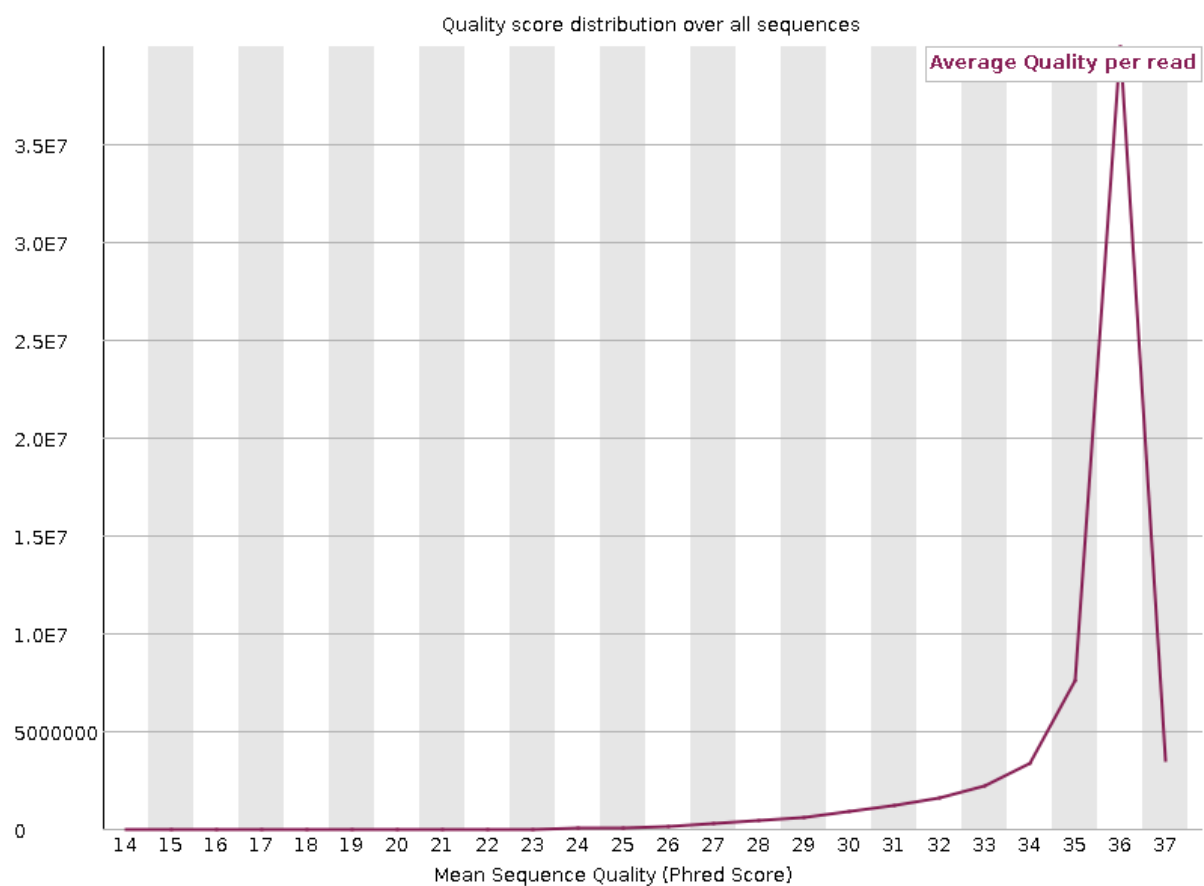
Length	Count	Abundance
151	62721090	100.00%

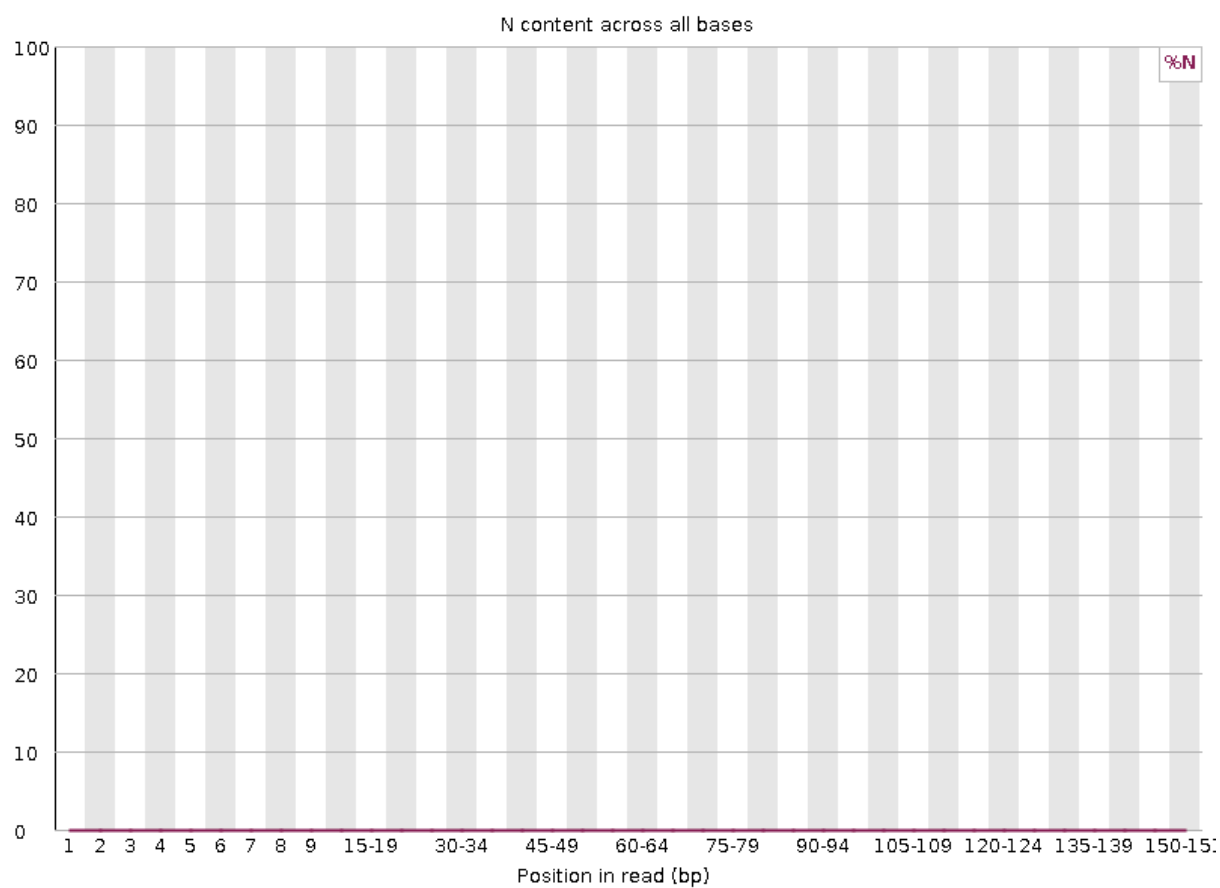
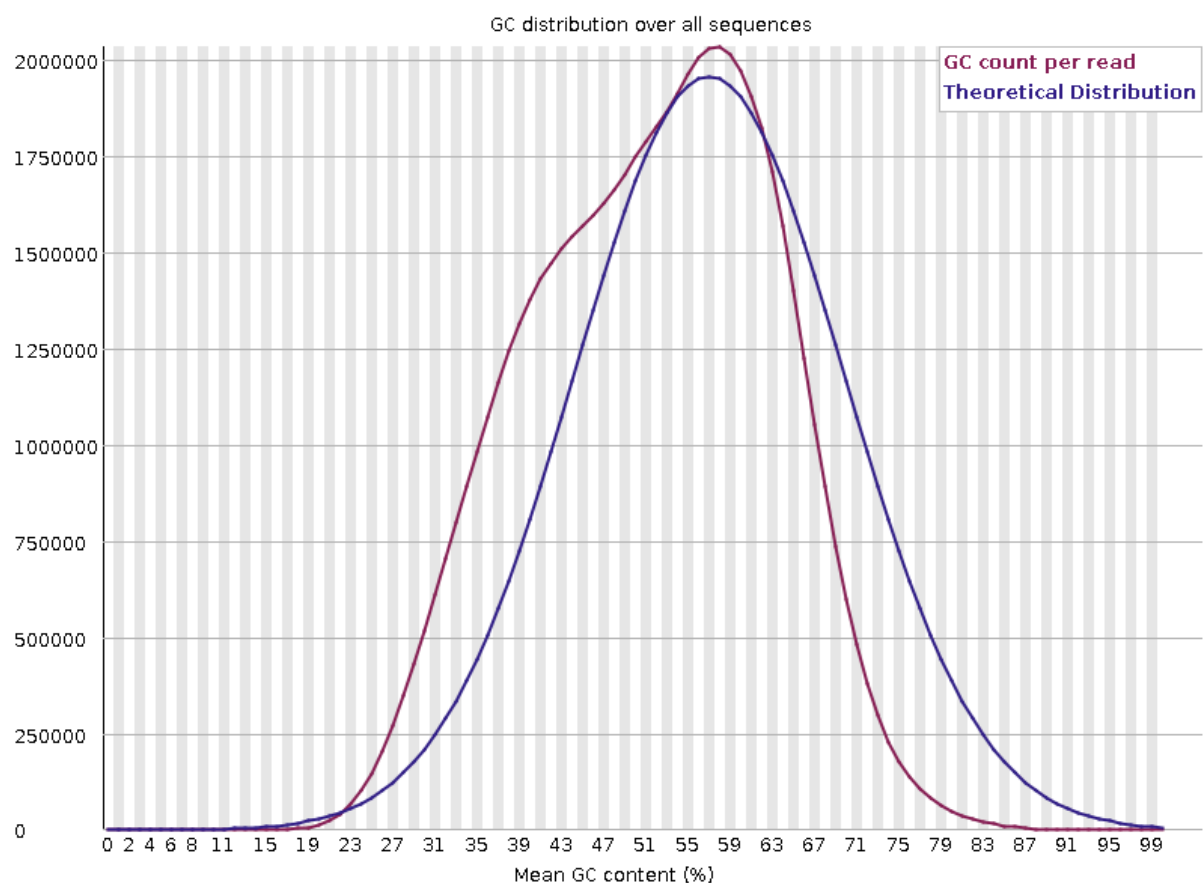
<Graphic distribution of read lengths>

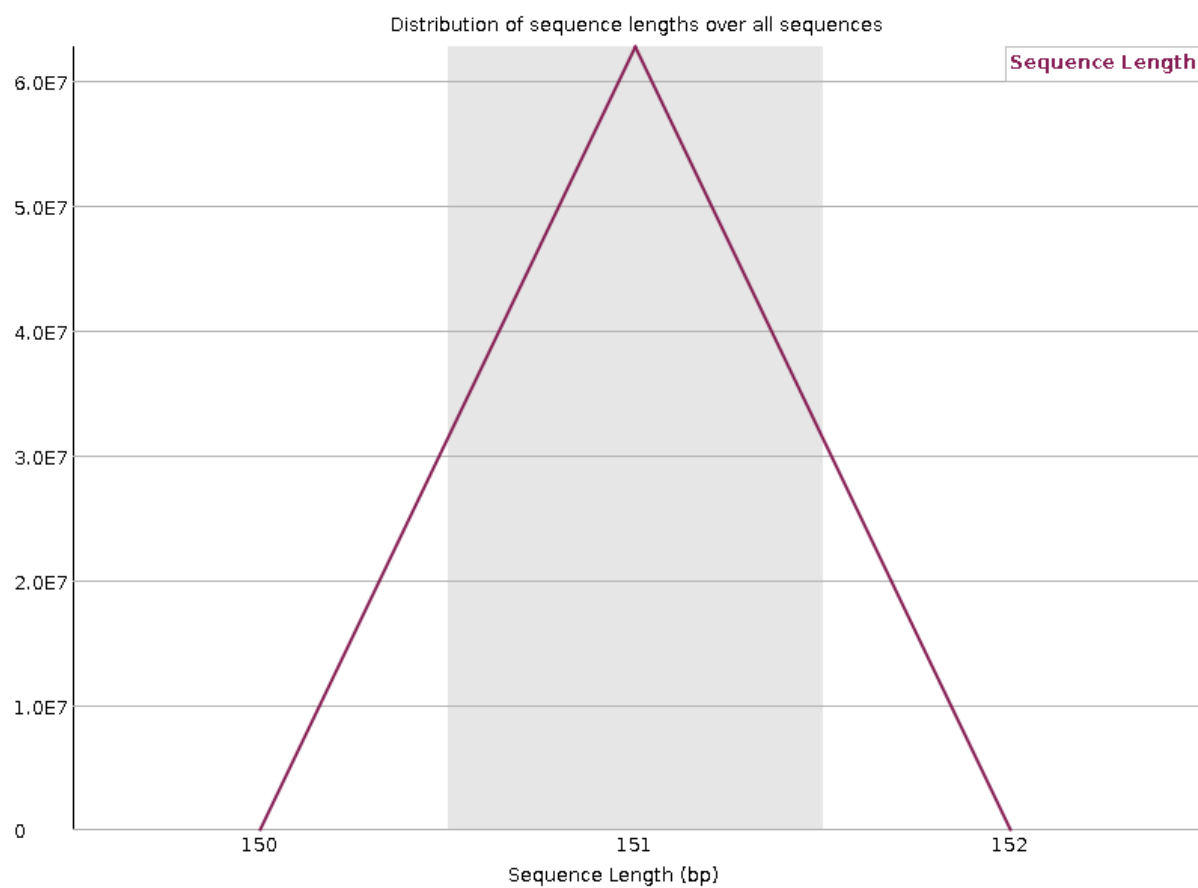
Length	Distribution	Abundance
151	■■■■■■■■■■■■■	100.00%

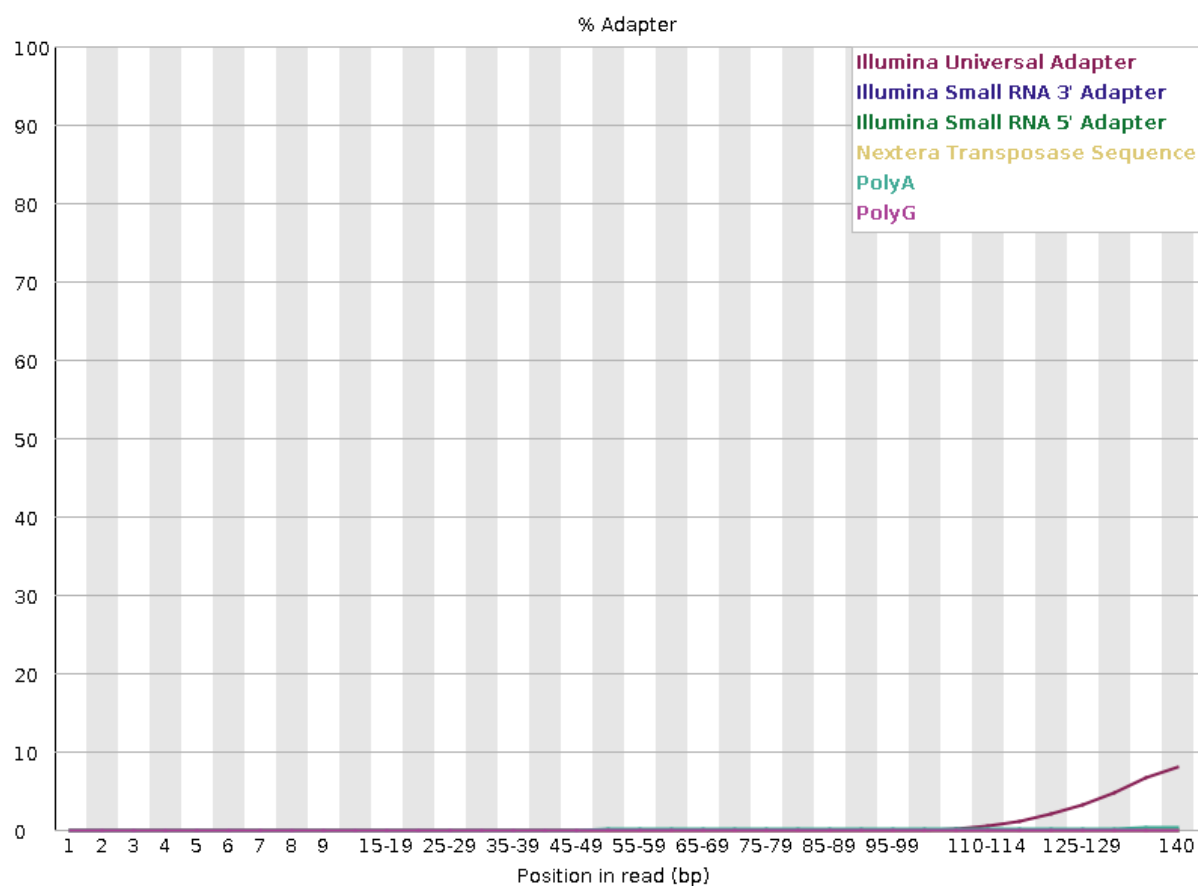
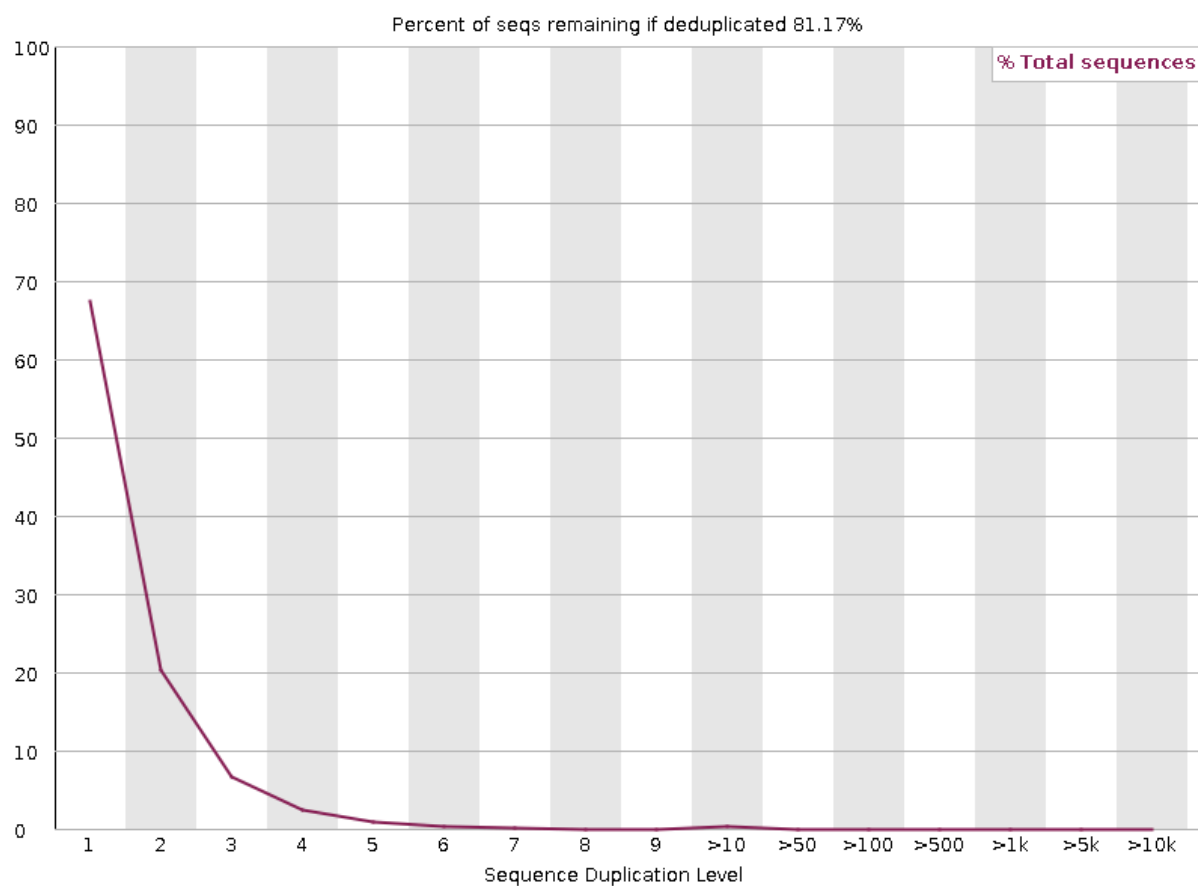
SRR37843243_1.fastq.gz	
Measure	Value
Filename	SRR37843243_1.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	62,721,090
Total Bases	9.4 Gbp
Sequences flagged as poor quality	0
Sequence length	151
%GC	51





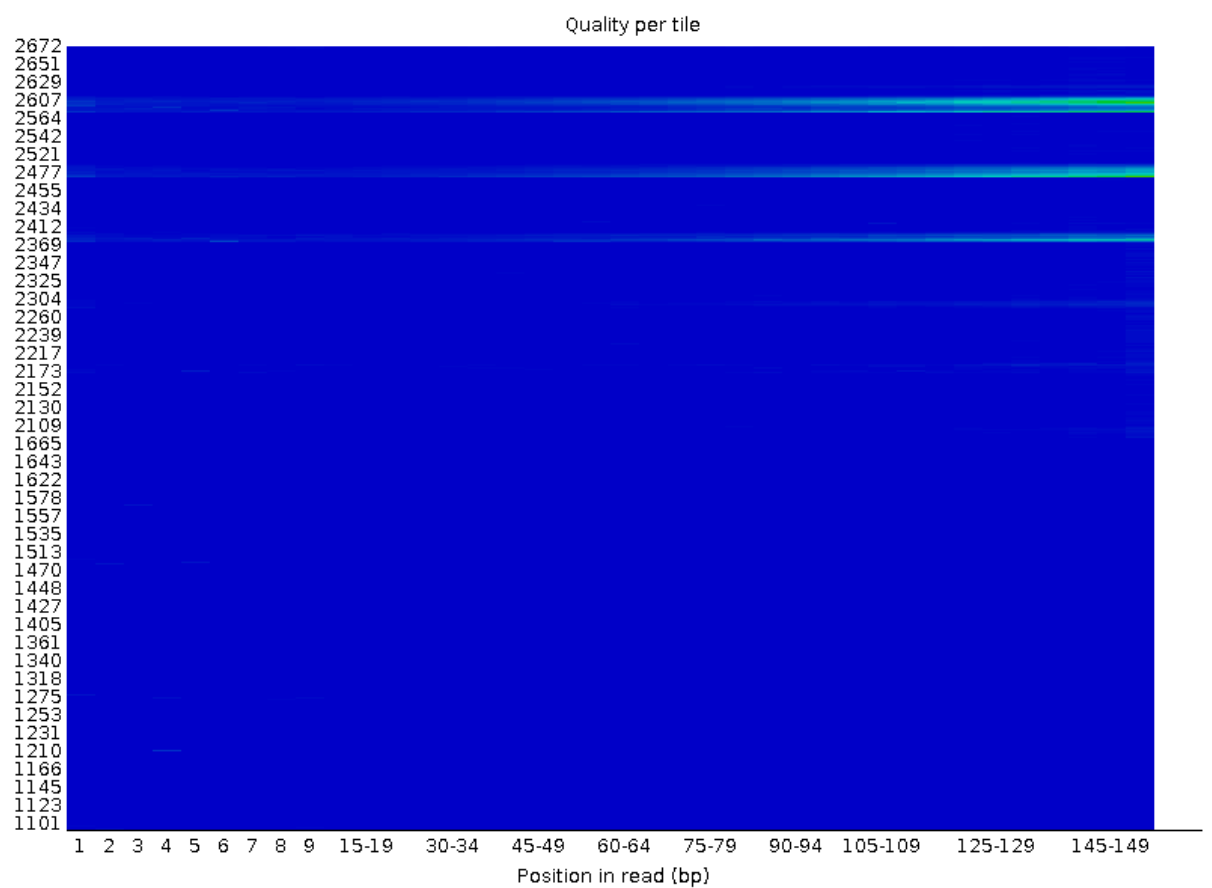
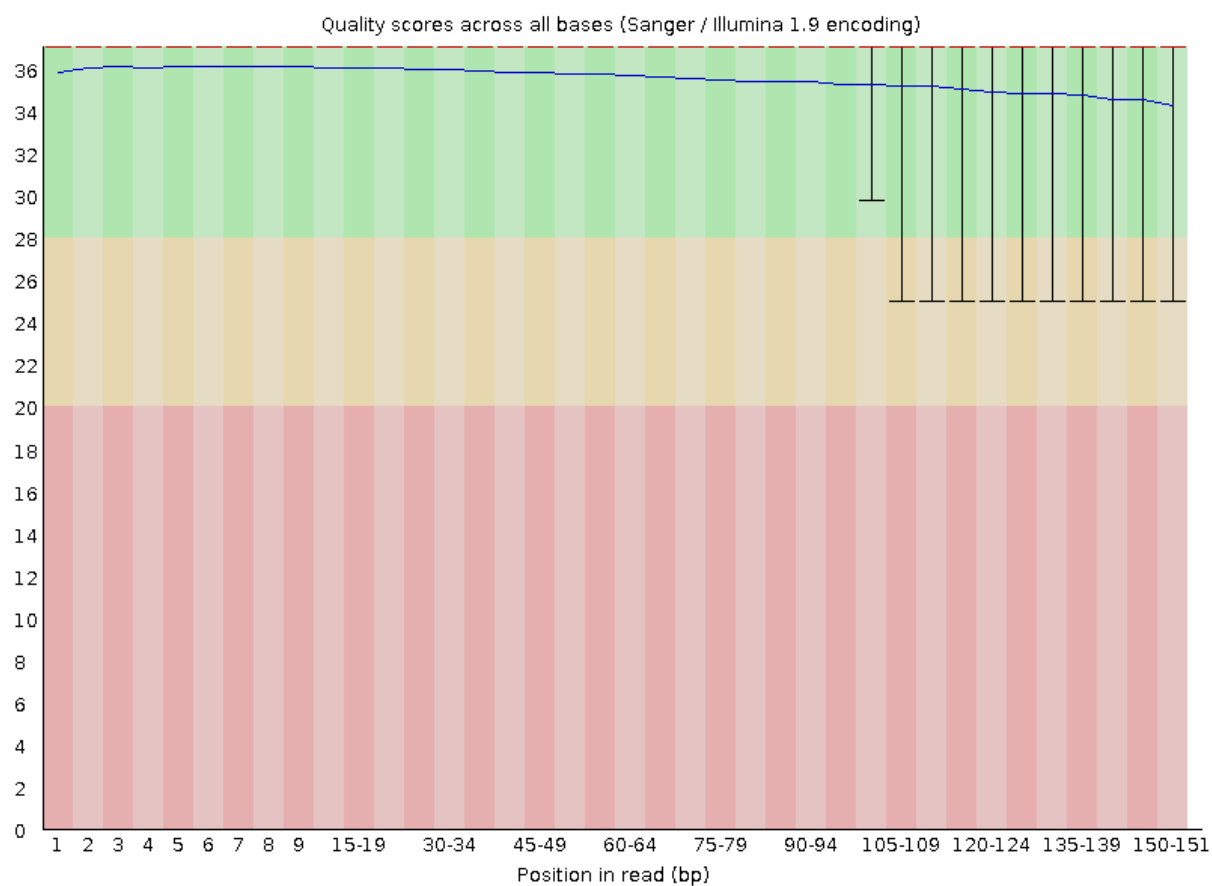


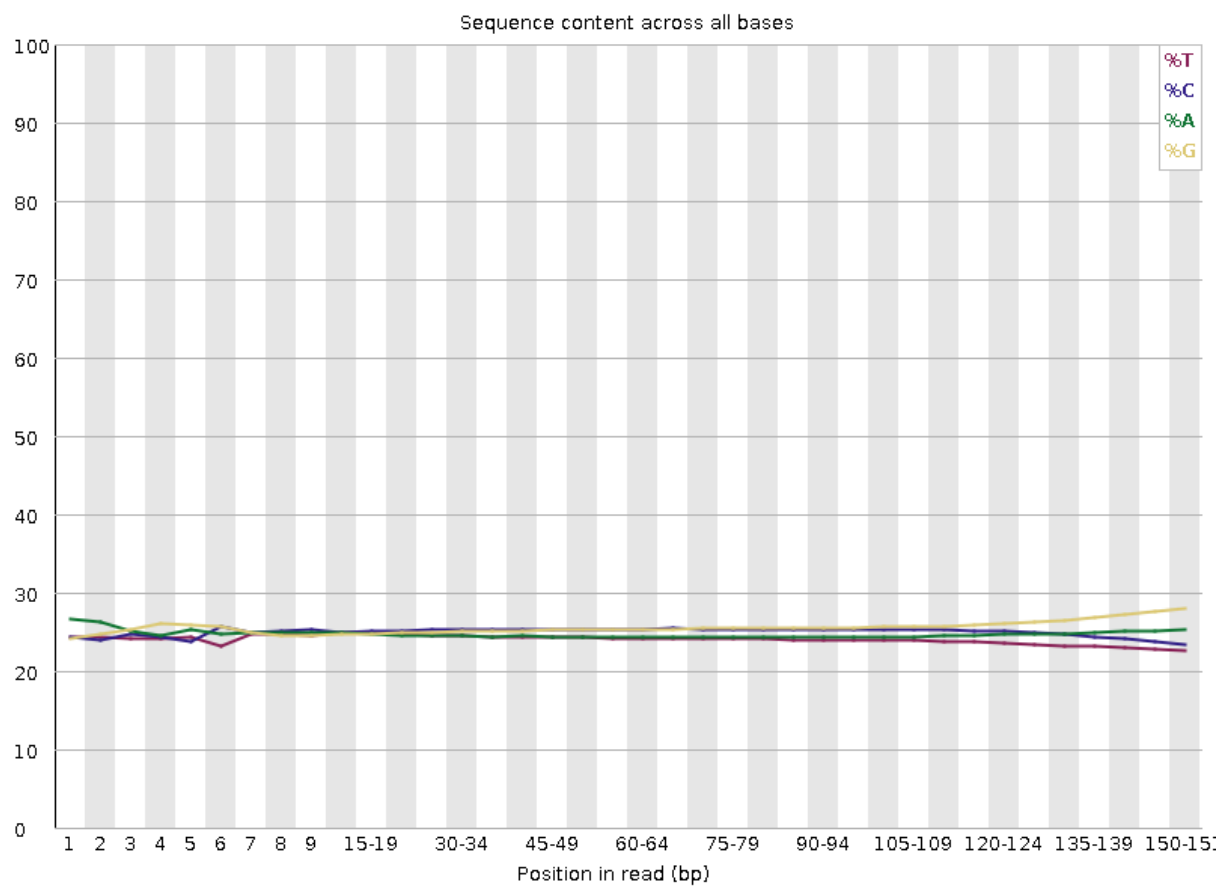
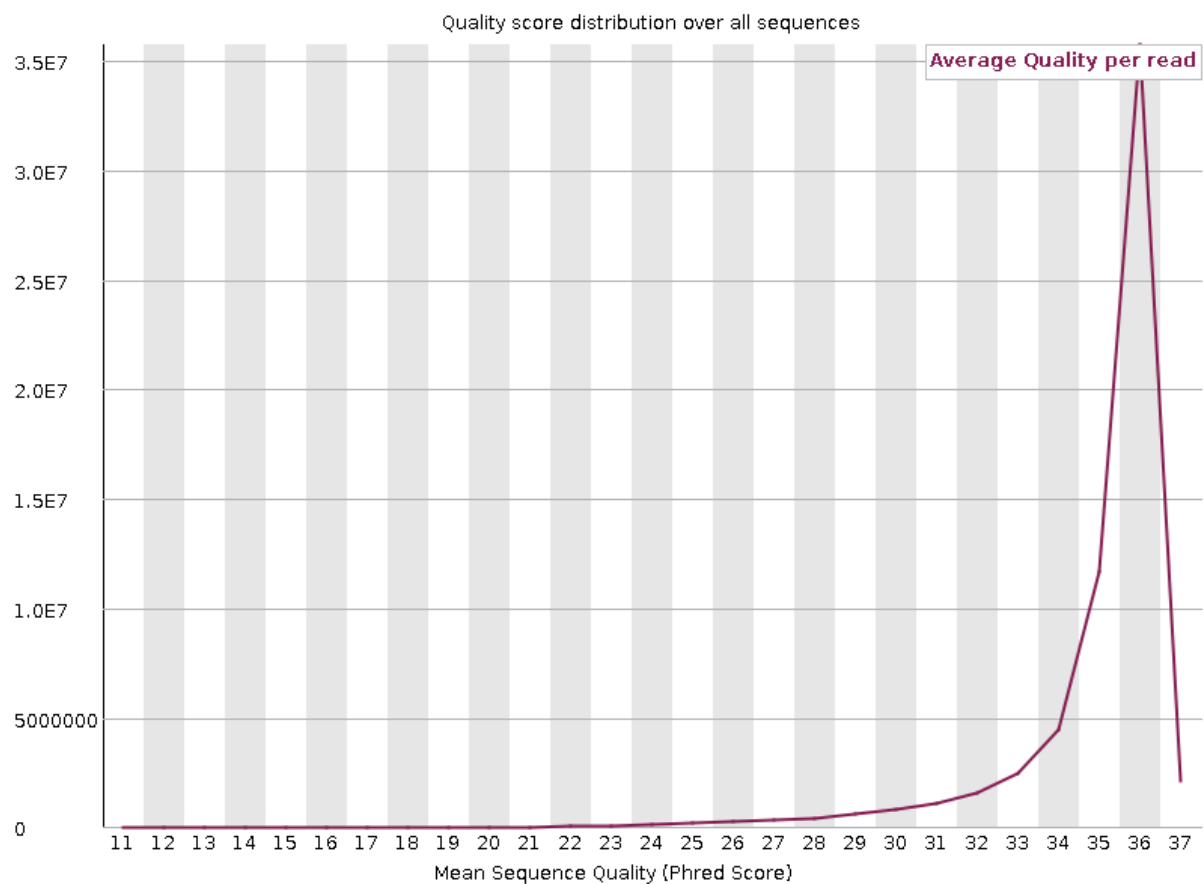


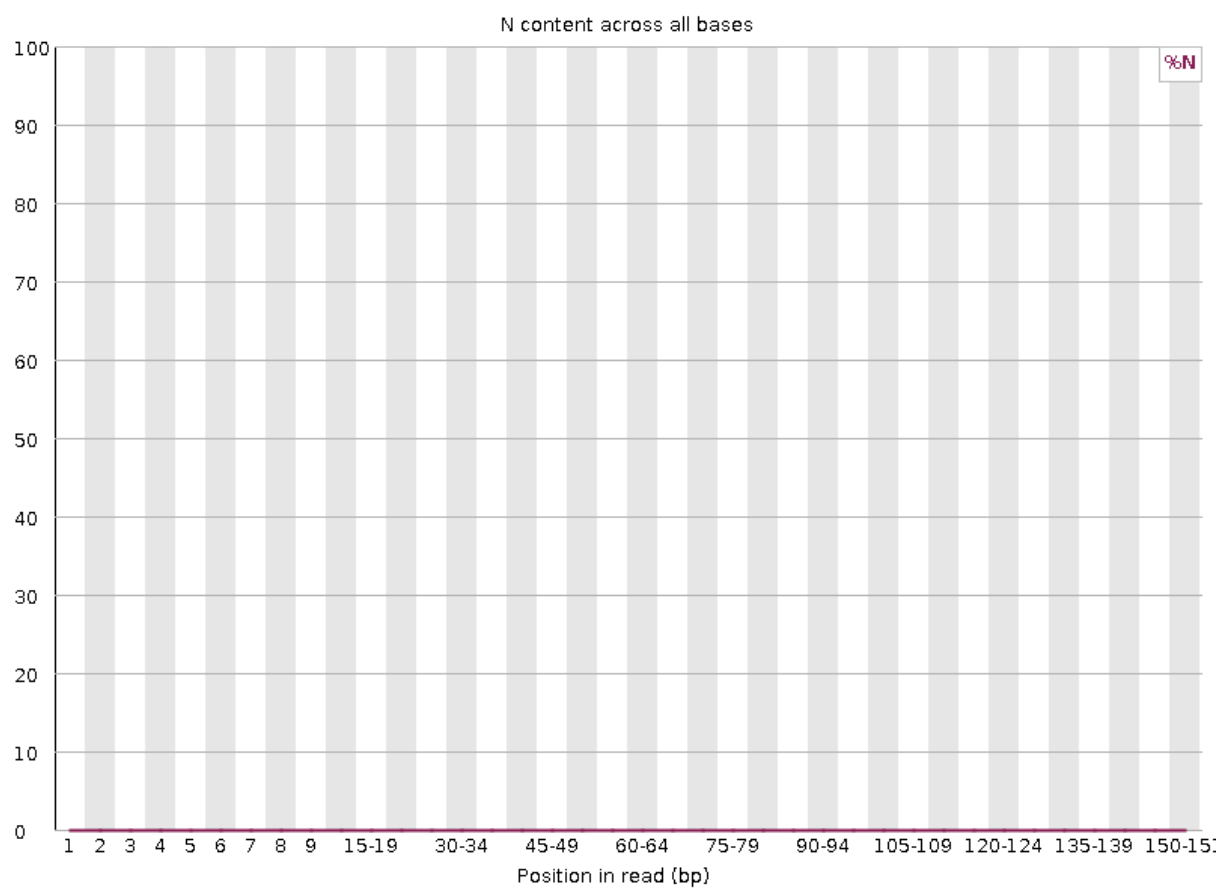
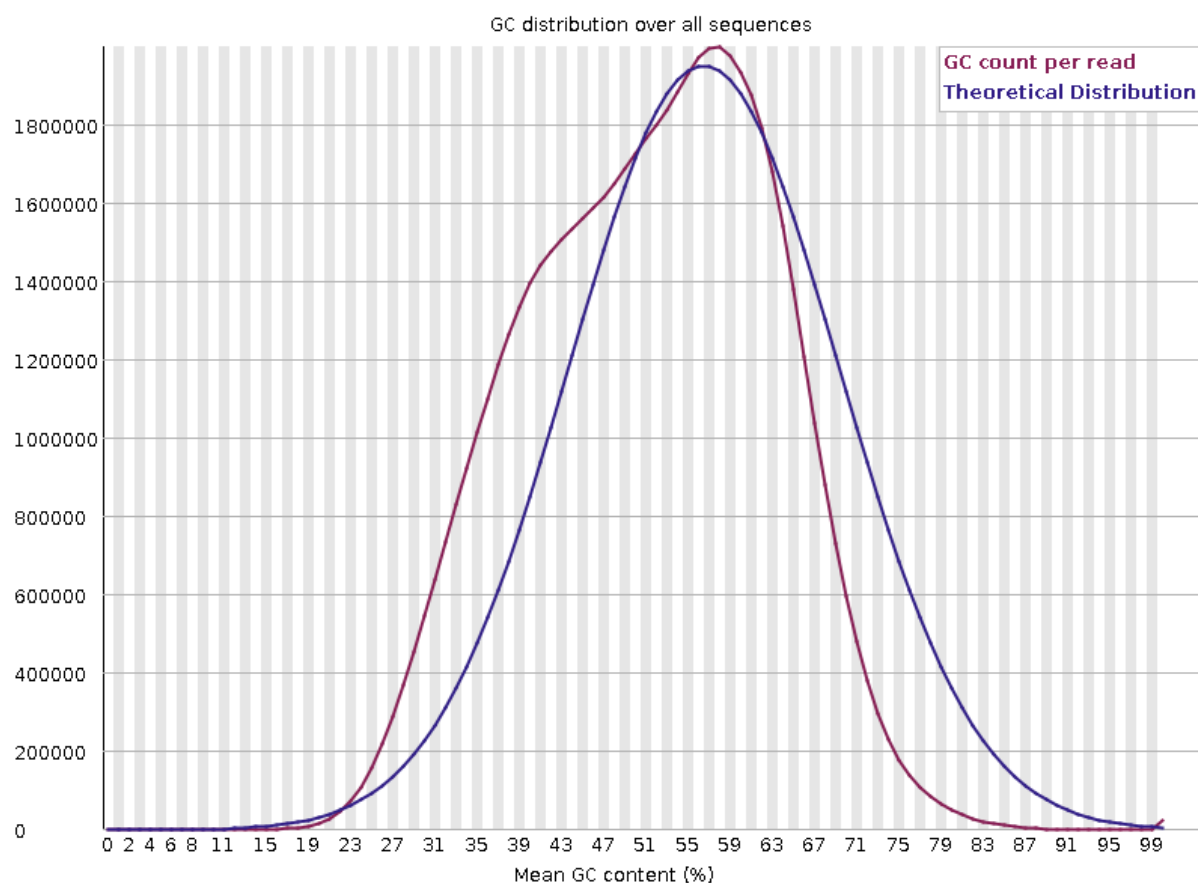


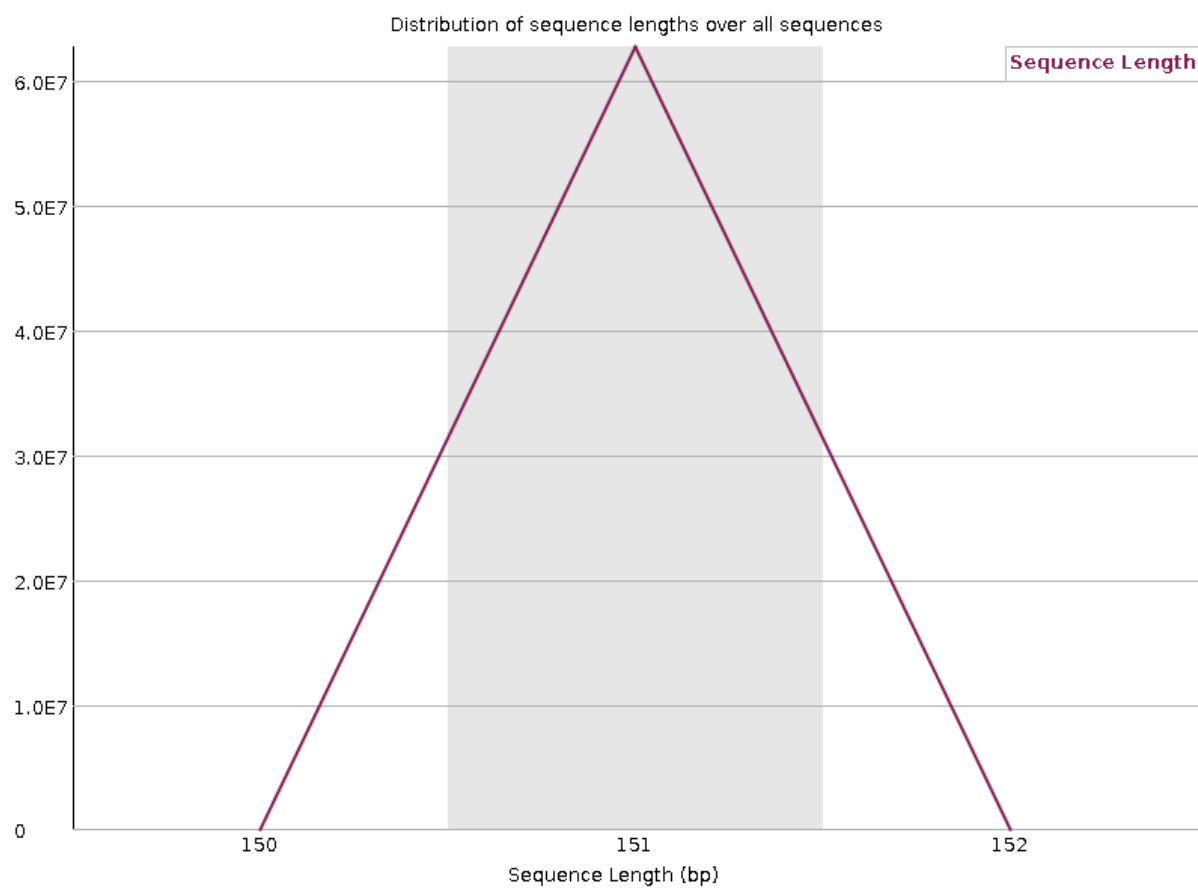
SRR37843243_2.fastq

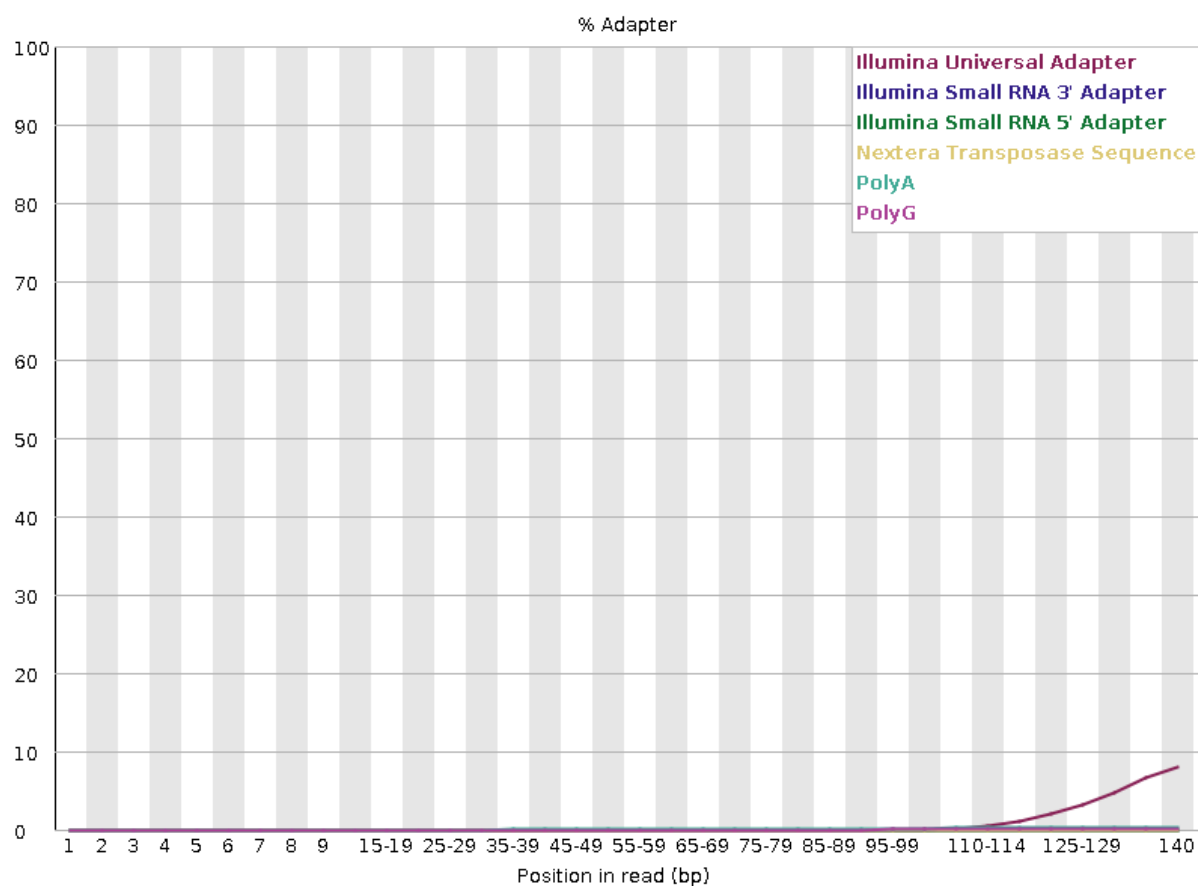
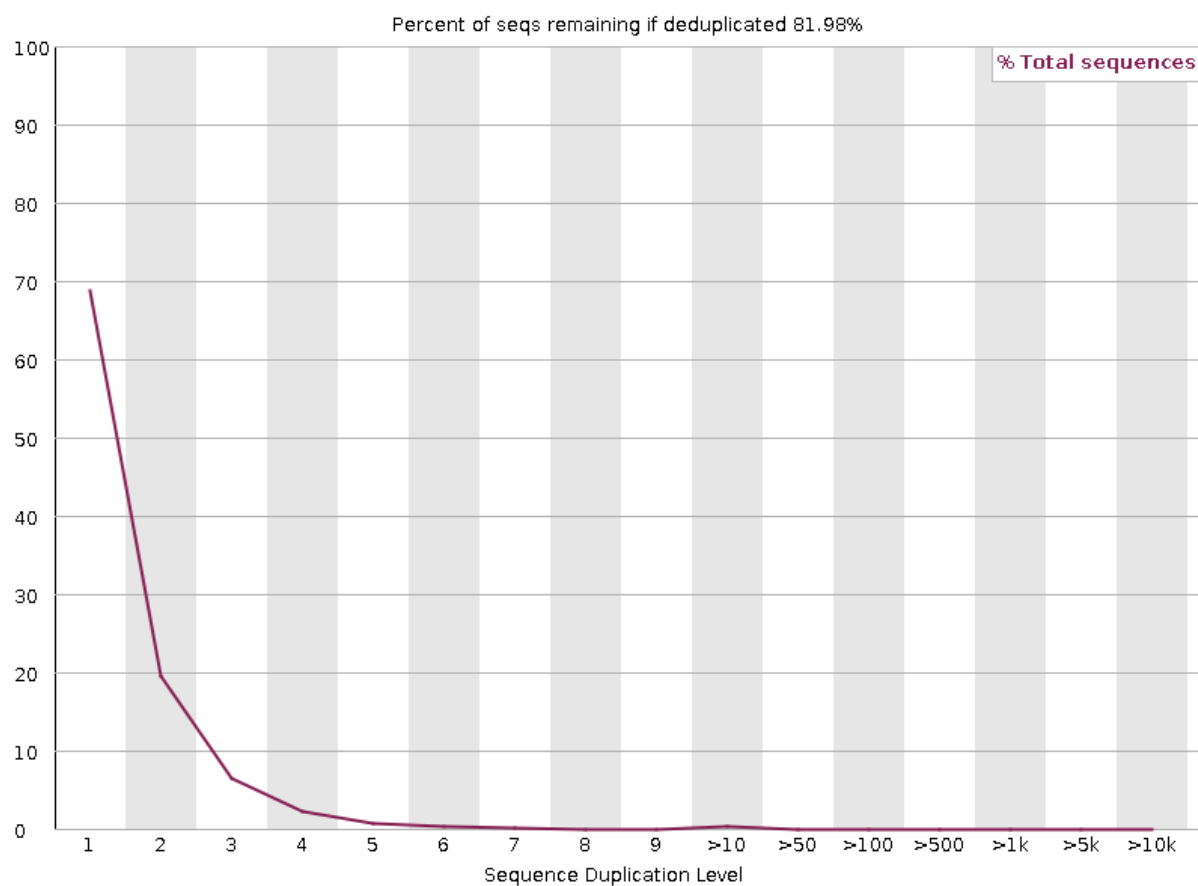
SRR37843243_2.fastq.gz	
Measure	Value
Filename	SRR37843243_2.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	62,721,090
Total Bases	9.4 Gbp
Sequences flagged as poor quality	0
Sequence length	151
%GC	51











Overrepresented sequences

No overrepresented sequences in both 1 and 2.

After Trimming

TrimmomaticPE: Started with arguments:

SRR37843243_1.fastq SRR37843243_2.fastq SRR37843243_1.p.fastq SRR37843243_1.up.fastq

SRR37843243_2.p.fastq SRR37843243_2.up.fastq

ILLUMINACLIP:/mss/home_student/202112379/programs/Trimmomatic-0.39/adapters/TruSeq3-PE.

fa:2:30:10:2:True SLIDINGWINDOW:4:20 LEADING:10 TRAILING:10 MINLEN:60

Using PrefixPair: 'TACACTCTTCCCTACACGACGCTCTTCCGATCT' and

'GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT'

ILLUMINACLIP: Using 1 prefix pairs, 0 forward/reverse sequences, 0 forward only sequences, 0 reverse only sequences

Quality encoding detected as phred33

Input Read Pairs: 62721090 **Both Surviving: 58329914 (93.00%)** Forward Only Surviving:

1910868 (3.05%) Reverse Only Surviving: 1603175 (2.56%) Dropped: 877133 (1.40%)

TrimmomaticPE: Completed successfully

SRR37843243_1.p.fastq

The number of read sequences is 58329914

Total length of read sequences is 8528951521

<Text distribution of read lengths>

Length	Count	Abundance
60	32651	0.06%
61	33197	0.06%
62	32130	0.06%
63	33279	0.06%
64	32282	0.06%
65	34054	0.06%
66	32701	0.06%
67	33758	0.06%
68	33097	0.06%
69	32900	0.06%
70	33077	0.06%
71	32459	0.06%
72	32444	0.06%
73	33042	0.06%
74	33109	0.06%
75	32999	0.06%
76	33160	0.06%
77	32422	0.06%
78	34372	0.06%

79	34020	0.06%
80	34390	0.06%
81	34978	0.06%
82	35573	0.06%
83	34212	0.06%
84	34101	0.06%
85	34208	0.06%
86	34504	0.06%
87	35002	0.06%
88	34977	0.06%
89	35337	0.06%
90	36358	0.06%
91	35936	0.06%
92	36258	0.06%
93	36338	0.06%
94	39060	0.07%
95	37644	0.06%
96	38854	0.07%
97	41744	0.07%
98	40576	0.07%
99	43571	0.07%
100	44971	0.08%
101	47714	0.08%
102	50452	0.09%
103	52899	0.09%
104	55028	0.09%
105	57773	0.10%
106	60860	0.10%
107	65186	0.11%
108	69903	0.12%
109	74403	0.13%
110	78467	0.13%
111	83730	0.14%
112	89119	0.15%
113	96144	0.16%
114	101625	0.17%
115	109702	0.19%
116	116666	0.20%
117	124958	0.21%
118	133071	0.23%
119	141470	0.24%
120	149685	0.26%
121	155490	0.27%
122	166309	0.29%
123	174279	0.30%
124	183927	0.32%
125	191025	0.33%
126	199065	0.34%

127	205867	0.35%
128	217684	0.37%
129	225785	0.39%
130	235588	0.40%
131	242777	0.42%
132	251664	0.43%
133	257893	0.44%
134	265418	0.46%
135	271407	0.47%
136	279677	0.48%
137	284046	0.49%
138	295422	0.51%
139	294954	0.51%
140	303022	0.52%
141	308232	0.53%
142	312336	0.54%
143	319611	0.55%
144	326030	0.56%
145	327911	0.56%
146	331524	0.57%
147	335023	0.57%
148	338957	0.58%
149	412597	0.71%
150	1484225	2.54%
151	45933569	78.75%

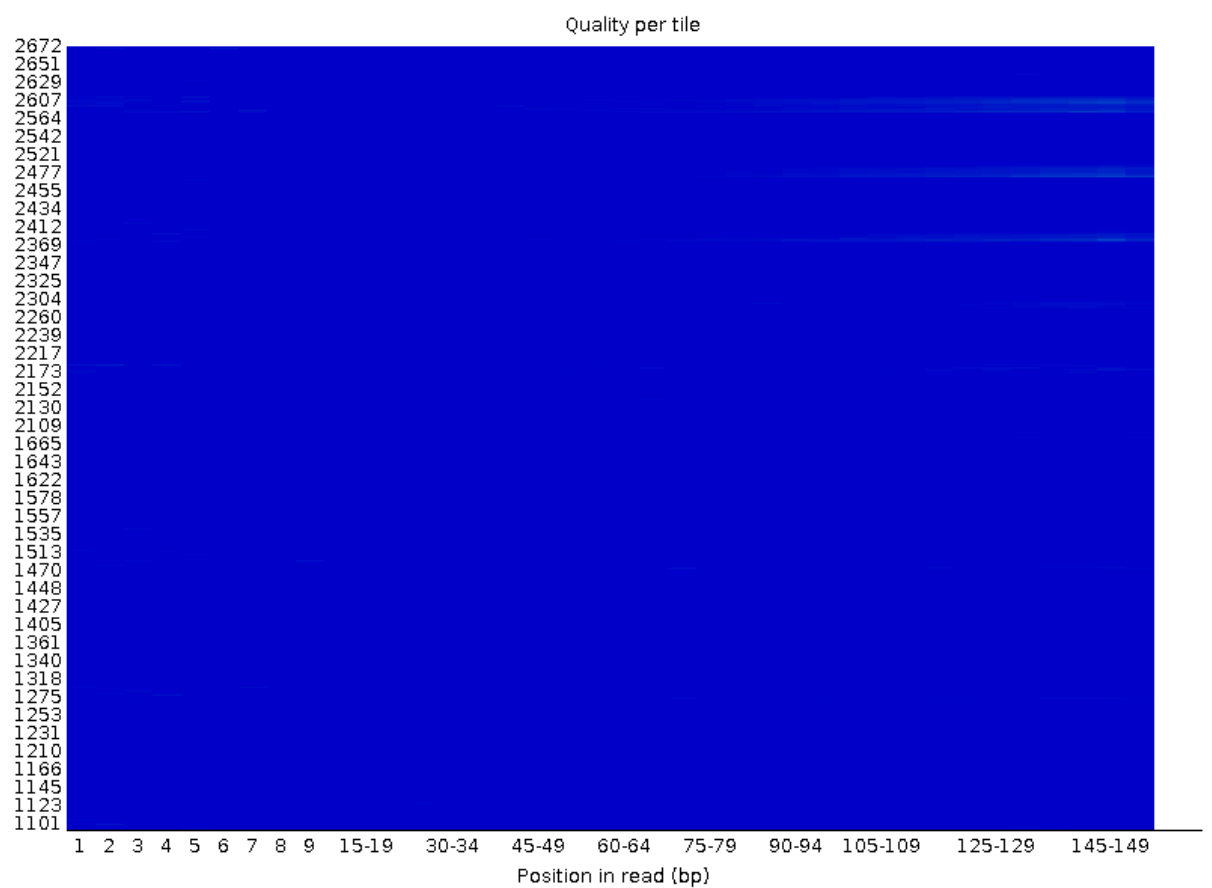
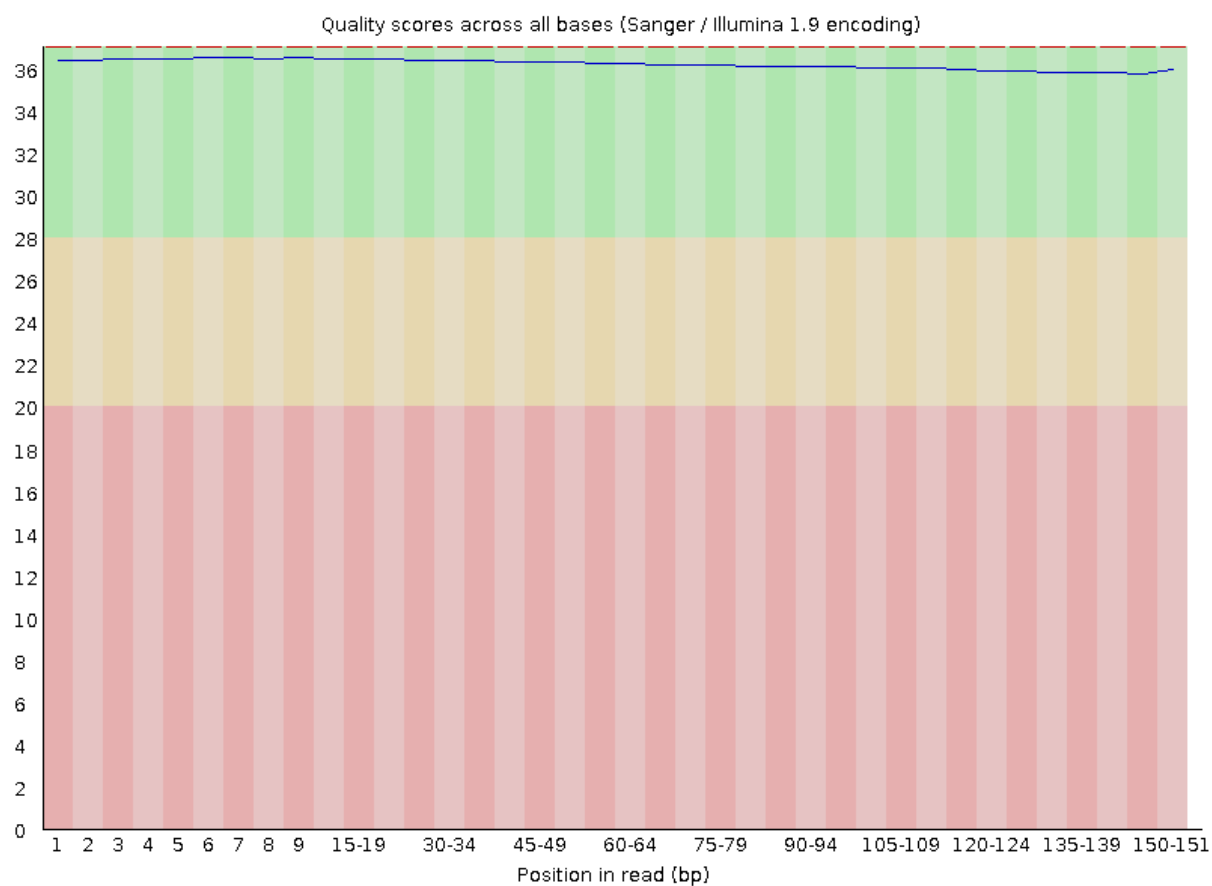
<Graphic distribution of read lengths>

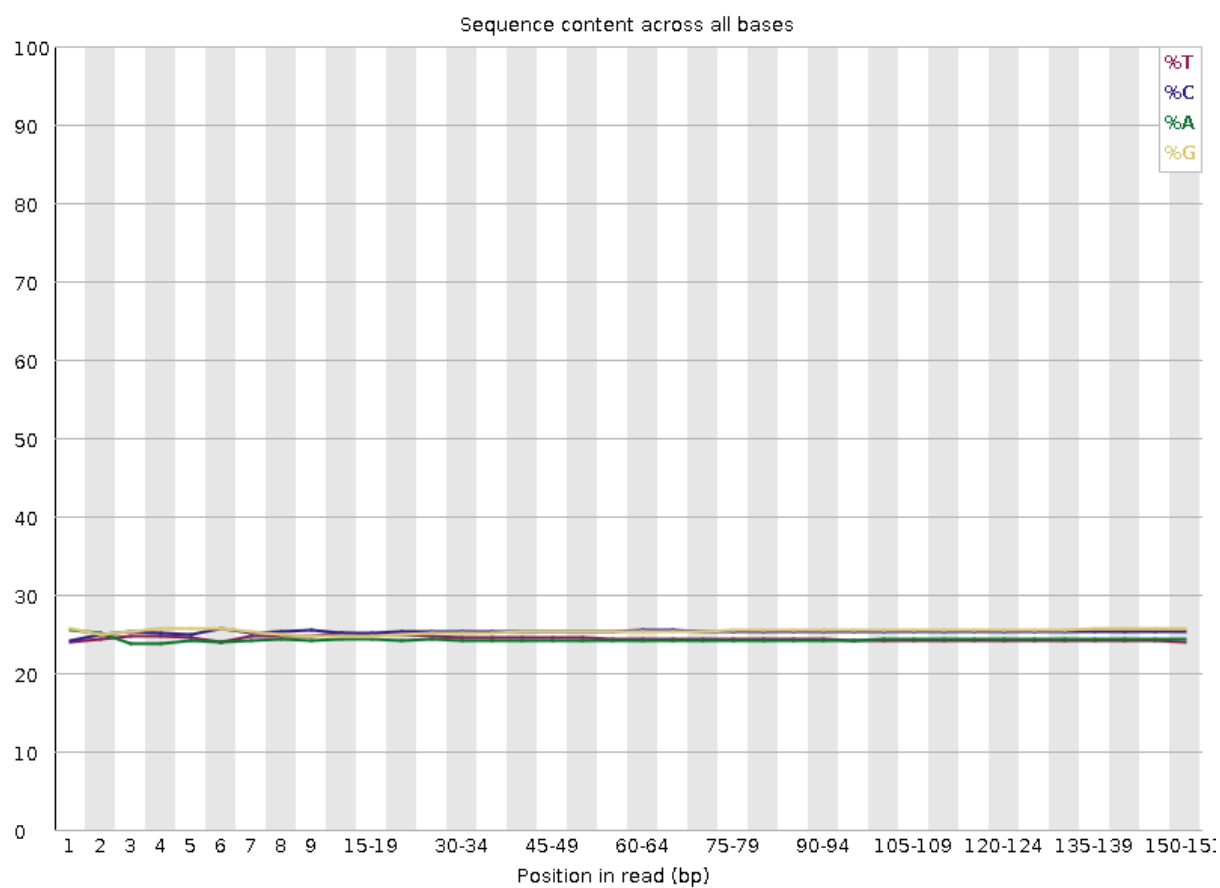
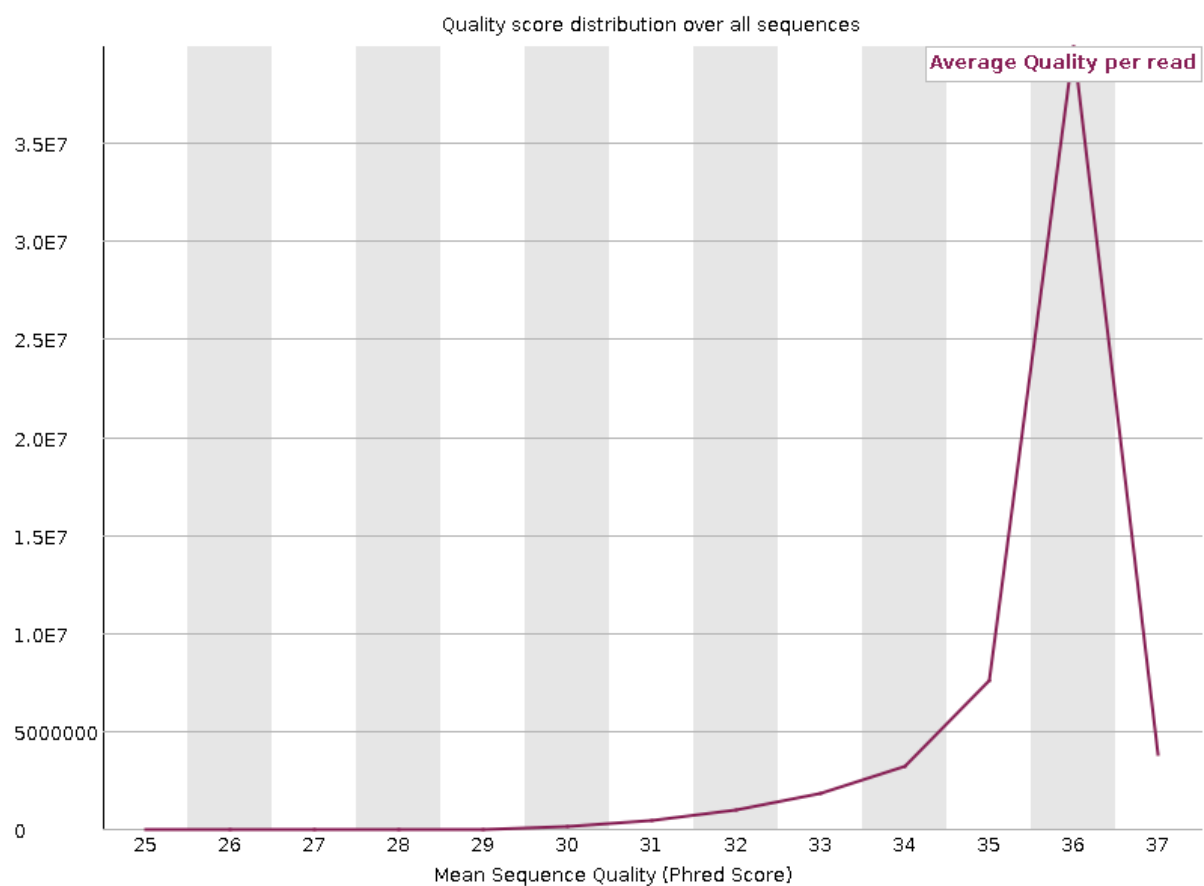
Length	Distribution	Abundance
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62	□□□□□□□□□□	0.06%
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64	□□□□□□□□□□	0.06%
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66	□□□□□□□□□□	0.06%
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74	□□□□□□□□□□	0.06%
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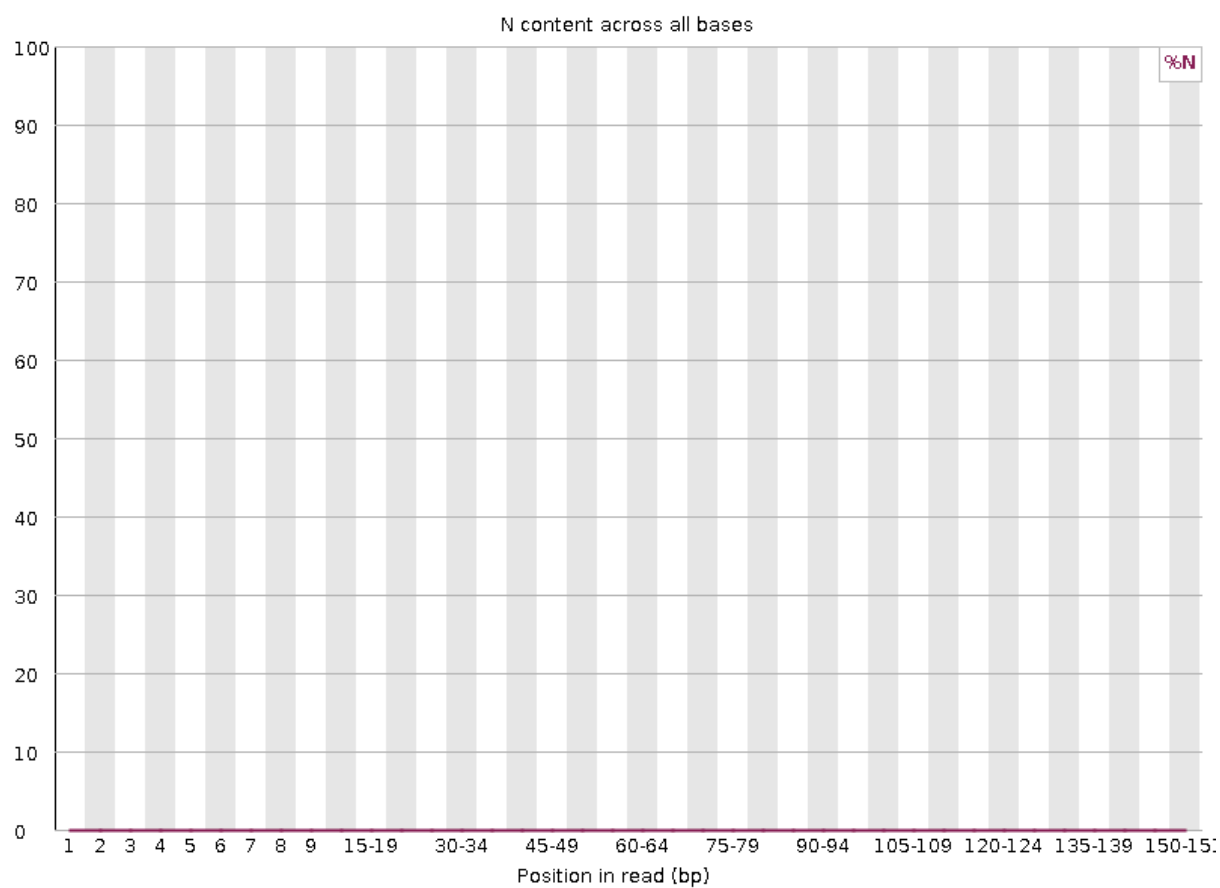
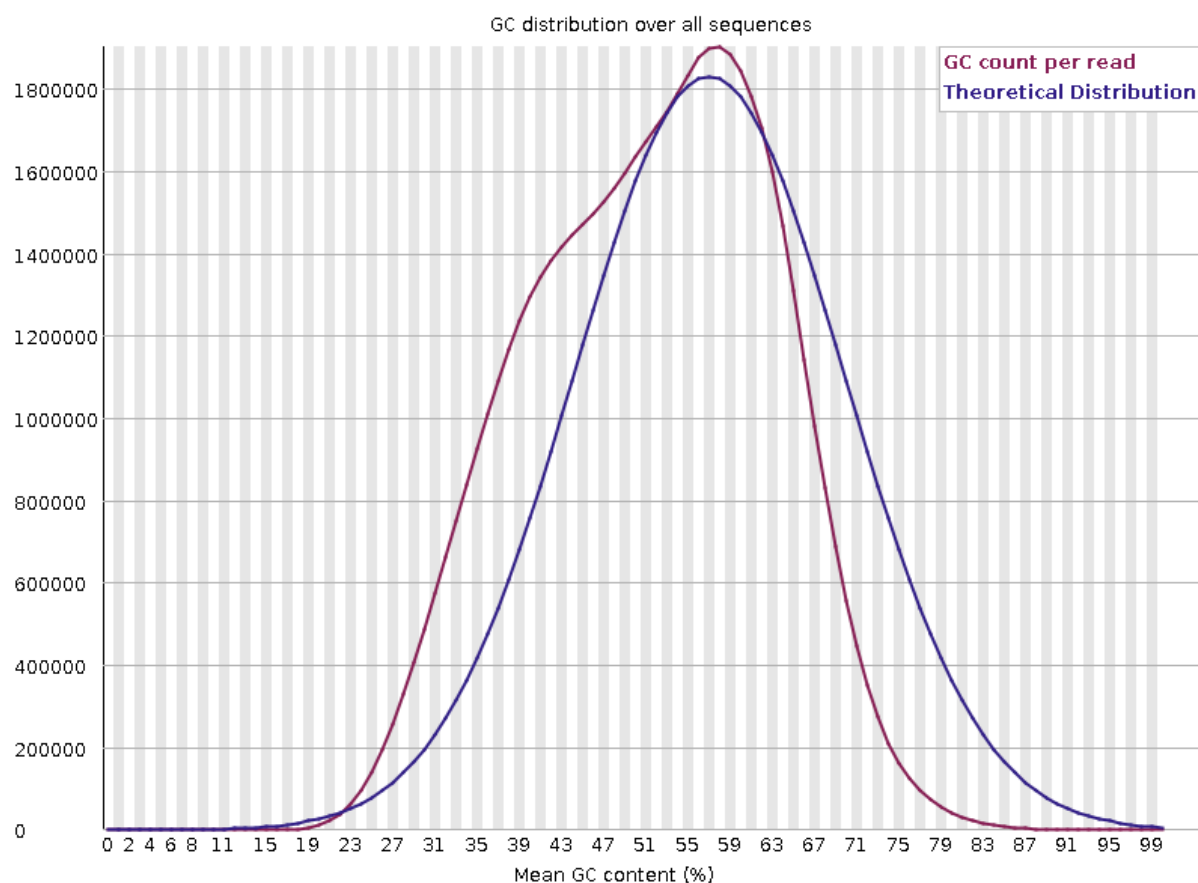
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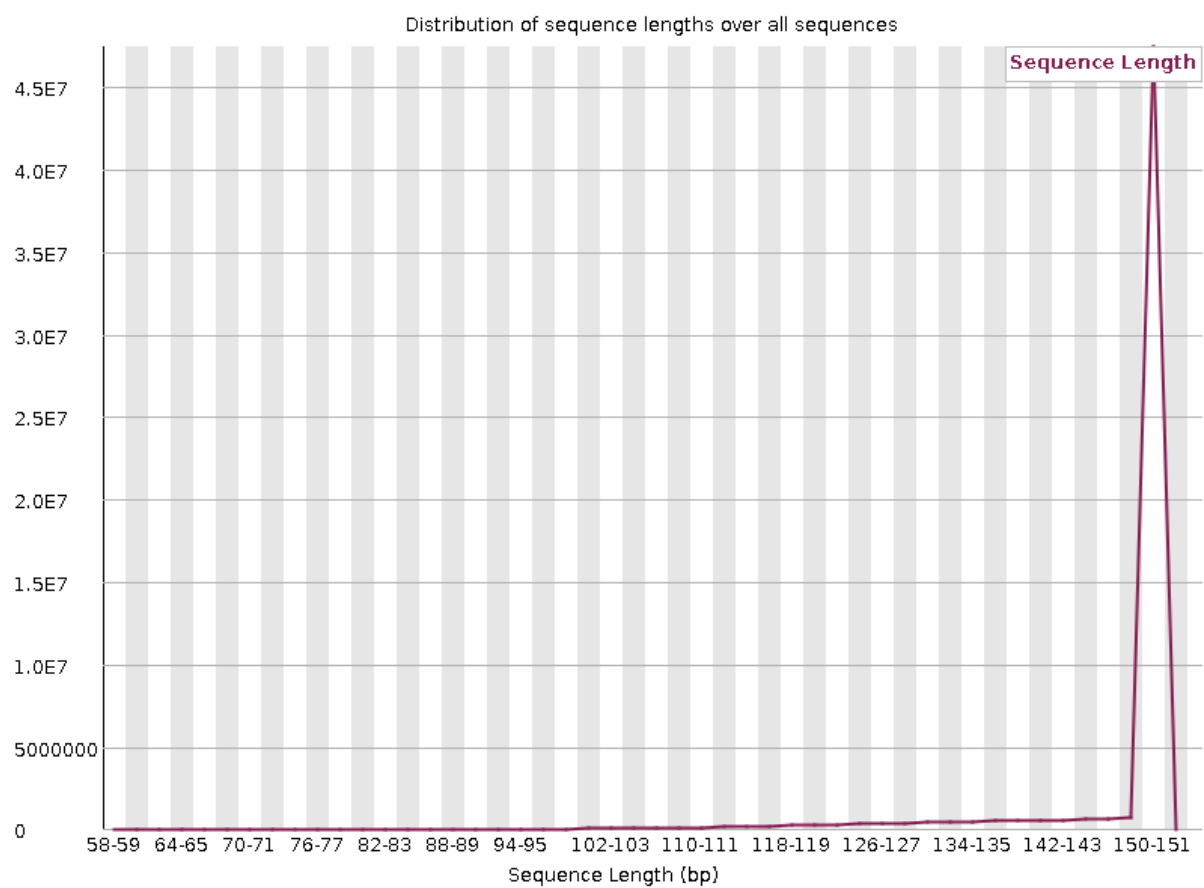
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140	□□□□□□□□□□	0.52%
141	□□□□□□□□□□	0.53%
142	□□□□□□□□□□	0.54%
143	□□□□□□□□□□	0.55%
144	□□□□□□□□□□	0.56%
145	□□□□□□□□□□	0.56%
146	□□□□□□□□□□	0.57%
147	□□□□□□□□□□	0.57%
148	□□□□□□□□□□	0.58%
149	□□□□□□□□□□	0.71%
150	□□□□□□□□□□	2.54%
151	■ ■ ■ ■ ■ ■ ■ ■ □ □ □ □	78.75%

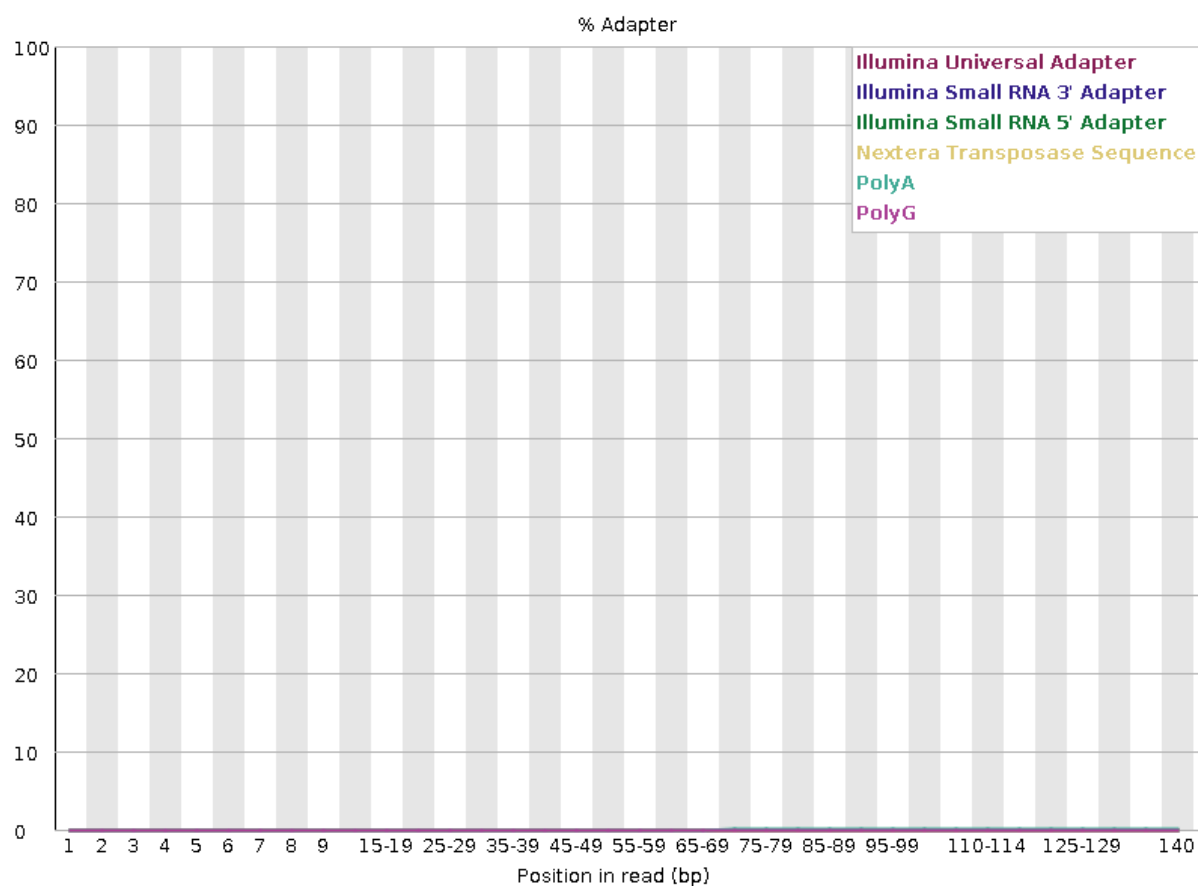
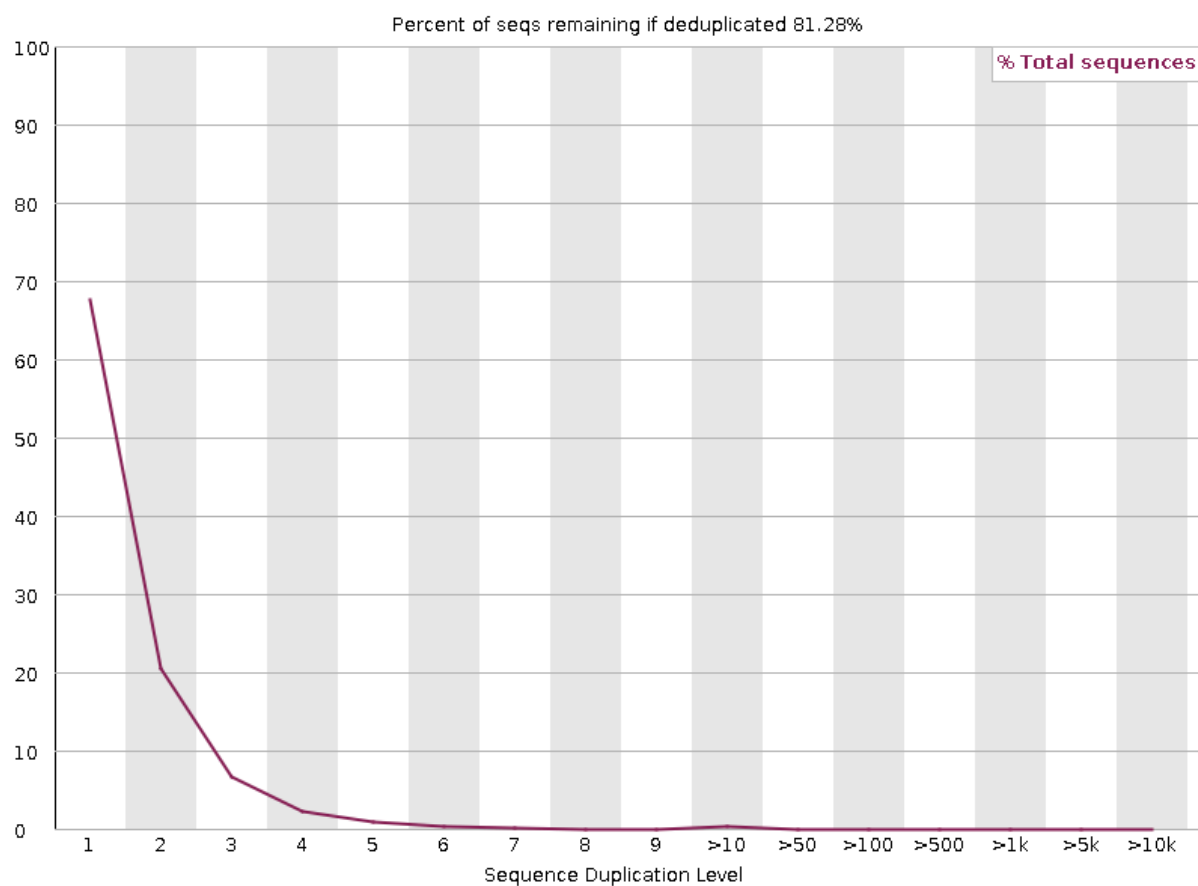
SRR37843243_1.p.fastq	
Measure	Value
Filename	SRR37843243_1.p.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	58,329,914
Total Bases	8.5 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	50





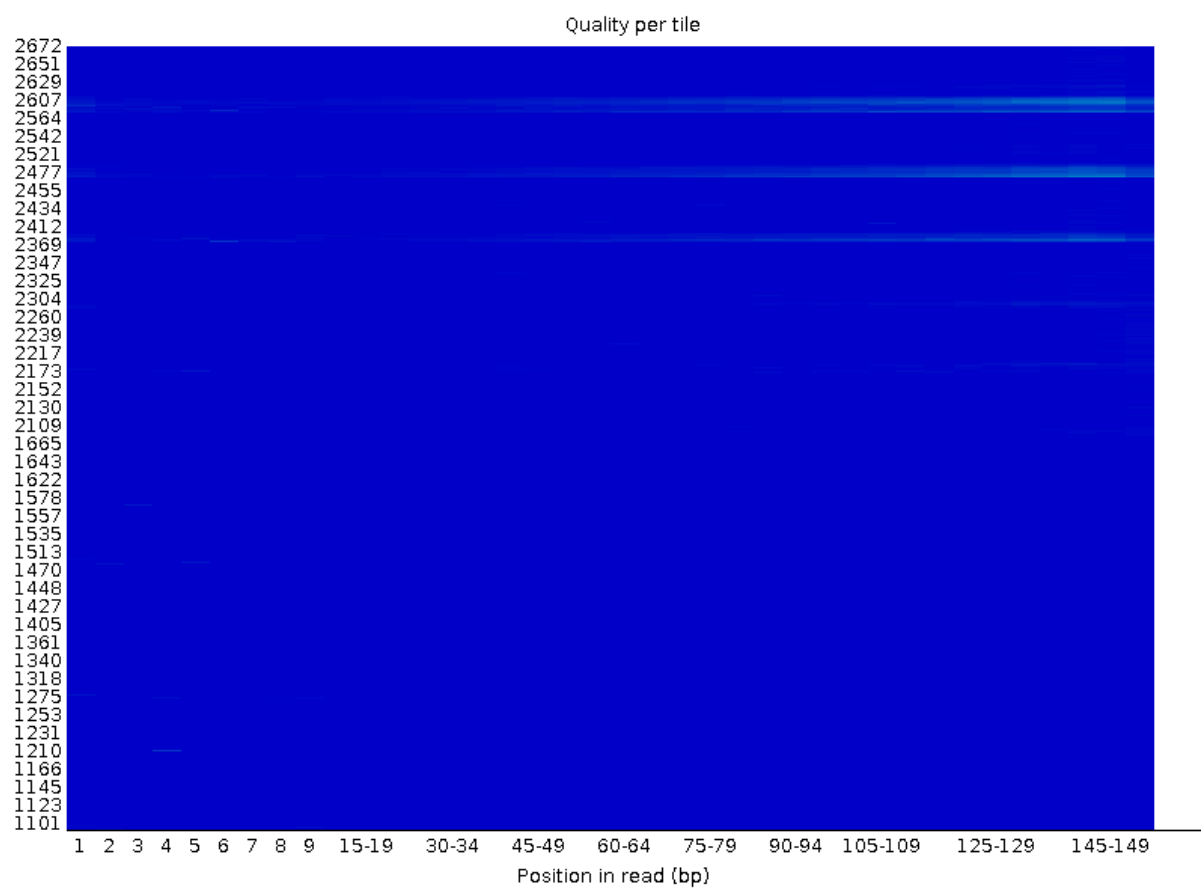
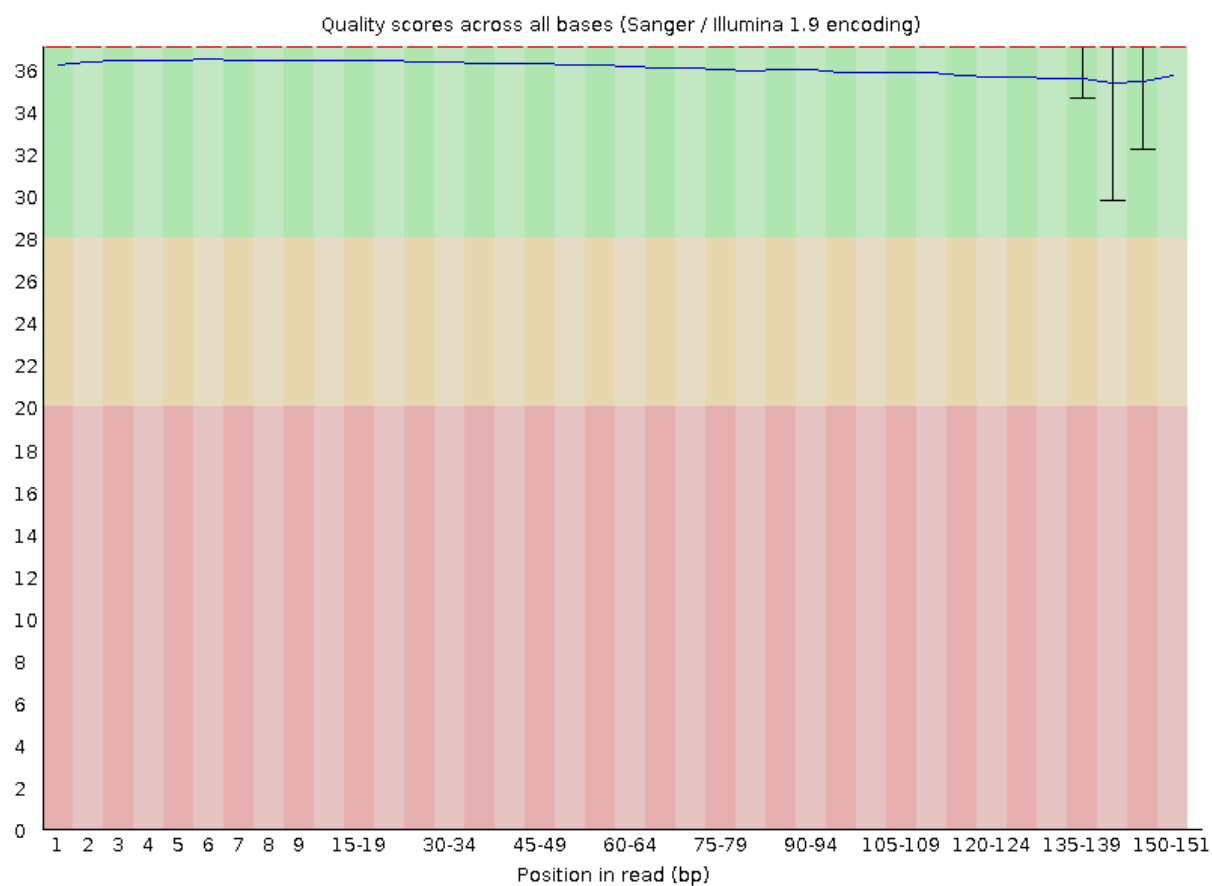


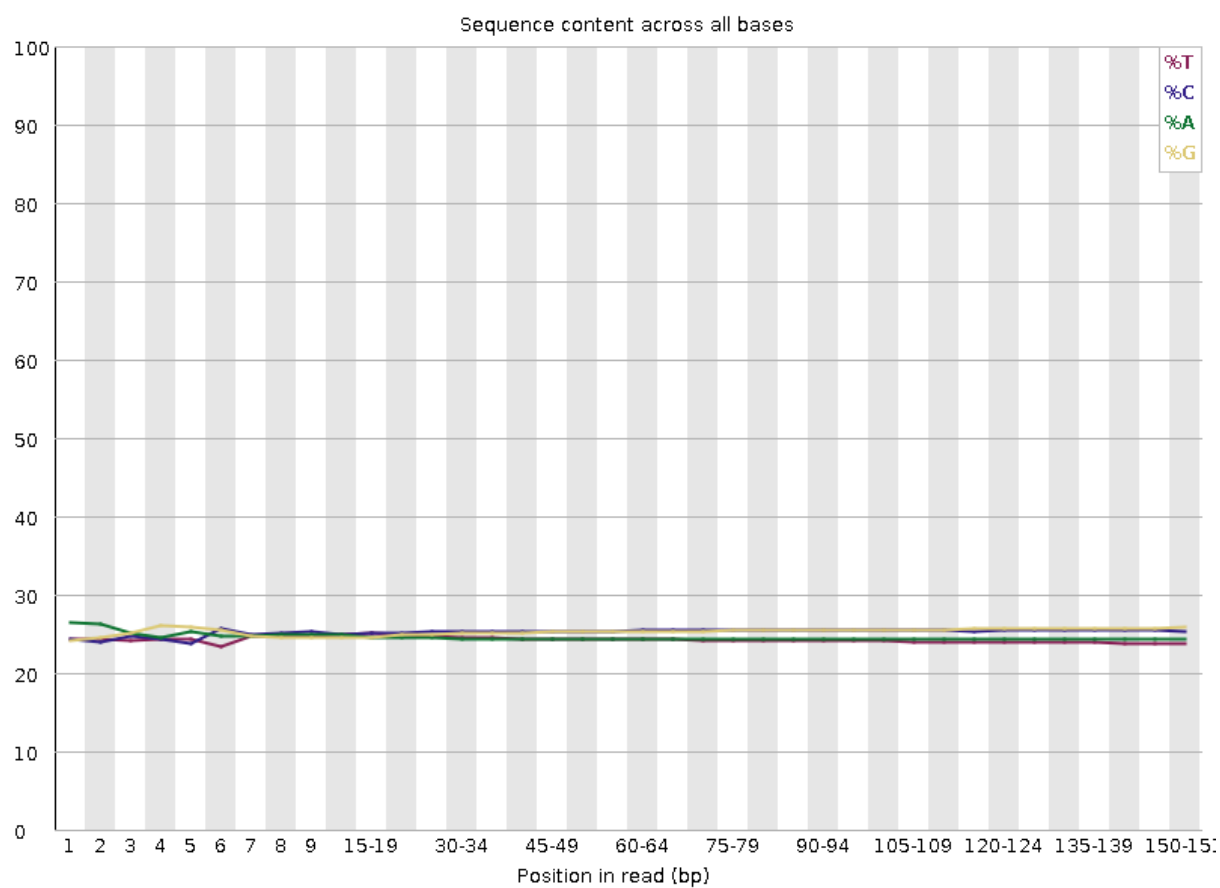
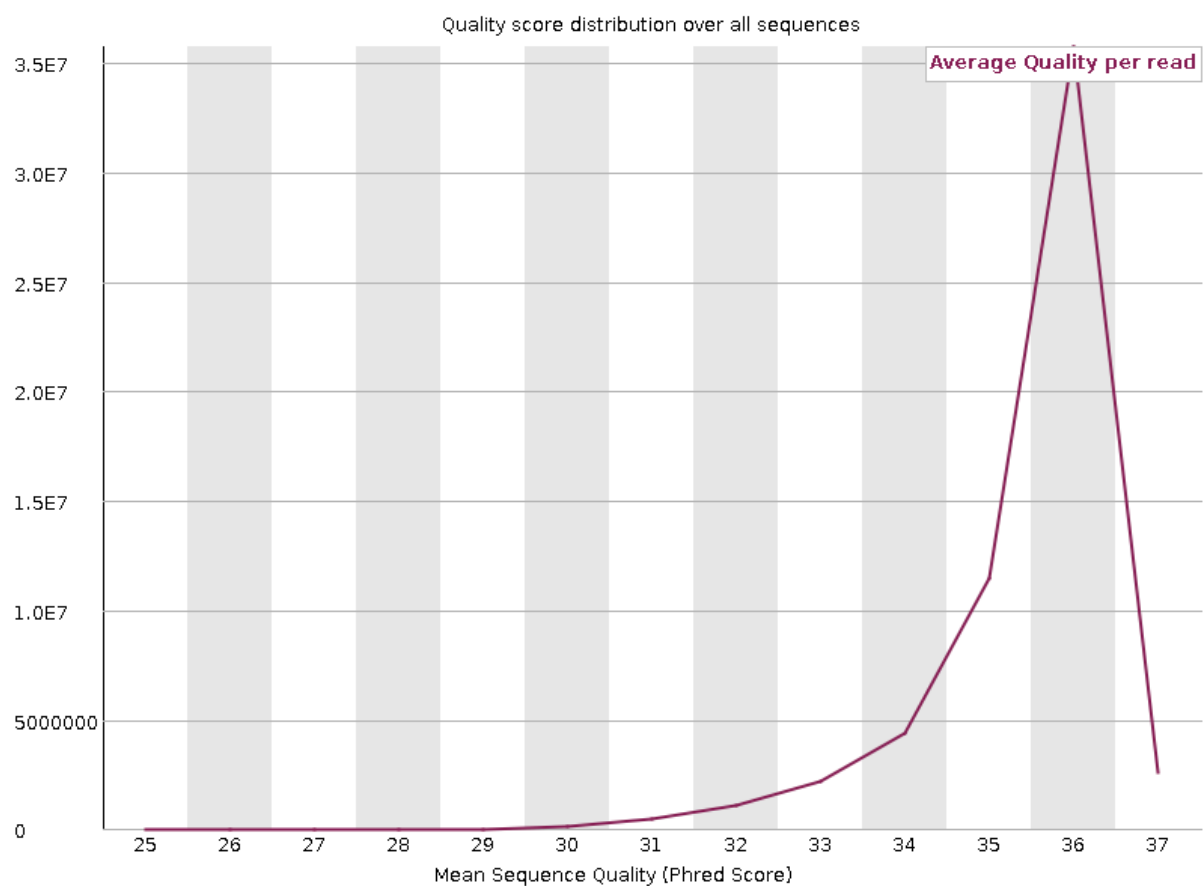


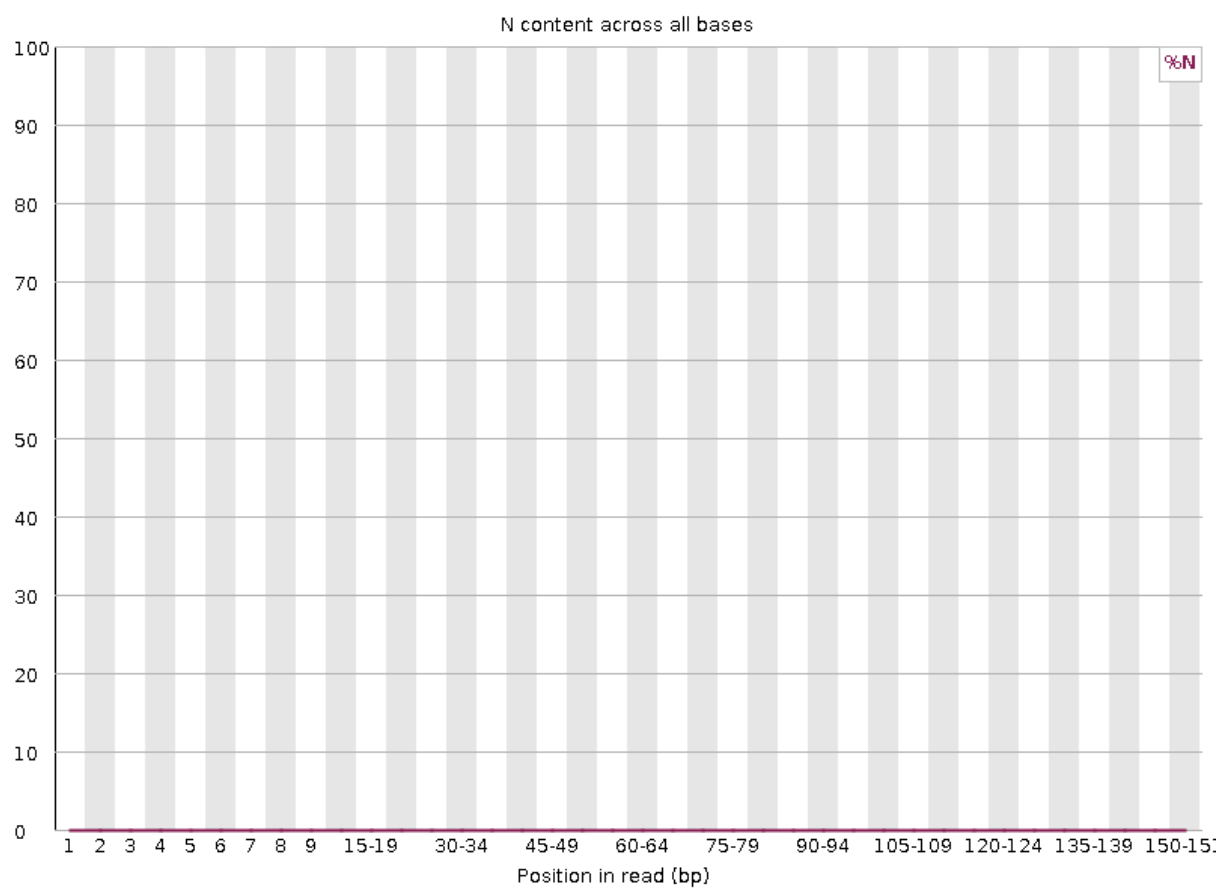
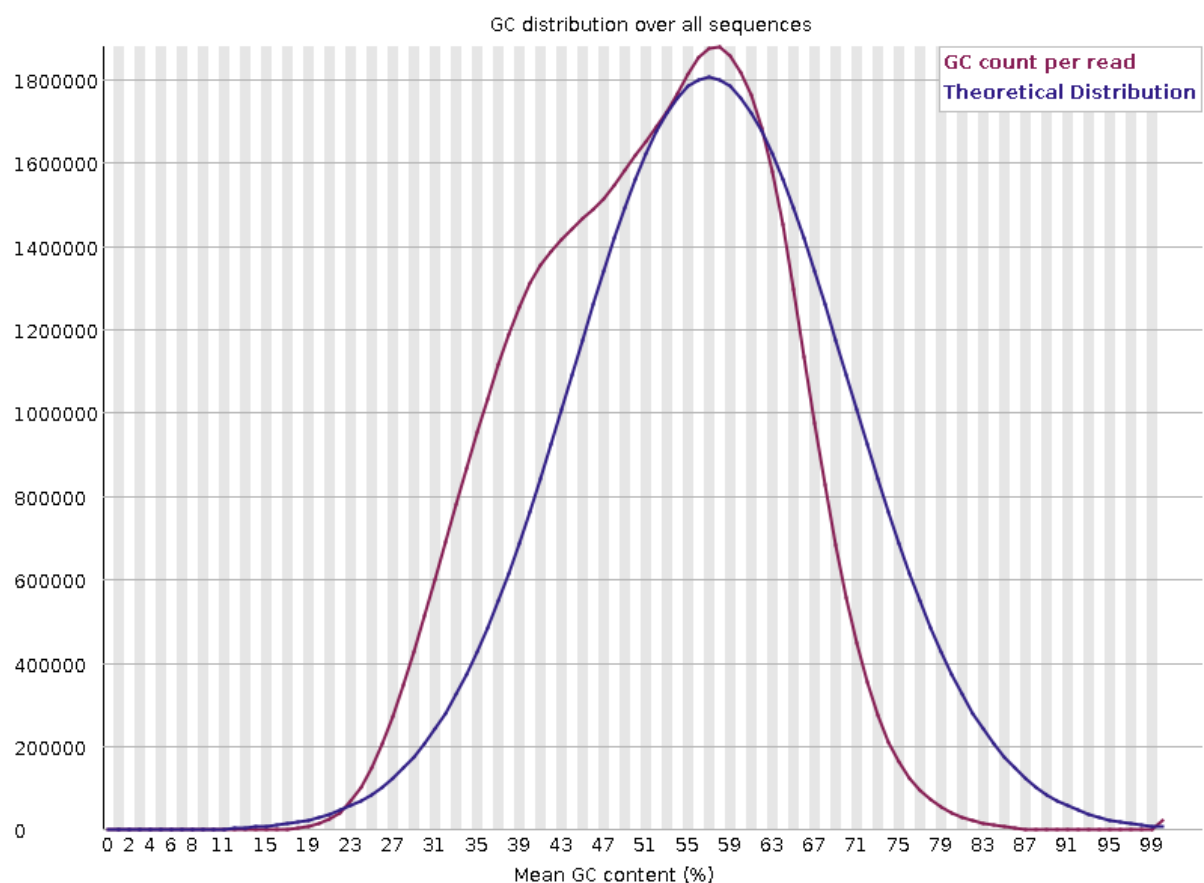


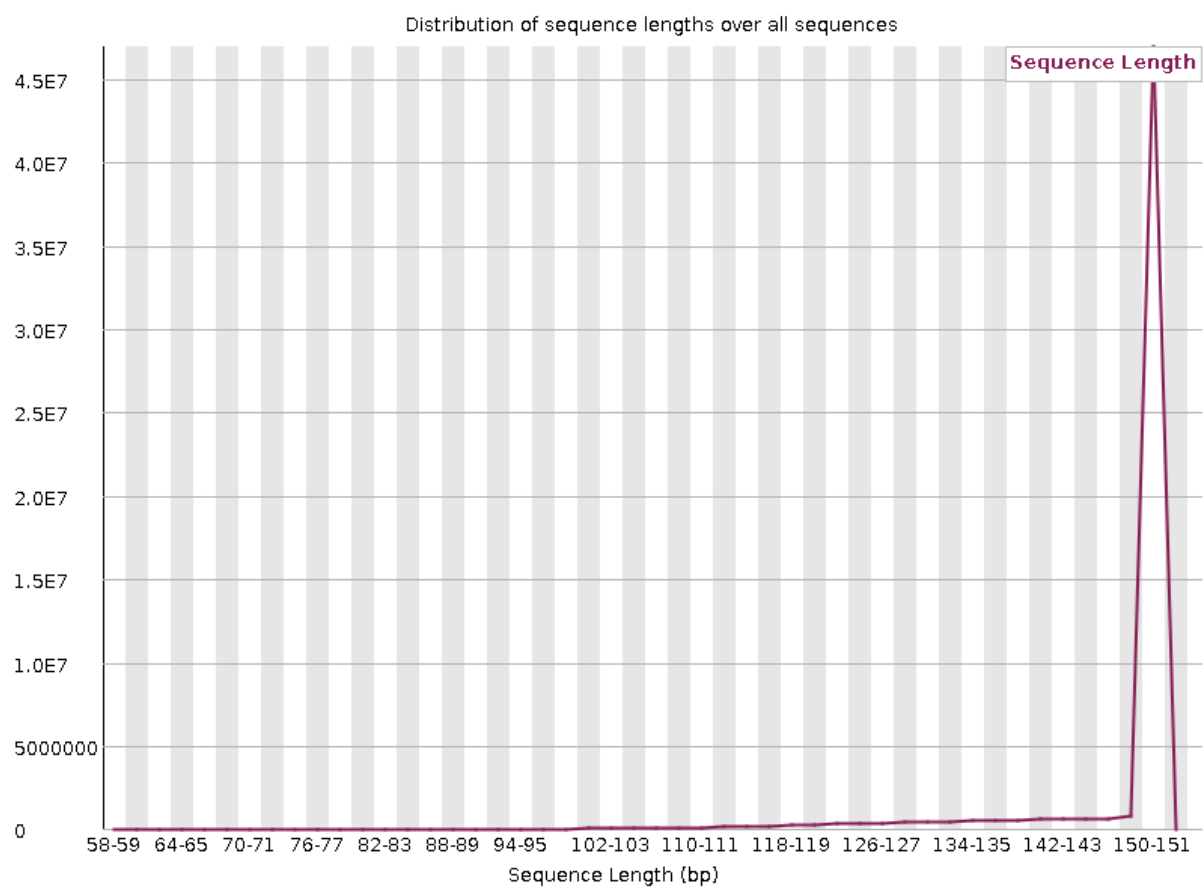
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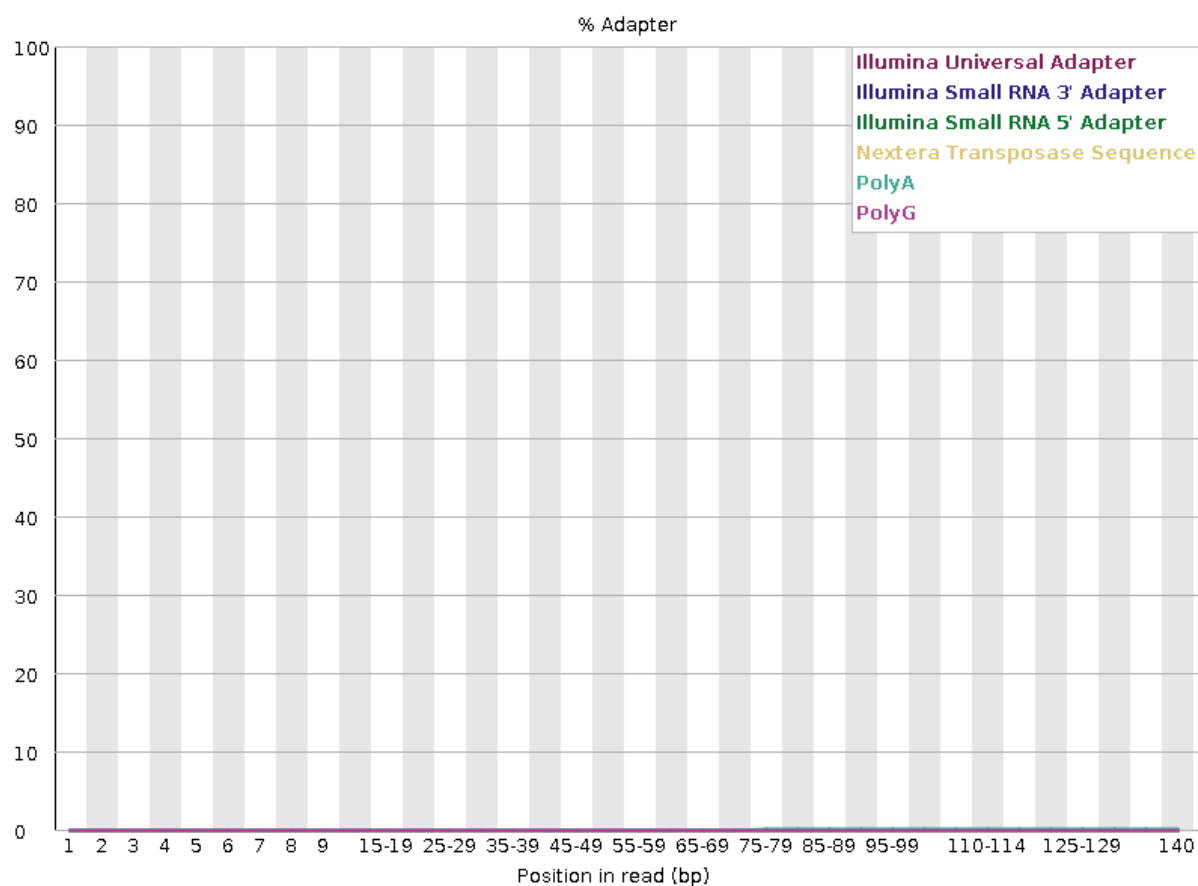
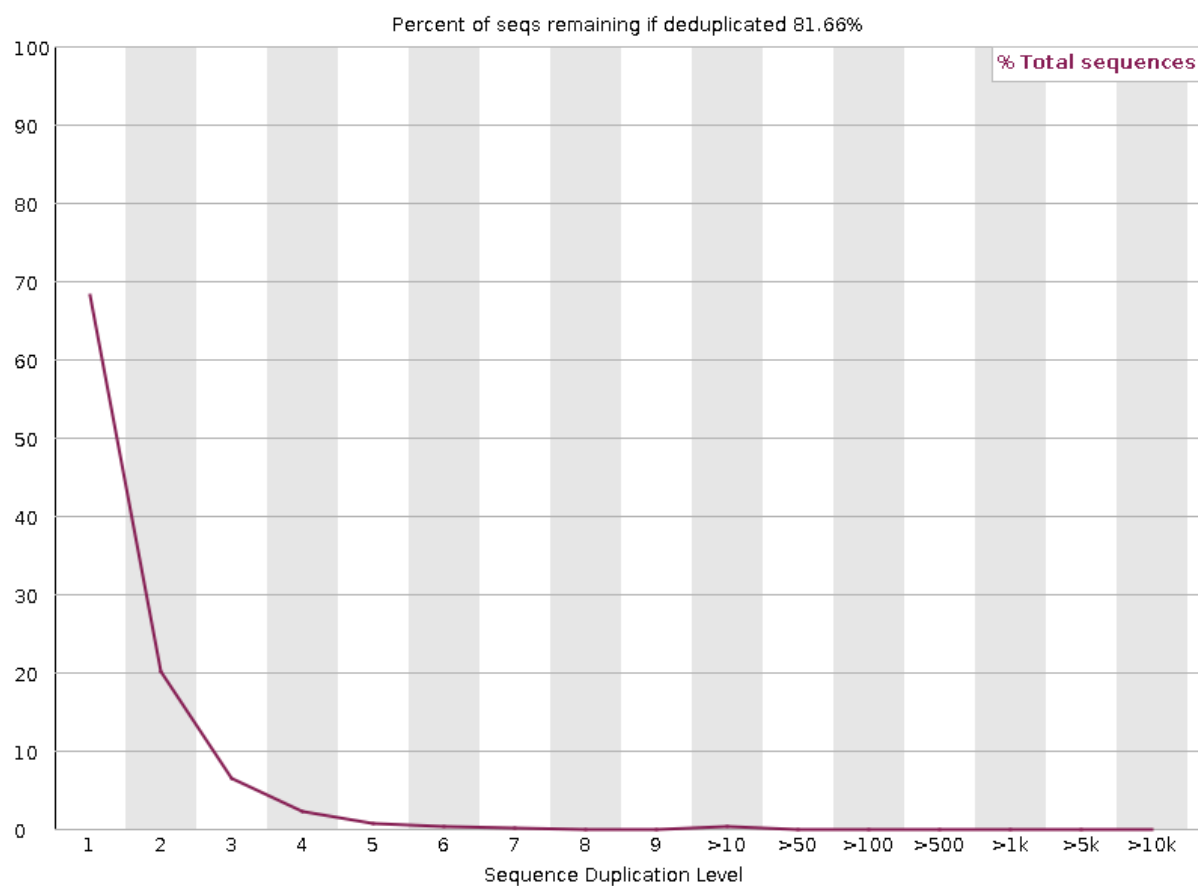
SRR37843243_2.p.fastq	
Measure	Value
Filename	SRR37843243_2.p.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	58,329,914
Total Bases	8.5 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	50





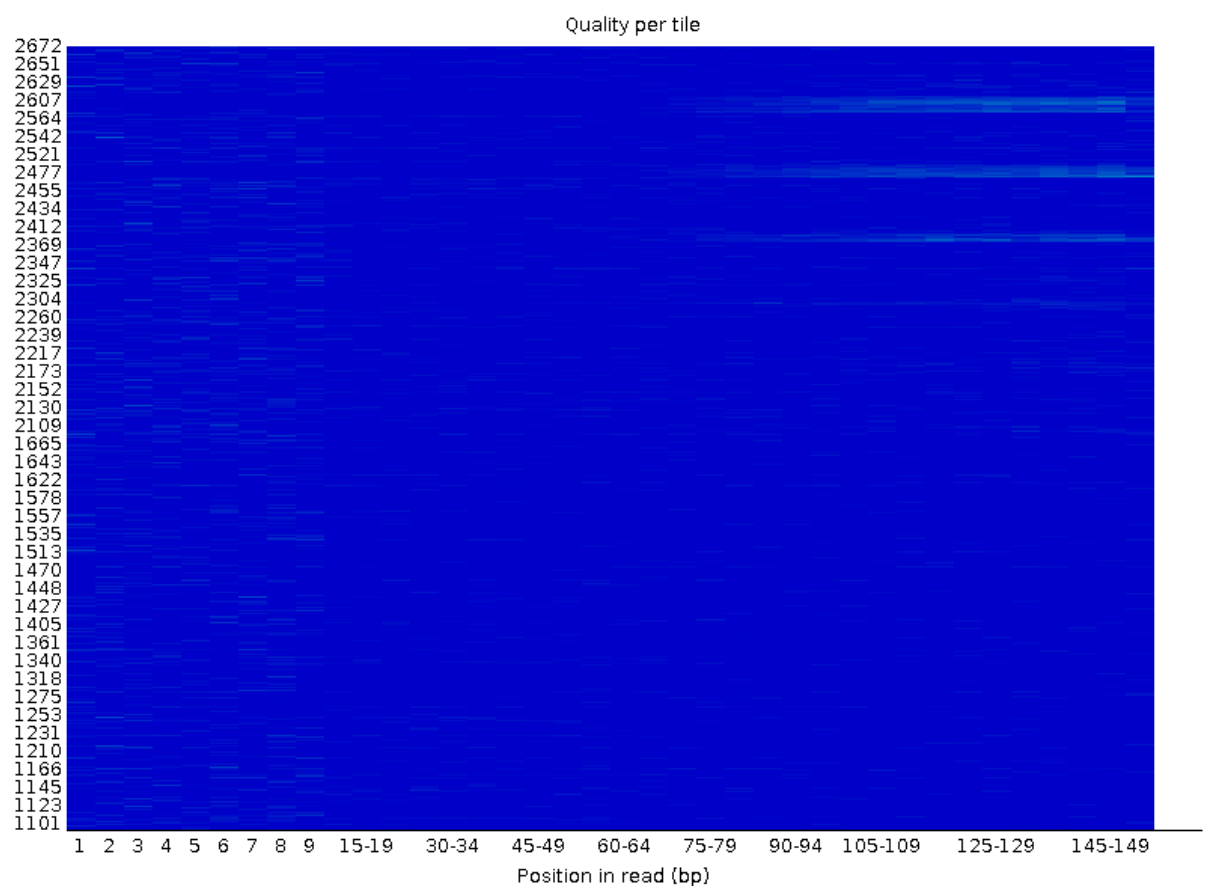
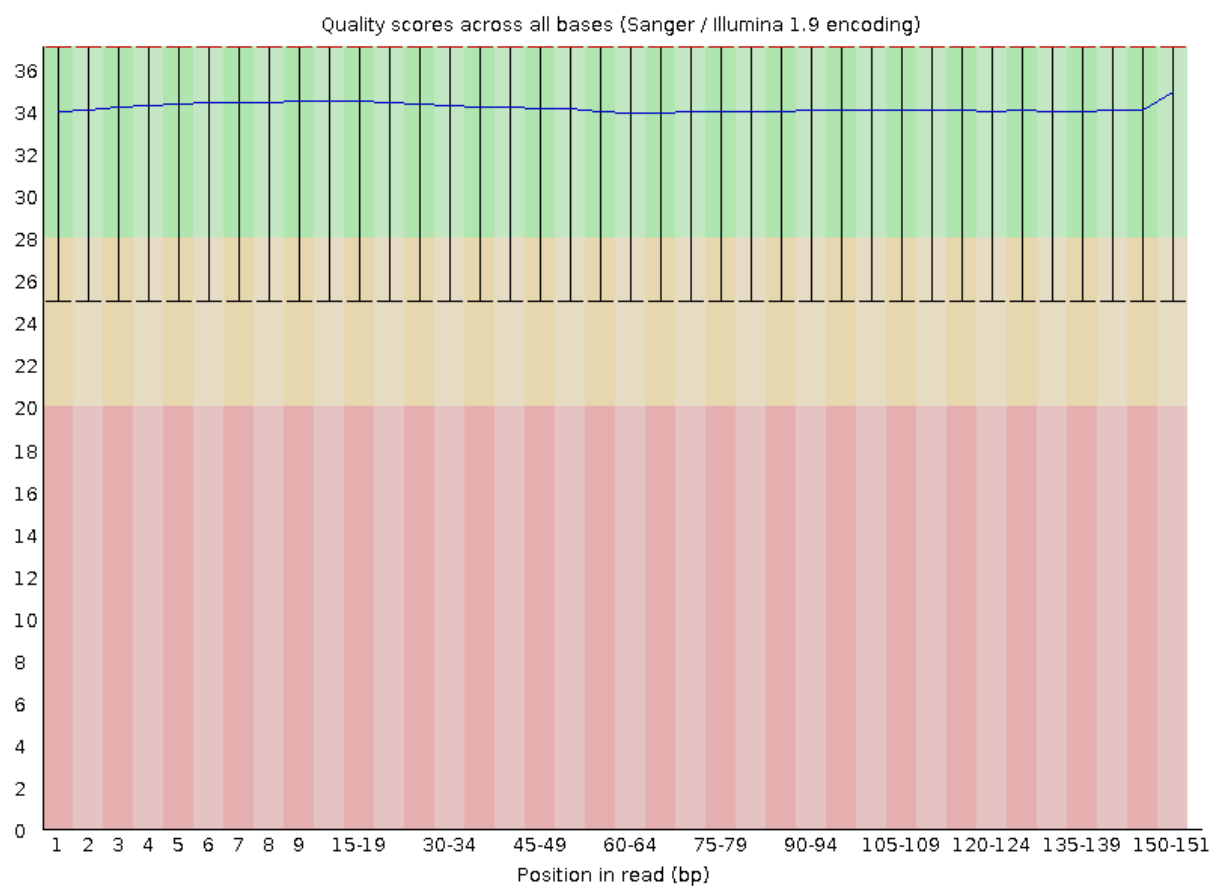


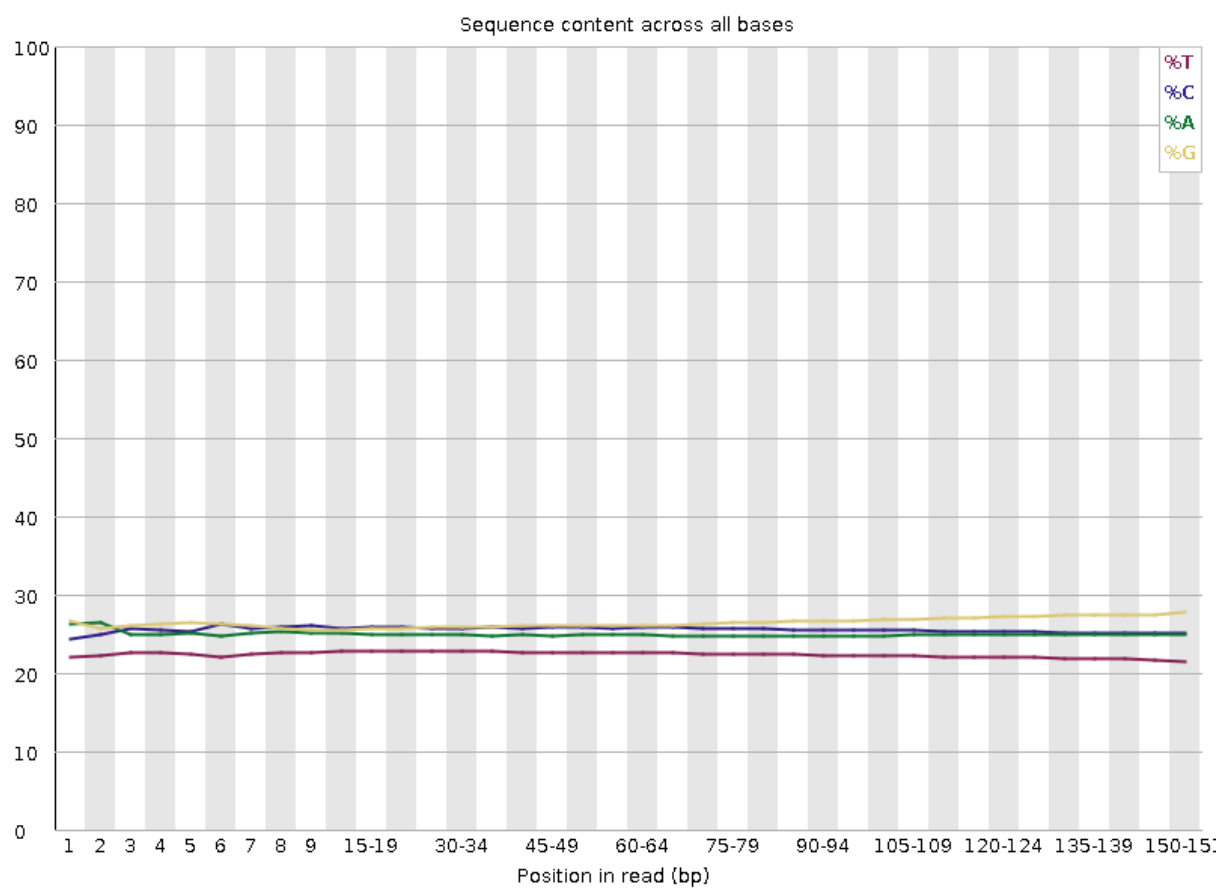
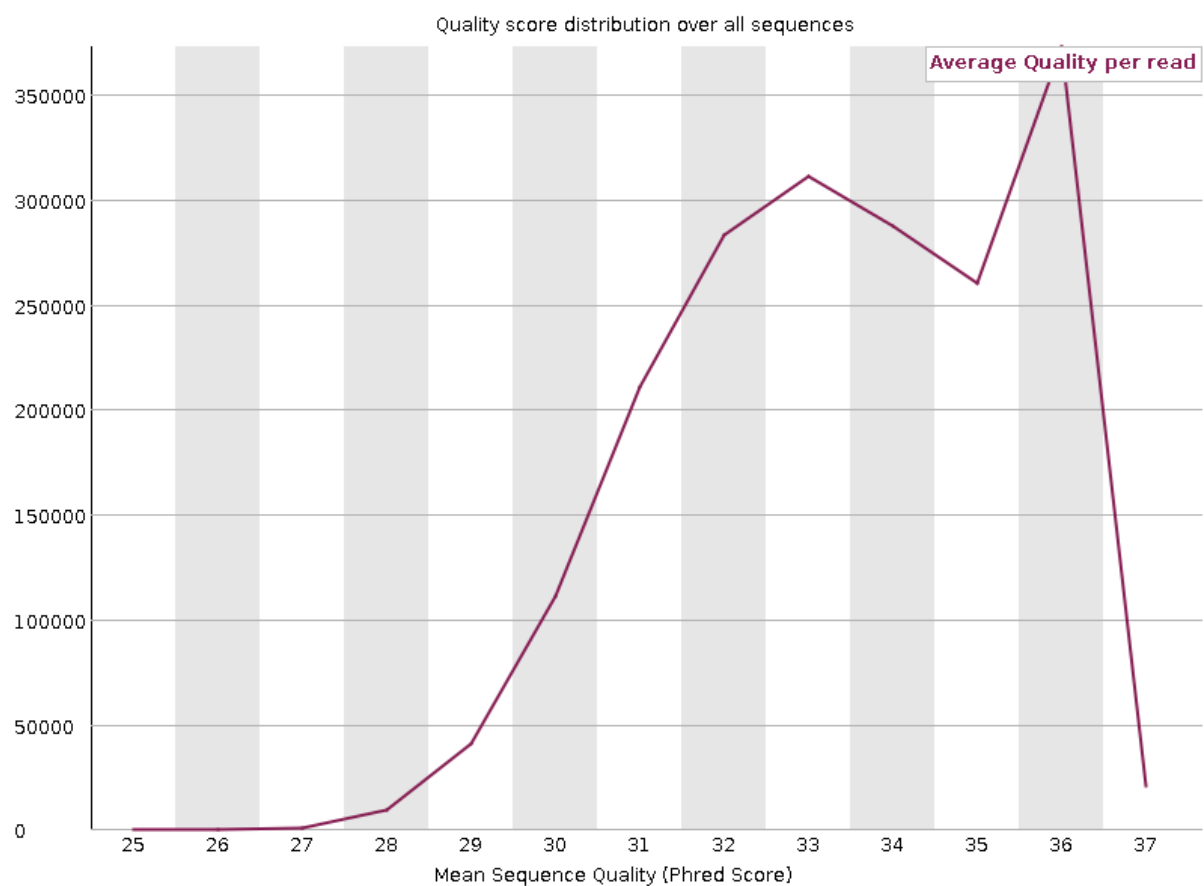


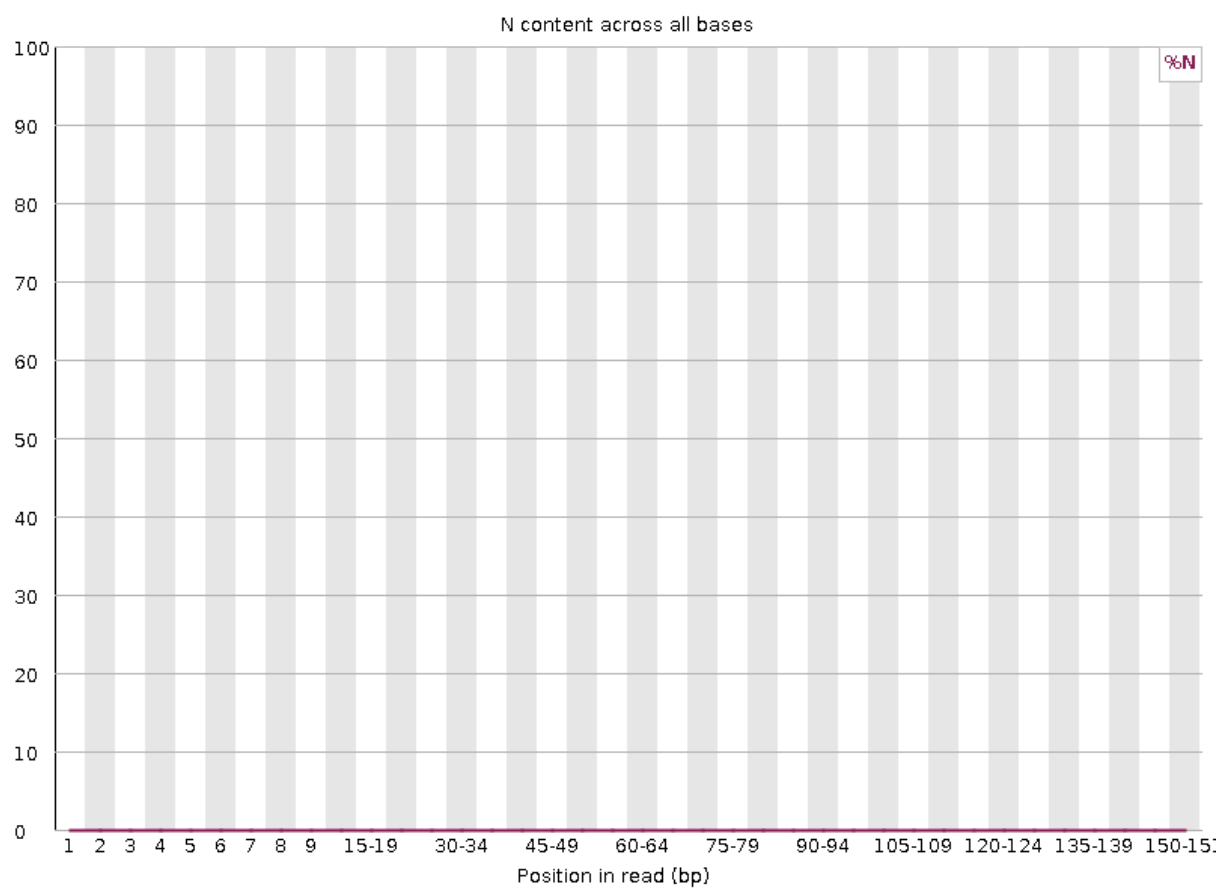
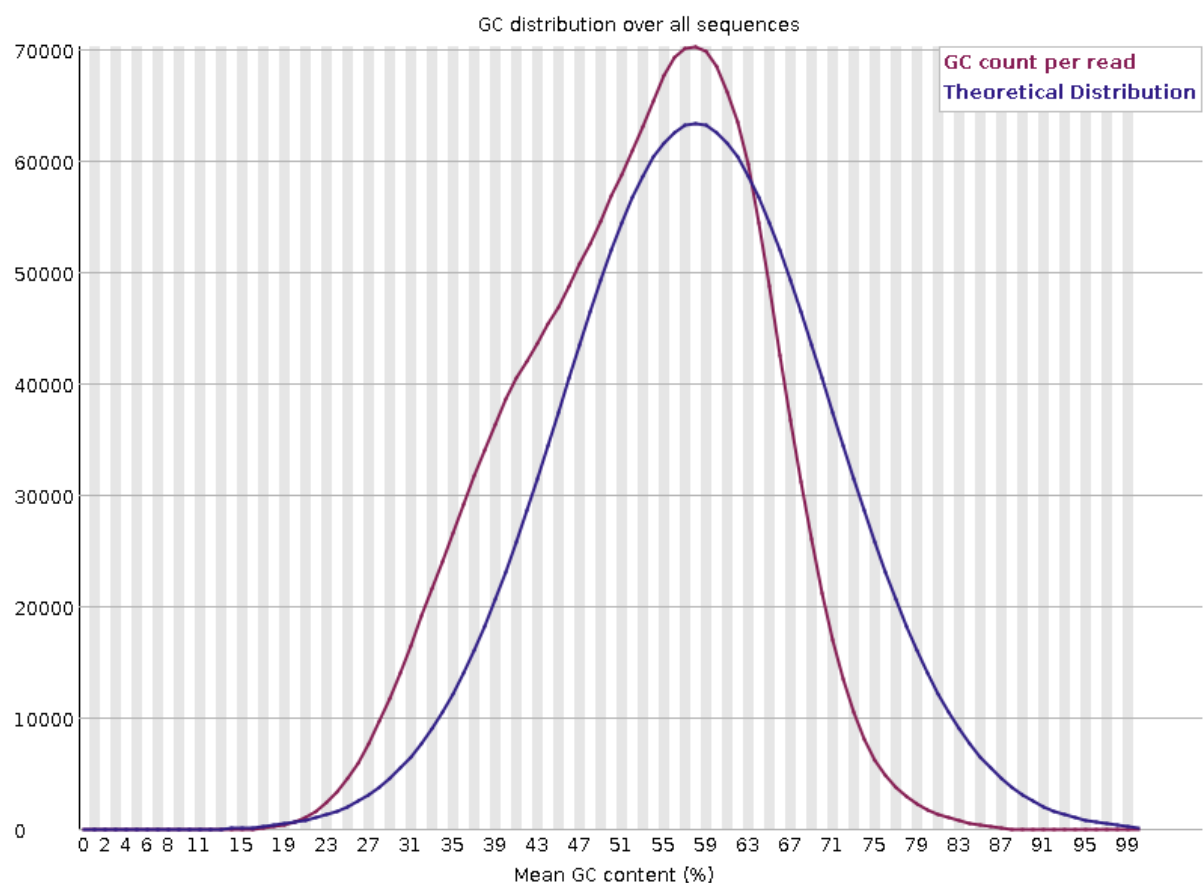


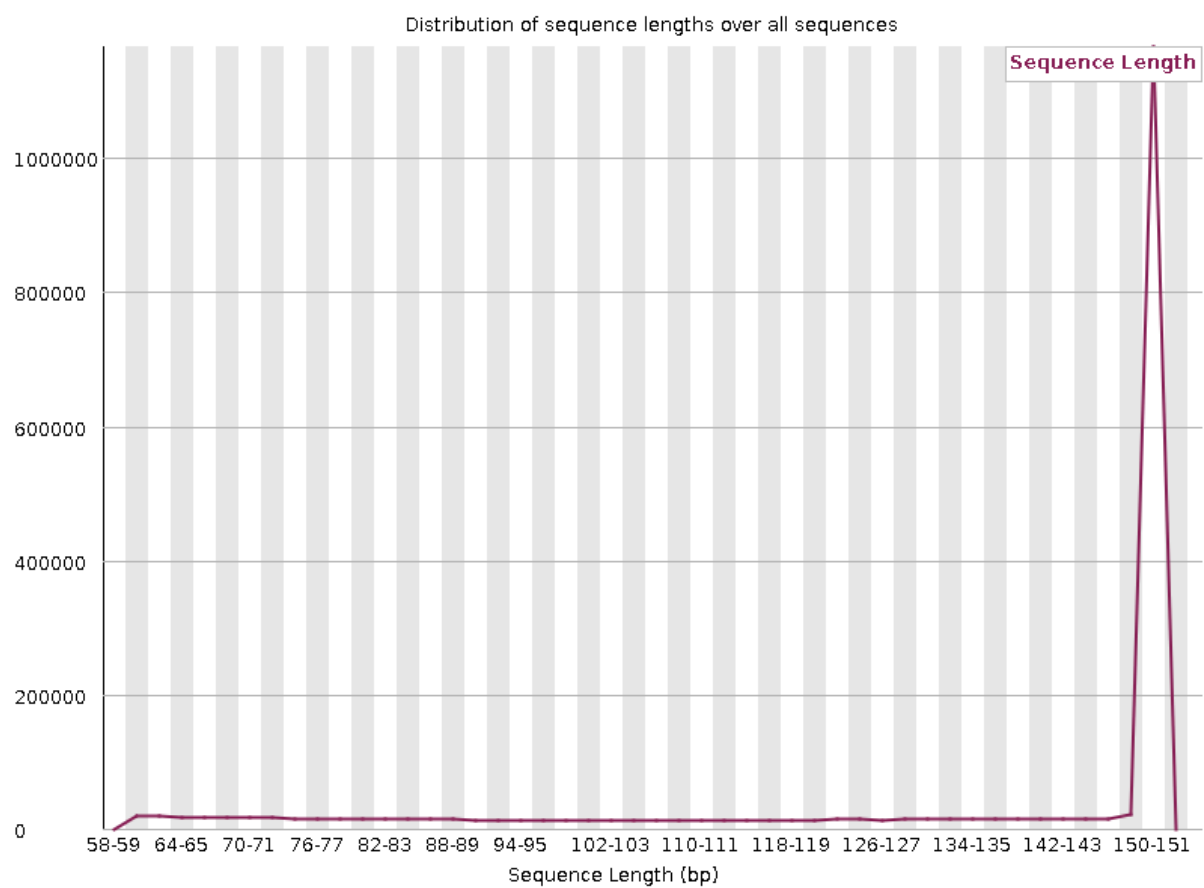
SRR37843243_1.up.fastq

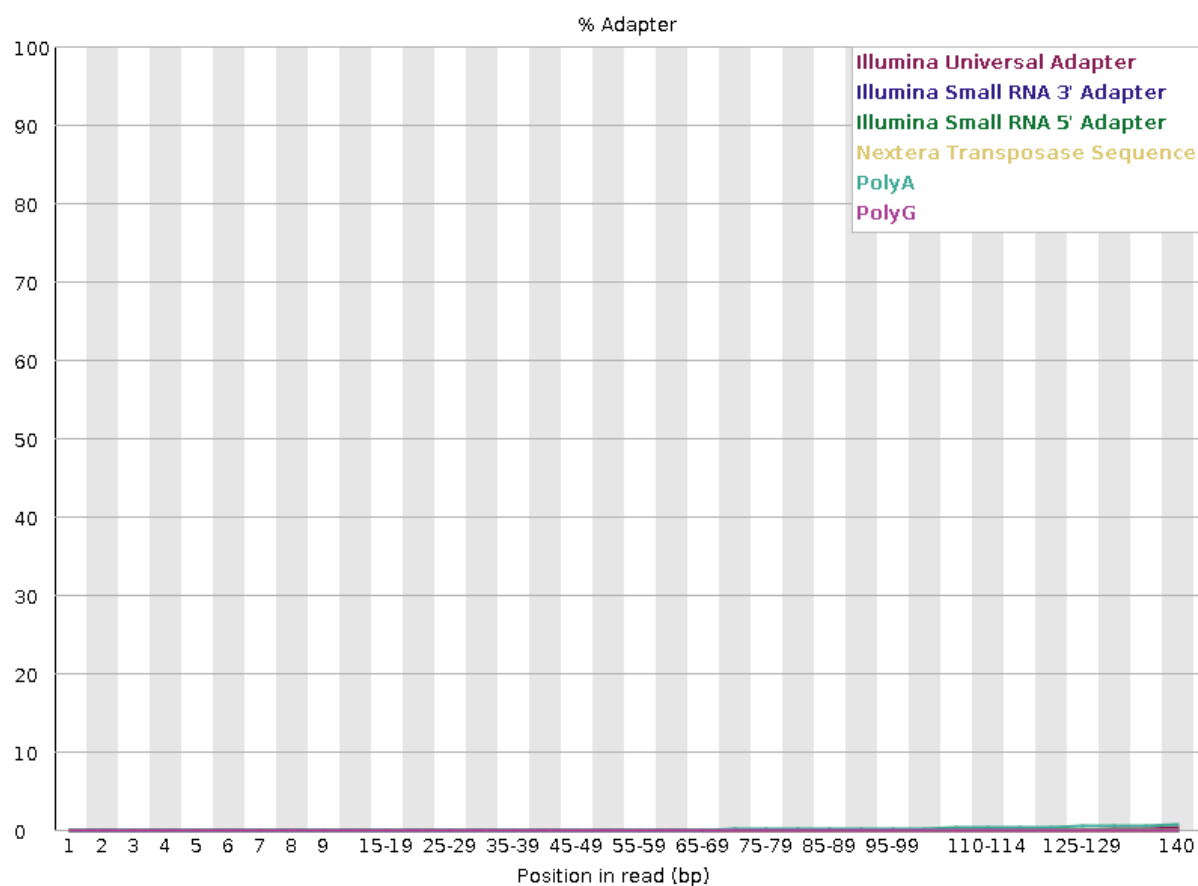
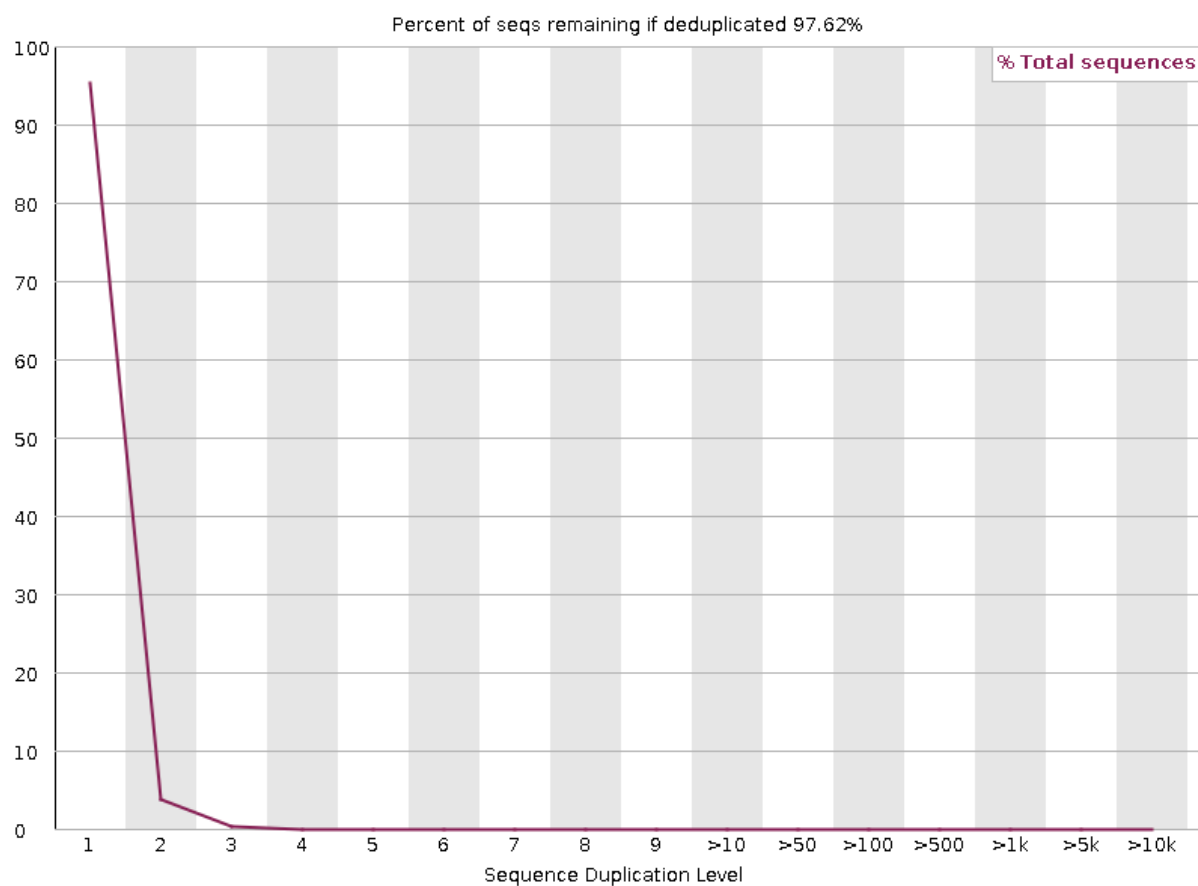
SRR37843243_1.up.fastq	
Measure	Value
Filename	SRR37843243_1.up.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1,910,868
Total Bases	253 Mbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	52





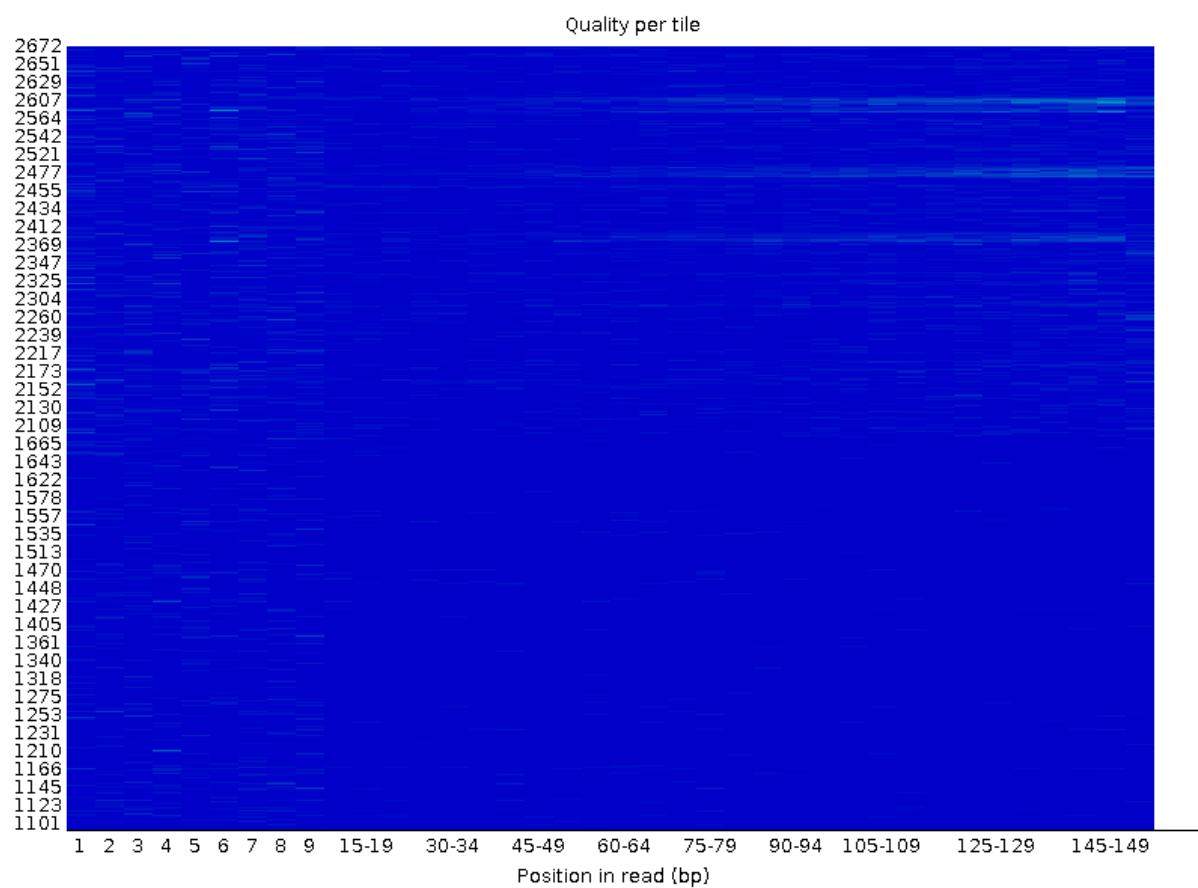
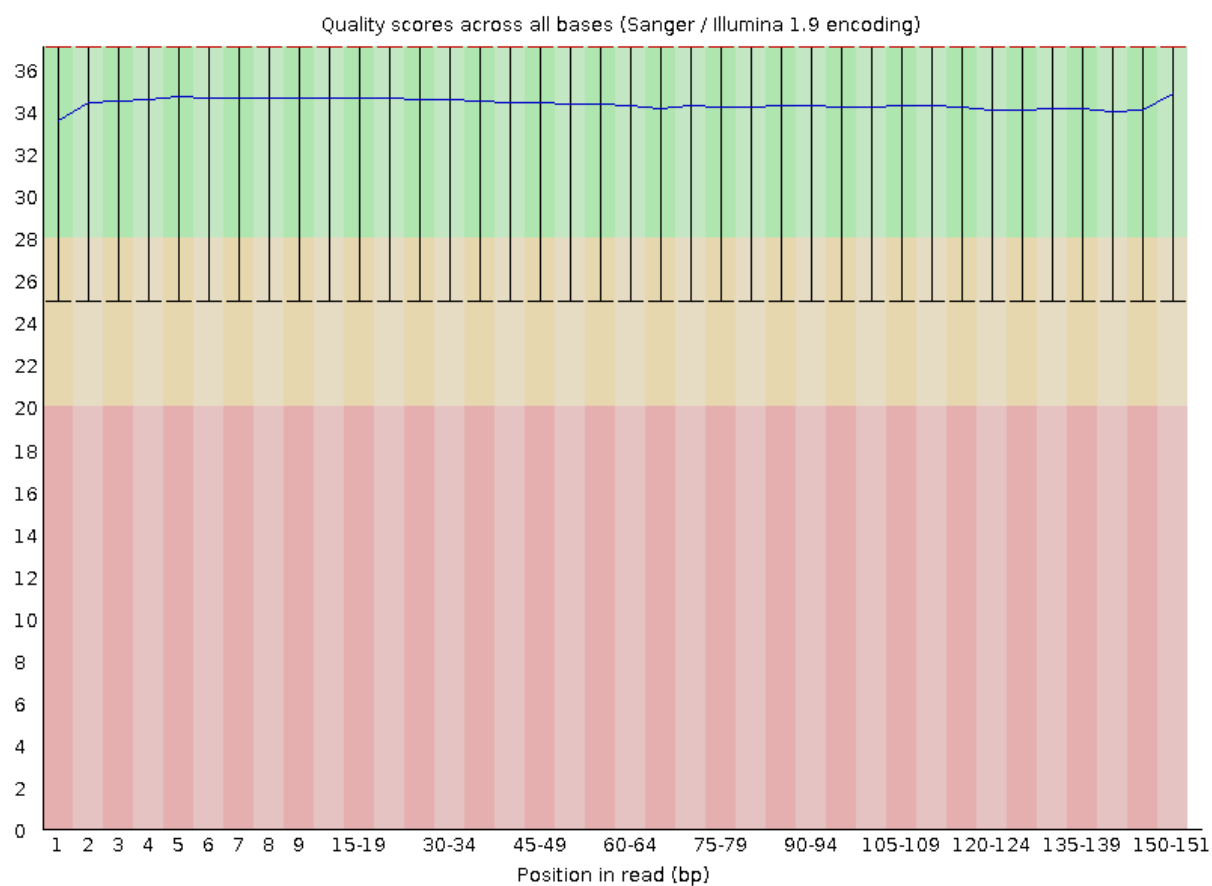


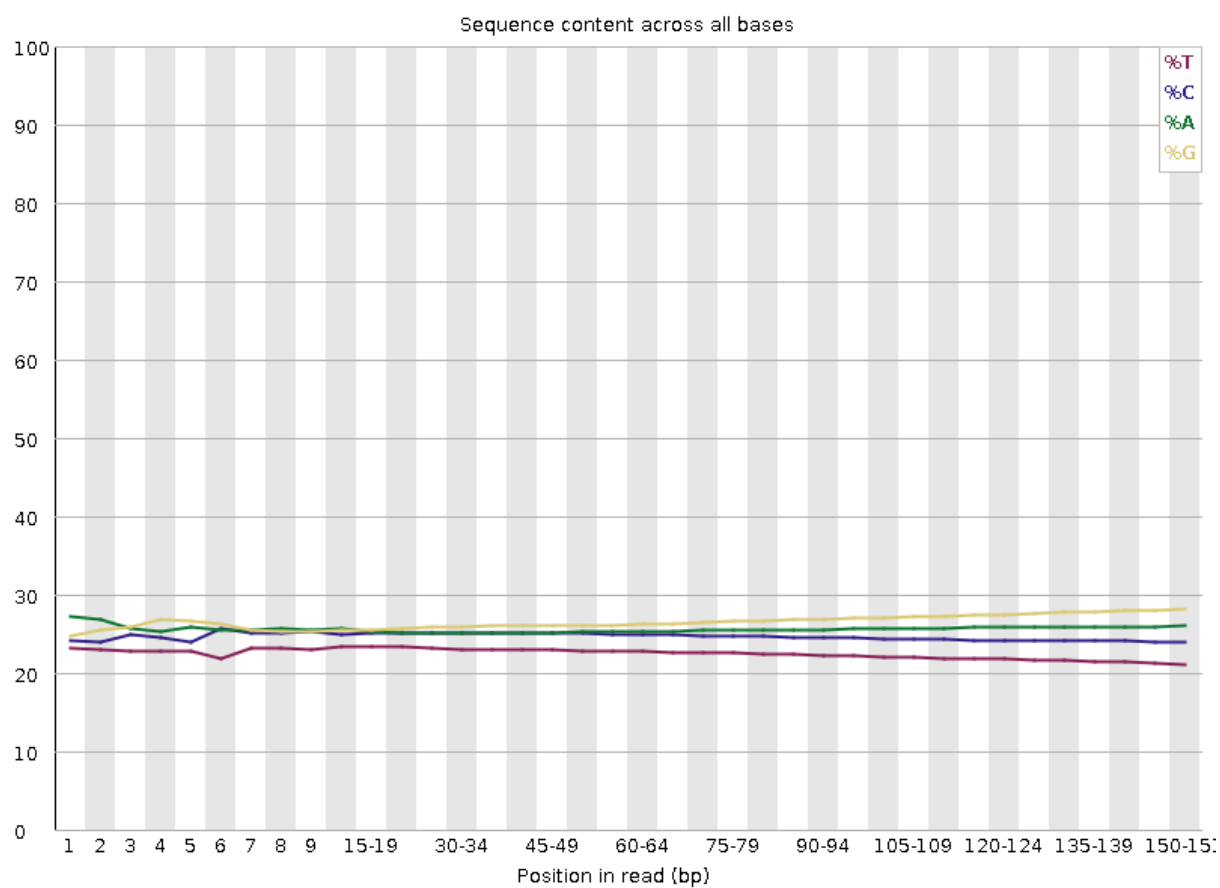
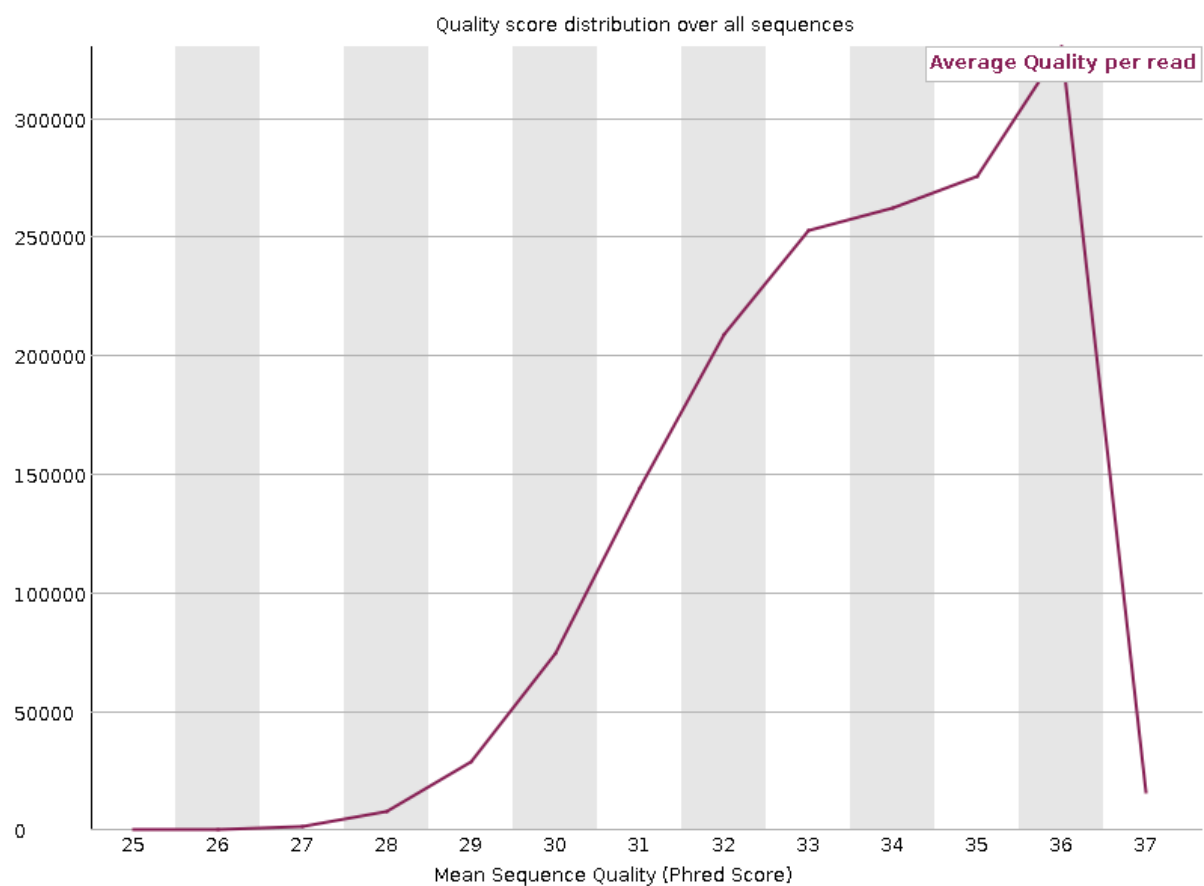


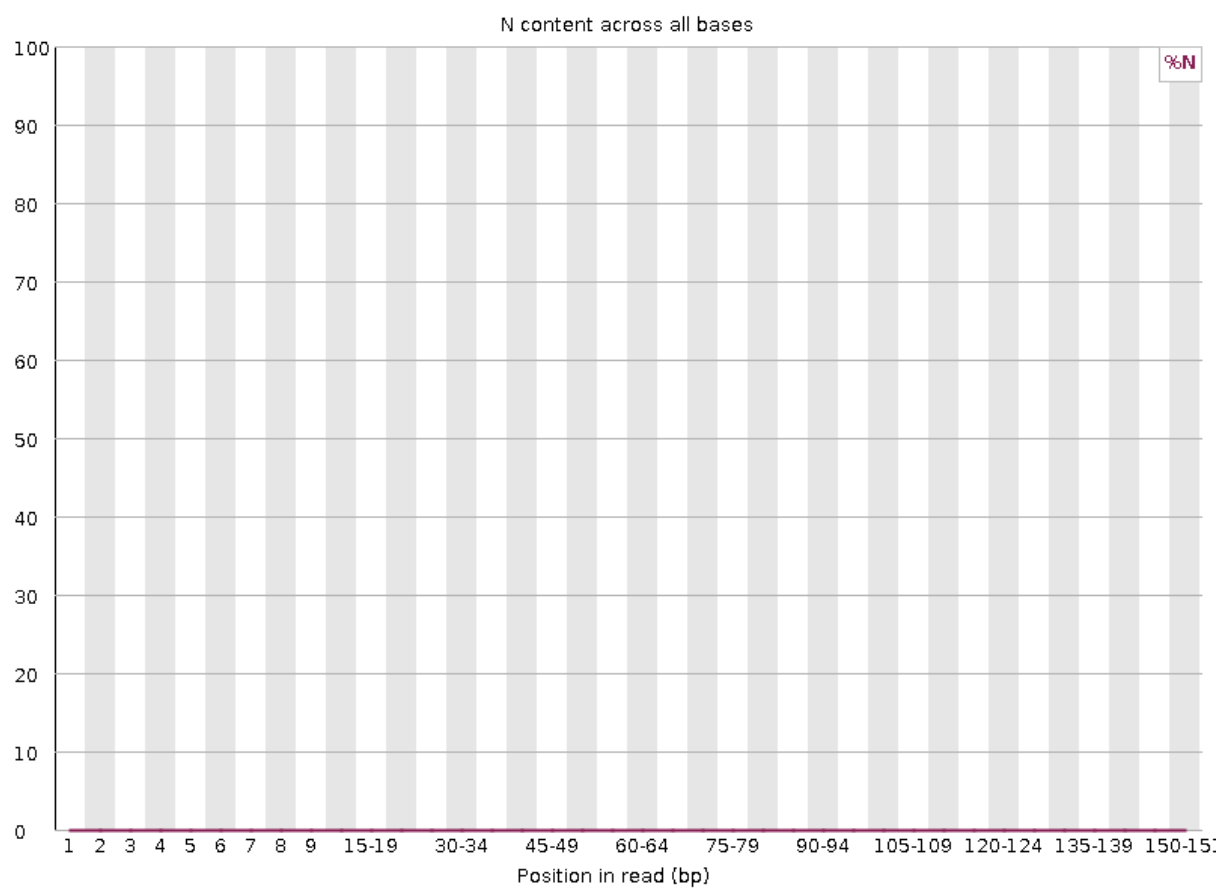
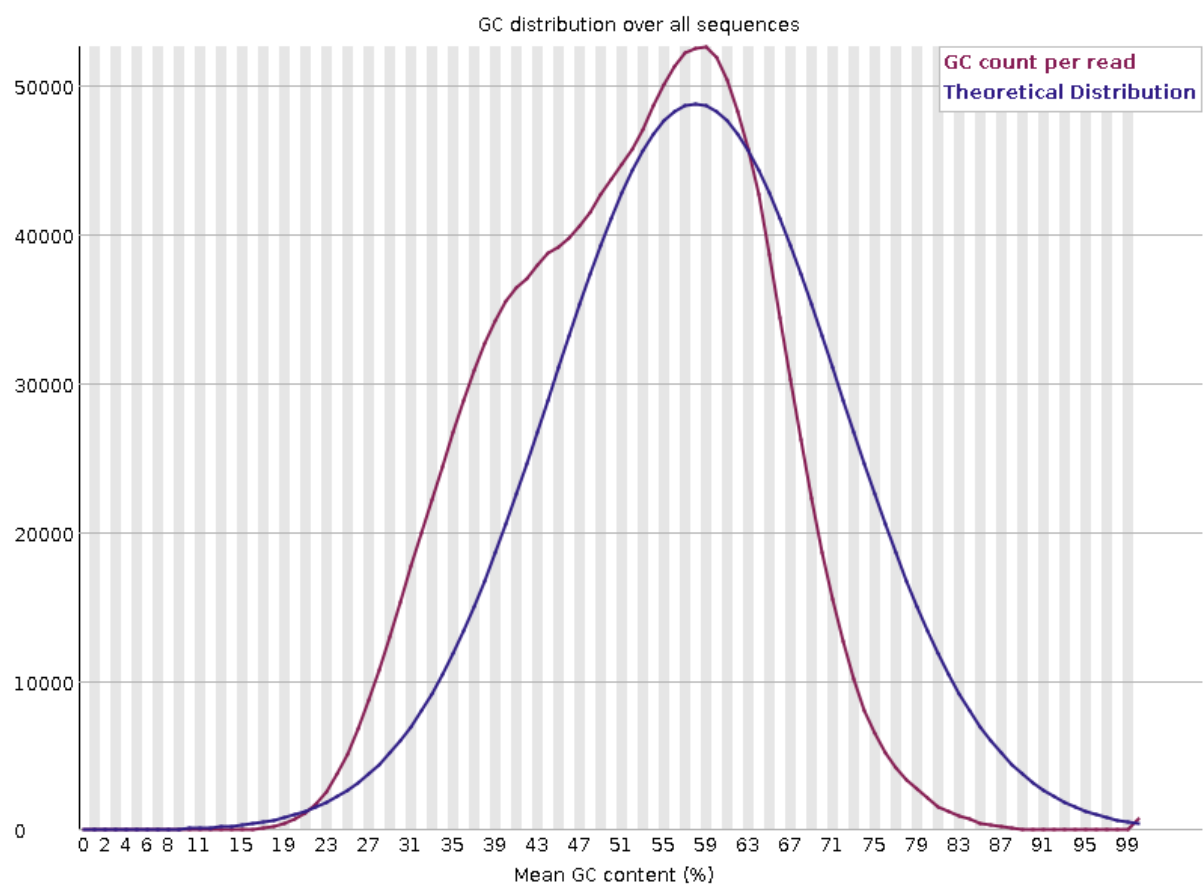


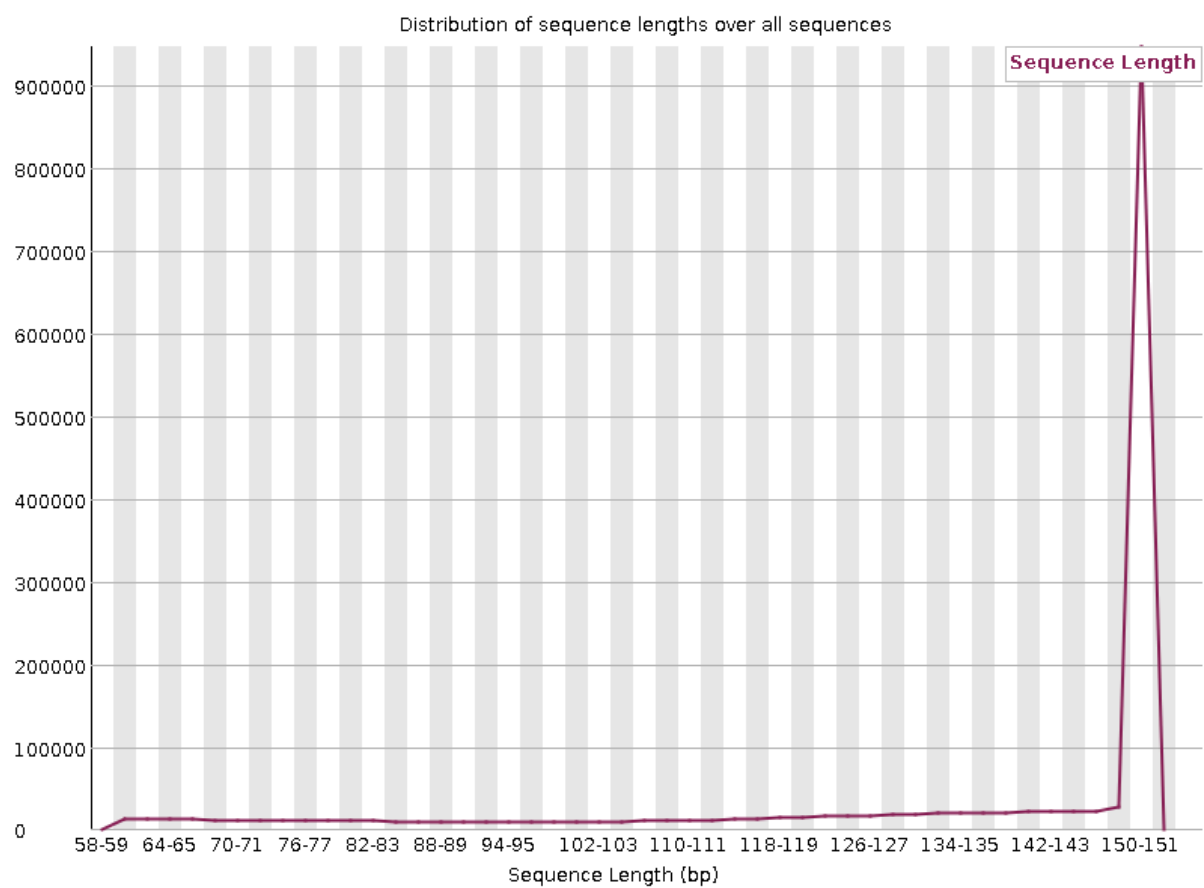
SRR37843243_2.up.fastq

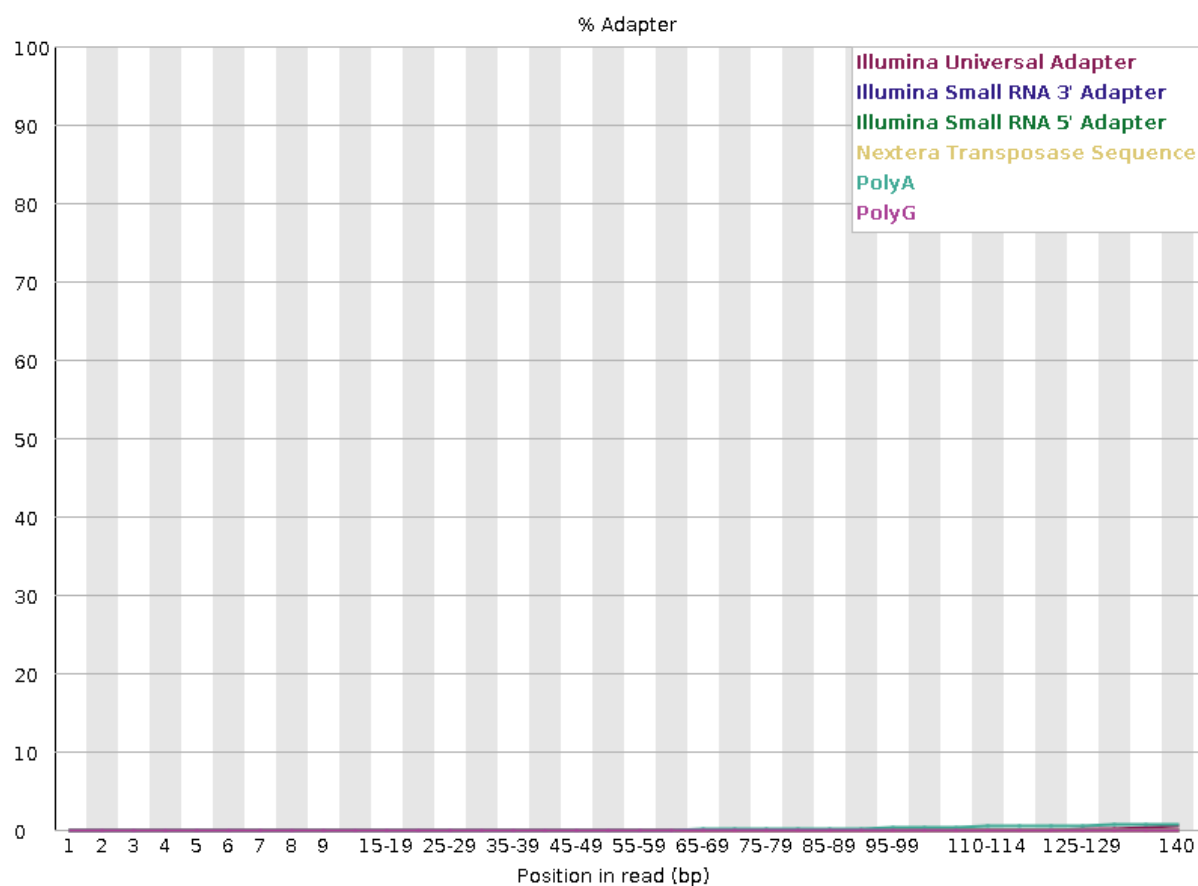
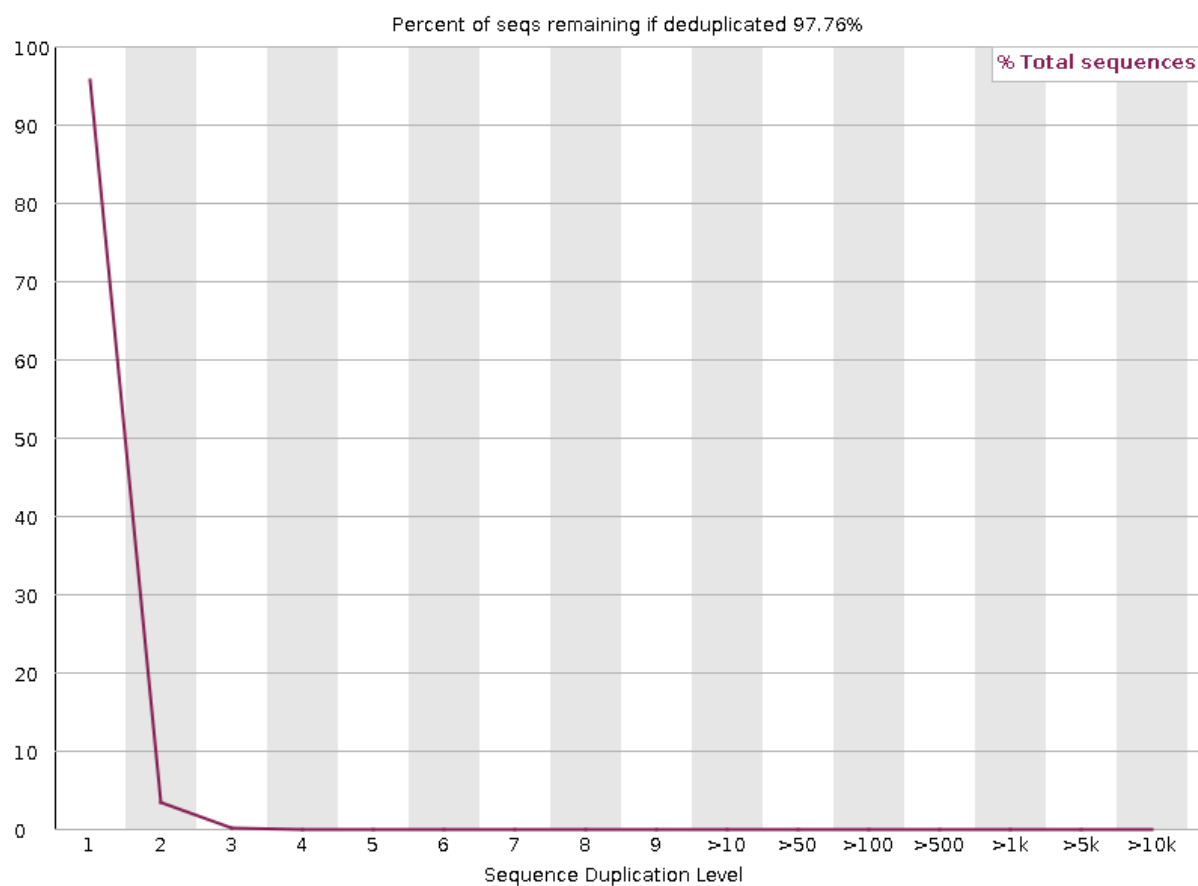
SRR37843243_2.up.fastq	
Measure	Value
Filename	SRR37843243_2.up.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1,603,175
Total Bases	215.6 Mbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51











Discussion

Per sequence GC content, Sequence Length Distribution을 제외하고 모든 항목이 양호하게 나왔다. Per base sequence content의 경우 100 bp부터 간극이 벌어지나, 그럼에도 모든 base가 20-30%에 분포하고 quality도 정상으로 나왔다. Overrepresented sequences 또한 모두 검출되지 않았다. 성공적인 trimming으로 판단했다.

SRR37843246

Description of Data

SRR37843246_1.fastq

The number of read sequences is 62288326
Total length of read sequences is 9405537226

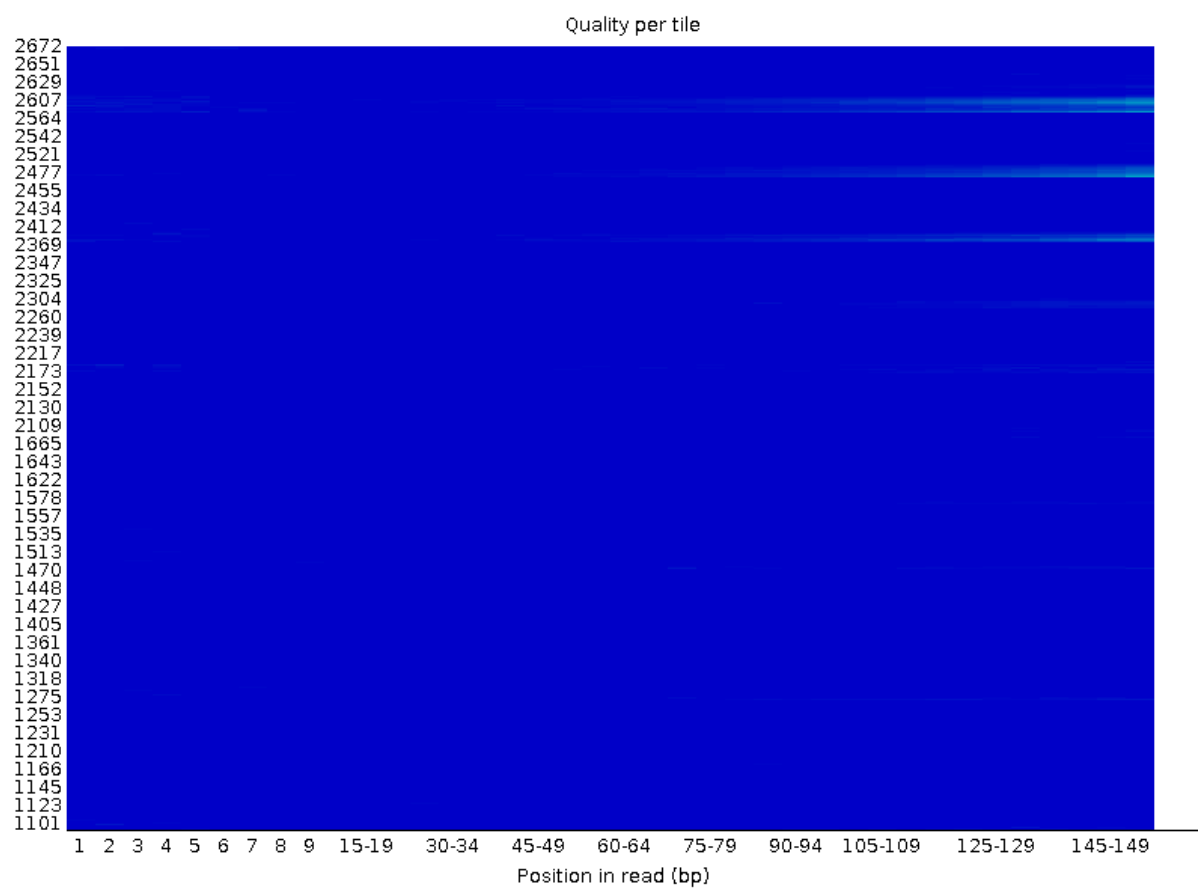
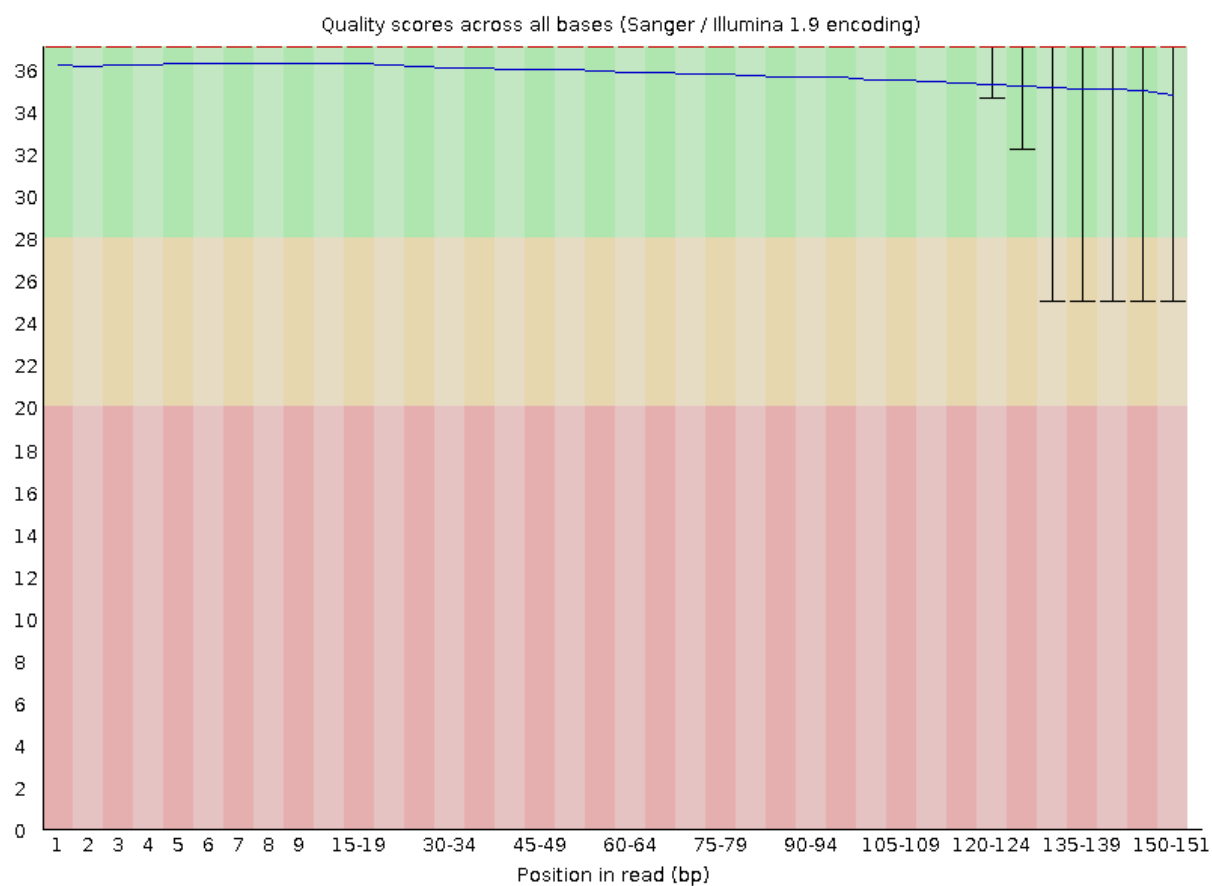
<Text distribution of read lengths>

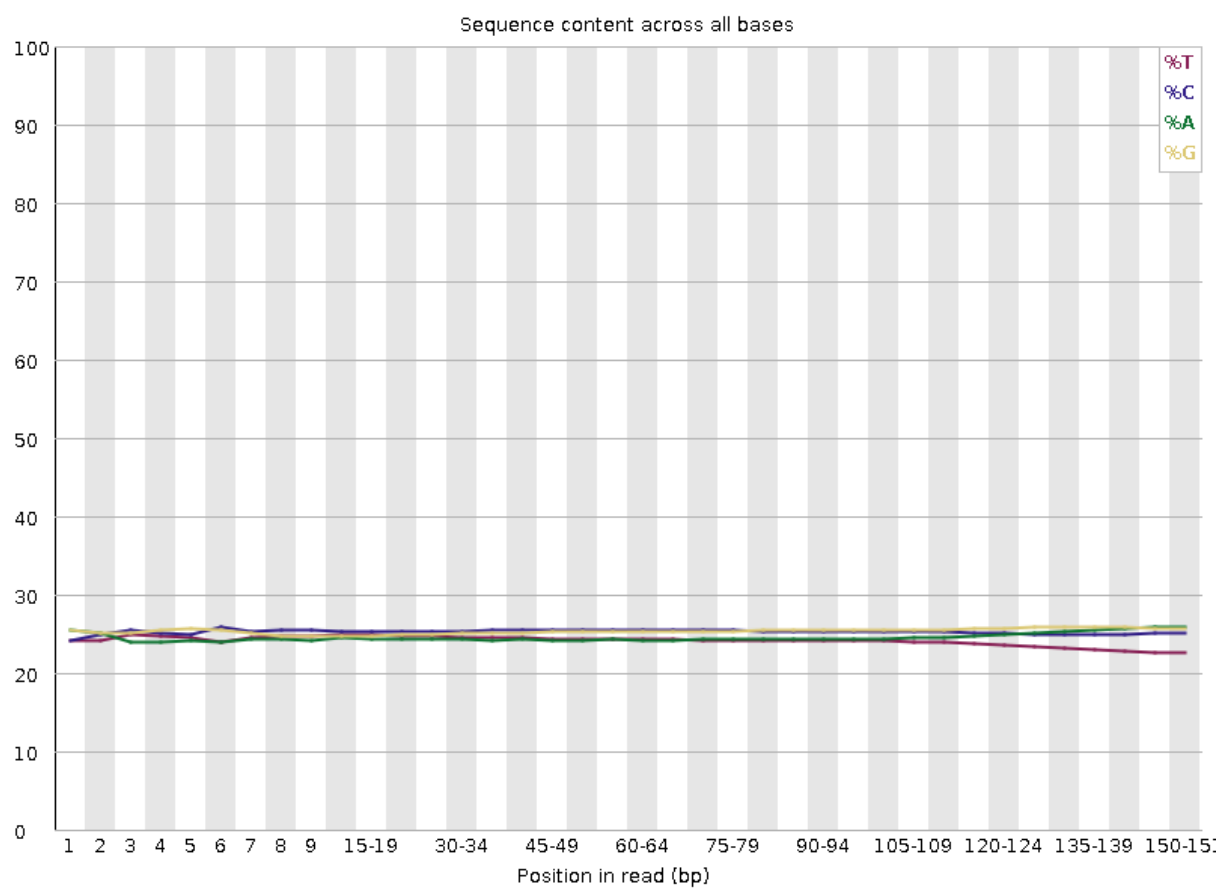
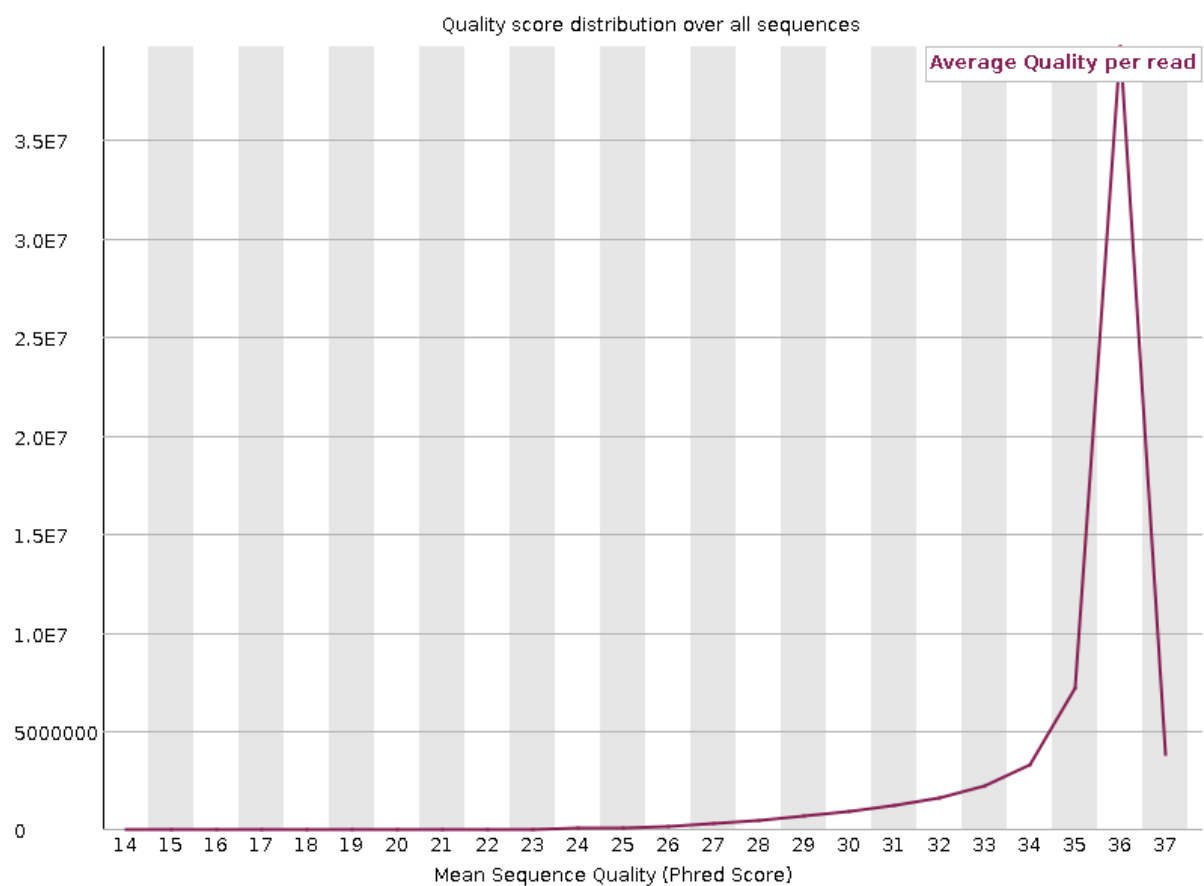
Length	Count	Abundance
151	62288326	100.00%

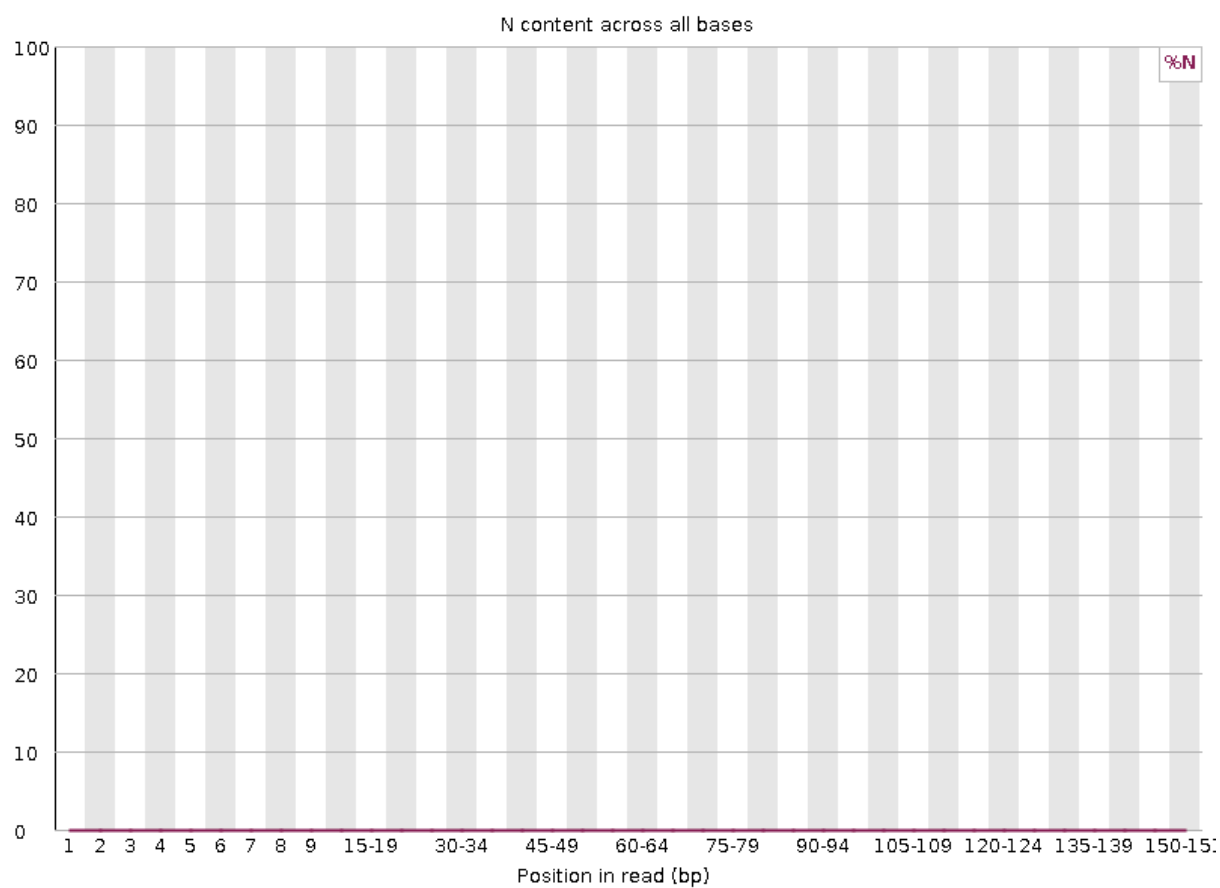
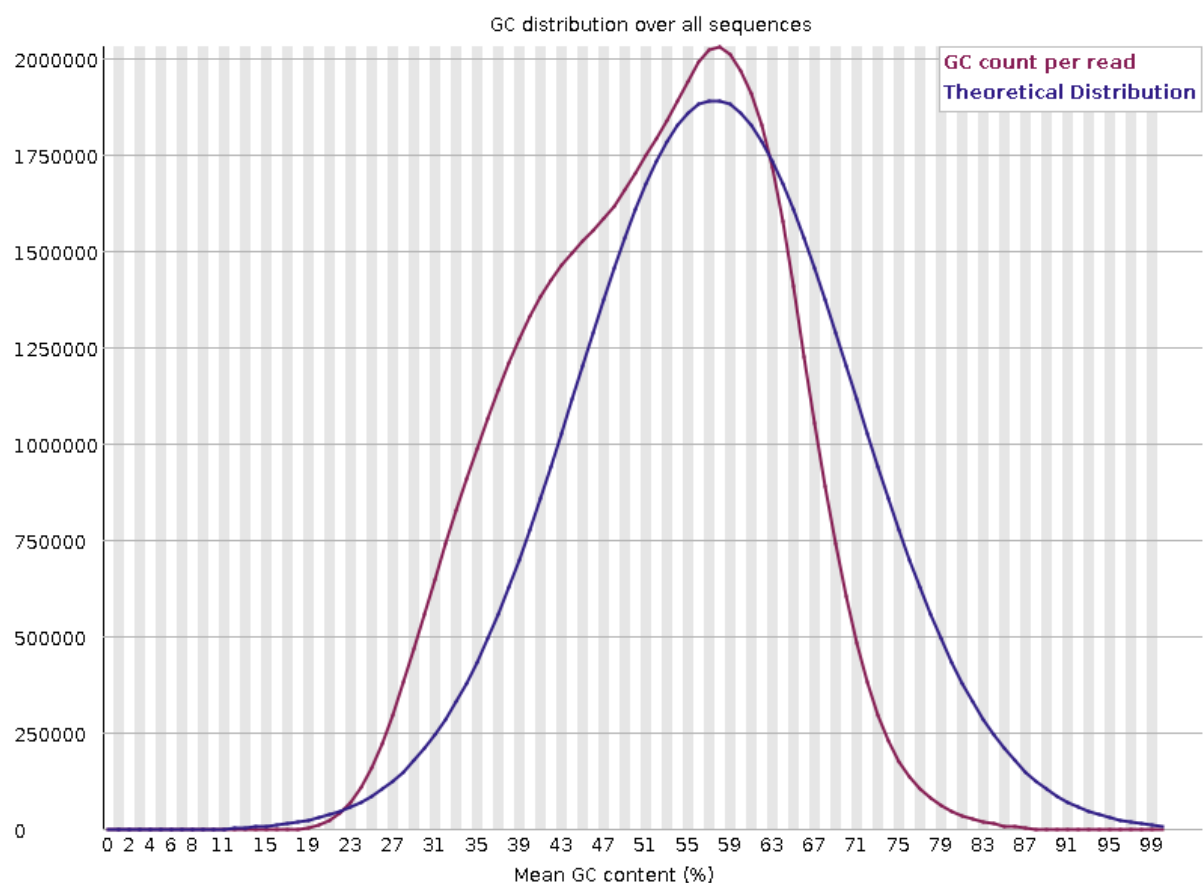
<Graphic distribution of read lengths>

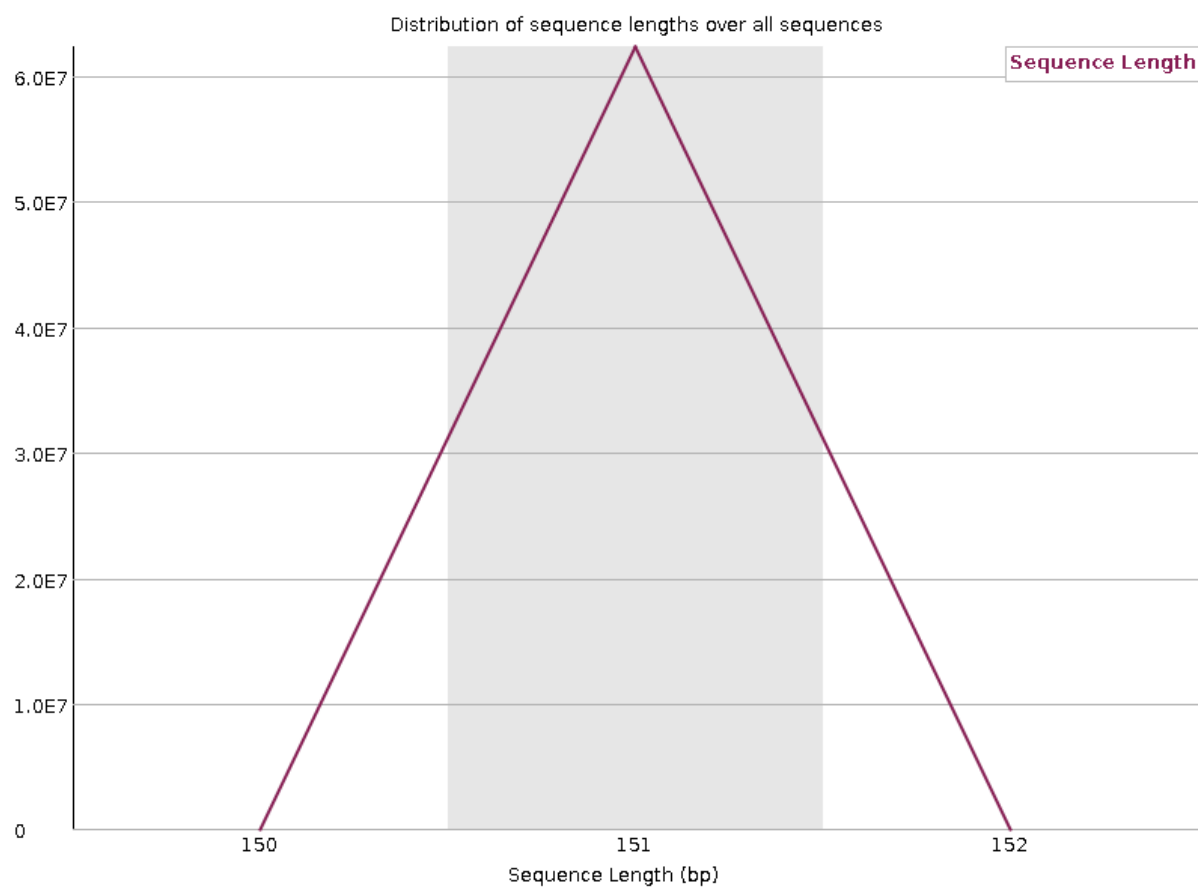
Length	Distribution	Abundance
151	■■■■■■■■■■■	100.00%

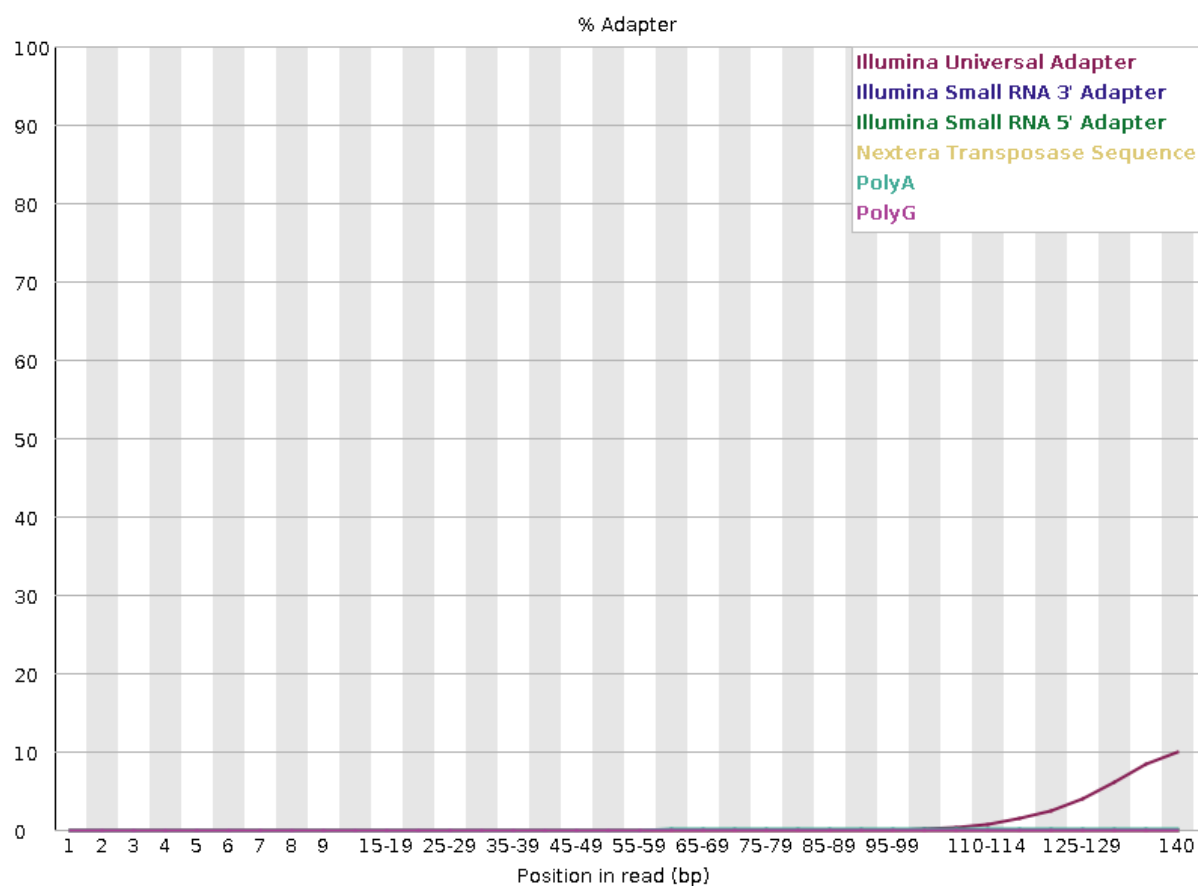
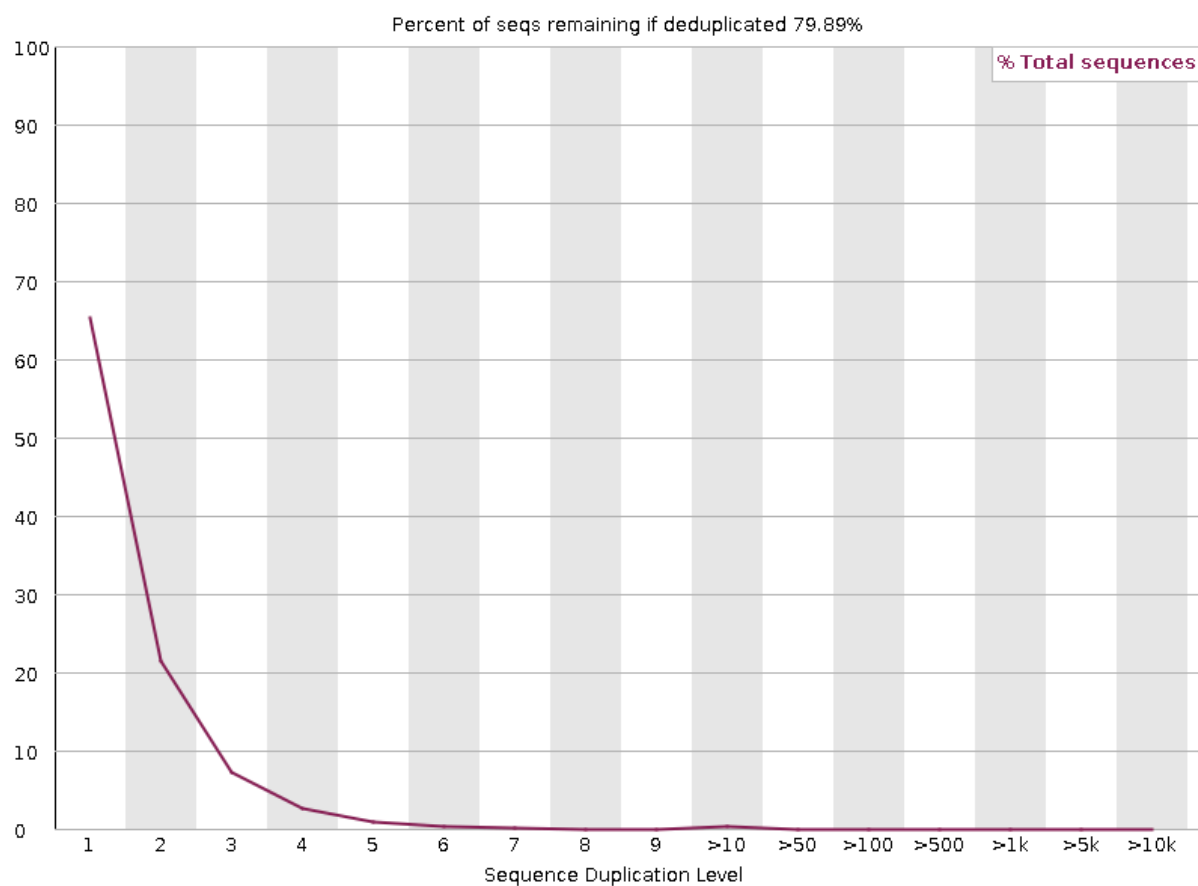
SRR37843243_1.fastq	
Measure	Value
Filename	SRR37843243_1.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	62,288,326
Total Bases	9.4 Gbp
Sequences flagged as poor quality	0
Sequence length	151
%GC	51











SRR37843246_2.fastq

The number of read sequences is 62288326

Total length of read sequences is 9405537226

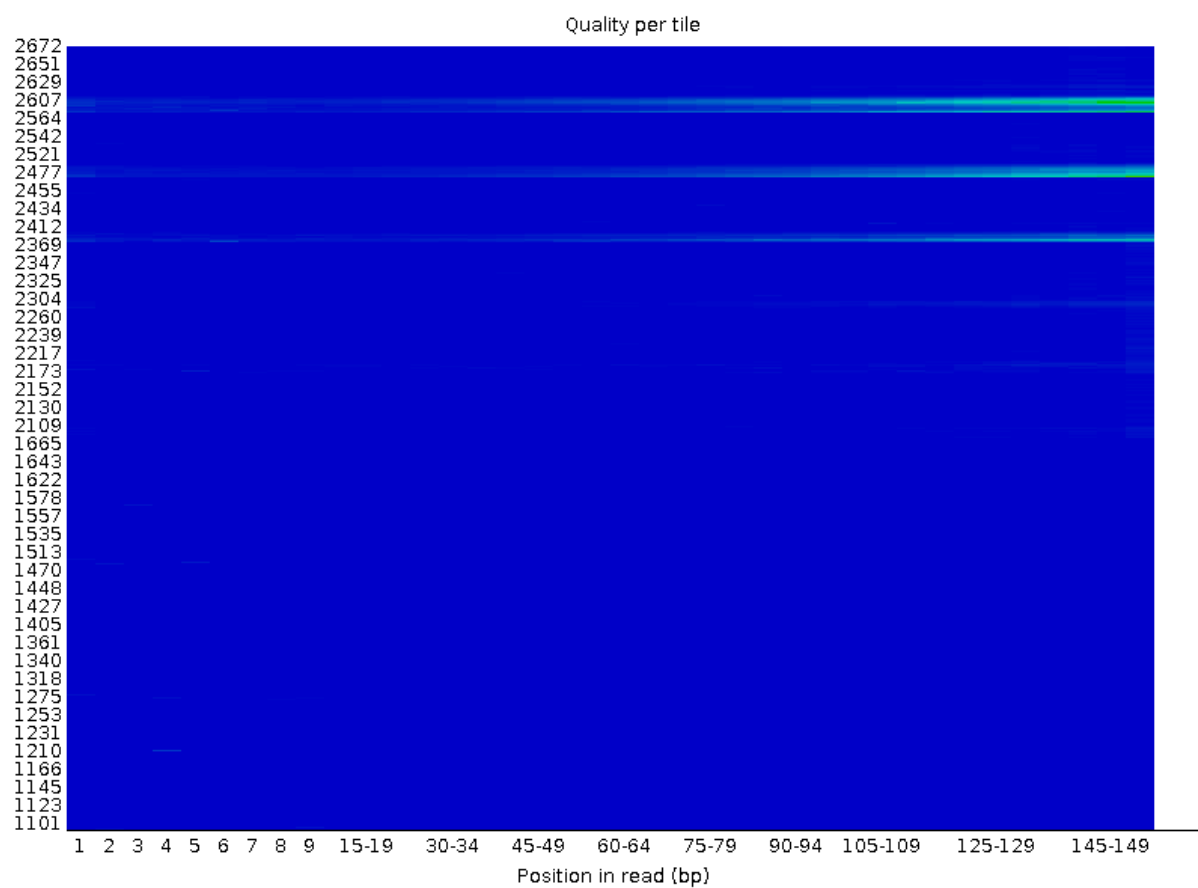
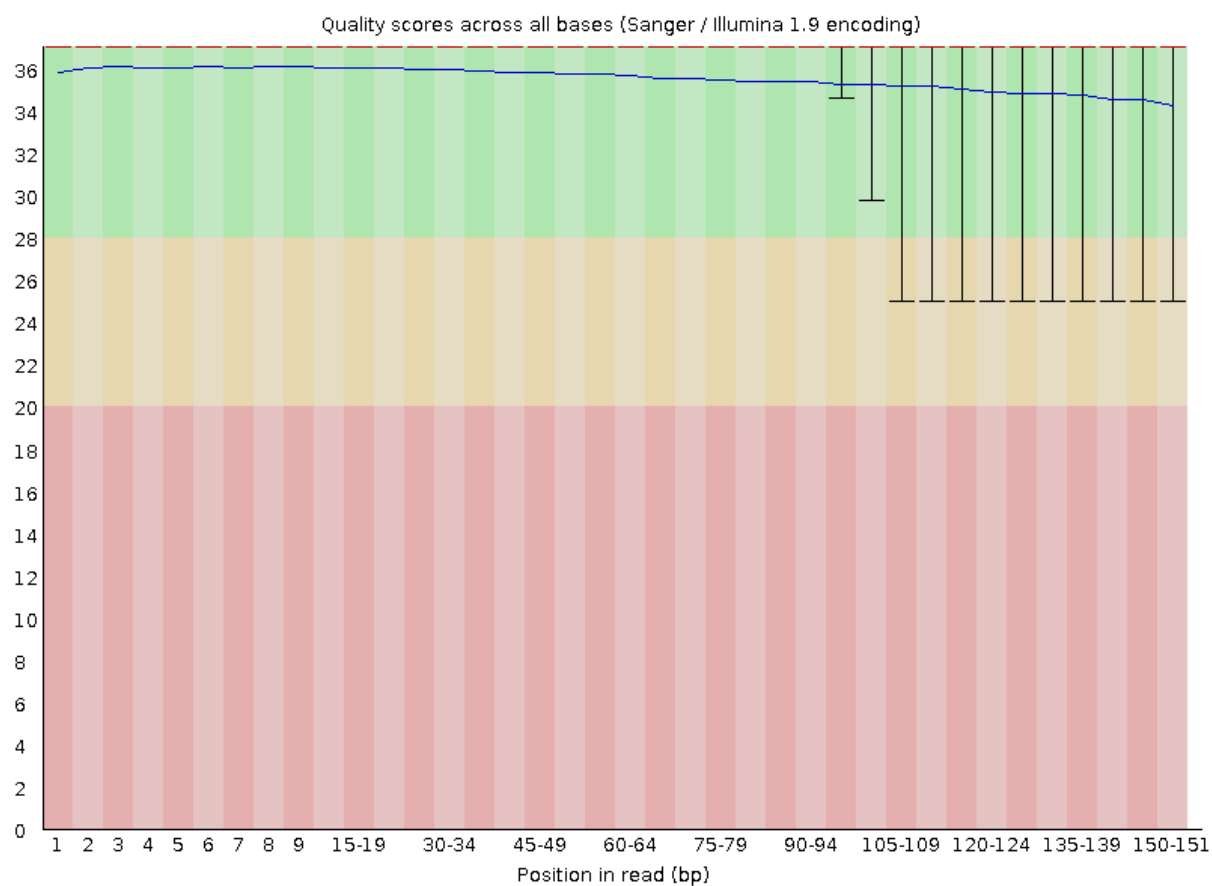
<Text distribution of read lengths>

Length	Count	Abundance
151	62288326	100.00%

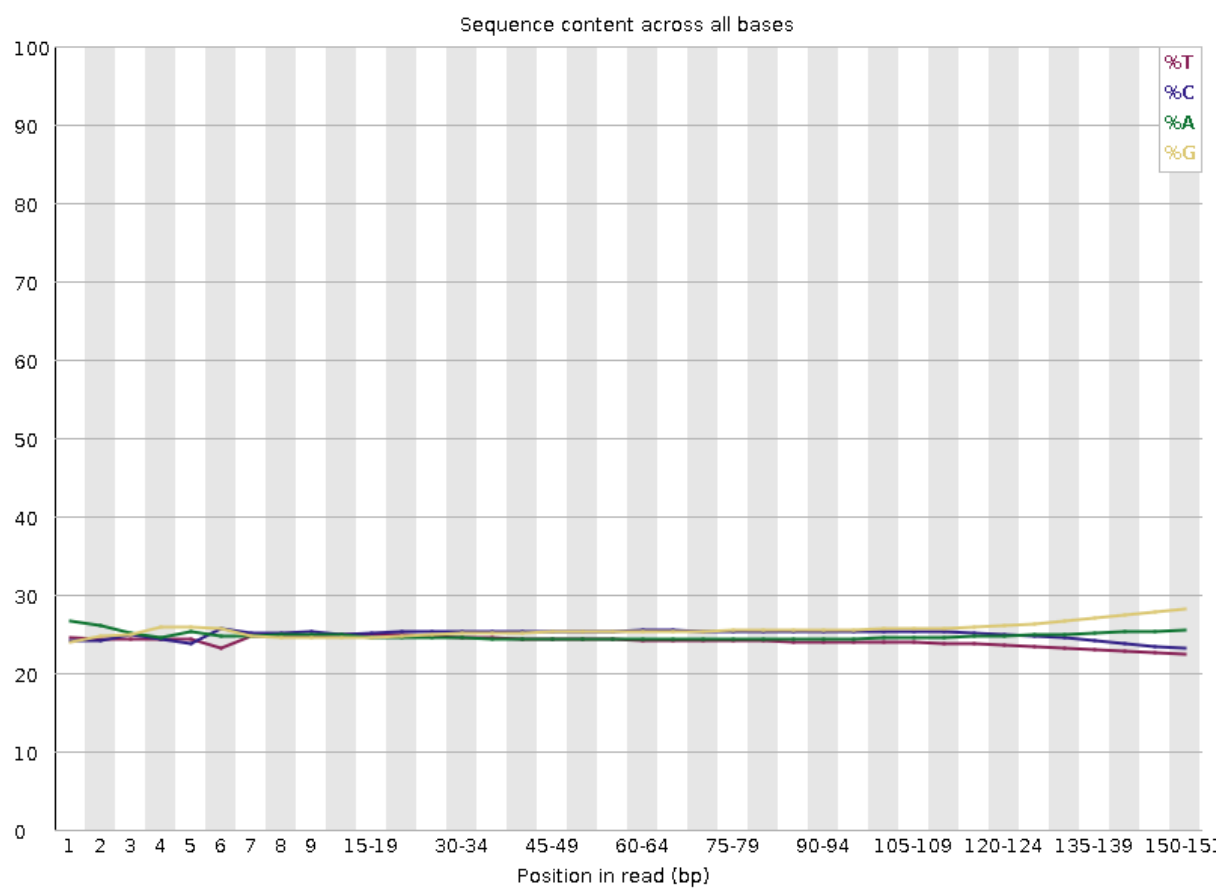
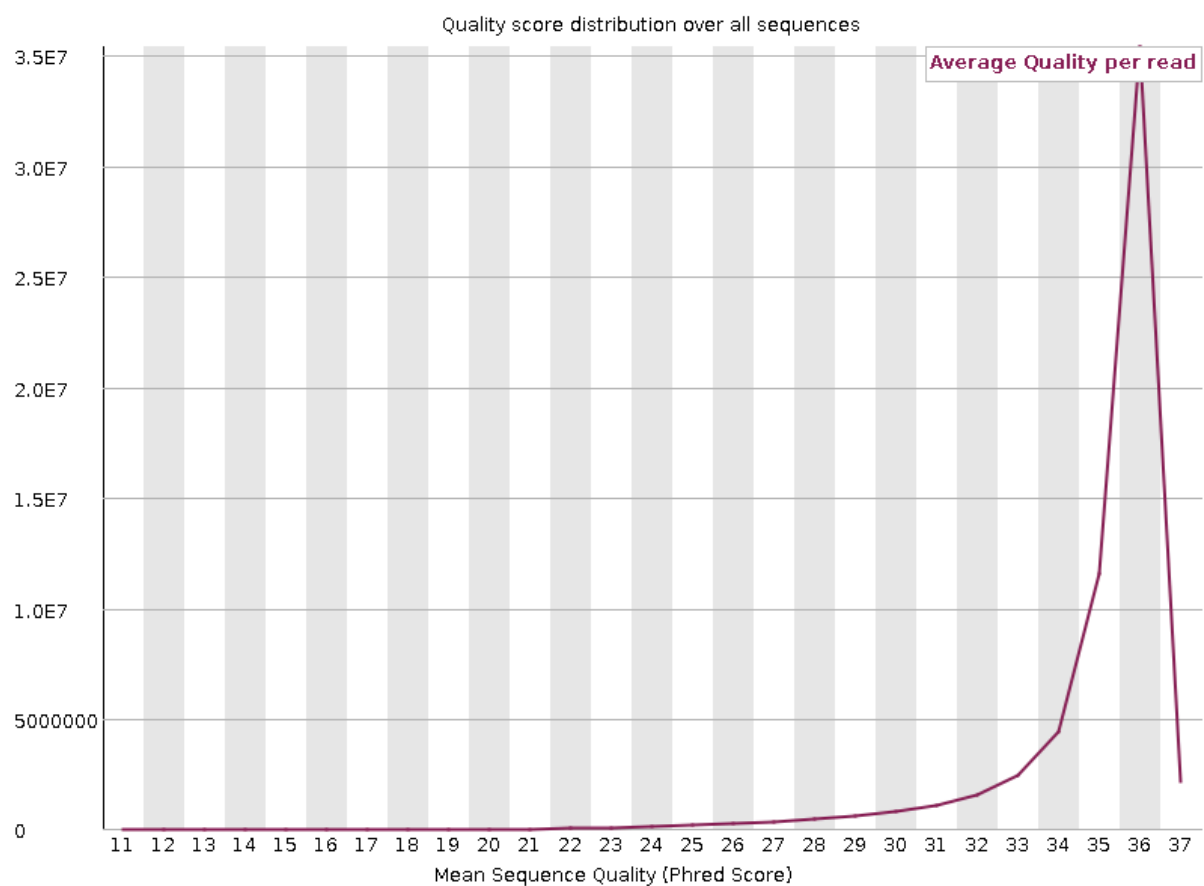
<Graphic distribution of read lengths>

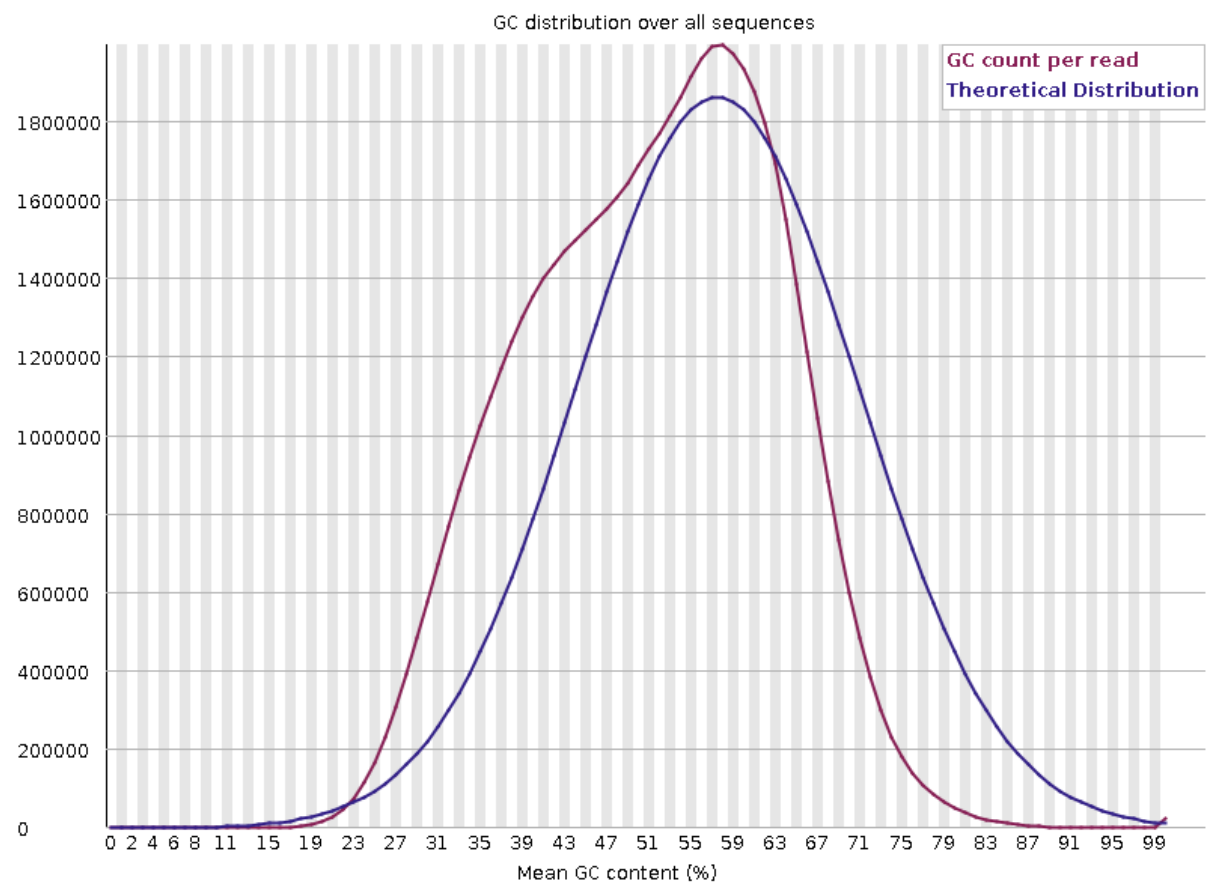
Length	Distribution	Abundance
151	■■■■■■■■■■	100.00%

SRR37843243_1.fastq	
Measure	Value
Filename	SRR37843243_1.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	62,288,326
Total Bases	9.4 Gbp
Sequences flagged as poor quality	0
Sequence length	151
%GC	51

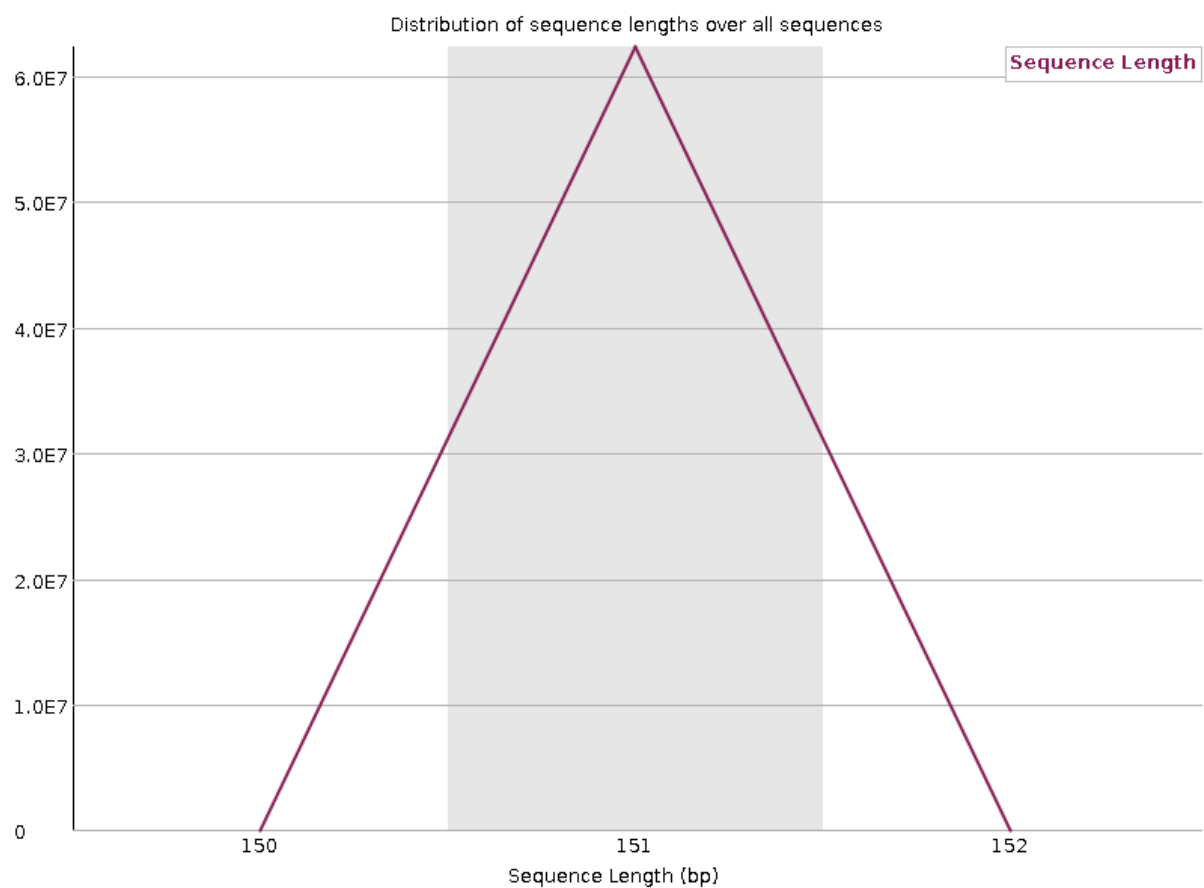
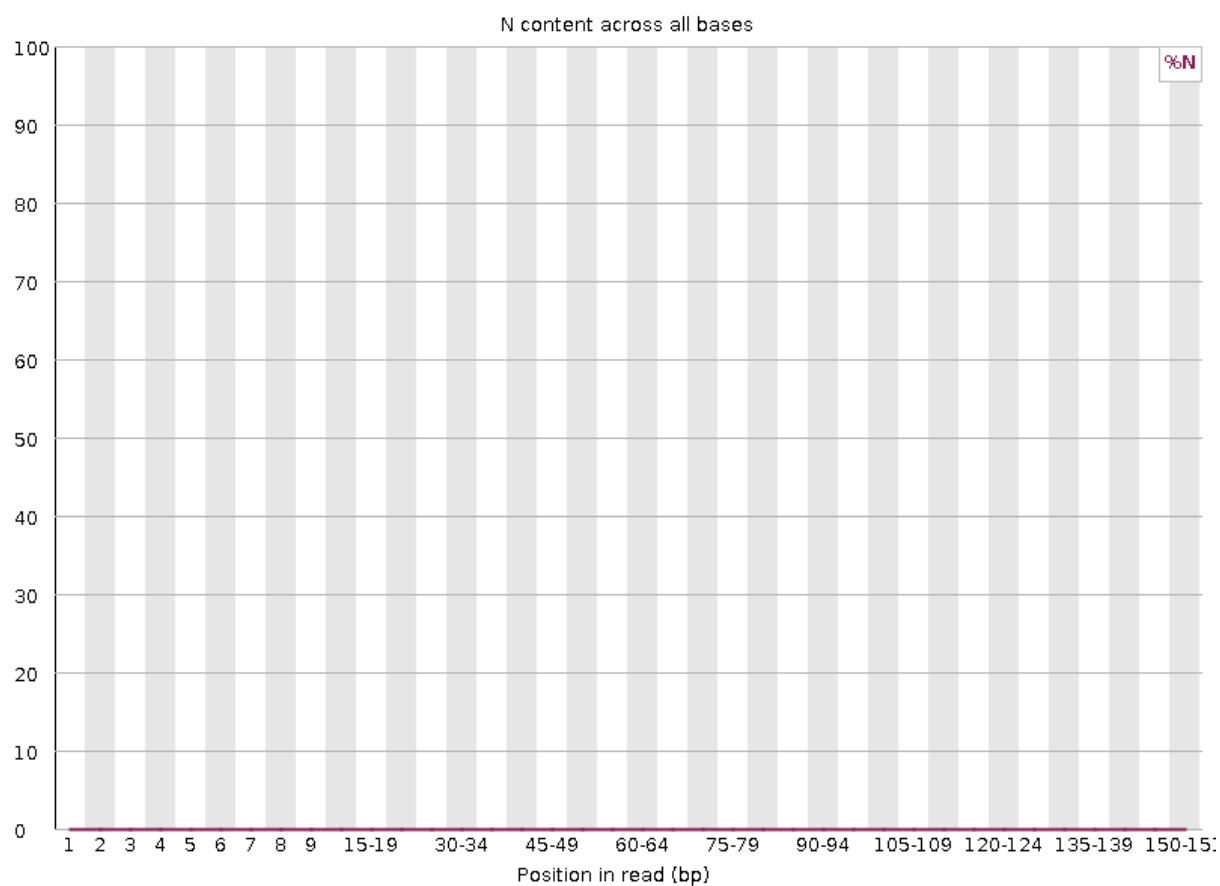


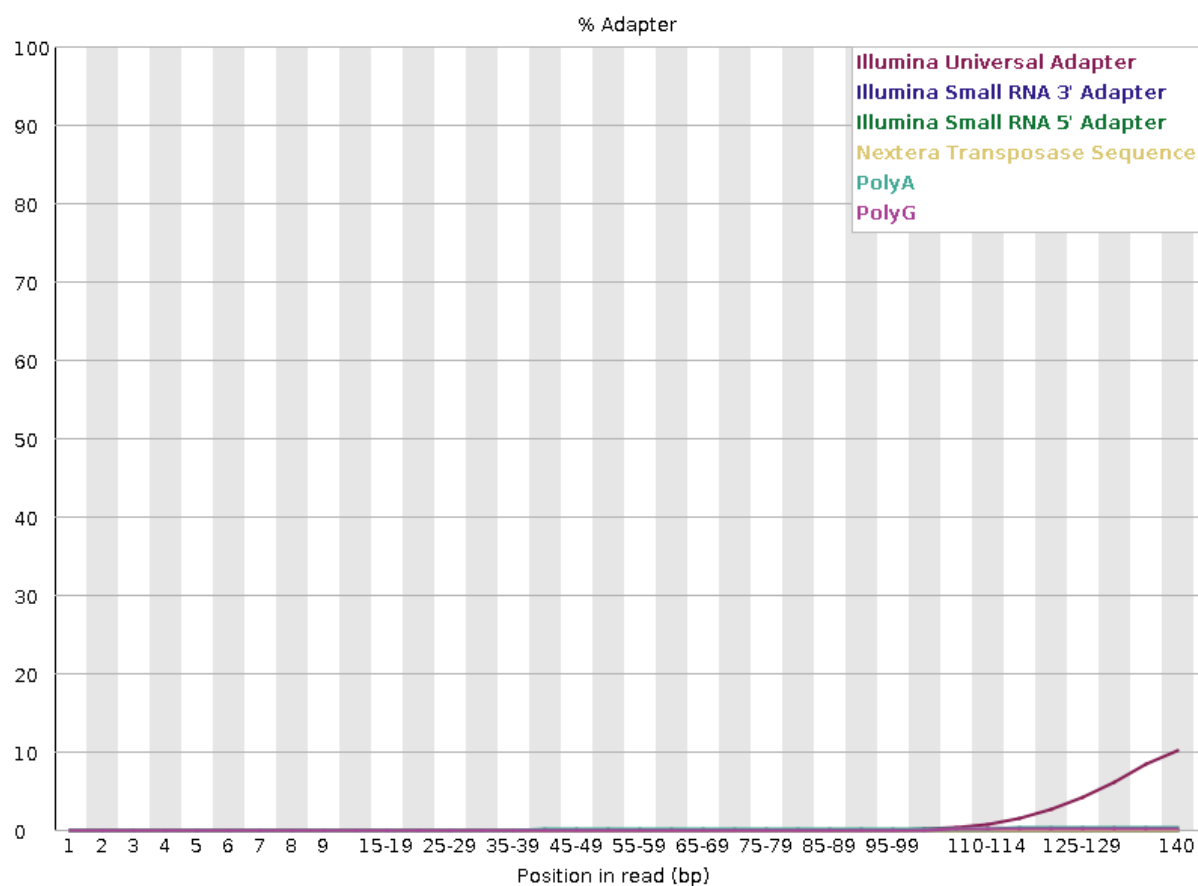
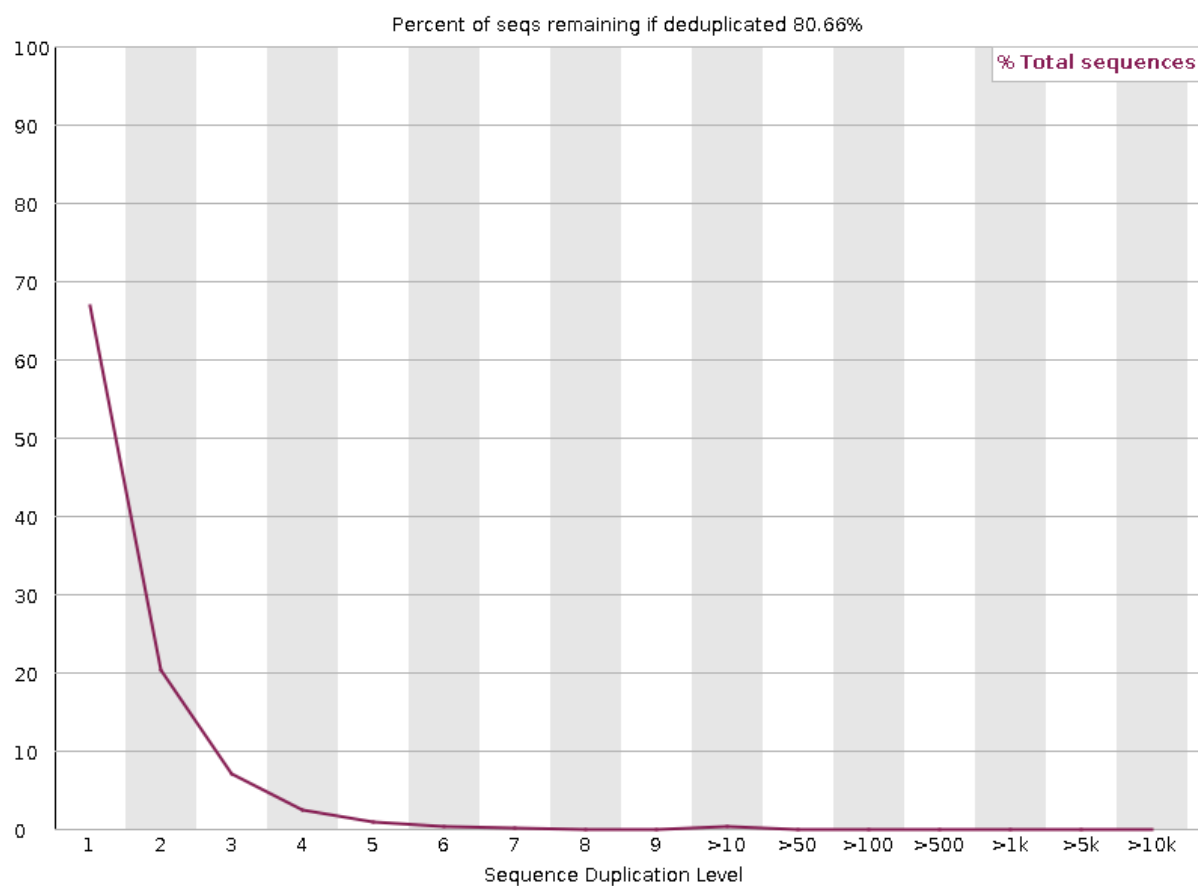
Warning





Failure





Failure

Overrepresented sequences

No overrepresented sequences in both 1 and 2.

After Trimming

TrimmomaticPE: Started with arguments:

SRR37843246_1.fastq SRR37843246_2.fastq SRR37843246_1.p.fastq SRR37843246_1.up.fastq

SRR37843246_2.p.fastq SRR37843246_2.up.fastq

ILLUMINACLIP:/mss/home_student/202112379/programs/Trimmomatic-0.39/adapters/TruSeq3-PE.

fa:2:30:10:2:True SLIDINGWINDOW:4:20 LEADING:10 TRAILING:10 MINLEN:60

Using PrefixPair: 'TACACTCTTCCCTACACGACGCTCTTCCGATCT' and

'GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT'

ILLUMINACLIP: Using 1 prefix pairs, 0 forward/reverse sequences, 0 forward only sequences, 0 reverse only sequences

Quality encoding detected as phred33

Input Read Pairs: 62288326 **Both Surviving: 57881966 (92.93%)** Forward Only Surviving:

1890021 (3.03%) Reverse Only Surviving: 1596819 (2.56%) Dropped: 919520 (1.48%)

TrimmomaticPE: Completed successfully

SRR37843246_1.p.fastq

The number of read sequences is 57881966

Total length of read sequences is 8431783799

<Text distribution of read lengths>

Length	Count	Abundance
60	32337	0.06%
61	32723	0.06%
62	32263	0.06%
63	33219	0.06%
64	32351	0.06%
65	33408	0.06%
66	32124	0.06%
67	32743	0.06%
68	32157	0.06%
69	32608	0.06%
70	32316	0.06%
71	32325	0.06%
72	32381	0.06%
73	32417	0.06%
74	32770	0.06%
75	31978	0.06%
76	32980	0.06%
77	31907	0.06%

78	33707	0.06%
79	33991	0.06%
80	33639	0.06%
81	35039	0.06%
82	35129	0.06%
83	34200	0.06%
84	34082	0.06%
85	34405	0.06%
86	34214	0.06%
87	34464	0.06%
88	34488	0.06%
89	34721	0.06%
90	36228	0.06%
91	35521	0.06%
92	35941	0.06%
93	36761	0.06%
94	38413	0.07%
95	38927	0.07%
96	39922	0.07%
97	42324	0.07%
98	41853	0.07%
99	44632	0.08%
100	47010	0.08%
101	49399	0.09%
102	52912	0.09%
103	56336	0.10%
104	58867	0.10%
105	62304	0.11%
106	66726	0.12%
107	71763	0.12%
108	77563	0.13%
109	83015	0.14%
110	87604	0.15%
111	94985	0.16%
112	102171	0.18%
113	110197	0.19%
114	117399	0.20%
115	127216	0.22%
116	136743	0.24%
117	145576	0.25%
118	156295	0.27%
119	166947	0.29%
120	177893	0.31%
121	185761	0.32%
122	196211	0.34%
123	207147	0.36%
124	219782	0.38%
125	229015	0.40%

126	238677	0.41%
127	247846	0.43%
128	261336	0.45%
129	271417	0.47%
130	282657	0.49%
131	291442	0.50%
132	300713	0.52%
133	310196	0.54%
134	318227	0.55%
135	325886	0.56%
136	335275	0.58%
137	341066	0.59%
138	352291	0.61%
139	351952	0.61%
140	362100	0.63%
141	368647	0.64%
142	372535	0.64%
143	379643	0.66%
144	386157	0.67%
145	389499	0.67%
146	392901	0.68%
147	396308	0.68%
148	399023	0.69%
149	466295	0.81%
150	1392354	2.41%
151	43873078	75.80%

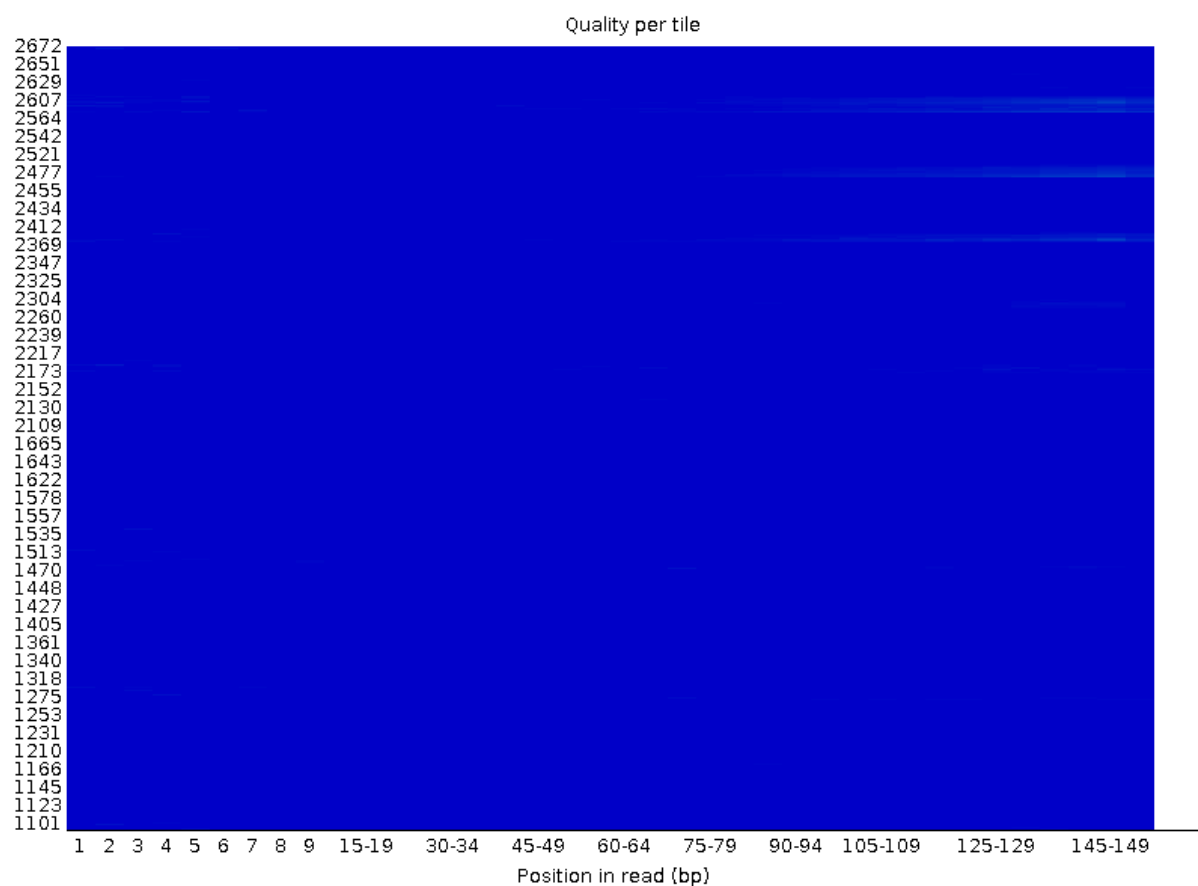
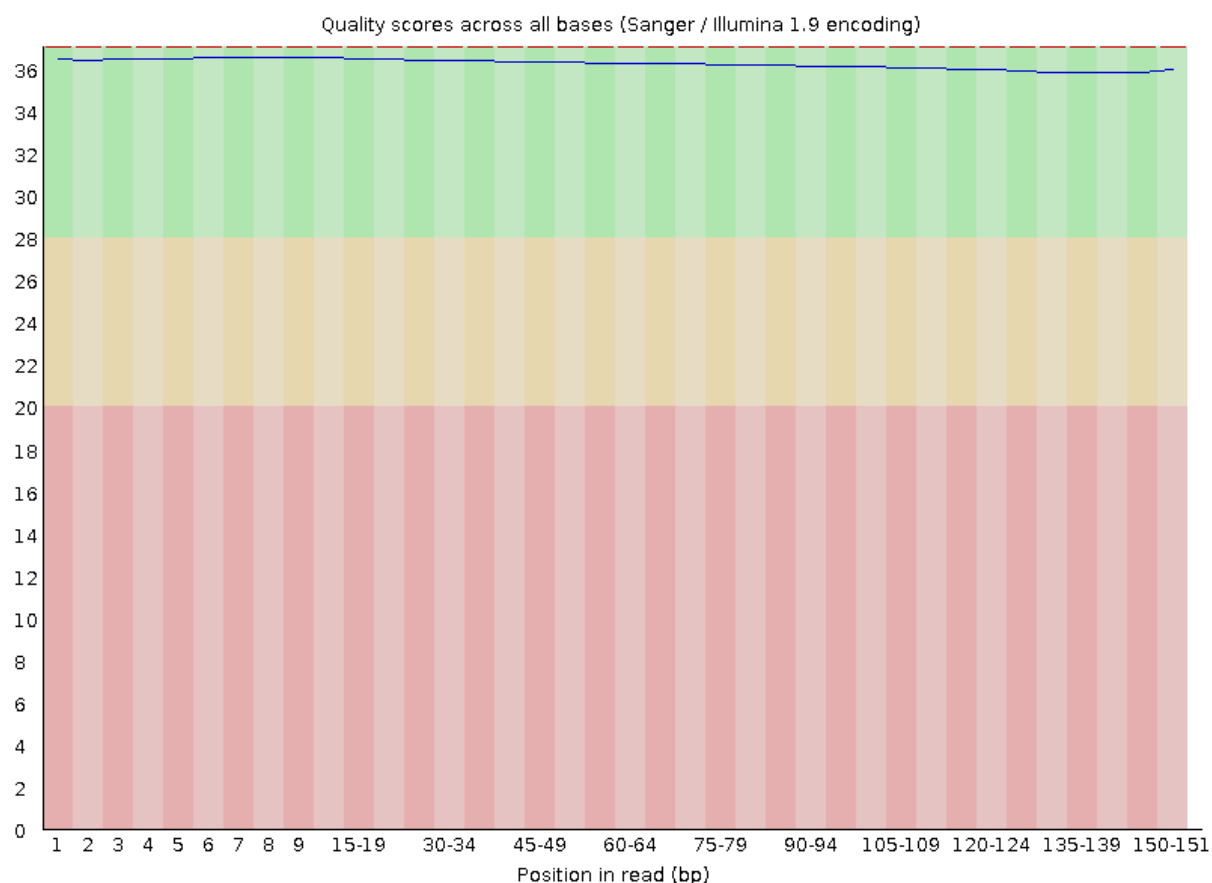
<Graphic distribution of read lengths>

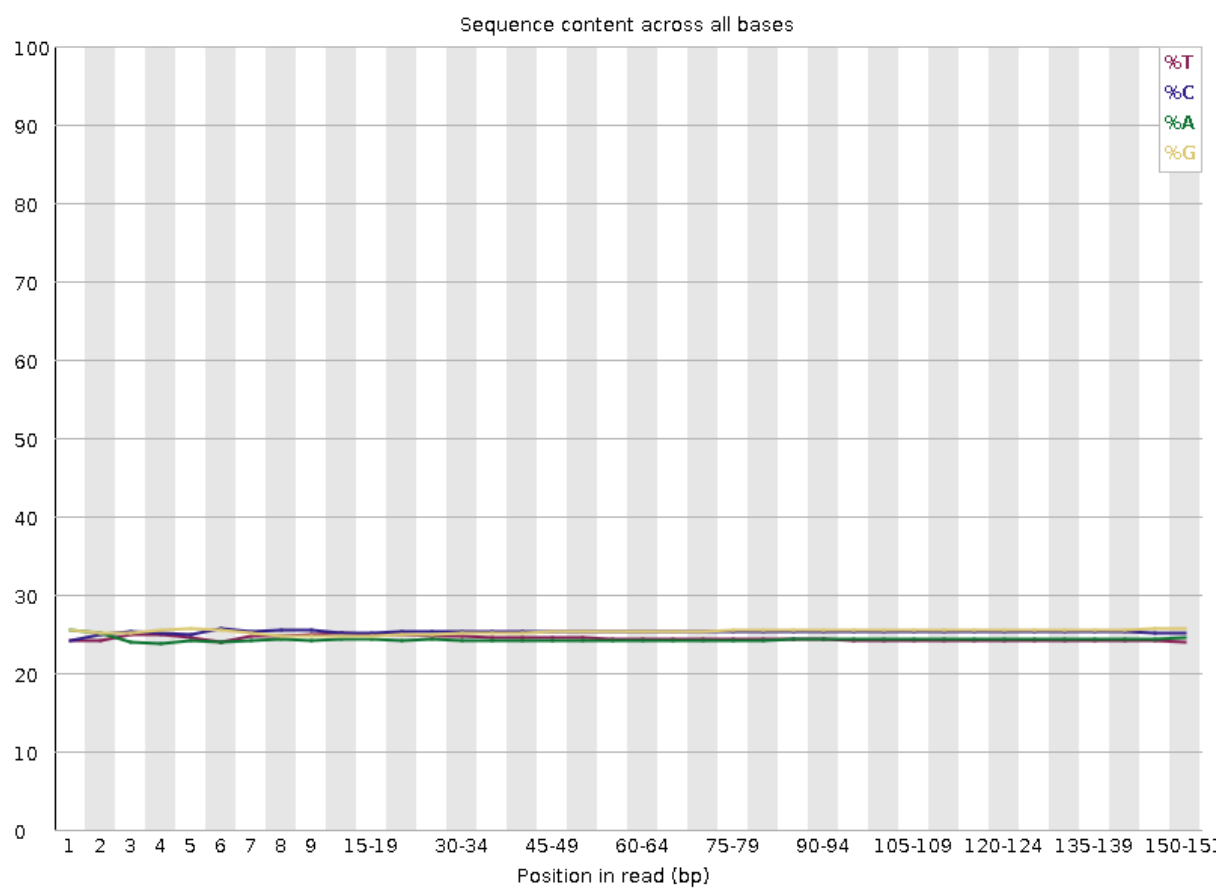
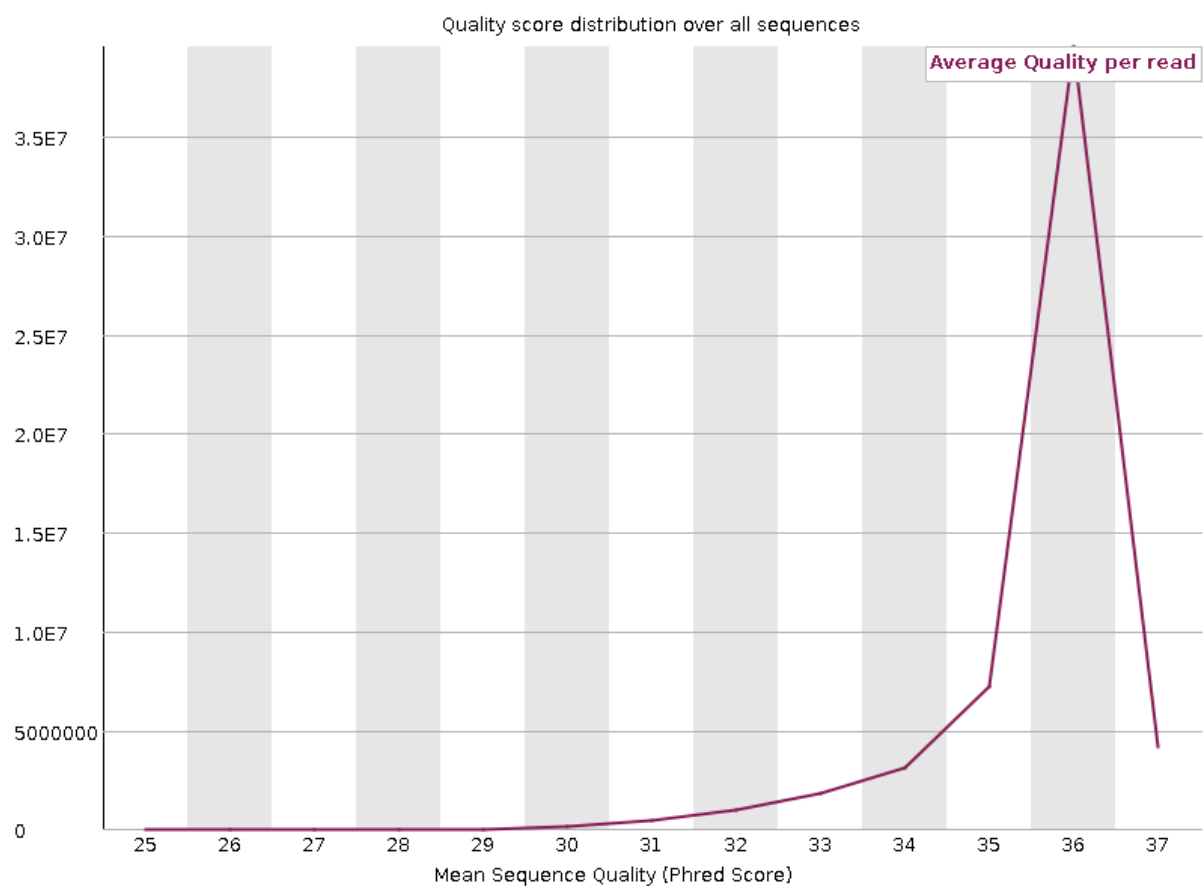
Length	Distribution	Abundance
60	□□□□□□□□□□	0.06%
61	□□□□□□□□□□	0.06%
62	□□□□□□□□□□	0.06%
63	□□□□□□□□□□	0.06%
64	□□□□□□□□□□	0.06%
65	□□□□□□□□□□	0.06%
66	□□□□□□□□□□	0.06%
67	□□□□□□□□□□	0.06%
68	□□□□□□□□□□	0.06%
69	□□□□□□□□□□	0.06%
70	□□□□□□□□□□	0.06%
71	□□□□□□□□□□	0.06%
72	□□□□□□□□□□	0.06%
73	□□□□□□□□□□	0.06%
74	□□□□□□□□□□	0.06%
75	□□□□□□□□□□	0.06%
76	□□□□□□□□□□	0.06%
77	□□□□□□□□□□	0.06%
78	□□□□□□□□□□	0.06%

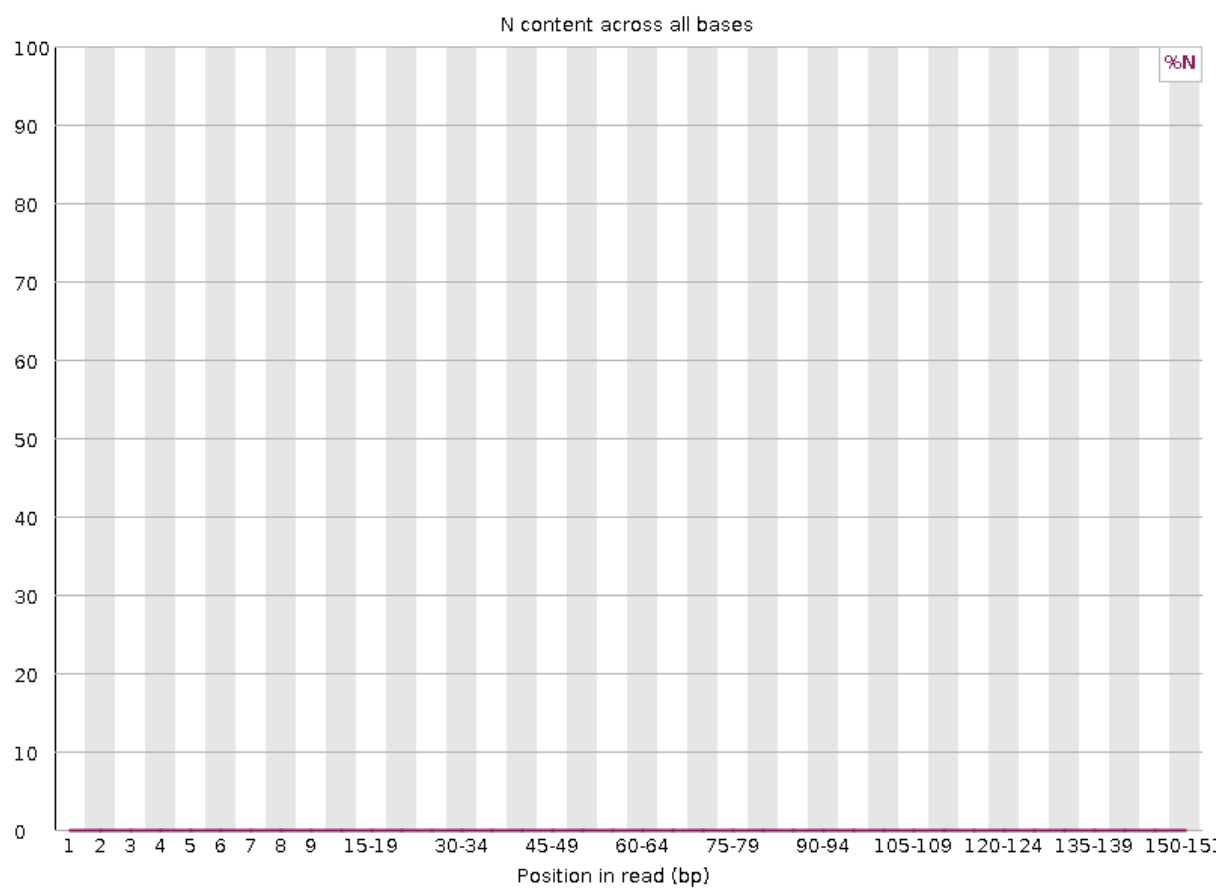
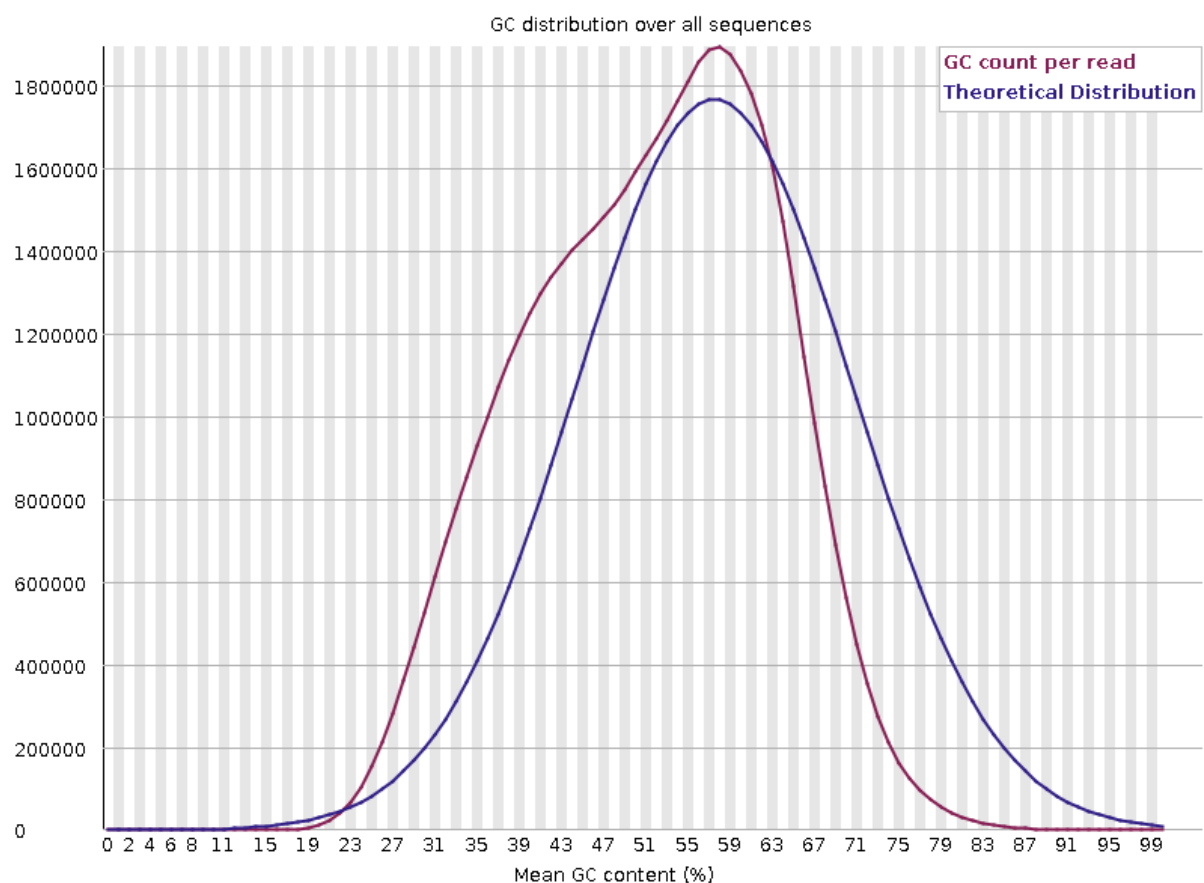
79	□□□□□□□□	0.06%
80	□□□□□□□□	0.06%
81	□□□□□□□□	0.06%
82	□□□□□□□□	0.06%
83	□□□□□□□□	0.06%
84	□□□□□□□□	0.06%
85	□□□□□□□□	0.06%
86	□□□□□□□□	0.06%
87	□□□□□□□□	0.06%
88	□□□□□□□□	0.06%
89	□□□□□□□□	0.06%
90	□□□□□□□□	0.06%
91	□□□□□□□□	0.06%
92	□□□□□□□□	0.06%
93	□□□□□□□□	0.06%
94	□□□□□□□□	0.07%
95	□□□□□□□□	0.07%
96	□□□□□□□□	0.07%
97	□□□□□□□□	0.07%
98	□□□□□□□□	0.07%
99	□□□□□□□□	0.08%
100	□□□□□□□□	0.08%
101	□□□□□□□□	0.09%
102	□□□□□□□□	0.09%
103	□□□□□□□□	0.10%
104	□□□□□□□□	0.10%
105	□□□□□□□□	0.11%
106	□□□□□□□□	0.12%
107	□□□□□□□□	0.12%
108	□□□□□□□□	0.13%
109	□□□□□□□□	0.14%
110	□□□□□□□□	0.15%
111	□□□□□□□□	0.16%
112	□□□□□□□□	0.18%
113	□□□□□□□□	0.19%
114	□□□□□□□□	0.20%
115	□□□□□□□□	0.22%
116	□□□□□□□□	0.24%
117	□□□□□□□□	0.25%
118	□□□□□□□□	0.27%
119	□□□□□□□□	0.29%
120	□□□□□□□□	0.31%
121	□□□□□□□□	0.32%
122	□□□□□□□□	0.34%
123	□□□□□□□□	0.36%
124	□□□□□□□□	0.38%
125	□□□□□□□□	0.40%
126	□□□□□□□□	0.41%

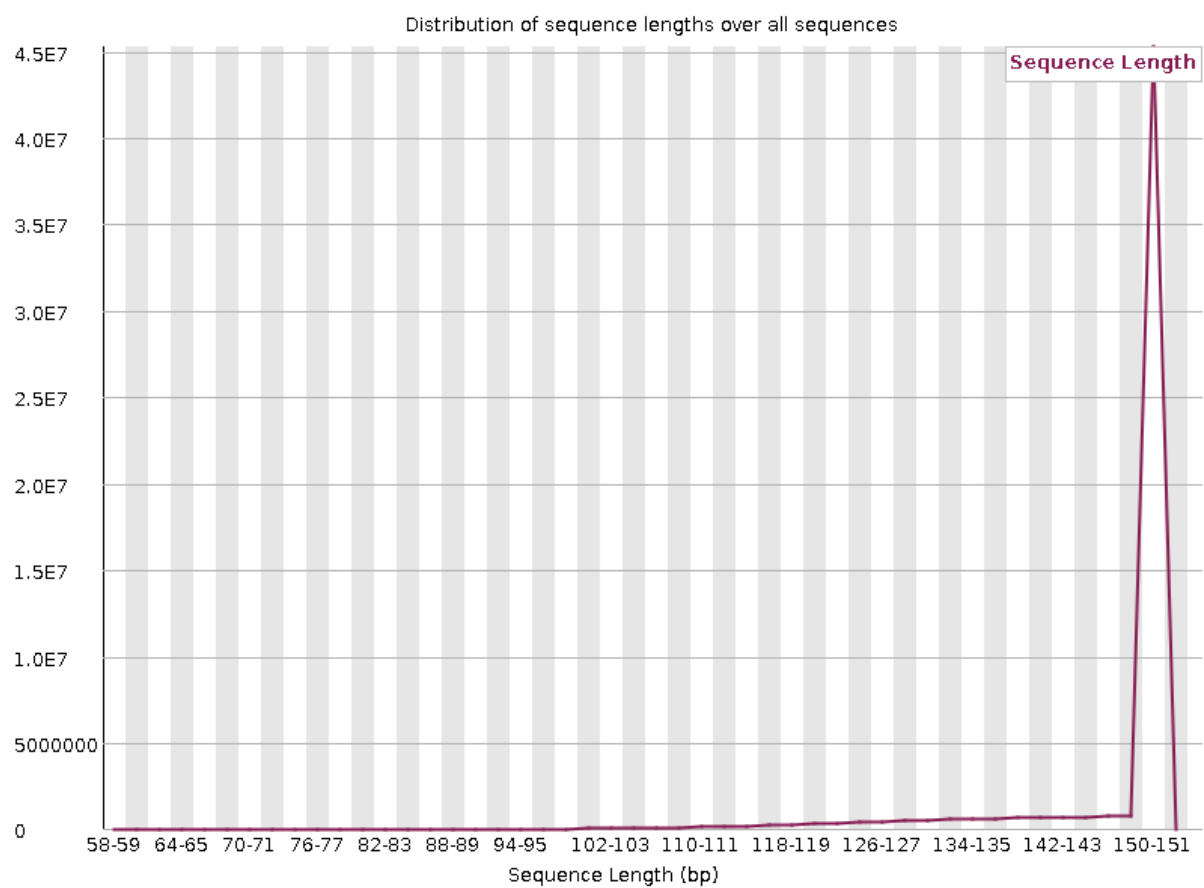
127	□□□□□□□□□□	0.43%
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129	□□□□□□□□□□	0.47%
130	□□□□□□□□□□	0.49%
131	□□□□□□□□□□	0.50%
132	□□□□□□□□□□	0.52%
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134	□□□□□□□□□□	0.55%
135	□□□□□□□□□□	0.56%
136	□□□□□□□□□□	0.58%
137	□□□□□□□□□□	0.59%
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141	□□□□□□□□□□	0.64%
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143	□□□□□□□□□□	0.66%
144	□□□□□□□□□□	0.67%
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146	□□□□□□□□□□	0.68%
147	□□□□□□□□□□	0.68%
148	□□□□□□□□□□	0.69%
149	□□□□□□□□□□	0.81%
150	□□□□□□□□□□	2.41%
151	■ ■ ■ ■ ■ ■ ■ □ □ □	75.80%

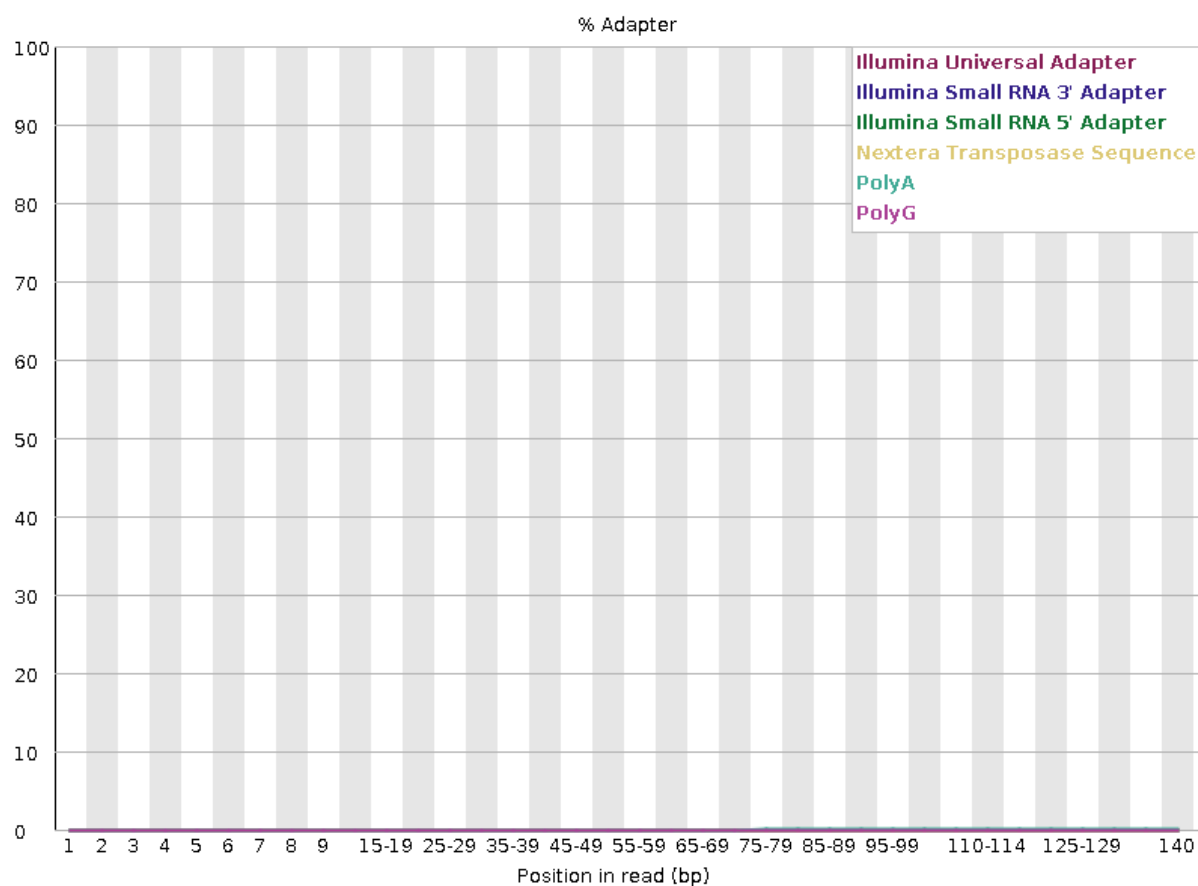
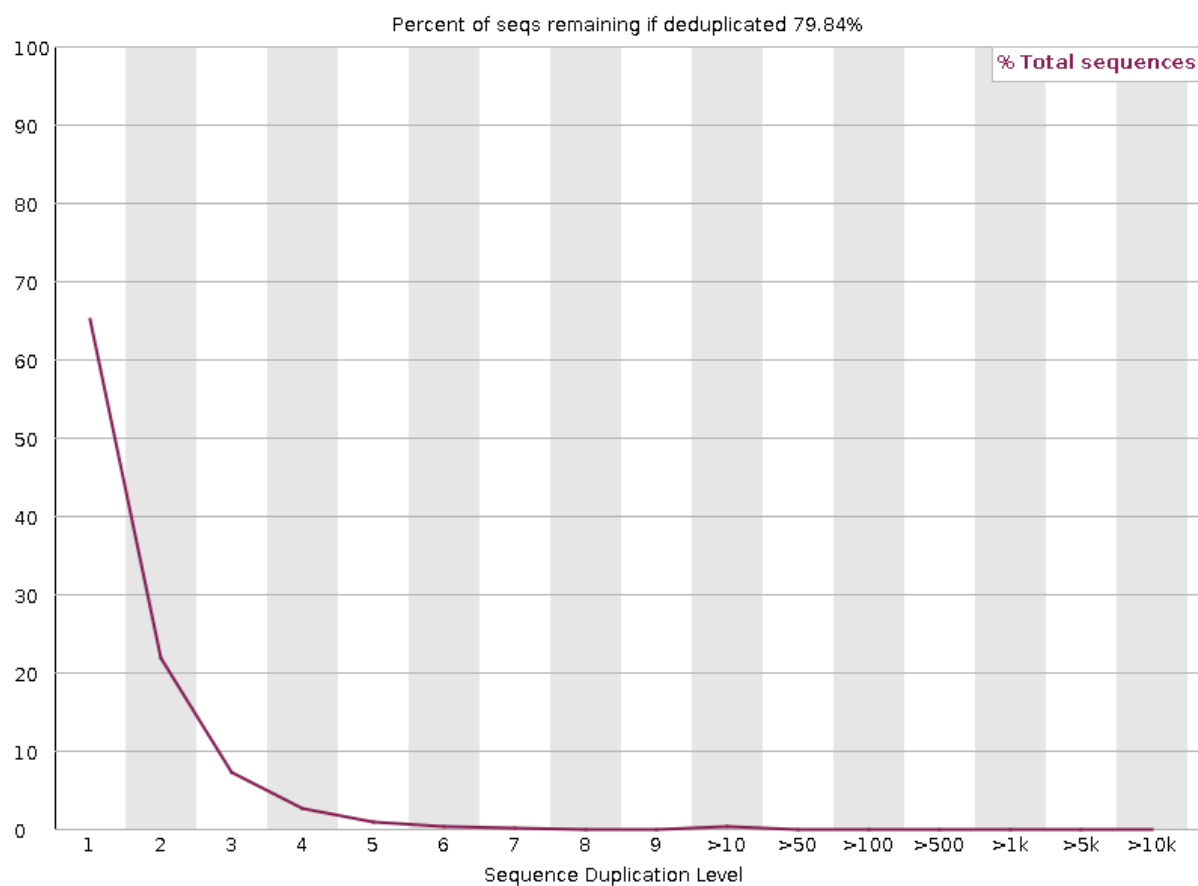
SRR37843243_1.p.fastq	
Measure	Value
Filename	SRR37843243_1.p.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	57,881,966
Total Bases	8.4 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51











SRR37843246_2.p.fastq

The number of read sequences is 57881966

Total length of read sequences is 8423671359

<Text distribution of read lengths>

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62	30224	0.05%
63	29177	0.05%
64	30510	0.05%
65	29630	0.05%
66	30867	0.05%
67	33370	0.06%
68	31426	0.05%
69	31505	0.05%
70	35345	0.06%
71	30439	0.05%
72	30929	0.05%
73	30099	0.05%
74	32245	0.06%
75	30506	0.05%
76	33167	0.06%
77	32885	0.06%
78	32561	0.06%
79	32577	0.06%
80	35983	0.06%
81	34112	0.06%
82	33913	0.06%
83	33092	0.06%
84	32954	0.06%
85	33612	0.06%
86	33347	0.06%
87	33003	0.06%
88	31980	0.06%
89	34472	0.06%
90	34902	0.06%
91	36110	0.06%
92	37964	0.07%
93	36891	0.06%
94	38066	0.07%
95	42928	0.07%
96	42706	0.07%
97	46097	0.08%
98	45333	0.08%
99	46297	0.08%
100	53197	0.09%

101	51828	0.09%
102	53977	0.09%
103	57950	0.10%
104	61841	0.11%
105	64635	0.11%
106	71421	0.12%
107	76054	0.13%
108	79766	0.14%
109	86447	0.15%
110	90330	0.16%
111	98251	0.17%
112	107147	0.19%
113	114137	0.20%
114	124442	0.21%
115	132882	0.23%
116	142397	0.25%
117	152074	0.26%
118	162541	0.28%
119	172840	0.30%
120	184229	0.32%
121	195366	0.34%
122	206577	0.36%
123	221268	0.38%
124	231767	0.40%
125	241307	0.42%
126	249291	0.43%
127	261274	0.45%
128	271038	0.47%
129	283043	0.49%
130	290186	0.50%
131	302230	0.52%
132	314247	0.54%
133	322574	0.56%
134	332565	0.57%
135	336768	0.58%
136	350136	0.60%
137	356350	0.62%
138	362185	0.63%
139	369308	0.64%
140	380929	0.66%
141	388290	0.67%
142	393752	0.68%
143	395966	0.68%
144	404551	0.70%
145	402184	0.69%
146	411404	0.71%
147	414371	0.72%
148	419604	0.72%

149	524655	0.91%
150	1869601	3.23%
151	42871316	74.07%

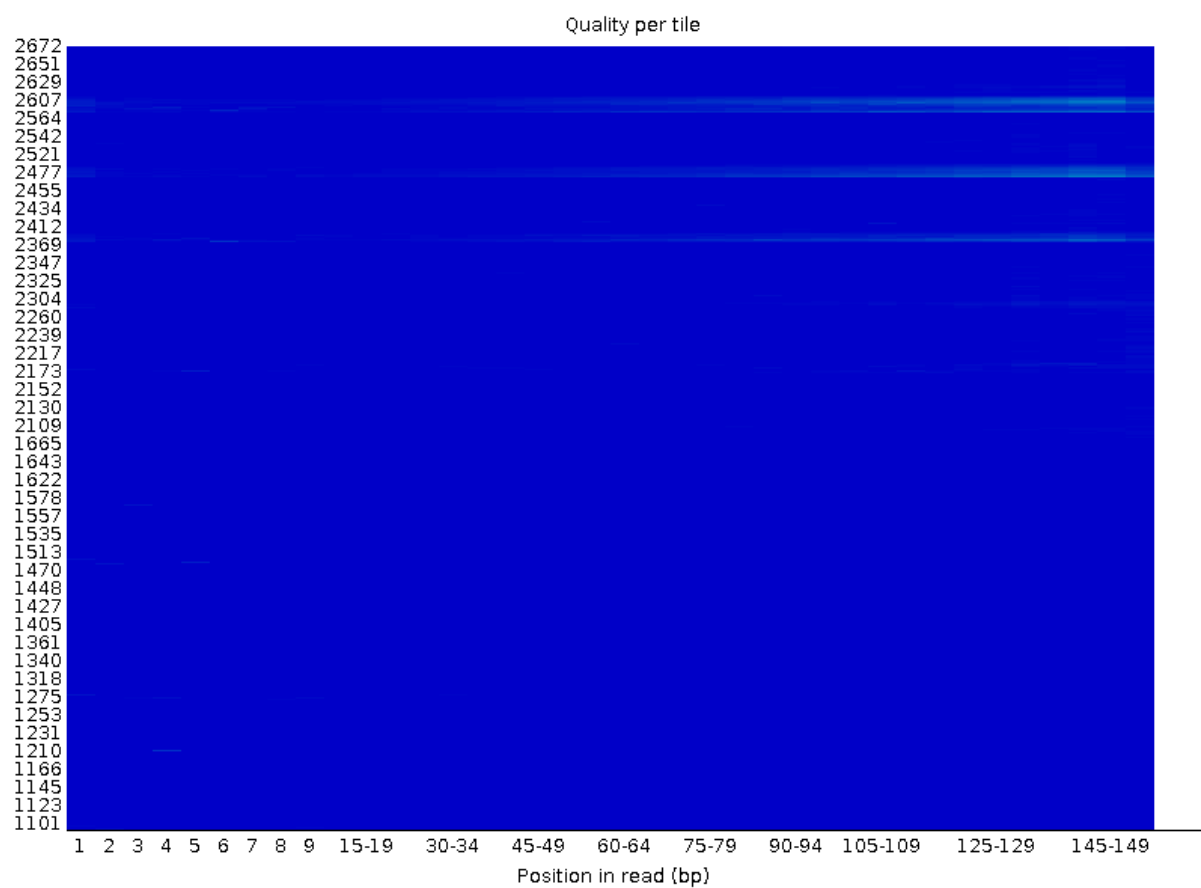
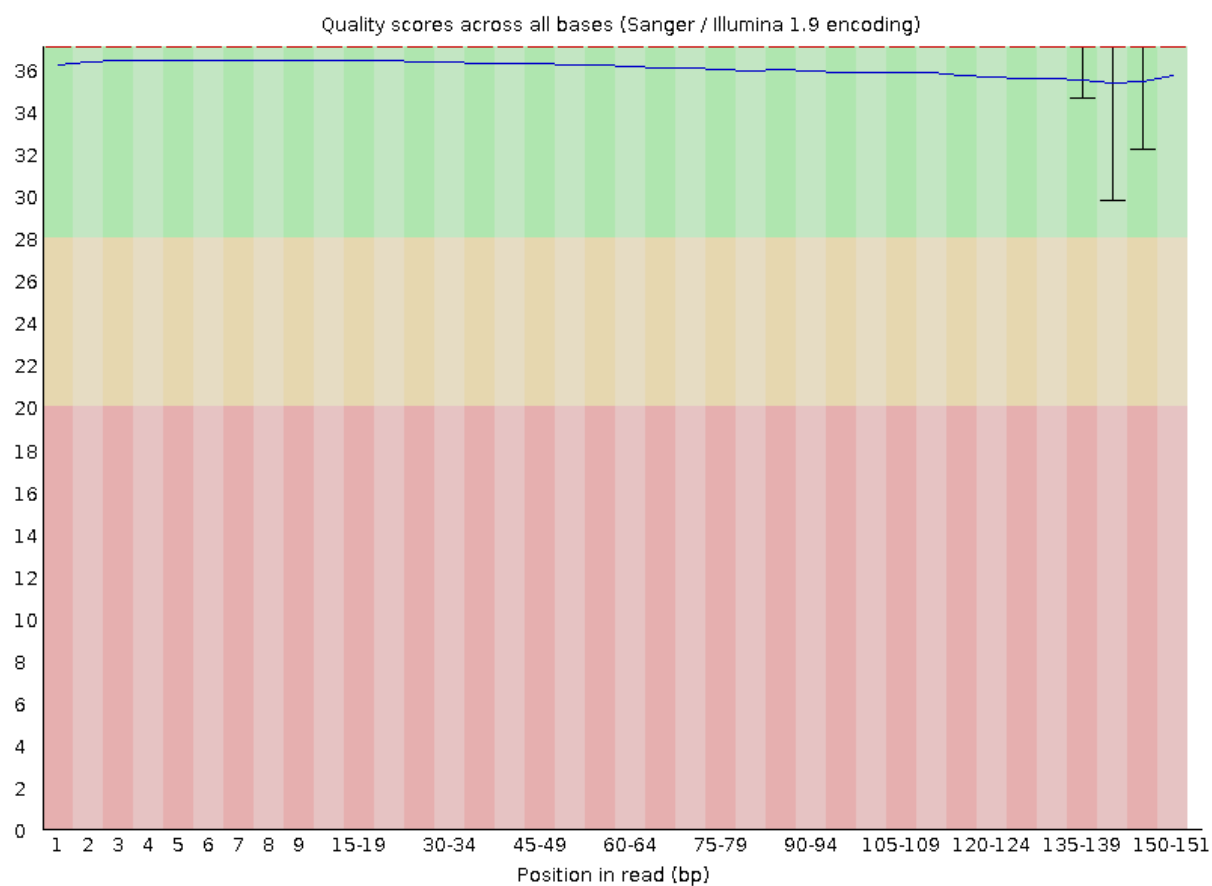
<Graphic distribution of read lengths>

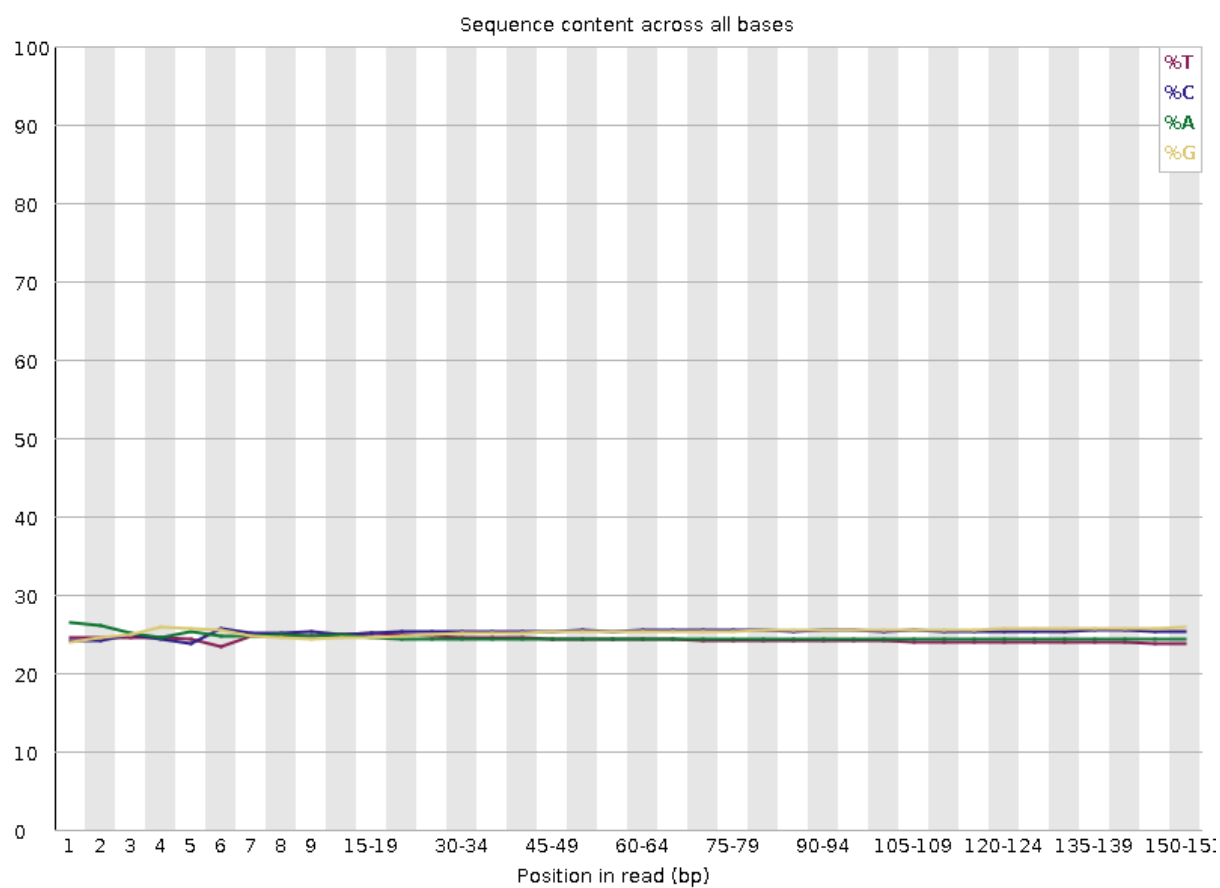
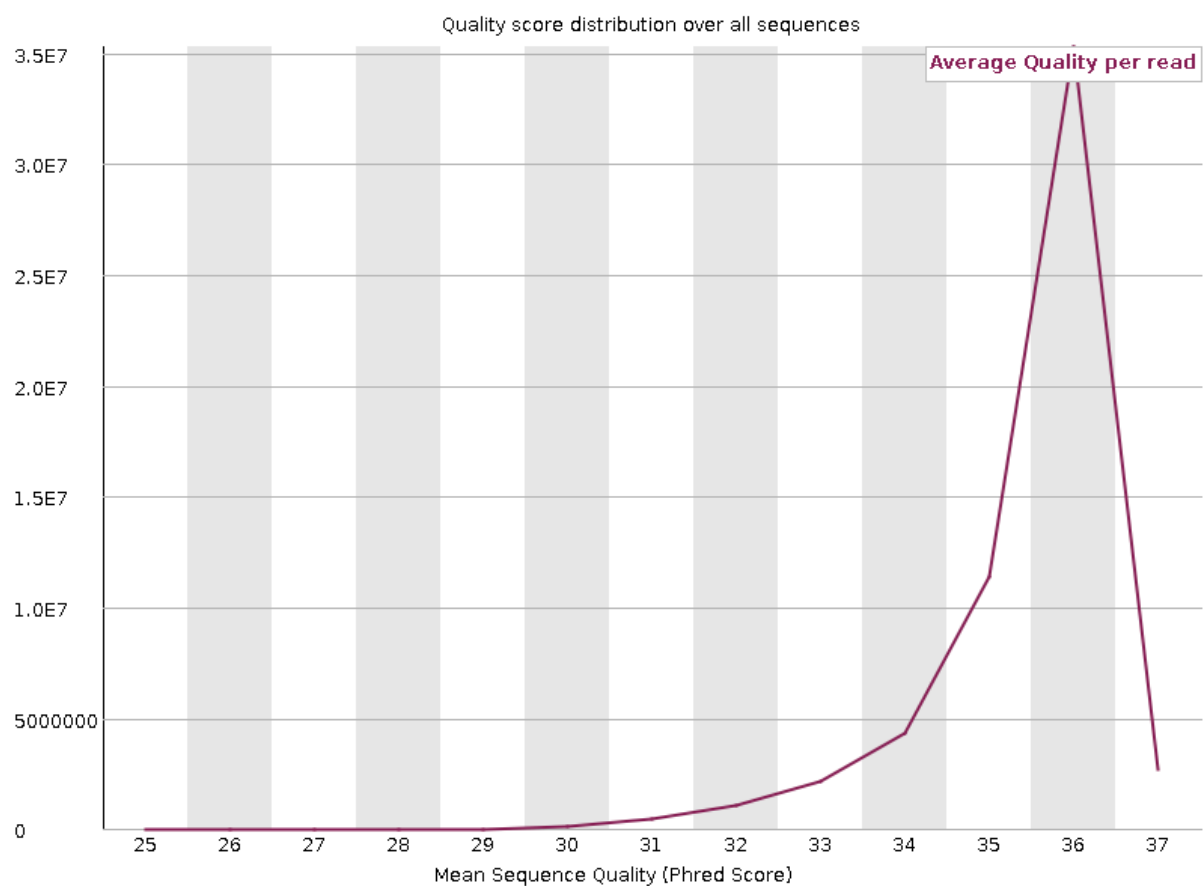
Length	Distribution	Abundance
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61	□□□□□□□□□□	0.05%
62	□□□□□□□□□□	0.05%
63	□□□□□□□□□□	0.05%
64	□□□□□□□□□□	0.05%
65	□□□□□□□□□□	0.05%
66	□□□□□□□□□□	0.05%
67	□□□□□□□□□□	0.06%
68	□□□□□□□□□□	0.05%
69	□□□□□□□□□□	0.05%
70	□□□□□□□□□□	0.06%
71	□□□□□□□□□□	0.05%
72	□□□□□□□□□□	0.05%
73	□□□□□□□□□□	0.05%
74	□□□□□□□□□□	0.06%
75	□□□□□□□□□□	0.05%
76	□□□□□□□□□□	0.06%
77	□□□□□□□□□□	0.06%
78	□□□□□□□□□□	0.06%
79	□□□□□□□□□□	0.06%
80	□□□□□□□□□□	0.06%
81	□□□□□□□□□□	0.06%
82	□□□□□□□□□□	0.06%
83	□□□□□□□□□□	0.06%
84	□□□□□□□□□□	0.06%
85	□□□□□□□□□□	0.06%
86	□□□□□□□□□□	0.06%
87	□□□□□□□□□□	0.06%
88	□□□□□□□□□□	0.06%
89	□□□□□□□□□□	0.06%
90	□□□□□□□□□□	0.06%
91	□□□□□□□□□□	0.06%
92	□□□□□□□□□□	0.07%
93	□□□□□□□□□□	0.06%
94	□□□□□□□□□□	0.07%
95	□□□□□□□□□□	0.07%
96	□□□□□□□□□□	0.07%
97	□□□□□□□□□□	0.08%
98	□□□□□□□□□□	0.08%
99	□□□□□□□□□□	0.08%
100	□□□□□□□□□□	0.09%
101	□□□□□□□□□□	0.09%

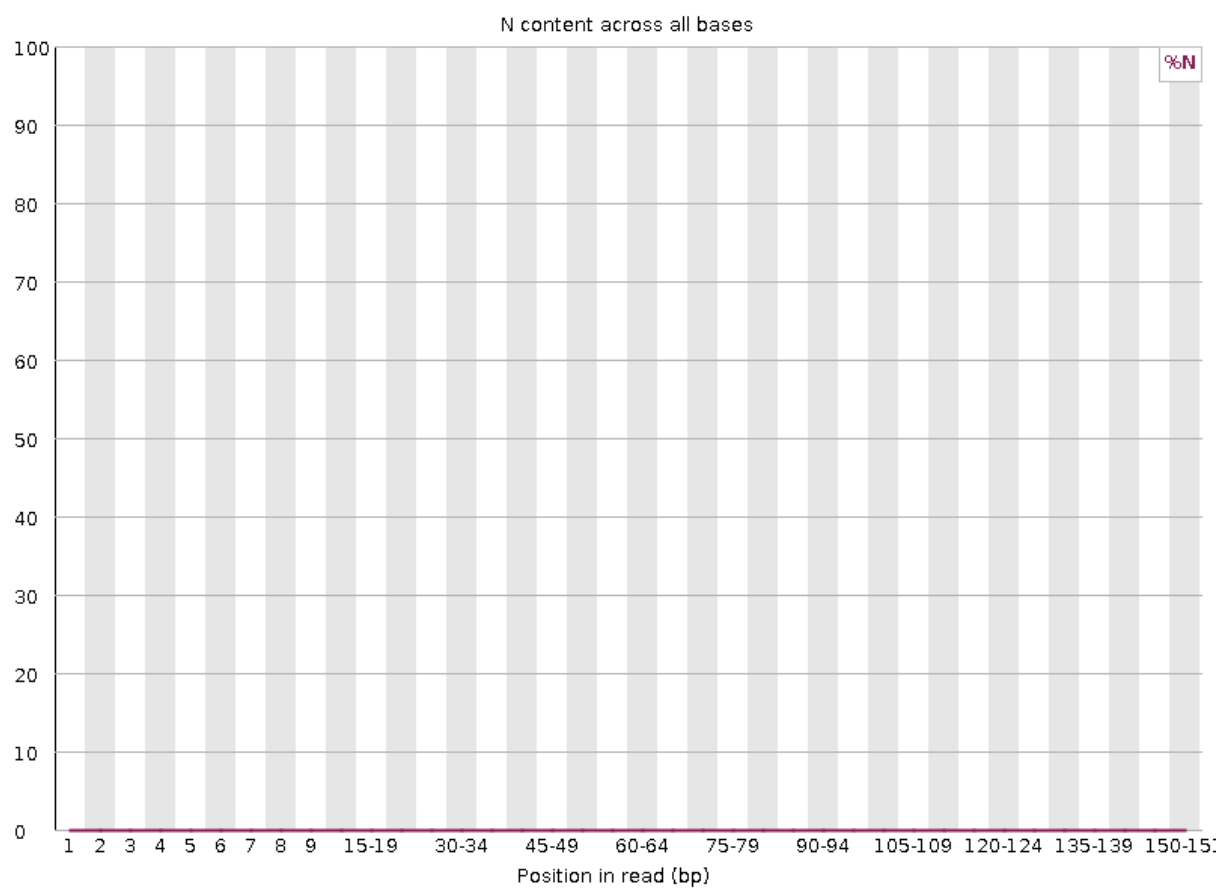
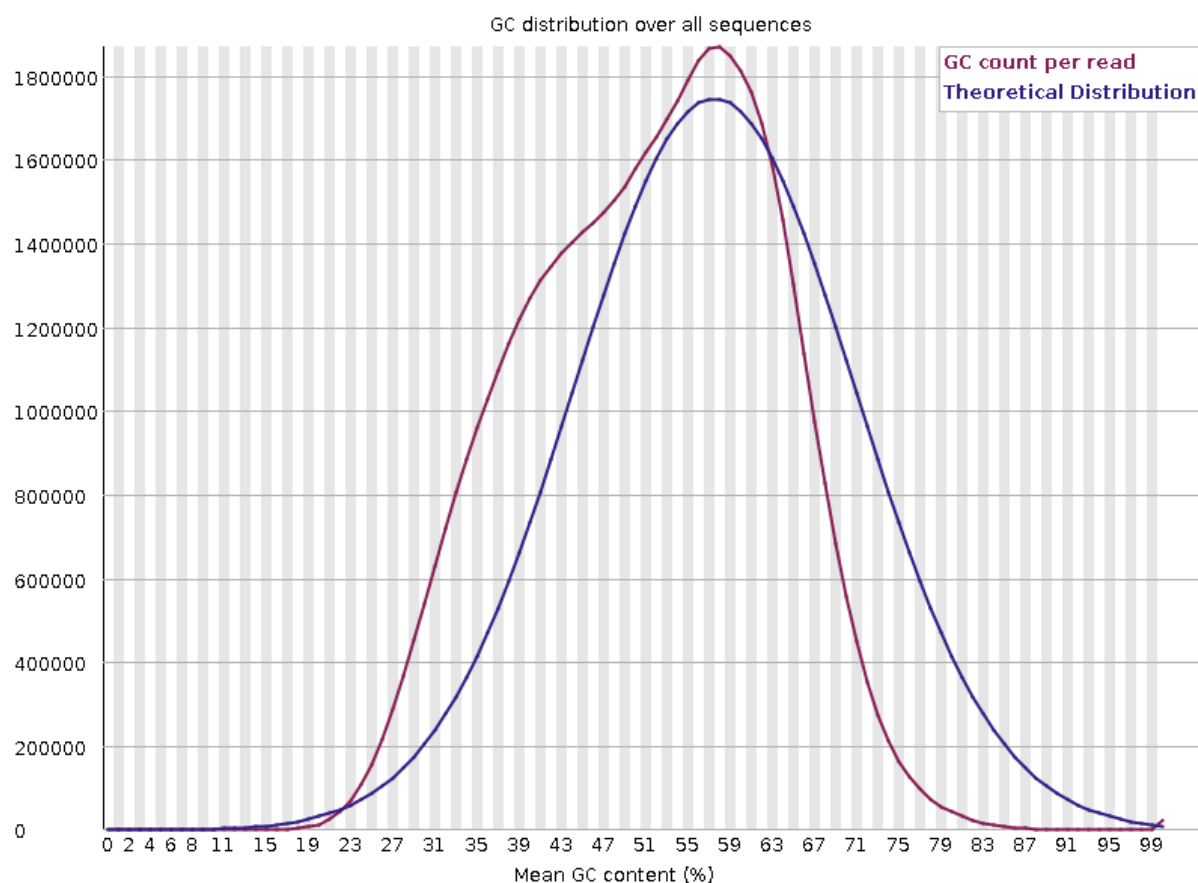
102	□□□□□□□□	0.09%
103	□□□□□□□□	0.10%
104	□□□□□□□□	0.11%
105	□□□□□□□□	0.11%
106	□□□□□□□□	0.12%
107	□□□□□□□□	0.13%
108	□□□□□□□□	0.14%
109	□□□□□□□□	0.15%
110	□□□□□□□□	0.16%
111	□□□□□□□□	0.17%
112	□□□□□□□□	0.19%
113	□□□□□□□□	0.20%
114	□□□□□□□□	0.21%
115	□□□□□□□□	0.23%
116	□□□□□□□□	0.25%
117	□□□□□□□□	0.26%
118	□□□□□□□□	0.28%
119	□□□□□□□□	0.30%
120	□□□□□□□□	0.32%
121	□□□□□□□□	0.34%
122	□□□□□□□□	0.36%
123	□□□□□□□□	0.38%
124	□□□□□□□□	0.40%
125	□□□□□□□□	0.42%
126	□□□□□□□□	0.43%
127	□□□□□□□□	0.45%
128	□□□□□□□□	0.47%
129	□□□□□□□□	0.49%
130	□□□□□□□□	0.50%
131	□□□□□□□□	0.52%
132	□□□□□□□□	0.54%
133	□□□□□□□□	0.56%
134	□□□□□□□□	0.57%
135	□□□□□□□□	0.58%
136	□□□□□□□□	0.60%
137	□□□□□□□□	0.62%
138	□□□□□□□□	0.63%
139	□□□□□□□□	0.64%
140	□□□□□□□□	0.66%
141	□□□□□□□□	0.67%
142	□□□□□□□□	0.68%
143	□□□□□□□□	0.68%
144	□□□□□□□□	0.70%
145	□□□□□□□□	0.69%
146	□□□□□□□□	0.71%
147	□□□□□□□□	0.72%
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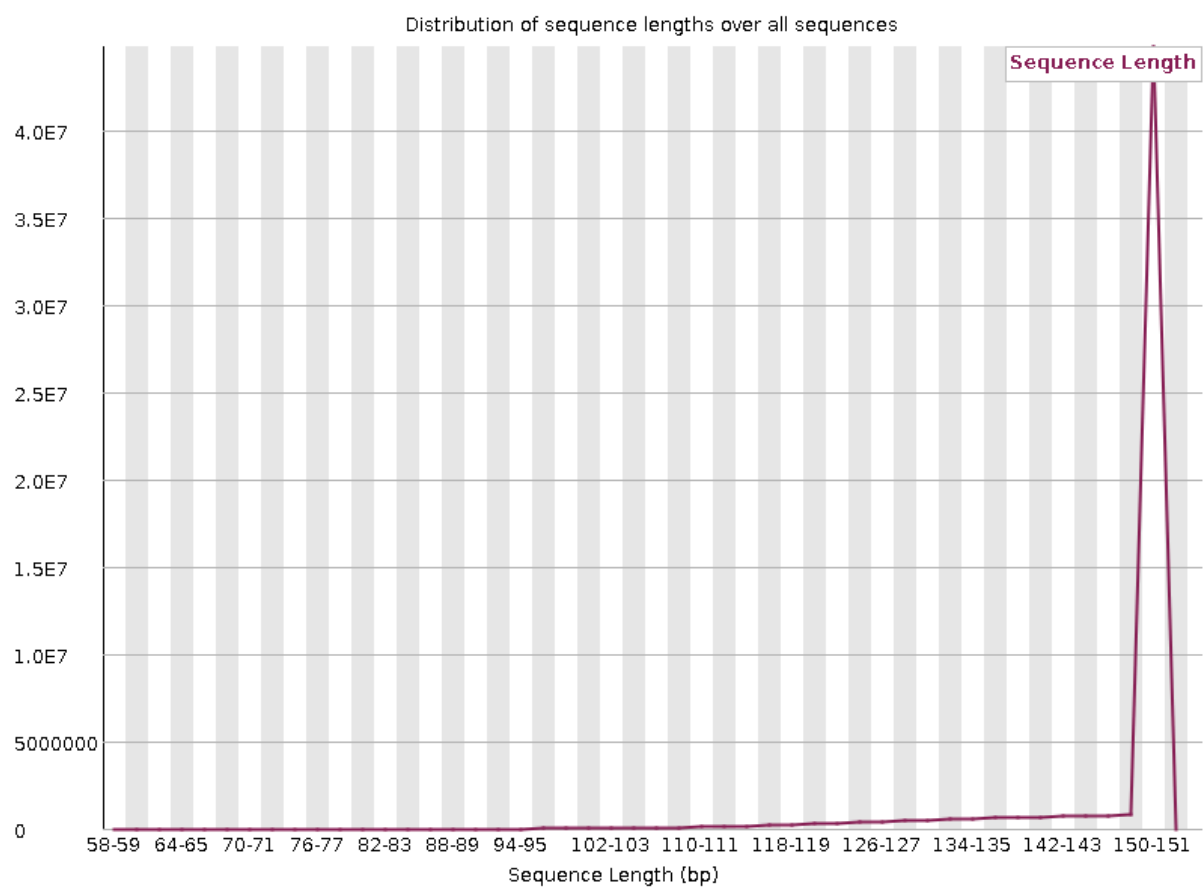
150 □□□□□□□□ 3.23%
 151 ■■■■■■■■□□ 74.07%

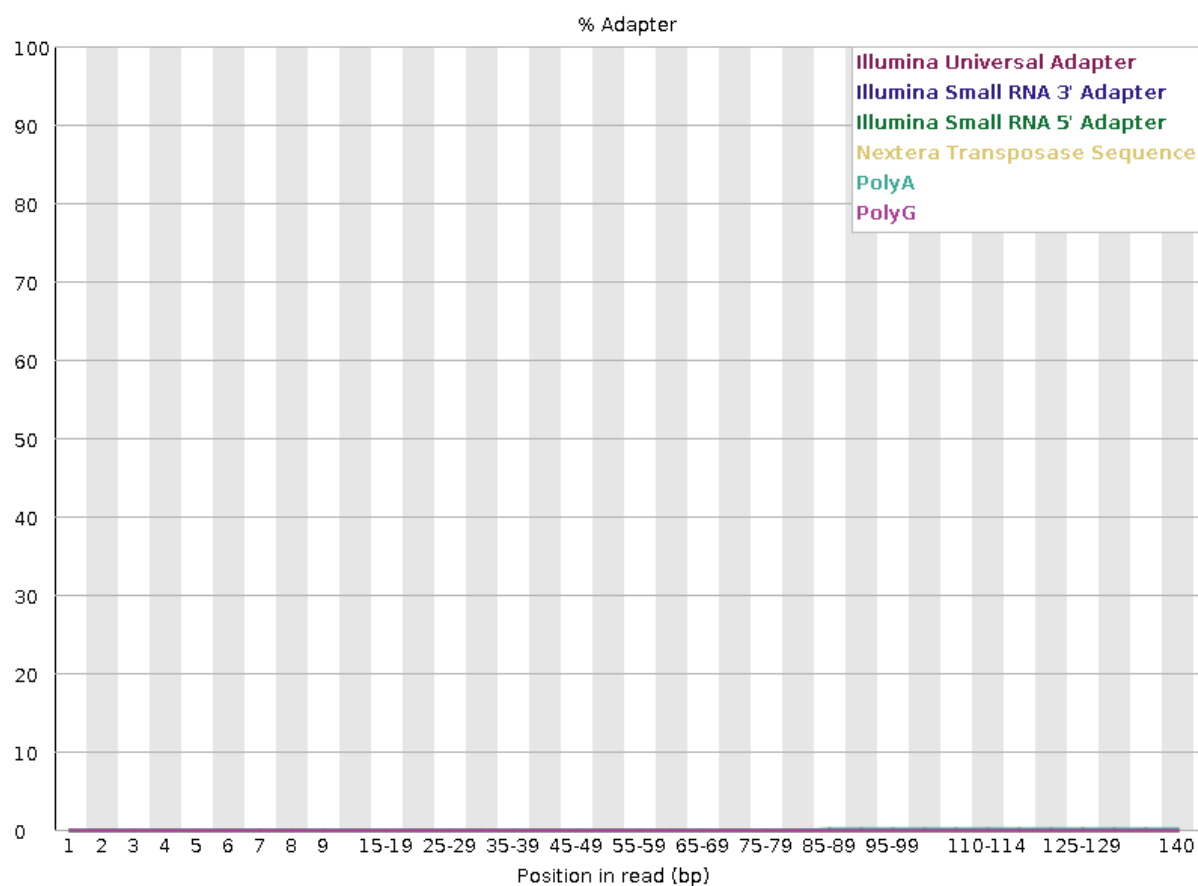
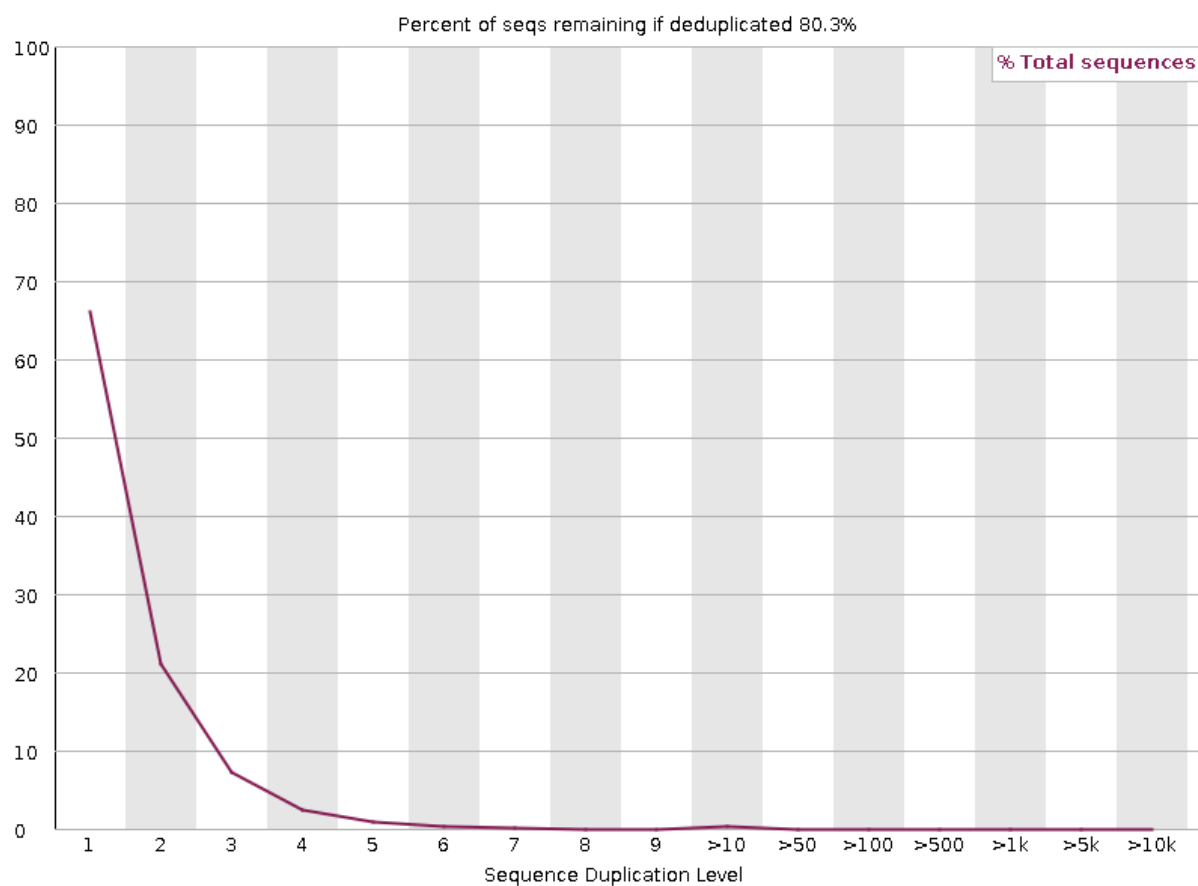
SRR37843243_2.p.fastq	
Measure	Value
Filename	SRR37843243_2.p.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	57,881,966
Total Bases	8.4 Gbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	50











SRR37843246_1.up.fastq

The number of read sequences is 1890021

Total length of read sequences is 248859231

<Text distribution of read lengths>

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63	10737	0.57%
64	10414	0.55%
65	10457	0.55%
66	9725	0.51%
67	10308	0.55%
68	9664	0.51%
69	9740	0.52%
70	9597	0.51%
71	9418	0.50%
72	9154	0.48%
73	9231	0.49%
74	9384	0.50%
75	8953	0.47%
76	9118	0.48%
77	8670	0.46%
78	8912	0.47%
79	8920	0.47%
80	8892	0.47%
81	8741	0.46%
82	9003	0.48%
83	8539	0.45%
84	8310	0.44%
85	8188	0.43%
86	8304	0.44%
87	8127	0.43%
88	7944	0.42%
89	8066	0.43%
90	7835	0.41%
91	7996	0.42%
92	7778	0.41%
93	7642	0.40%
94	7903	0.42%
95	7816	0.41%
96	7506	0.40%
97	7842	0.41%
98	7478	0.40%
99	7372	0.39%
100	7476	0.40%

101	7646	0.40%
102	7620	0.40%
103	7558	0.40%
104	7621	0.40%
105	7461	0.39%
106	7293	0.39%
107	7543	0.40%
108	7443	0.39%
109	7507	0.40%
110	7304	0.39%
111	7508	0.40%
112	7518	0.40%
113	7572	0.40%
114	7641	0.40%
115	7754	0.41%
116	7733	0.41%
117	7832	0.41%
118	7932	0.42%
119	7981	0.42%
120	8179	0.43%
121	7788	0.41%
122	8077	0.43%
123	8108	0.43%
124	8417	0.45%
125	8250	0.44%
126	8379	0.44%
127	7964	0.42%
128	8399	0.44%
129	8590	0.45%
130	8683	0.46%
131	8585	0.45%
132	8767	0.46%
133	8537	0.45%
134	8722	0.46%
135	8762	0.46%
136	9031	0.48%
137	8668	0.46%
138	9378	0.50%
139	8663	0.46%
140	8684	0.46%
141	8801	0.47%
142	8680	0.46%
143	9018	0.48%
144	9043	0.48%
145	8859	0.47%
146	8961	0.47%
147	9079	0.48%
148	8885	0.47%

149	15992	0.85%
150	87175	4.61%
151	1028611	54.42%

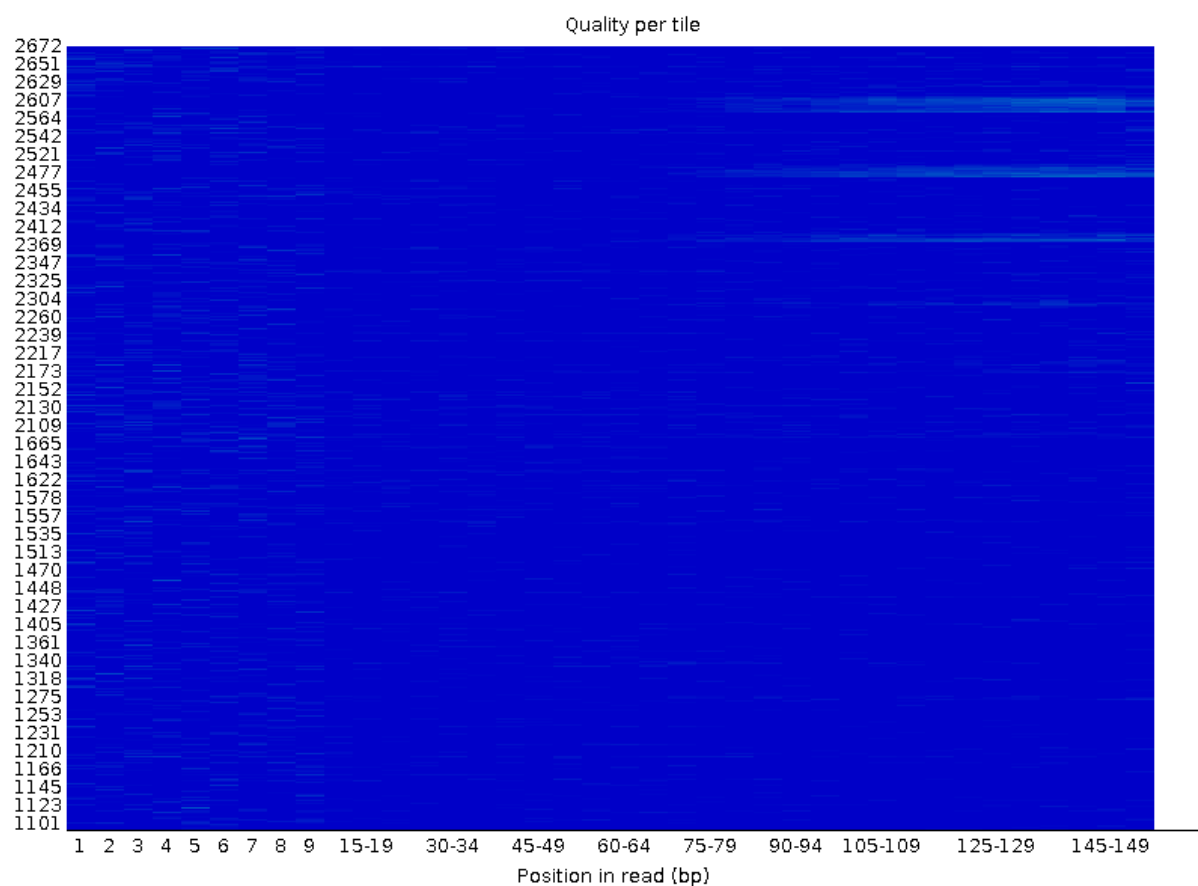
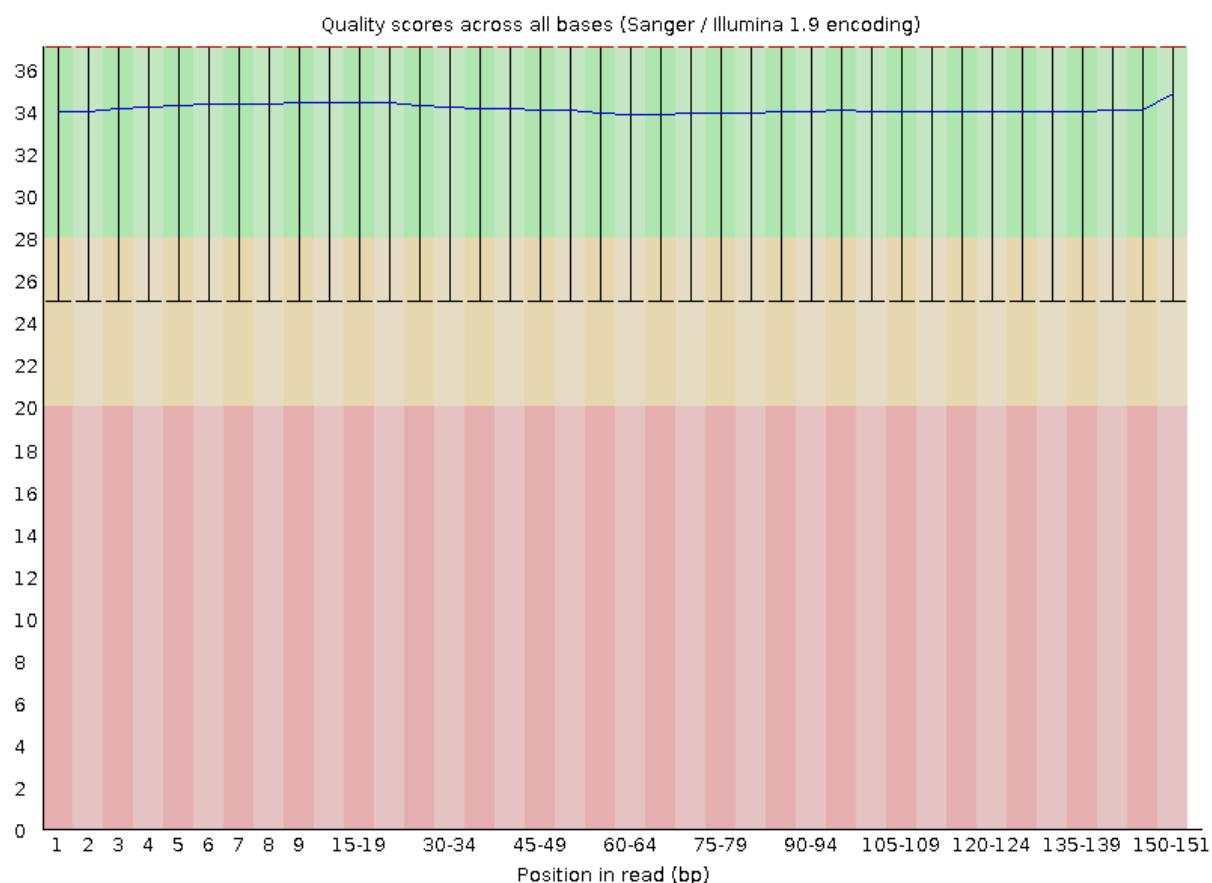
<Graphic distribution of read lengths>

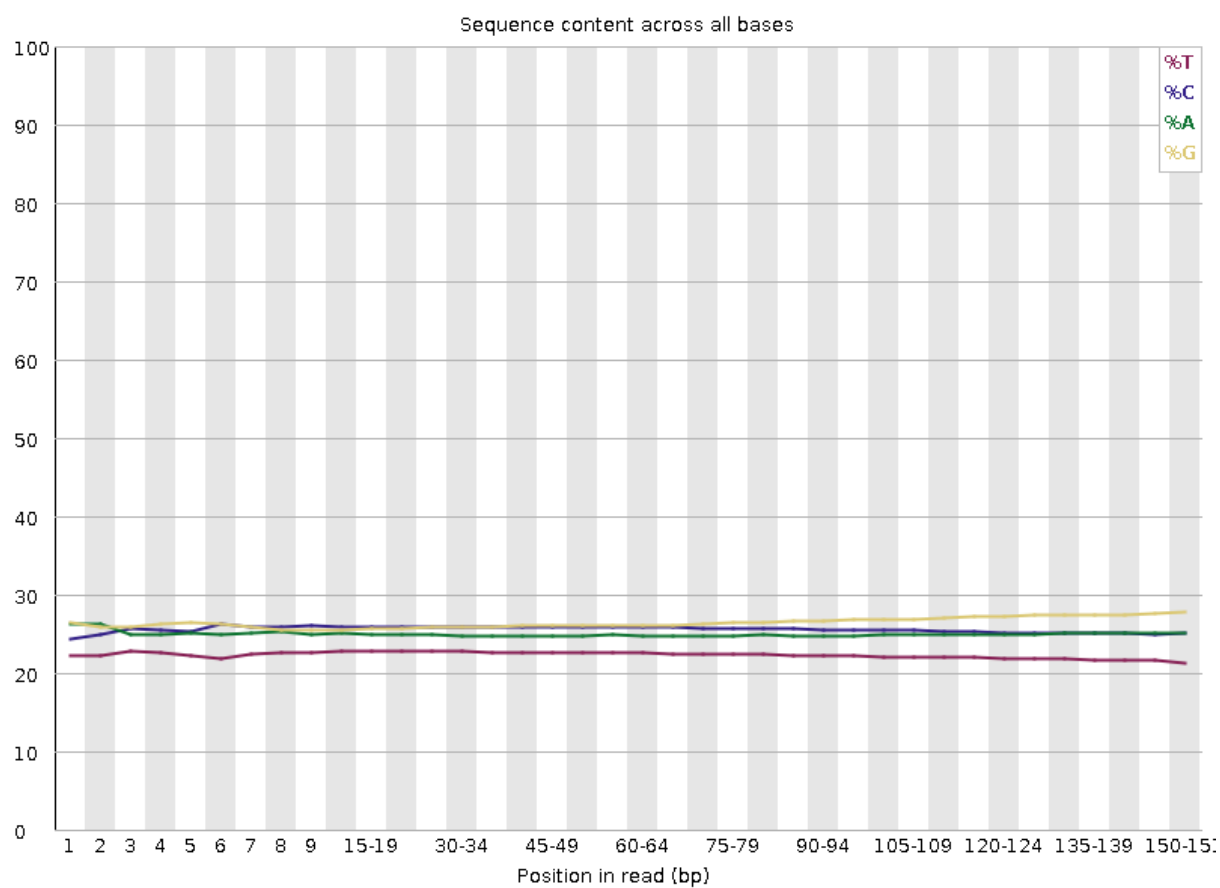
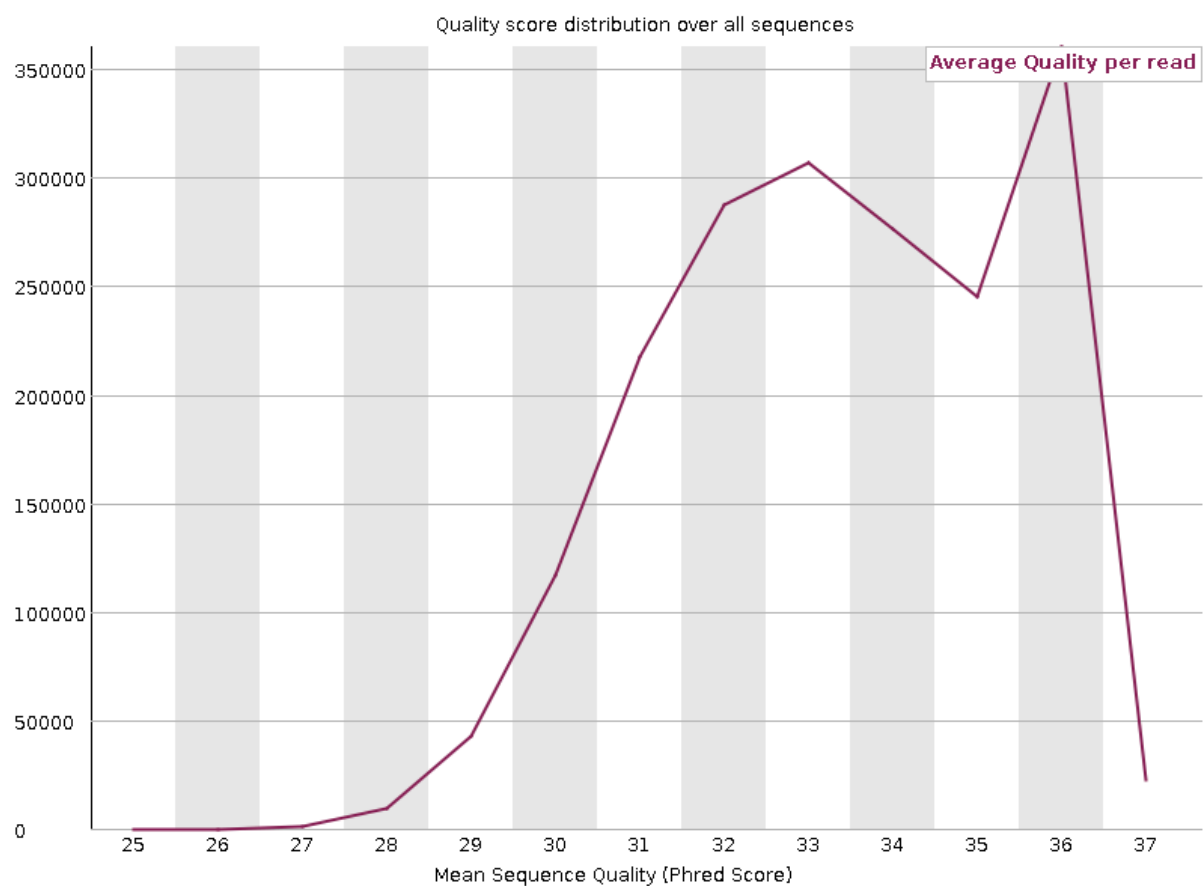
Length	Distribution	Abundance
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62	□□□□□□□□□□	0.55%
63	□□□□□□□□□□	0.57%
64	□□□□□□□□□□	0.55%
65	□□□□□□□□□□	0.55%
66	□□□□□□□□□□	0.51%
67	□□□□□□□□□□	0.55%
68	□□□□□□□□□□	0.51%
69	□□□□□□□□□□	0.52%
70	□□□□□□□□□□	0.51%
71	□□□□□□□□□□	0.50%
72	□□□□□□□□□□	0.48%
73	□□□□□□□□□□	0.49%
74	□□□□□□□□□□	0.50%
75	□□□□□□□□□□	0.47%
76	□□□□□□□□□□	0.48%
77	□□□□□□□□□□	0.46%
78	□□□□□□□□□□	0.47%
79	□□□□□□□□□□	0.47%
80	□□□□□□□□□□	0.47%
81	□□□□□□□□□□	0.46%
82	□□□□□□□□□□	0.48%
83	□□□□□□□□□□	0.45%
84	□□□□□□□□□□	0.44%
85	□□□□□□□□□□	0.43%
86	□□□□□□□□□□	0.44%
87	□□□□□□□□□□	0.43%
88	□□□□□□□□□□	0.42%
89	□□□□□□□□□□	0.43%
90	□□□□□□□□□□	0.41%
91	□□□□□□□□□□	0.42%
92	□□□□□□□□□□	0.41%
93	□□□□□□□□□□	0.40%
94	□□□□□□□□□□	0.42%
95	□□□□□□□□□□	0.41%
96	□□□□□□□□□□	0.40%
97	□□□□□□□□□□	0.41%
98	□□□□□□□□□□	0.40%
99	□□□□□□□□□□	0.39%
100	□□□□□□□□□□	0.40%
101	□□□□□□□□□□	0.40%

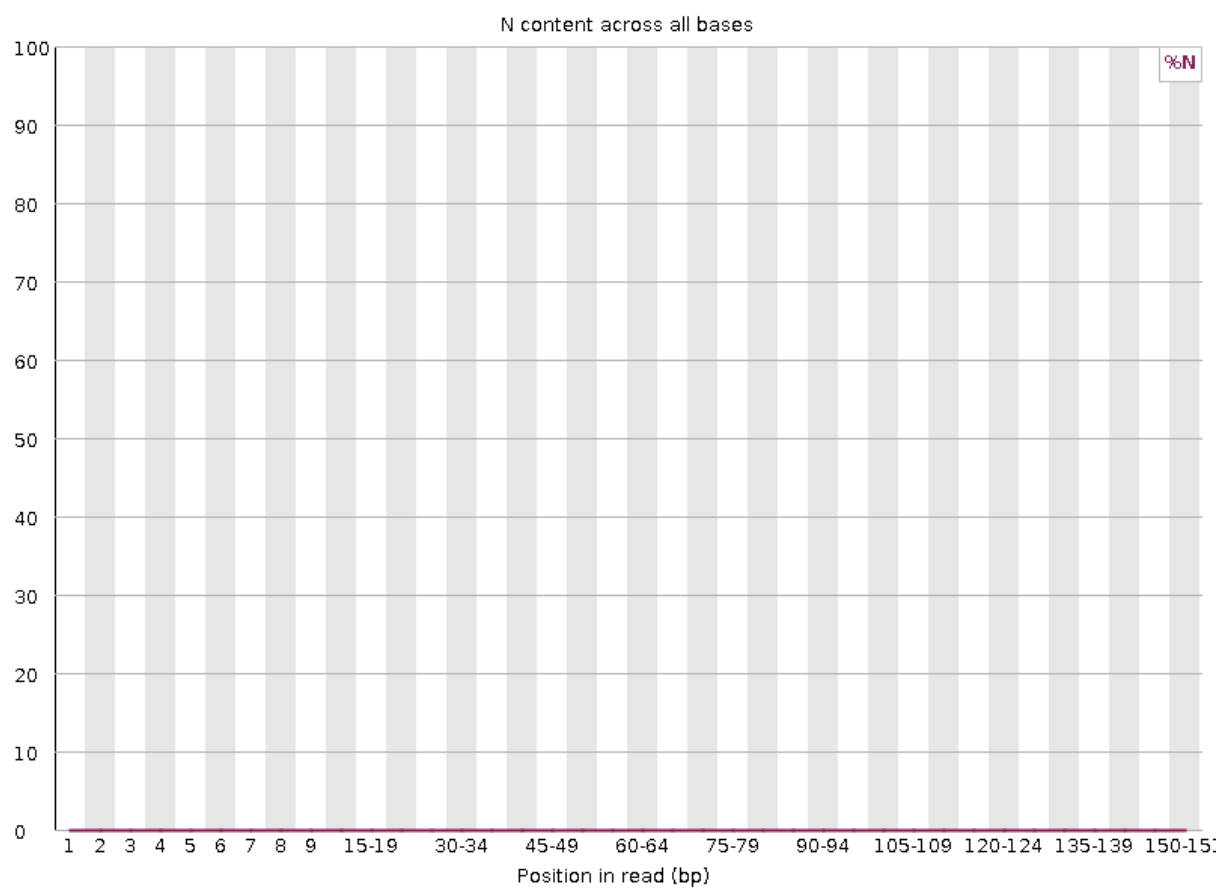
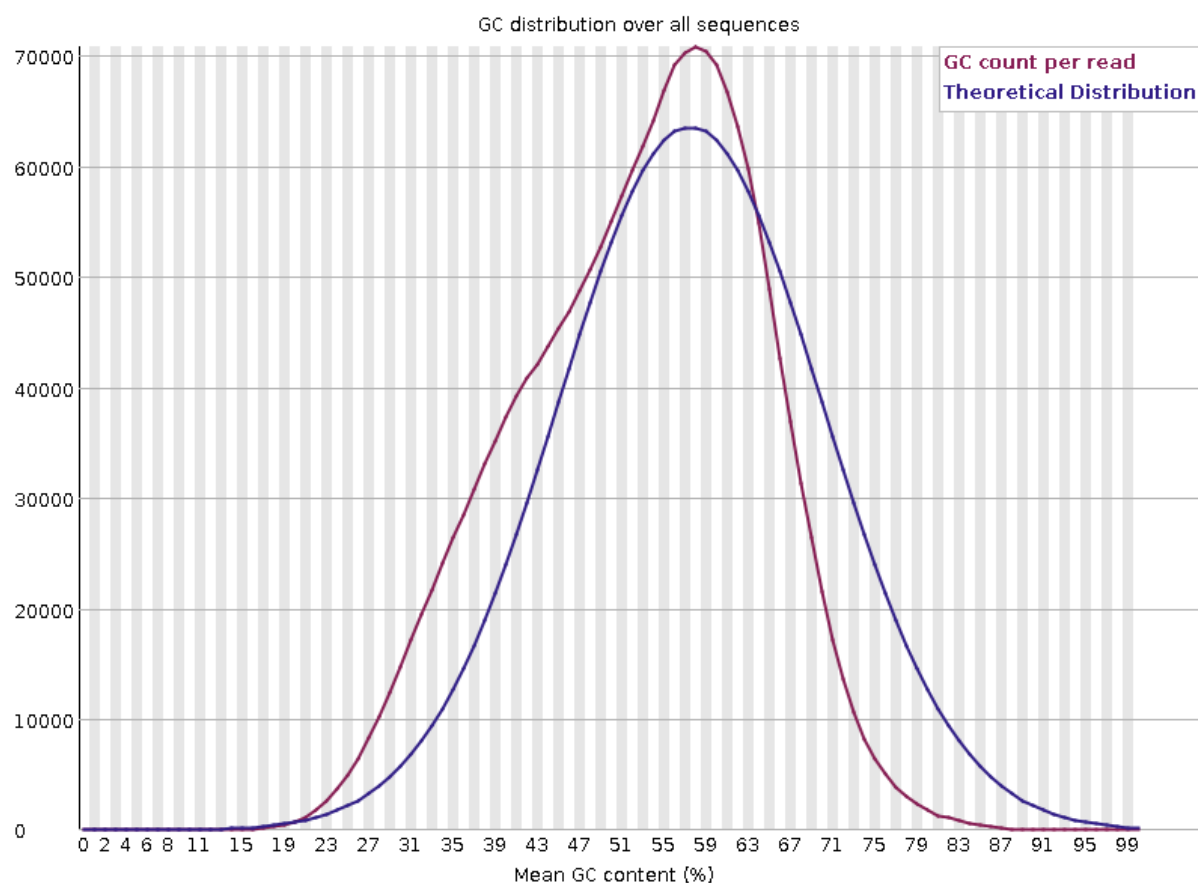
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103	□□□□□□□□	0.40%
104	□□□□□□□□	0.40%
105	□□□□□□□□	0.39%
106	□□□□□□□□	0.39%
107	□□□□□□□□	0.40%
108	□□□□□□□□	0.39%
109	□□□□□□□□	0.40%
110	□□□□□□□□	0.39%
111	□□□□□□□□	0.40%
112	□□□□□□□□	0.40%
113	□□□□□□□□	0.40%
114	□□□□□□□□	0.40%
115	□□□□□□□□	0.41%
116	□□□□□□□□	0.41%
117	□□□□□□□□	0.41%
118	□□□□□□□□	0.42%
119	□□□□□□□□	0.42%
120	□□□□□□□□	0.43%
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122	□□□□□□□□	0.43%
123	□□□□□□□□	0.43%
124	□□□□□□□□	0.45%
125	□□□□□□□□	0.44%
126	□□□□□□□□	0.44%
127	□□□□□□□□	0.42%
128	□□□□□□□□	0.44%
129	□□□□□□□□	0.45%
130	□□□□□□□□	0.46%
131	□□□□□□□□	0.45%
132	□□□□□□□□	0.46%
133	□□□□□□□□	0.45%
134	□□□□□□□□	0.46%
135	□□□□□□□□	0.46%
136	□□□□□□□□	0.48%
137	□□□□□□□□	0.46%
138	□□□□□□□□	0.50%
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140	□□□□□□□□	0.46%
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142	□□□□□□□□	0.46%
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144	□□□□□□□□	0.48%
145	□□□□□□□□	0.47%
146	□□□□□□□□	0.47%
147	□□□□□□□□	0.48%
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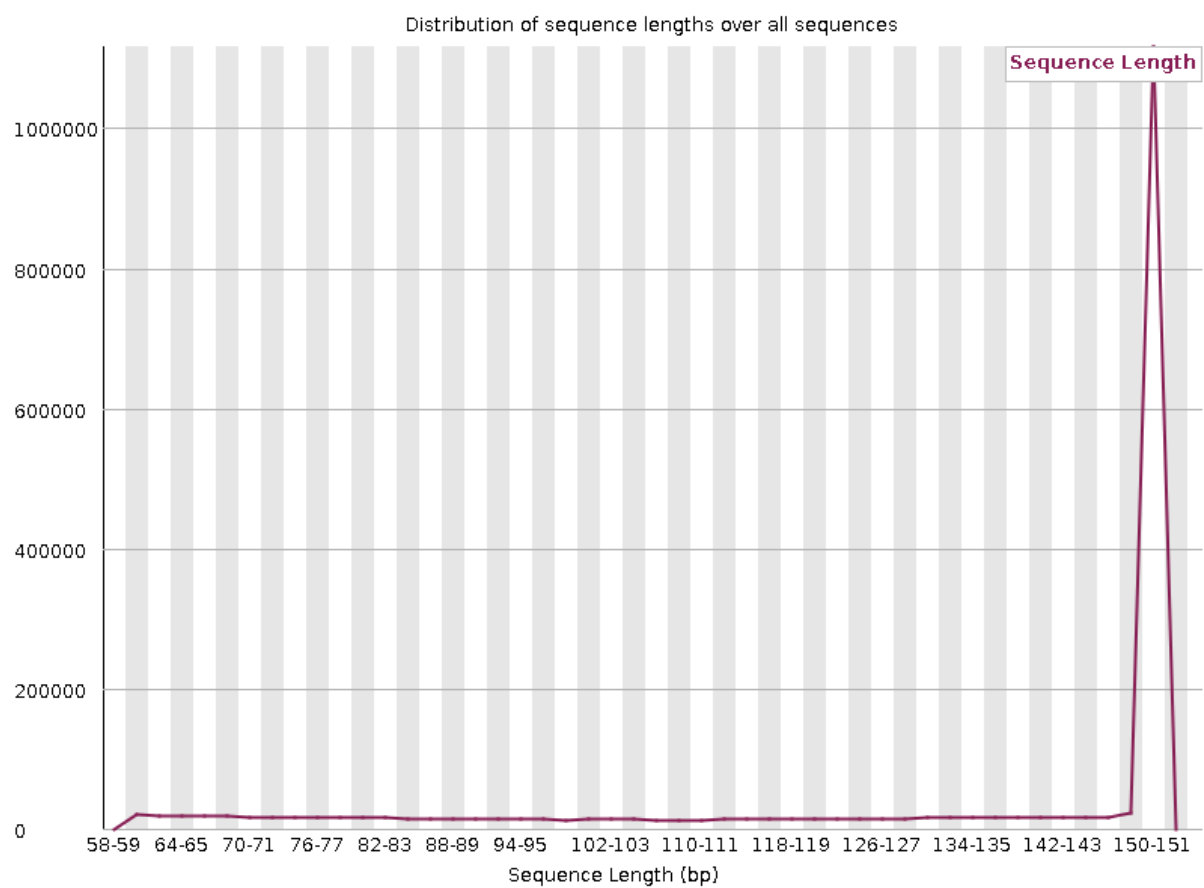
150 □□□□□□□□ 4.61%
 151 ■■■■■□□□□□ 54.42%

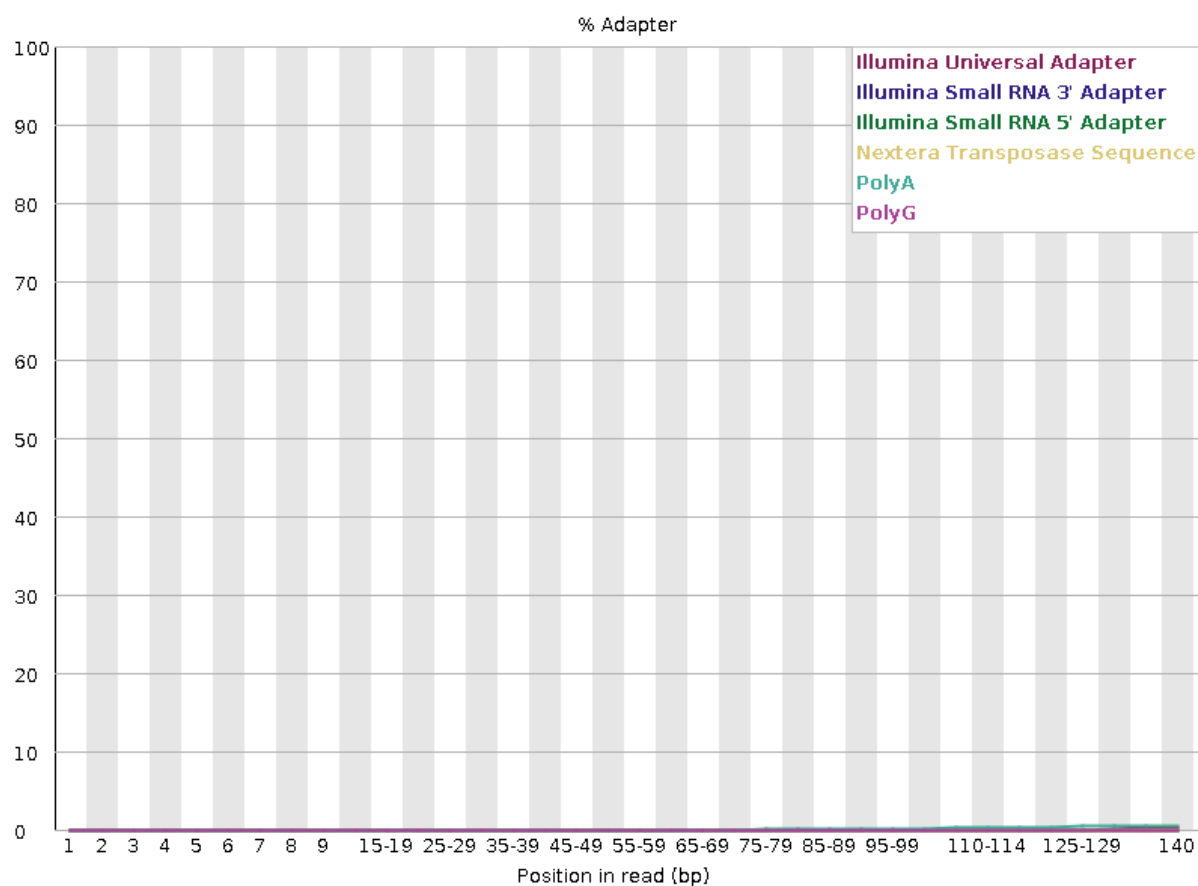
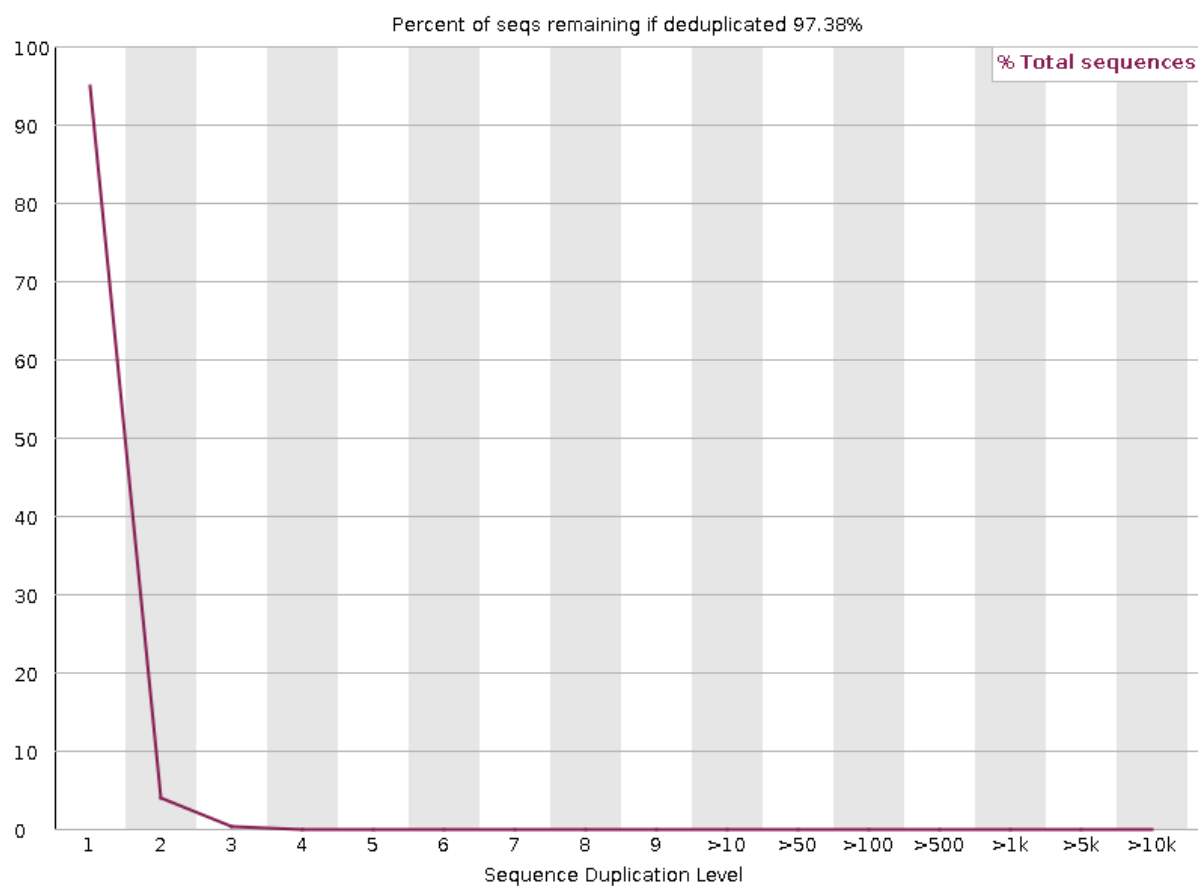
SRR37843243_1.up.fastq	
Measure	Value
Filename	SRR37843243_1.up.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1,890,021
Total Bases	248.8 Mbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	52











SRR37843246_2.up.fastq

The number of read sequences is 1596819

Total length of read sequences is 213573488

<Text distribution of read lengths>

Length	Count	Abundance
60	6964	0.44%
61	6819	0.43%
62	6916	0.43%
63	6648	0.42%
64	6878	0.43%
65	6576	0.41%
66	6601	0.41%
67	7180	0.45%
68	6591	0.41%
69	6460	0.40%
70	6676	0.42%
71	6116	0.38%
72	6091	0.38%
73	5918	0.37%
74	6030	0.38%
75	5943	0.37%
76	6162	0.39%
77	5868	0.37%
78	5906	0.37%
79	5661	0.35%
80	6274	0.39%
81	5885	0.37%
82	5703	0.36%
83	5476	0.34%
84	5467	0.34%
85	5440	0.34%
86	5279	0.33%
87	5011	0.31%
88	4959	0.31%
89	5191	0.33%
90	5194	0.33%
91	5082	0.32%
92	5261	0.33%
93	5095	0.32%
94	5235	0.33%
95	5389	0.34%
96	5221	0.33%
97	5613	0.35%
98	5327	0.33%
99	5224	0.33%
100	5665	0.35%

101	5431	0.34%
102	5295	0.33%
103	5573	0.35%
104	5602	0.35%
105	5568	0.35%
106	5977	0.37%
107	5795	0.36%
108	5935	0.37%
109	6044	0.38%
110	5944	0.37%
111	6178	0.39%
112	6627	0.42%
113	6737	0.42%
114	7127	0.45%
115	7023	0.44%
116	7511	0.47%
117	7705	0.48%
118	7905	0.50%
119	8115	0.51%
120	8429	0.53%
121	8971	0.56%
122	9088	0.57%
123	9490	0.59%
124	9578	0.60%
125	9724	0.61%
126	9914	0.62%
127	10103	0.63%
128	10338	0.65%
129	10383	0.65%
130	10498	0.66%
131	10896	0.68%
132	11119	0.70%
133	11083	0.69%
134	11124	0.70%
135	11236	0.70%
136	11679	0.73%
137	11624	0.73%
138	11618	0.73%
139	11703	0.73%
140	12153	0.76%
141	12221	0.77%
142	12269	0.77%
143	12037	0.75%
144	12175	0.76%
145	11784	0.74%
146	12103	0.76%
147	12223	0.77%
148	12167	0.76%

149	17142	1.07%
150	69750	4.37%
151	829110	51.92%

<Graphic distribution of read lengths>

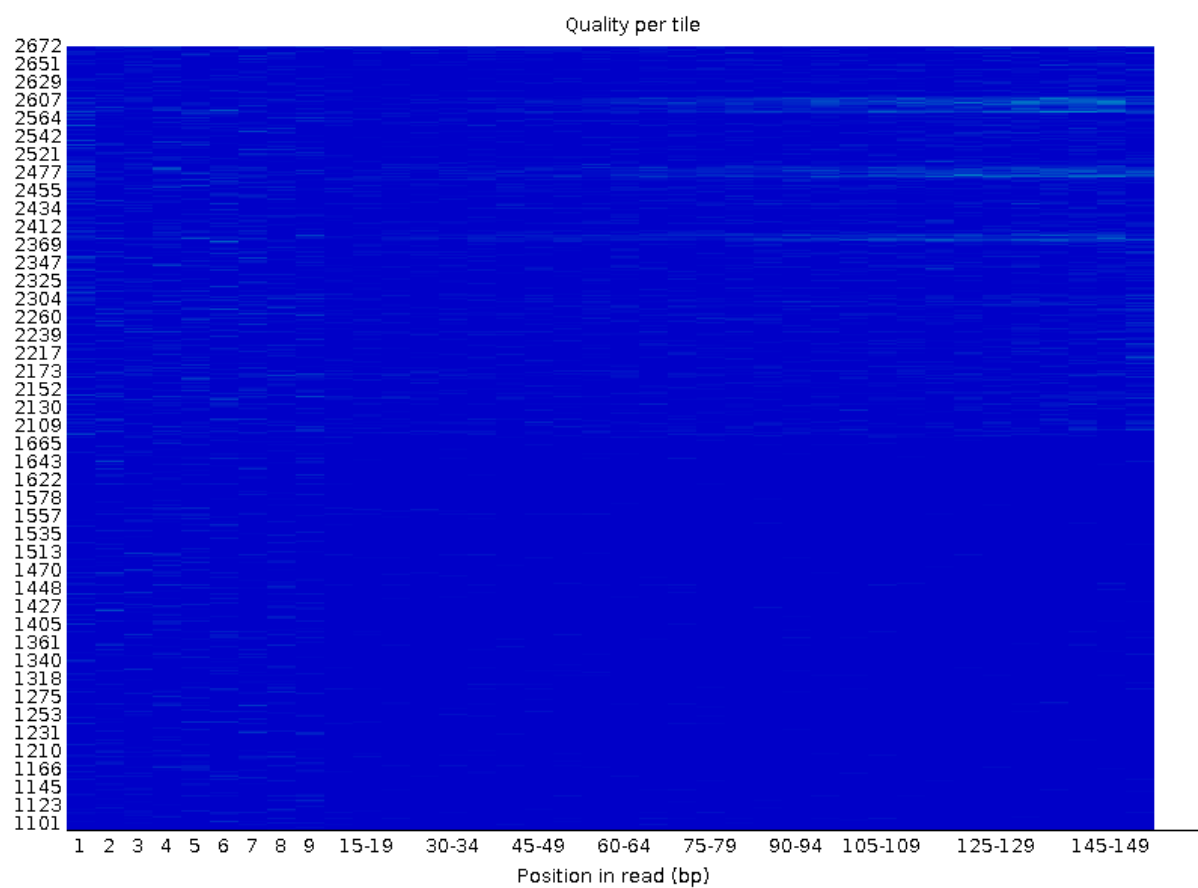
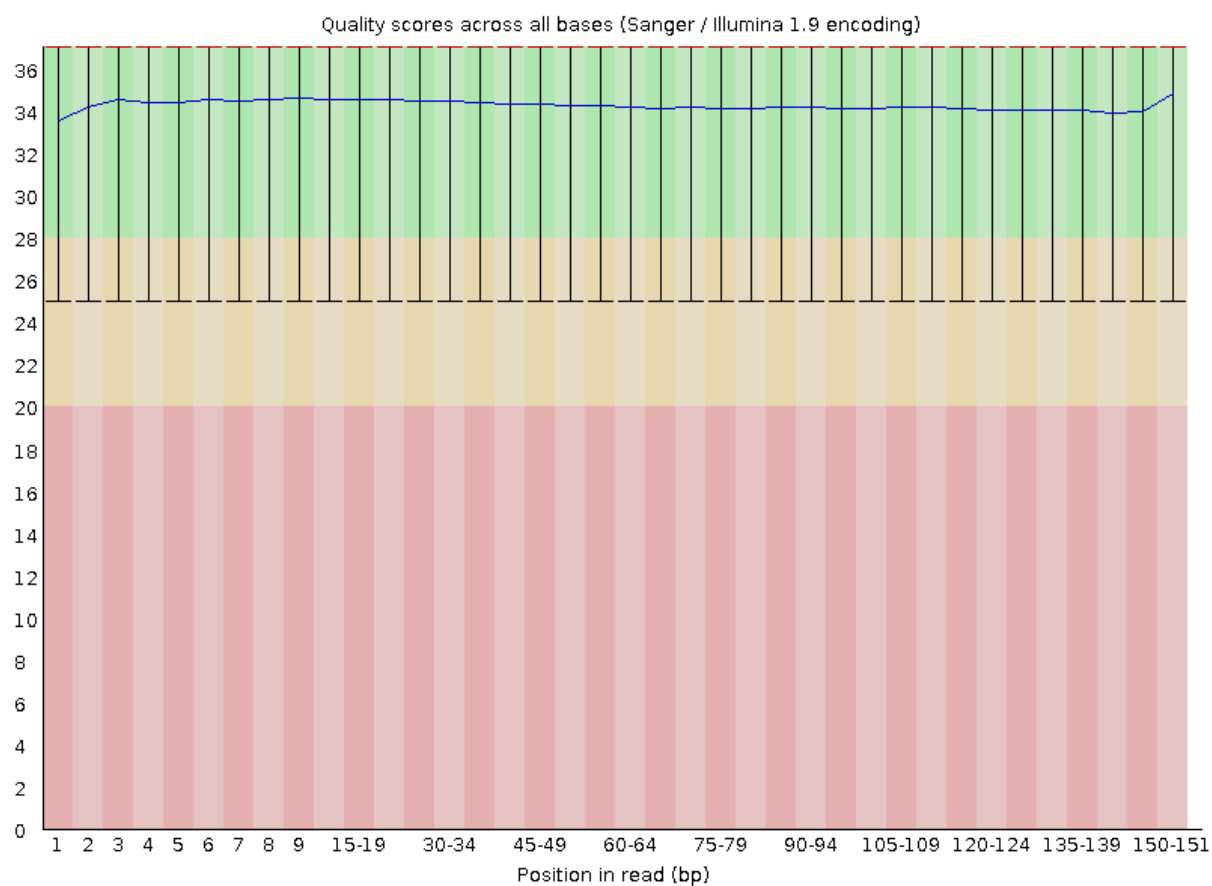
Length	Distribution	Abundance
60	□□□□□□□□□□	0.44%
61	□□□□□□□□□□	0.43%
62	□□□□□□□□□□	0.43%
63	□□□□□□□□□□	0.42%
64	□□□□□□□□□□	0.43%
65	□□□□□□□□□□	0.41%
66	□□□□□□□□□□	0.41%
67	□□□□□□□□□□	0.45%
68	□□□□□□□□□□	0.41%
69	□□□□□□□□□□	0.40%
70	□□□□□□□□□□	0.42%
71	□□□□□□□□□□	0.38%
72	□□□□□□□□□□	0.38%
73	□□□□□□□□□□	0.37%
74	□□□□□□□□□□	0.38%
75	□□□□□□□□□□	0.37%
76	□□□□□□□□□□	0.39%
77	□□□□□□□□□□	0.37%
78	□□□□□□□□□□	0.37%
79	□□□□□□□□□□	0.35%
80	□□□□□□□□□□	0.39%
81	□□□□□□□□□□	0.37%
82	□□□□□□□□□□	0.36%
83	□□□□□□□□□□	0.34%
84	□□□□□□□□□□	0.34%
85	□□□□□□□□□□	0.34%
86	□□□□□□□□□□	0.33%
87	□□□□□□□□□□	0.31%
88	□□□□□□□□□□	0.31%
89	□□□□□□□□□□	0.33%
90	□□□□□□□□□□	0.33%
91	□□□□□□□□□□	0.32%
92	□□□□□□□□□□	0.33%
93	□□□□□□□□□□	0.32%
94	□□□□□□□□□□	0.33%
95	□□□□□□□□□□	0.34%
96	□□□□□□□□□□	0.33%
97	□□□□□□□□□□	0.35%
98	□□□□□□□□□□	0.33%
99	□□□□□□□□□□	0.33%
100	□□□□□□□□□□	0.35%
101	□□□□□□□□□□	0.34%

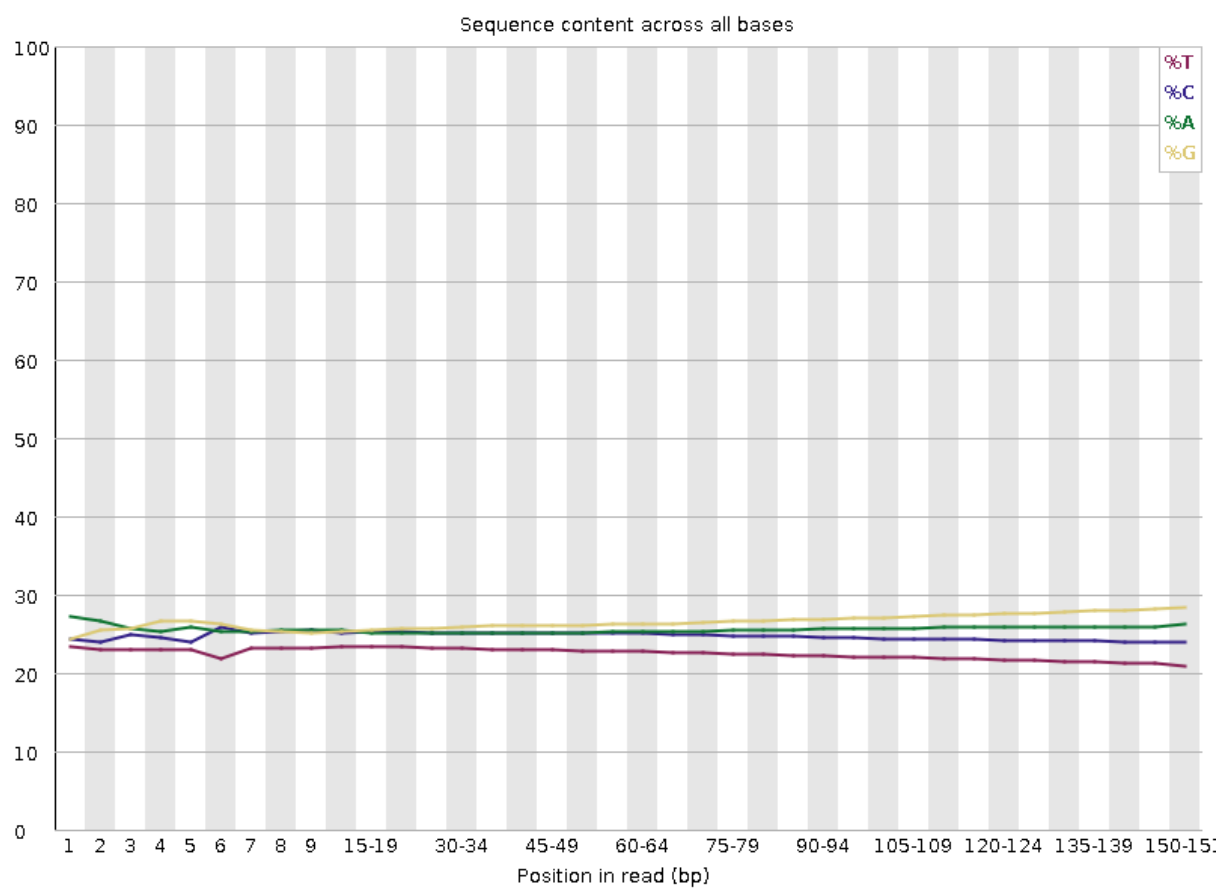
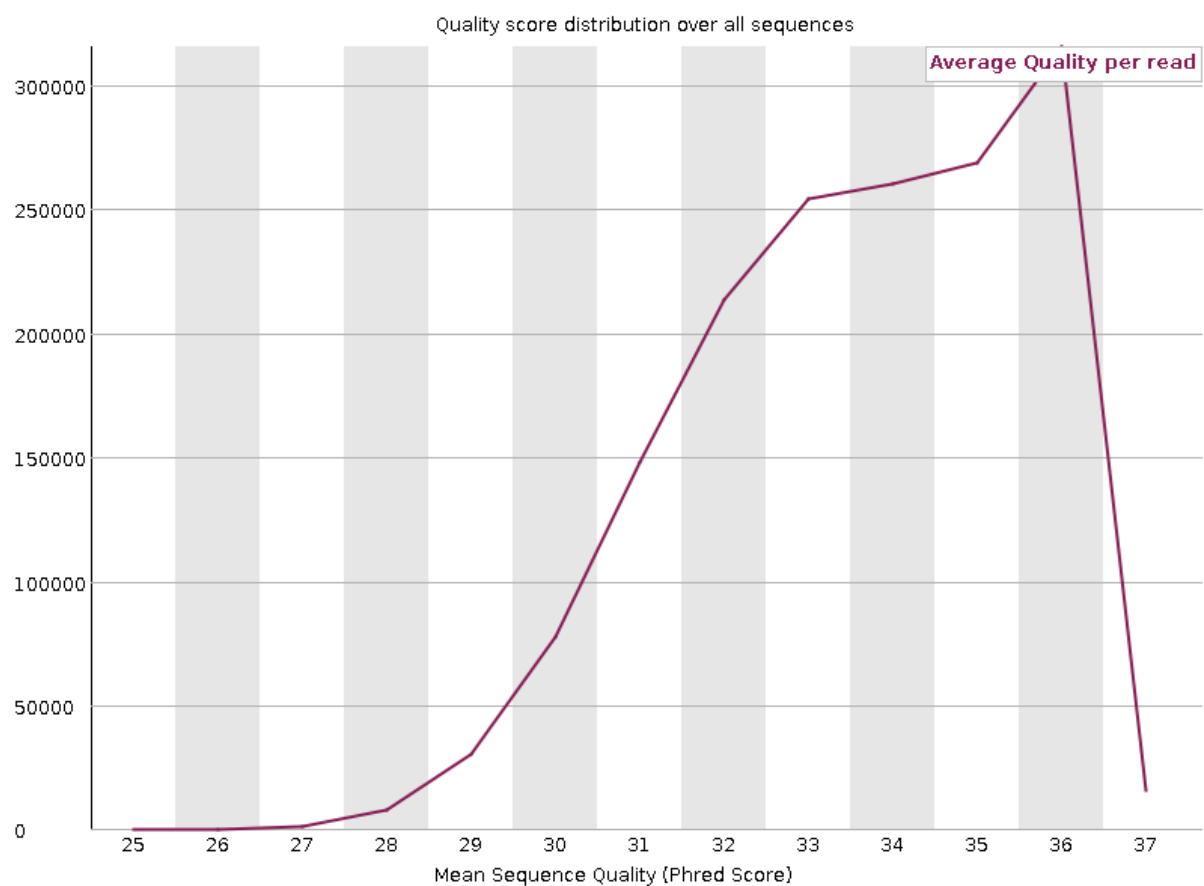
102	□□□□□□□□	0.33%
103	□□□□□□□□	0.35%
104	□□□□□□□□	0.35%
105	□□□□□□□□	0.35%
106	□□□□□□□□	0.37%
107	□□□□□□□□	0.36%
108	□□□□□□□□	0.37%
109	□□□□□□□□	0.38%
110	□□□□□□□□	0.37%
111	□□□□□□□□	0.39%
112	□□□□□□□□	0.42%
113	□□□□□□□□	0.42%
114	□□□□□□□□	0.45%
115	□□□□□□□□	0.44%
116	□□□□□□□□	0.47%
117	□□□□□□□□	0.48%
118	□□□□□□□□	0.50%
119	□□□□□□□□	0.51%
120	□□□□□□□□	0.53%
121	□□□□□□□□	0.56%
122	□□□□□□□□	0.57%
123	□□□□□□□□	0.59%
124	□□□□□□□□	0.60%
125	□□□□□□□□	0.61%
126	□□□□□□□□	0.62%
127	□□□□□□□□	0.63%
128	□□□□□□□□	0.65%
129	□□□□□□□□	0.65%
130	□□□□□□□□	0.66%
131	□□□□□□□□	0.68%
132	□□□□□□□□	0.70%
133	□□□□□□□□	0.69%
134	□□□□□□□□	0.70%
135	□□□□□□□□	0.70%
136	□□□□□□□□	0.73%
137	□□□□□□□□	0.73%
138	□□□□□□□□	0.73%
139	□□□□□□□□	0.73%
140	□□□□□□□□	0.76%
141	□□□□□□□□	0.77%
142	□□□□□□□□	0.77%
143	□□□□□□□□	0.75%
144	□□□□□□□□	0.76%
145	□□□□□□□□	0.74%
146	□□□□□□□□	0.76%
147	□□□□□□□□	0.77%
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149	□□□□□□□□	1.07%

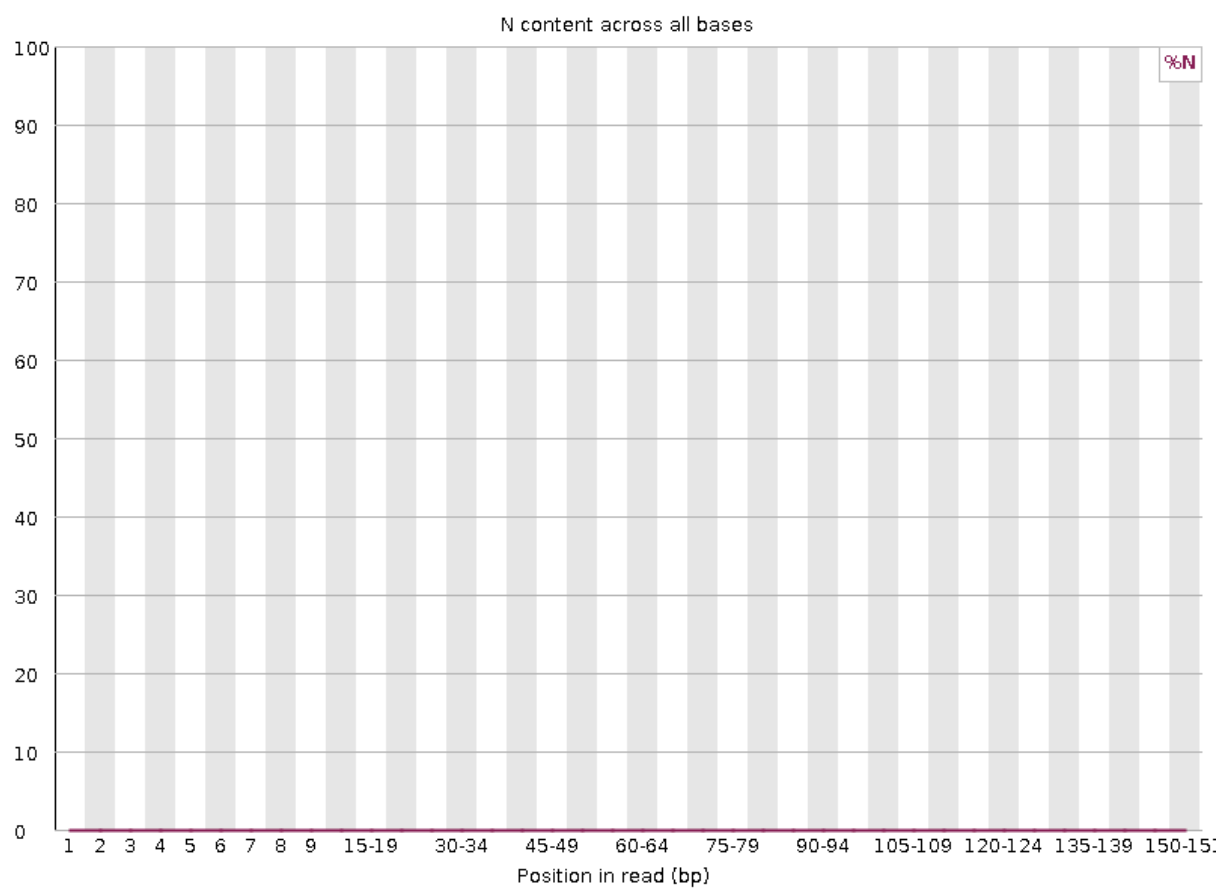
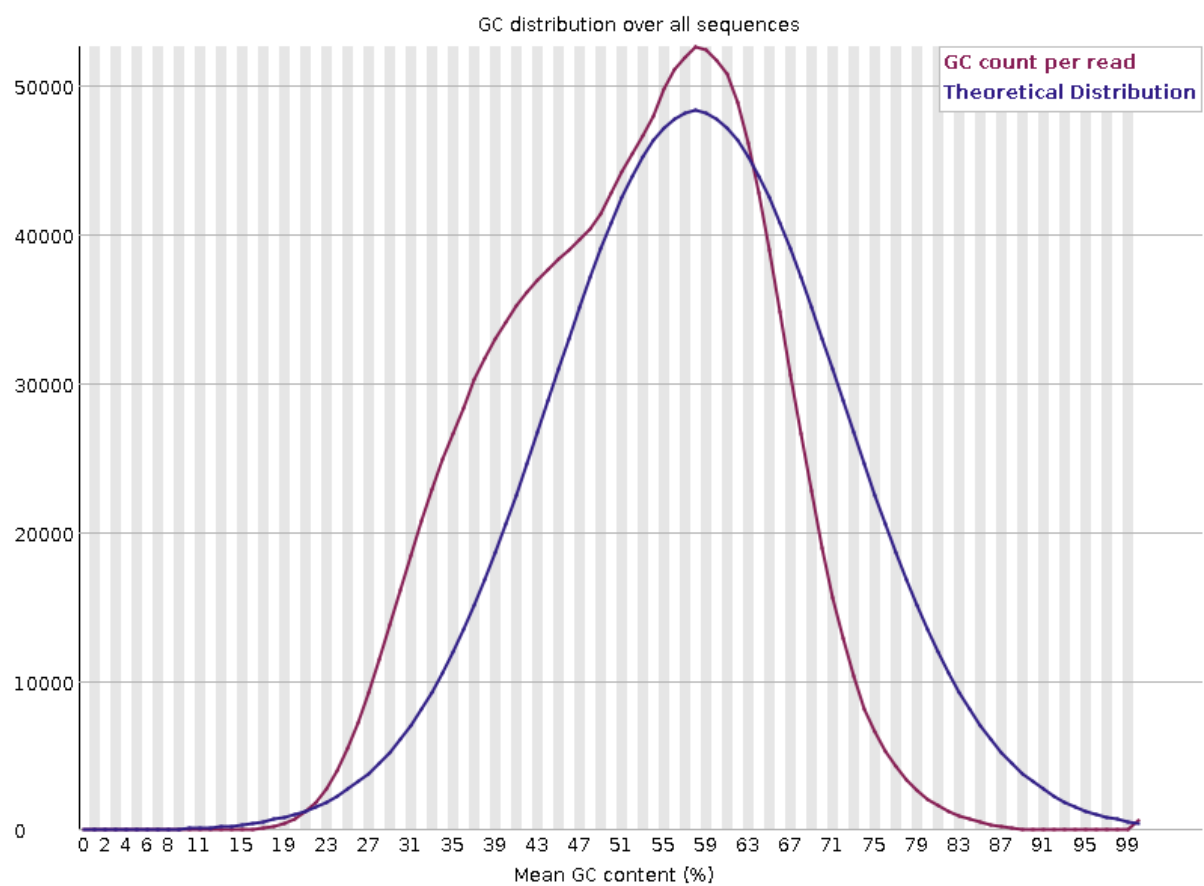
150 □□□□□□□□ 4.37%

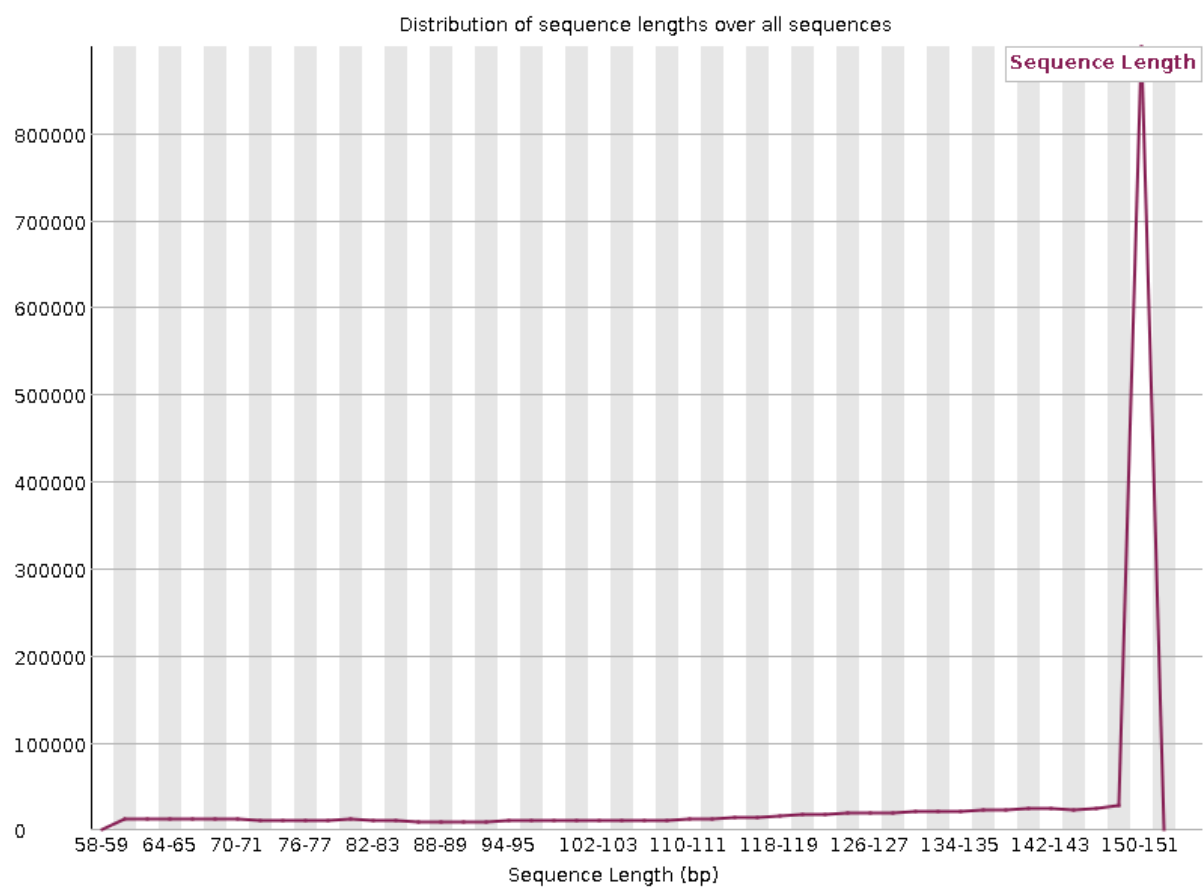
151 ■■■■■□□□□□ 51.92%

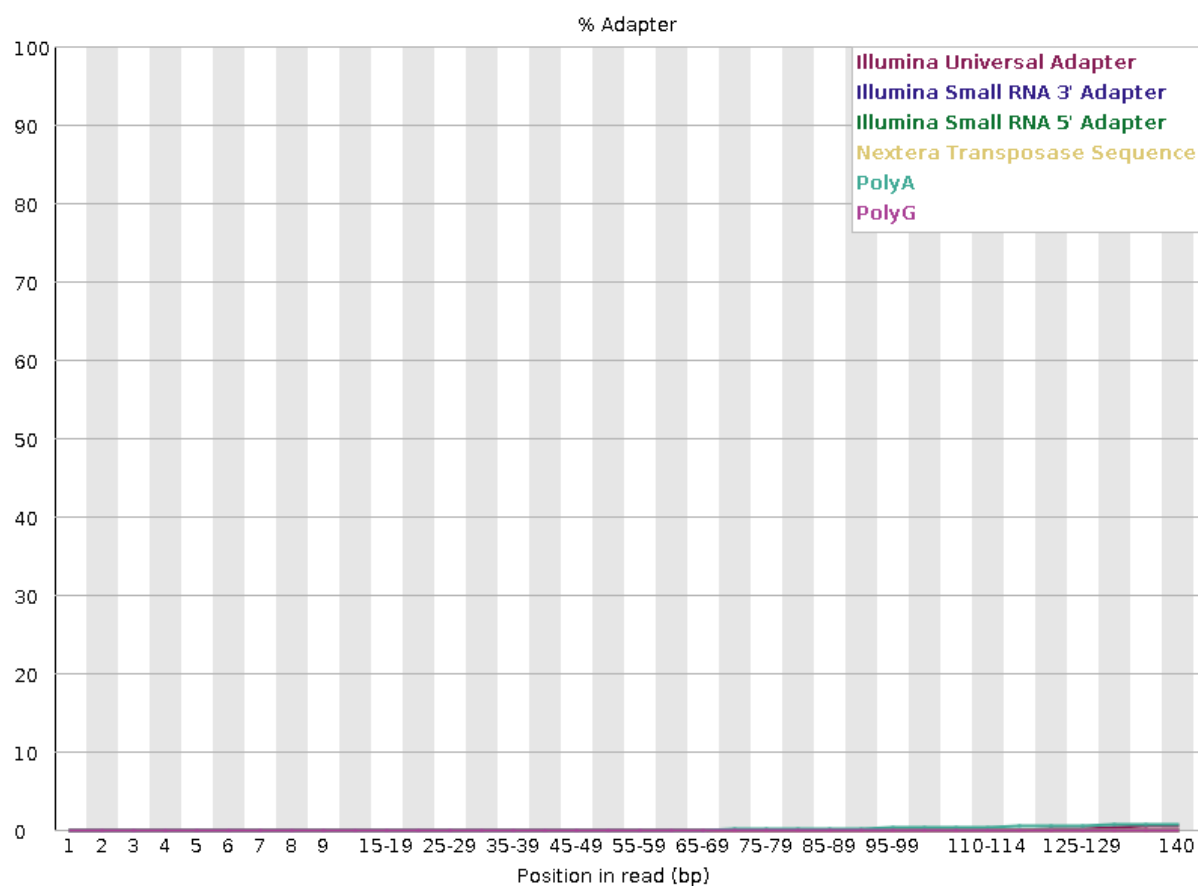
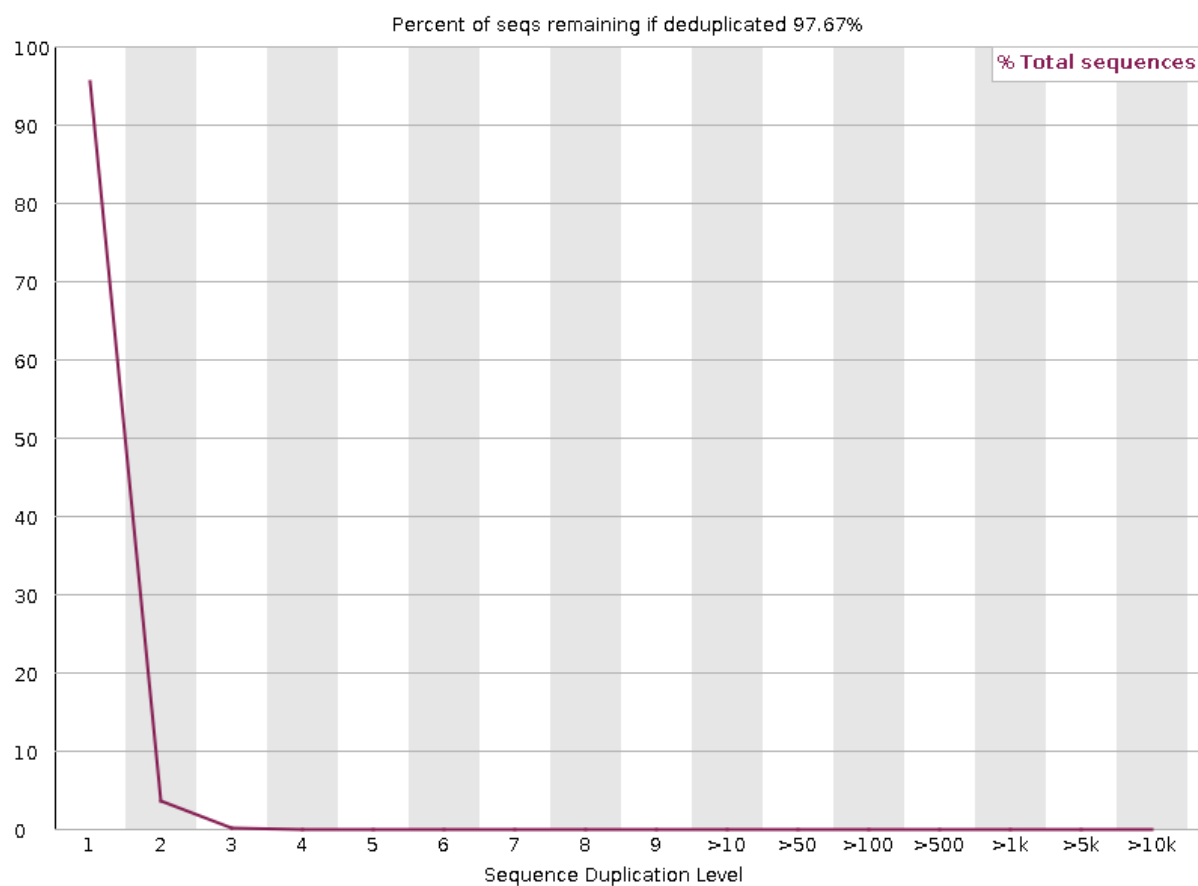
SRR37843243_2.up.fastq	
Measure	Value
Filename	SRR37843243_2.up.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1,596,819
Total Bases	213.5 Mbp
Sequences flagged as poor quality	0
Sequence length	60-151
%GC	51











Discussion

2.fastq에서 관측되던 Per tile sequence quality의 Warning, 그리고 공통적으로 관찰되던 Adapter Content 문제가 모두 해결되었다. Overrepresented sequences도 모든 파일에서 관측되지 않았다. 이후 downstream 분석에 큰 문제 없다 판단하였다.

Reference

1. Usadel Lab. Trimmomatic [Computer software]. GitHub.
<https://github.com/usadellab/Trimmomatic> (accessed 2026-04-11)
2. Andrews, S. (2010). FastQC: A quality control tool for high throughput sequence data. Babraham Bioinformatics. Retrieved April 11, 2026, from
<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>